Certified Lean Six Sigma Black Belt (CLSSBB) Training

WSQ Course Code: TGS-2024051900

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

TSC: Quality Process Control · ELE-QUA-5006-1.1

Version v4 · 23 September 2026 · Trainer: Dr. Alfred Ang

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR code with your phone camera
3. Key in your NRIC/FIN and submit
4. Repeat for AM, PM and the assessment
5. Keep 75% attendance to stay eligible for funding

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD; Certified Lean Six Sigma Black Belt practitioner and trainer.

DELIVERS
WSQ courses on Lean Six Sigma, quality management and data analytics.

EXPERIENCE
Process improvement across manufacturing, service and technology sectors.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your Green Belt background and the projects you have run or supported.
- Your comfort level with statistics — regression, ANOVA, design of experiments.
- A REAL process at work you could improve — this becomes your capstone project.

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HOW THIS COURSE WORKS

Your Capstone Project Starts Today

You will choose a real process from your own workplace in Lab 1, and every lab across the next five days adds one artifact to it. On Day 5 you present it to a steering committee - in addition to the WSQ assessment.

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THE CAPSTONE THREAD

How Your Capstone Is Built

1

Day 1 - select and charter your own workplace project

2

Days 2-4 - every lab adds one artifact to that project

3

Day 5 - consolidate the artifacts into an A3 storyboard

4

Day 5 - present and defend it to the steering committee

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HOUSEKEEPING

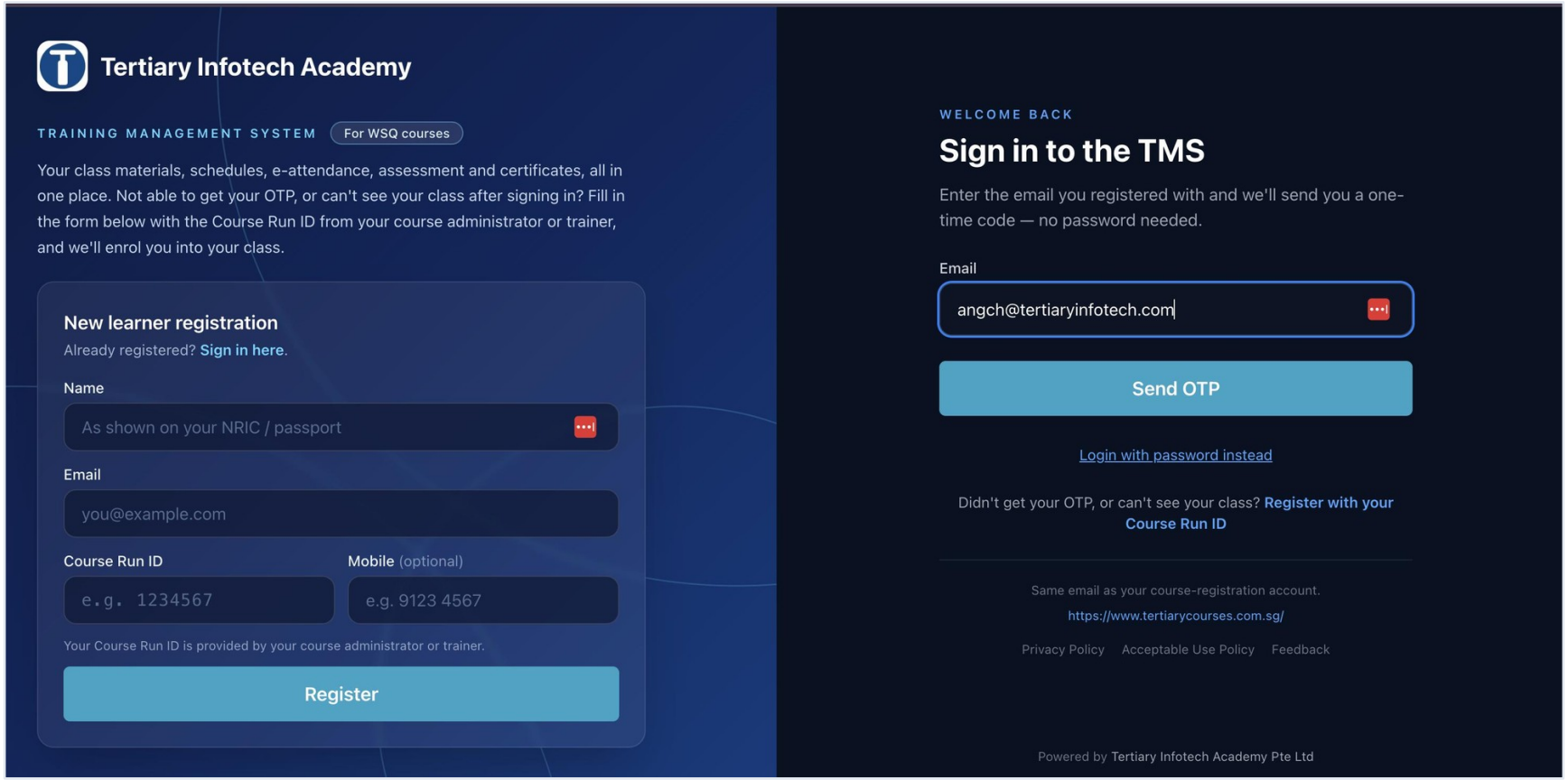
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for certification.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Download Your Course Material

1. Go to lms-tms.tertiaryinfotech.com
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3. Open this course from your dashboard
4. Download the slides, Learner Guide and lab files
5. Keep them open — the assessment is open book

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BROWSER-BASED TOOLS USED IN THE LABS

Your Interactive Toolkit

1

SIPOC

Build a SIPOC map and agree process boundaries — alfredang.github.io/sipoc

2

Fishbone Diagram

Build an Ishikawa cause-and-effect diagram — alfredang.github.io/fishbone

3

5 Whys

Build and share a 5 Whys root-cause chain — alfredang.github.io/5whys

4

Pareto Chart

Team brainstorms and votes; the chart builds itself live — alfredang.github.io/paretochart

5

NovaSPC

Run charts, SPC charts and process capability from your CSV — alfredang.github.io/novaspc

6

No installation

All run in the browser. No licence, no setup — use them in the labs.

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SCHEDULE · 9:30am-6:30pm with a 1-hour lunch

Lesson Plan — Days 1 & 2

Day 1

- Day 1 — The Black Belt role, portfolio selection, capstone chartering and Define
  - Digital attendance (AM) · Introductions
  - The Black Belt role vs the Green Belt role
  - Project portfolio selection and strategy alignment
  - DMAIC recap at depth · Y = f(X) · sigma level
  - Labs 1-3 — SELECT AND CHARTER YOUR CAPSTONE
  - DEFINE: VOC, Kano, CTQ trees and CTQ flowdown
  - Stakeholders, resistance and change leadership
  - Labs 4-7 — VOC/Kano, CTQ flowdown, SIPOC, stakeholders

Day 2

- Day 2 — Measure — advanced measurement systems, attribute MSA and capability
  - Digital attendance (AM)
  - MEASURE recap: mapping, VSM, takt, eight wastes
  - Data types, sampling and rational subgrouping
  - Continuous Gage R&R at depth
  - ATTRIBUTE MSA and Kappa analysis (Black Belt only)
  - Nested and destructive gauge studies
  - Non-normal capability, Cpk vs Ppk, Box-Cox
  - Labs 8-14 — VSM, sampling, Gage R&R, Kappa, capability

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SCHEDULE · 9:30am-6:30pm with a 1-hour lunch

Lesson Plan — Days 3 & 4

Day 3

- Day 3 — Analyze — multiple regression, ANOVA, non-parametric and multi-vari
  - Digital attendance (AM)
  - ANALYZE recap: variation, run charts, Pareto, fishbone
  - Hypothesis testing at depth and test selection
  - MULTIPLE REGRESSION and model diagnostics (VIF, residuals)
  - ANOVA, two-way ANOVA and interaction effects
  - Non-parametric tests and multi-vari studies
  - Chi-square and categorical association
  - Labs 15-21 — regression, ANOVA, multi-vari, tollgate

Day 4

- Day 4 — Improve and Control — DOE, RSM and advanced SPC
  - Digital attendance (AM)
  - IMPROVE recap: solutions, Lean tools, FMEA
  - FULL FACTORIAL DOE — main effects and interactions
  - Fractional factorial, confounding and resolution
  - Response surface methodology and robust design
  - CONTROL recap: SPC, chart selection, Cp/Cpk
  - CUSUM, EWMA, short-run and multivariate SPC
  - Labs 22-30 — DOE, RSM, FMEA, SPC, control plan

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SCHEDULE · 9:30am-6:30pm with a 1-hour lunch

Lesson Plan — Day 5

Day 5 AM

- Day 5 — Capstone project consolidation and steering committee presentation
  - Digital attendance (AM)
  - Consolidating your capstone artifacts from Labs 1-30
  - The A3 storyboard — one page, seven sections
  - Lab 31 — build your A3 storyboard
  - Building the steering committee presentation
  - Lab 32 — build and rehearse your presentation

Day 5 PM

- Day 5 (continued)
  - Lab 33 — CAPSTONE PRESENTATION AND DEFENCE
  - Defending the analysis under challenge
  - Peer presentations and observation
  - Lessons learned and knowledge transfer
  - Mentoring Green Belts — the coaching model
  - Lab 34 — your Black Belt journey and 90-day plan
  - TRAQOM survey · Course close

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WSQ SKILLS FRAMEWORK

Skills Framework — TSC: Quality Process Control

- TSC Code: ELE-QUA-5006-1.1
- A1: Lead improvement projects to meet process performance requirements.
- A2: Establish project scope of work and resourcing based on organisational requirements.
- A3: Analyse process performance data using advanced statistical methods to identify root causes of variation.
- A4: Measure process performance against defined quality standards and validate the measurement system.
- A5: Recommend, pilot and control improvement actions to sustain process performance.
- K1: DMAIC methodology and the advanced Lean Six Sigma body of knowledge.
- K2: Advanced statistical tools, experimental design and problem-solving techniques for process improvement.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 — Lead a portfolio

Select, charter and govern projects aligned to business strategy.

2

LO2 — Validate measurement

Gage R&R for continuous data AND Kappa for attribute data.

3

LO3 — Advanced analysis

Multiple regression, ANOVA, non-parametric and multi-vari studies.

4

LO4 — Design experiments

Full and fractional factorial DOE and response surface methods.

5

LO5 — De-risk & validate

FMEA, piloting and financial benefits with Finance sign-off.

6

LO6 — Advanced SPC

CUSUM, EWMA, short-run and multivariate control charts.

7

LO7 — Lead change

Resistance, stakeholder management and mentoring Green Belts.

8

LO8 — Deliver the capstone

Present and defend a real project to a steering committee.

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COURSE ROADMAP · 5 DAYS · 34 HANDS-ON LABS · CAPSTONE

Course Outline — DMAIC Reviewed in Depth, Then Extended



Define

- VOC, Kano, CTQ recap
- CTQ FLOWDOWN
- Resistance & change
- Labs 4-7

Measure

- Gage R&R recap
- ATTRIBUTE MSA / KAPPA
- Nested & destructive
- Labs 8-14

Analyze

- Hypothesis recap
- MULTIPLE REGRESSION
- ANOVA, multi-vari
- Labs 15-21

Improve

- FMEA & Lean recap
- FULL & FRACTIONAL DOE
- RSM, robust design
- Labs 22-26

Control

- SPC recap
- CUSUM, EWMA, T-squared
- Benefits validation
- Labs 27-30

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THE STEP UP

How This Course Steps Up From Green Belt

GREEN BELT

Leads one scoped project

- Simple linear regression, one X
- Continuous Gage R&R only
- DOE mentioned in passing
- Shewhart control charts
- Supports the change effort

BLACK BELT

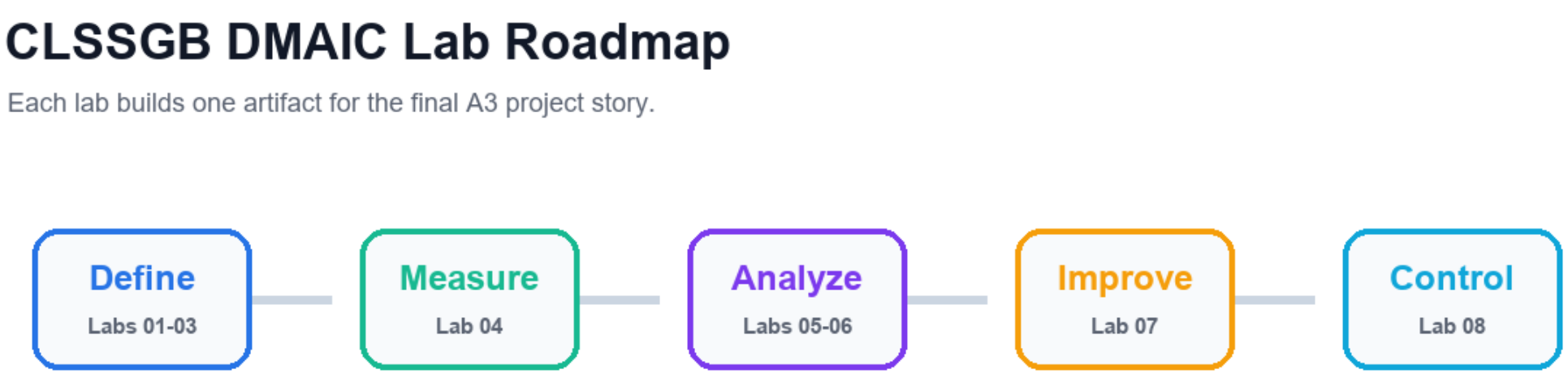
Leads a portfolio and mentors belts

- Multiple regression, ANOVA, multi-vari
- Attribute MSA, Kappa, nested studies
- Full and fractional factorial DOE, RSM
- CUSUM, EWMA, short-run, multivariate
- Leads change and validates the money

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COURSE ROADMAP

The DMAIC Roadmap



Every topic and lab in this course sits inside one of these five phases.

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BEFORE YOU SIT THE ASSESSMENT

Briefing for Assessment

1

Clear your desk

Phones and other materials go under the table or on the floor.

2

No photos or recording

Assessment scripts must not be photographed or recorded.

3

No discussion

The assessment is individual work — no talking once it starts.

4

Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

No correction fluid

No liquid paper or correction tape — strike through instead.

6

Open book

Slides, Learner Guide and approved materials only.

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FINAL ASSESSMENT

Assessment

1. **Written Assessment (WA)**
   Short-Answer Questions (SAQ) — 2 questions, 1 hour, open book (K1, K2).

2. **Case Study (CS)**
   An applied DMAIC case study — 90 minutes, open book (A1-A5).

3. **Capstone (additional)**
   Your project + steering committee presentation, on top of the WSQ assessment.

4. **Built from Day 1**
   Every lab from Lab 1 onwards adds one artifact to your capstone.

5. **Attendance**
   A minimum of 75% attendance is required to be eligible for assessment and funding.

6. **Appeals**
   An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ, 1 hour), then the Case Study (90 minutes)
4. Submit your capstone report and present it to the steering committee
5. Sign the Assessment Summary Record

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FOUNDATIONS

The Black Belt Role, Portfolio Selection & Your Capstone

BB vs GB - Portfolio selection - Strategy alignment - COPQ - Capstone chartering - DMAIC at altitude

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WHY YOU ARE HERE

You already know how to run a project.

This week is about running a PORTFOLIO of them, owning the statistics nobody else in the room can defend, and coaching the Green Belts who report their work to you.

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What This Opening Module Delivers

1

A shared technical baseline

Six Sigma, Lean, Y = f(X), sigma level and DPMO re-established at Black Belt precision.

2

A clear role boundary

Exactly what a Black Belt does that a Green Belt does not - and is paid to do.

3

A portfolio method

Criteria, weighted scoring and strategy alignment for choosing which projects live.

4

A money language

COPQ rolled up from one project to a whole programme, in dollars Finance accepts.

5

Your capstone project

You pick your own workplace process today and build it across every lab this week.

6

DMAIC and tollgates

The five phases and the five gates, at the depth needed to mentor others through them.

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THE RUNNING SCENARIO

Meridian Medical Devices

Our running case all week: infusion pump assemblies built on 6 lines across 3 plants - Singapore, Penang and Suzhou. OTIF, first-pass yield and warranty returns are the metrics the board watches. Every worked example lands here.

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MERIDIAN MEDICAL DEVICES

Meridian at a Glance - The Numbers You Will Reuse

1

Volume

412,000 infusion pump assemblies shipped last financial year across the three plants.

2

First-pass yield

Singapore 96.1 pct, Penang 91.4 pct, Suzhou 88.7 pct - a 7.4 point spread nobody has explained.

3

OTIF

On-time in-full sits at 87.2 pct against a customer contract commitment of 96 pct.

4

Warranty returns

2,140 field returns last year, average settlement cost SGD 780 per unit.

5

Rework

One line in Penang runs a permanent 4-person rework cell that is not in the standard cost.

6

Board pressure

The CEO has asked for a 30 pct reduction in cost of poor quality within 12 months.

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WHAT SIX SIGMA ACTUALLY IS

Six Sigma is three things at once.

A statistical measure of capability, a structured improvement method, and a management philosophy about running the business on data. A Black Belt has to be fluent in all three.

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THREE FACES

The Three Faces of Six Sigma

A metric

Sigma level and DPMO

- Counts how many standard deviations fit between mean and spec
- Six sigma performance = 3.4 defects per million opportunities
- Lets you compare a soldering line to an invoicing process
- This is the face you defend in front of engineers

A method

DMAIC and DMADV

- Define, Measure, Analyze, Improve, Control - in that order
- Each phase closes at a tollgate with evidence, not opinion
- Prevents jumping from symptom straight to pet solution
- This is the face you run projects with

A philosophy

Management by data

- Decisions are made from measured process data, not seniority
- Every process has a customer whose voice sets the target
- Variation is treated as a cost, not as bad luck
- This is the face you sell to the steering committee

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CLEARING THE MYTHS

What Six Sigma Is Not

1

Not a quality department

It is a line-management discipline. Quality assurance audits; Six Sigma changes the process.

2

Not statistics for its own sake

Statistics is the evidence, never the deliverable. The deliverable is a changed, controlled process.

3

Not a training programme

Belts without live projects produce certificates and no benefit. Projects are the point.

4

Not a substitute for strategy

It executes strategy faster. It cannot tell you which market to be in.

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A SHORT HISTORY

How We Got Here

1924

Shewhart invents the control chart at Western Electric

1950s

Deming and Juran rebuild Japanese manufacturing quality

1986

Bill Smith coins Six Sigma at Motorola

1995

Jack Welch makes it GE's central initiative

2002

Lean and Six Sigma formally merge into Lean Six Sigma

Today

Applied across services, healthcare, finance and software

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THE INHERITANCE

What Each Era Contributed - And Why It Still Matters to You

1

Shewhart, 1924

Separated common cause from special cause. Without this distinction you tamper and make things worse.

2

Deming, 1950s

94 pct of problems belong to the system, not the worker. This is why you attack process, not people.

3

Juran, 1950s

The vital few versus the trivial many - the logic behind every Pareto chart you will draw.

4

Motorola, 1986

Turned capability into a single comparable number so plants could be ranked honestly.

5

GE, 1995

Tied belt certification to hard financial benefit signed by Finance. Still the standard.

6

Toyota, ongoing

Flow, pull and respect for people - the Lean half of what you now lead.

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Three Definitions of Quality - Only One Survives Scrutiny

Defect free

The zero defect answer

- Free of what defect, defined by whom?
- A pump with no cosmetic flaw can still arrive late
- Silent on everything the buyer actually valued
- Verdict: incomplete

Conforms to spec

The engineering answer

- Measurable and auditable - a real improvement
- But who wrote the spec, and from what evidence?
- A wrong spec met perfectly is still a failed product
- Verdict: necessary but not sufficient

Meets customer needs

The Six Sigma answer

- The customer defines what good looks like
- The spec is then derived from that voice, and traceable to it
- Covers delivery, service and cost, not only the part
- Verdict: this is the working definition

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THE COMBINATION

Lean plus Six Sigma equals Lean Six Sigma

Lean

- Speed, flow and lead time
- Attacks waste and non-value-add steps
- Fast, visual, shop-floor tools
- Weak on statistical proof

+

Six Sigma

- Accuracy, capability and variation
- Attacks defects and inconsistency
- Rigorous data and inferential statistics
- Slow on waste that is already obvious

▼

LEAN SIX SIGMA

Right things, done fast AND done right - waste removal carrying statistical proof, run through DMAIC

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CHOOSING YOUR EMPHASIS

Which Half Leads - A Black Belt Judgement Call

1

Long lead time, few defects

Lean leads. Map the value stream, attack queues and batch size before touching statistics.

2

Short lead time, many defects

Six Sigma leads. Find the Xs driving variation and bring them under control.

3

Both at once

Lean first to expose the true process, then Six Sigma on the variation that remains.

4

Neither is clear

Measure first. Choosing a toolkit before baseline data is the classic Green Belt error.

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THE COST OF POOR QUALITY

Most of what poor quality costs you is invisible.

Scrap and rework sit above the waterline where accounting can see them. The larger cost - expediting, lost customers, excess inventory, management attention - never appears in a cost code.

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WHERE THE MONEY HIDES

The COPQ Iceberg

Above the waterline

Visible and already booked

- Scrap and material write-off
- Rework labour and rework cell headcount
- Warranty settlements and field returns
- Inspection and sorting labour
- Typically only 15 to 20 pct of true COPQ

Just below

Traceable if you look

- Expedited freight and premium shipping
- Excess safety stock held to cover variation
- Line downtime and unplanned changeovers
- Engineering time on recurring deviations
- Recoverable with effort from the ERP

Deep below

Real but never coded

- Lost customers and orders never quoted
- Price concessions granted to keep an account
- Management hours in escalation meetings
- Reputation and audit findings with regulators
- Estimated, not extracted - but still real

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COPQ ARITHMETIC

Quantifying COPQ at Meridian

COPQ as pct of revenue	**COPQ / Revenue x 100**	Meridian: SGD 11.4m COPQ on SGD 142m revenue = 8.0 pct. World class is under 1 pct.
Cost per defect	**Total failure cost / Defect count**	2,140 warranty returns x SGD 780 settlement = SGD 1.67m external failure alone.
Hidden factory ratio	**Rework hours / Total direct hours**	Penang rework cell: 4 staff x 1,800 h = 7,200 h against 96,000 direct h = 7.5 pct.

Rule of thumb: a 3 sigma organisation loses 25 to 40 pct of revenue to COPQ; a 5 sigma one loses under 5 pct.

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The Four COPQ Categories - Know Which You Are Attacking

1

Internal failure

Defects caught before the customer - scrap, rework, re-test, downgrade. Cheap to find, expensive to repeat.

2

External failure

Defects the customer caught - warranty, returns, recalls, credits. 10x the cost of internal failure.

3

Appraisal

The cost of checking - inspection, testing, audit. Spending here treats symptoms, not causes.

4

Prevention

The cost of designing defects out - training, capability studies, poka-yoke. The only spend that pays back.

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THE CORE MODEL

Y = f(X)

The single most important equation in Six Sigma. You cannot manage the Y. You can only find, control and improve the Xs that produce it. Every phase of DMAIC exists to move you from the Y to the vital few Xs.

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Y = f(X)

Reading the Equation Properly

Y - the output

The dependent variable

- What the customer feels and the board measures
- Always a result, always downstream, always lagging
- Cannot be changed directly - only through the Xs
- Meridian Y: on-time in-full at 87.2 pct

f - the transfer function

The relationship

- The process itself: how the Xs combine to make the Y
- May be linear, curved, or full of interactions
- Discovering f IS the Analyze phase
- Regression, ANOVA and DOE are how you estimate it

X - the inputs

The independent variables

- Materials, methods, machines, people, measures, environment
- Controllable, noise, or critical-to-process
- Leading indicators - they move before Y does
- Meridian Xs: supplier lot, oven profile, operator shift

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THE FUNNEL

How DMAIC Walks You From Y to the Vital Few Xs

1. DEFINE - name the Y precisely and set its target
2. MEASURE - trust the Y measurement, then baseline it
3. ANALYZE - list 30 candidate Xs, prove which 3 matter
4. IMPROVE - change those Xs and confirm the Y moved
5. CONTROL - hold the Xs so the Y never drifts back

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X CLASSIFICATION

Classifying Your Xs - A Black Belt Habit

1

Controllable X

You can set it and hold it - oven temperature, torque setting, approved supplier list.

2

Noise X

Real but not economically controllable - ambient humidity, incoming lot variation, shift fatigue.

3

Critical X

Proven by data to drive the Y. These earn a control plan entry. Usually only 3 to 5 exist.

4

Standard operating X

Already known and already controlled. Do not spend Analyze effort re-proving these.

5

Constraint X

Cannot be changed for regulatory or contractual reasons - design the solution around it.

6

Suspect X

Named by the team but unproven. Stays suspect until hypothesis testing promotes or kills it.

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FROM 27 XS TO 2

Y = f(X) at Meridian - Worked Through

1

Business Y

Warranty cost per 1,000 units shipped - the figure the board sees each quarter.

2

Process Y

First-pass yield at final electrical test - 88.7 pct at Suzhou versus 96.1 pct at Singapore.

3

Candidate Xs

Solder paste lot, reflow profile, operator certification level, board supplier, shift, humidity.

4

Proven X 1

Reflow zone-3 temperature - explains 41 pct of yield variation by regression.

5

Proven X 2

Solder paste age at point of use - beyond 8 hours yield drops 6 points.

6

Everything else

Twenty-one other suspected Xs tested and found statistically insignificant. That is a result too.

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MEASURING CAPABILITY

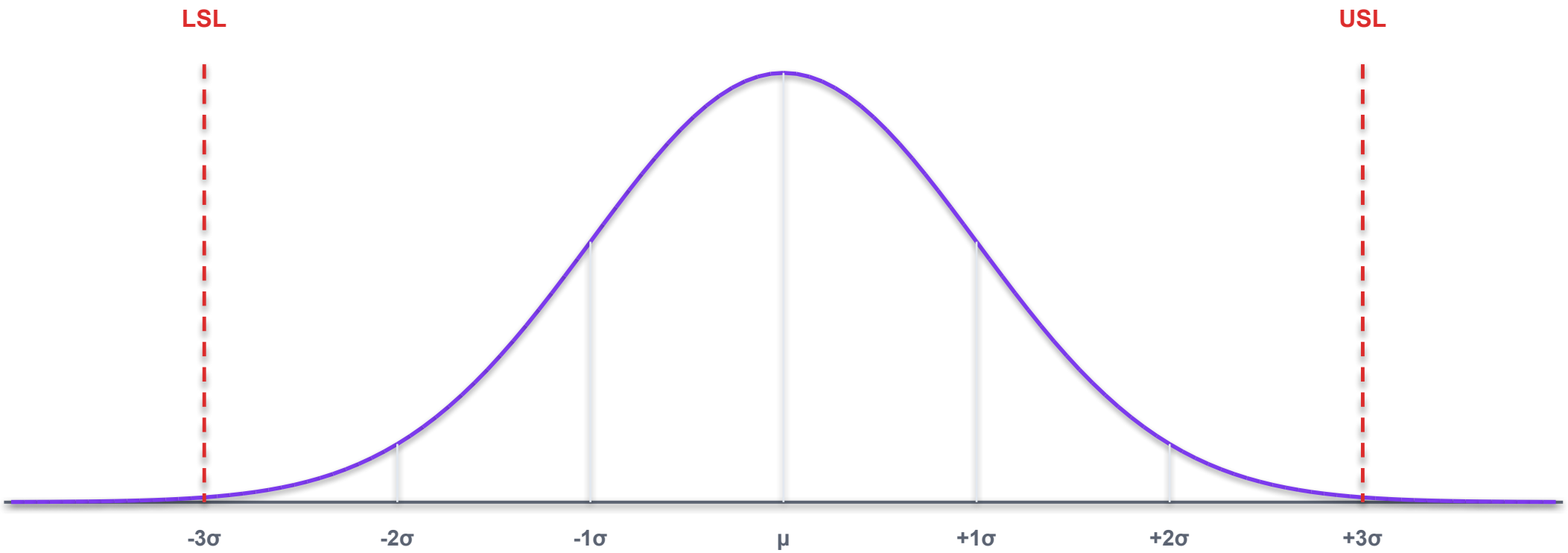
Sigma level is a common currency.

It lets you put a solder line, an invoice process and a discharge ward on the same scale. As a Black Belt you must be able to compute it, defend the 1.5 sigma shift, and explain both to a sceptical VP.

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THE BELL CURVE

Why It Is Called SIX Sigma



Six sigma capability means the nearest specification limit sits six standard deviations from the process mean.

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FORMULAS YOU MUST OWN

The Core Capability Formulas

Defects per unit	**DPU = Defects / Units**	Meridian Suzhou: 1,412 defects found on 12,400 boards = DPU 0.1139.
DPMO	**DPMO = (Defects / (Units x Opps)) x 1,000,000**	1,412 defects, 12,400 boards, 340 solder joints each = 335 DPMO.
First-pass yield	**FPY = Units passing first time / Units started**	11,001 of 12,400 boards passed at first test = 88.7 pct FPY.
Rolled throughput yield	**RTY = Y1 x Y2 x ... x Yn**	0.97 x 0.94 x 0.96 x 0.99 across four Suzhou stations = 86.6 pct RTY.

DPMO normalises for complexity - that is why it can compare a 340-joint board to a 3-field invoice.

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THE OPPORTUNITY TRAP

Defining an Opportunity - Where Most Sigma Numbers Go Wrong

1

An opportunity is a chance to fail

One measurable characteristic that can be judged conforming or not - nothing looser.

2

Be honest, not generous

Inflating opportunity count inflates your sigma level. Auditors and peers will find it.

3

Count only what you inspect

If you never test a joint, it is not an opportunity for your DPMO - it is a blind spot.

4

Keep the definition frozen

Change the opportunity definition mid-project and your baseline becomes meaningless.

5

Document it in the charter

The opportunity definition is a tollgate deliverable at Measure, not a footnote.

6

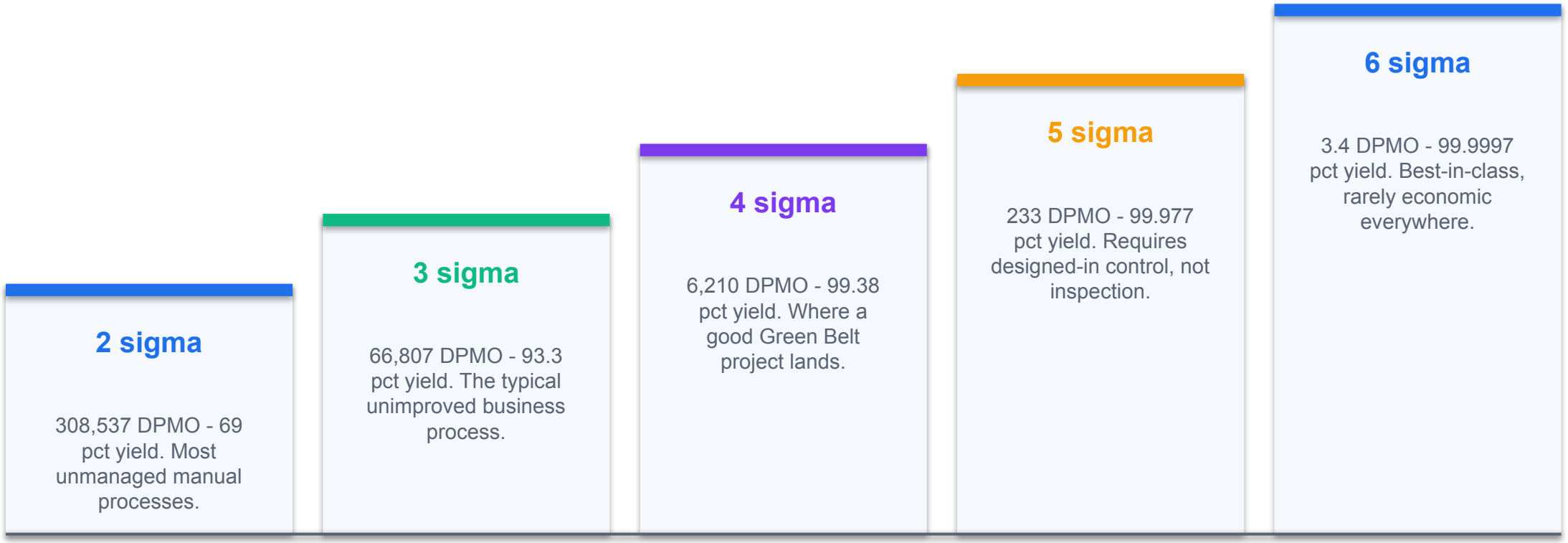
Same rule across sites

Meridian cannot compare Penang to Suzhou unless both count joints the same way.

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THE SIGMA SCALE

The Sigma Ladder - Capability Versus Defect Rate



Each step up is roughly a 10x reduction in defects - and roughly a 10x increase in the discipline required.

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YIELD -> DPMO -> SIGMA

Converting Yield to Sigma Level - The Meridian Worked Example

Step 1 - DPMO	**DPMO = (1 - Yield) x 1,000,000**	Suzhou final test yield 0.887 -> DPMO = 0.113 x 1,000,000 = 113,000.
Step 2 - Look up Z	**Z(short term) from the standard normal table**	113,000 DPMO -> long-term Z = 1.21 from the normal distribution.
Step 3 - Add the shift	**Sigma level = Z(long term) + 1.5**	1.21 + 1.5 = 2.71 sigma. Suzhou runs at roughly 2.7 sigma at final test.
Step 4 - Set the target	**Target DPMO from target sigma**	Moving Suzhou to 4 sigma means 6,210 DPMO - a 94 pct reduction in defects.

Report the assumption every time: sigma levels quoted here include the conventional 1.5 sigma long-term shift.

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The 1.5 Sigma Shift - What It Is and Why You Must Declare It

The observation

Motorola, 1980s

- Short-term studies looked far better than long-run results
- Over weeks the mean drifts with tooling, lots and shifts
- Empirically the drift averaged about 1.5 standard deviations
- So a 6 sigma short-term process yields 3.4 DPMO long term

The convention

How the industry uses it

- Published sigma tables already bake in the 1.5 shift
- Sigma level = long-term Z + 1.5, by convention
- Cp / Cpk are short term; Pp / Ppk are long term
- 6 sigma short term = 4.5 sigma long term = 3.4 DPMO

The Black Belt duty

Where it goes wrong

- It is an empirical convention, not a law of physics
- Some processes drift far more, some barely at all
- Never compare a shifted number to an unshifted one
- State the assumption on every capability slide you present

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CONVERSION TABLE

Sigma Conversion Table - The Numbers to Memorise

1

6.0 sigma

3.4 DPMO - 99.99966 pct yield - best in class

2

5.0 sigma

233 DPMO - 99.977 pct yield - excellent

3

4.5 sigma

1,350 DPMO - 99.87 pct yield - strong

4

4.0 sigma

6,210 DPMO - 99.38 pct yield - industry average plus

5

3.0 sigma

66,807 DPMO - 93.32 pct yield - industry average

6

2.0 sigma

308,537 DPMO - 69.15 pct yield - non-competitive

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THE COMPOUNDING PROBLEM

Why 3 Sigma Is Not Good Enough - The Compounding Argument

1

One step at 99.38 pct

Sounds excellent. A 4 sigma single operation looks like a solved problem.

2

Ten steps in series

0.9938 to the power 10 = 93.9 pct RTY. One in sixteen units touched rework.

3

Fifty steps in series

0.9938 to the power 50 = 73.2 pct RTY. Over a quarter of units need intervention.

4

Meridian reality

The pump assembly has 62 sequential process steps across three plants.

5

The lesson

Complexity multiplies defect exposure. Step-level yield flatters system-level truth.

6

The response

Always report RTY, never only final yield. RTY is what exposes the hidden factory.

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THE BELT ROLES

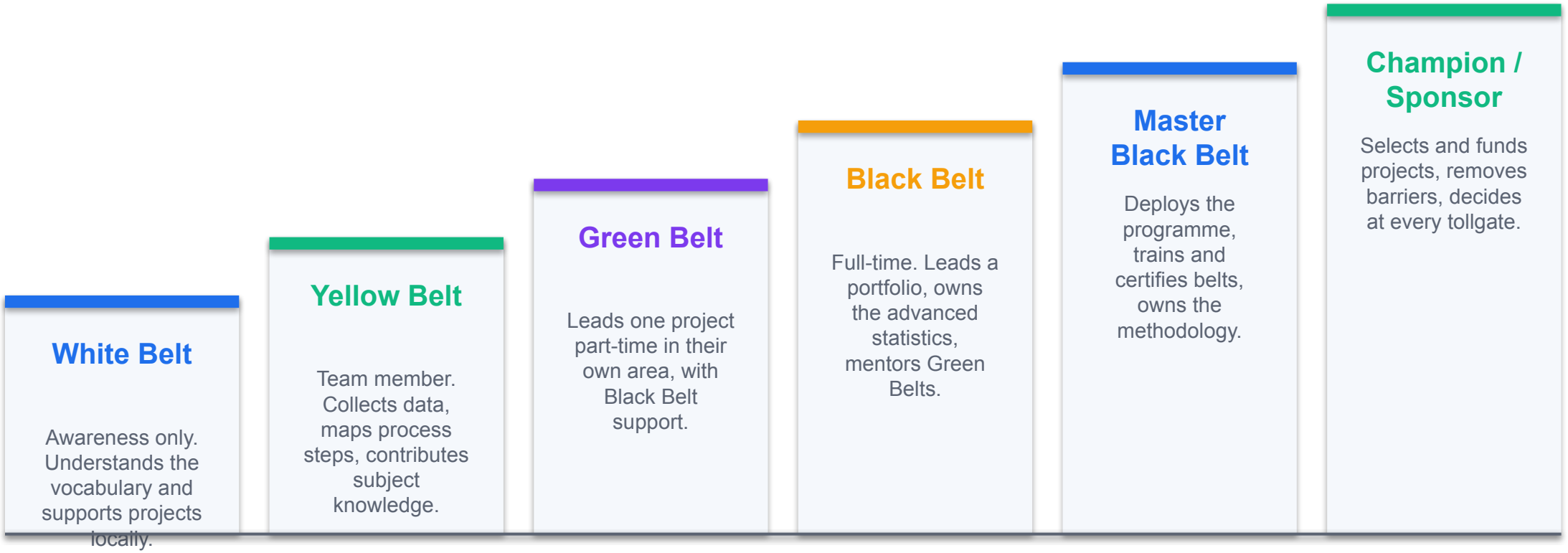
The belt system is a division of labour, not a hierarchy of intelligence.

Each belt owns a different scope, a different toolset and a different kind of accountability. Knowing exactly where your boundary sits is what makes a deployment work.

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SIX ROLES

The Belt Ladder



Time commitment rises with the rung: Yellow a few hours, Green 20 to 25 pct, Black Belt 100 pct.

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Green Belt Versus Black Belt - The Real Boundary

Green Belt

One project, part time

- Leads a single project inside their own department
- Uses descriptive stats, basic hypothesis tests, control charts
- Escalates anything needing DOE, MSA design or regression
- Reports at tollgates - does not run them
- Benefit typically SGD 50k to 150k per project

Black Belt

A portfolio, full time

- Leads 4 to 6 concurrent projects across functions and sites
- Owns DOE, regression, ANOVA, MSA design, capability analysis
- Mentors and technically reviews the Green Belts' work
- Runs the tollgate and prepares the sponsor's decision
- Benefit typically SGD 500k to 1m per year of portfolio

The five deltas

What you now add

- 1. Portfolio - selecting and sequencing, not just executing
- 2. Advanced statistics - you are the last technical authority
- 3. Mentoring - Green Belt capability is your deliverable too
- 4. Tollgate governance - you run the gate, sponsor decides
- 5. Finance validation - benefits signed off, not self-declared

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WHAT YOU ARE PAID FOR

The Five Black Belt Accountabilities in Detail

1

Own a portfolio

Select, sequence, resource and kill projects. Say no to work that will not deliver.

2

Be the statistical authority

When a result is challenged in a steering meeting, the defence stops with you.

3

Develop Green Belts

Coach through tools, review their analysis, catch their errors before the sponsor does.

4

Run disciplined tollgates

Assemble evidence, frame the decision, let the sponsor decide go, adjust or kill.

5

Validate benefit with Finance

No claimed saving is real until a finance controller signs the calculation.

6

Escalate honestly

Report a failing project early. Concealment is the fastest way to lose a programme.

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THE WIDER CAST

Who Else Is in the Room

1

Executive sponsor

Owns the business objective. Signs the charter, funds the resource, decides at gates.

2

Process owner

Runs the process day to day and inherits the control plan. Must be engaged from Define.

3

Master Black Belt

Your technical coach and the guardian of method. Escalate design-of-experiment doubts here.

4

Finance controller

Independently validates every benefit claim. Involve at Define, not at Control.

5

Team members

Yellow Belts and subject experts who know what actually happens on the floor.

6

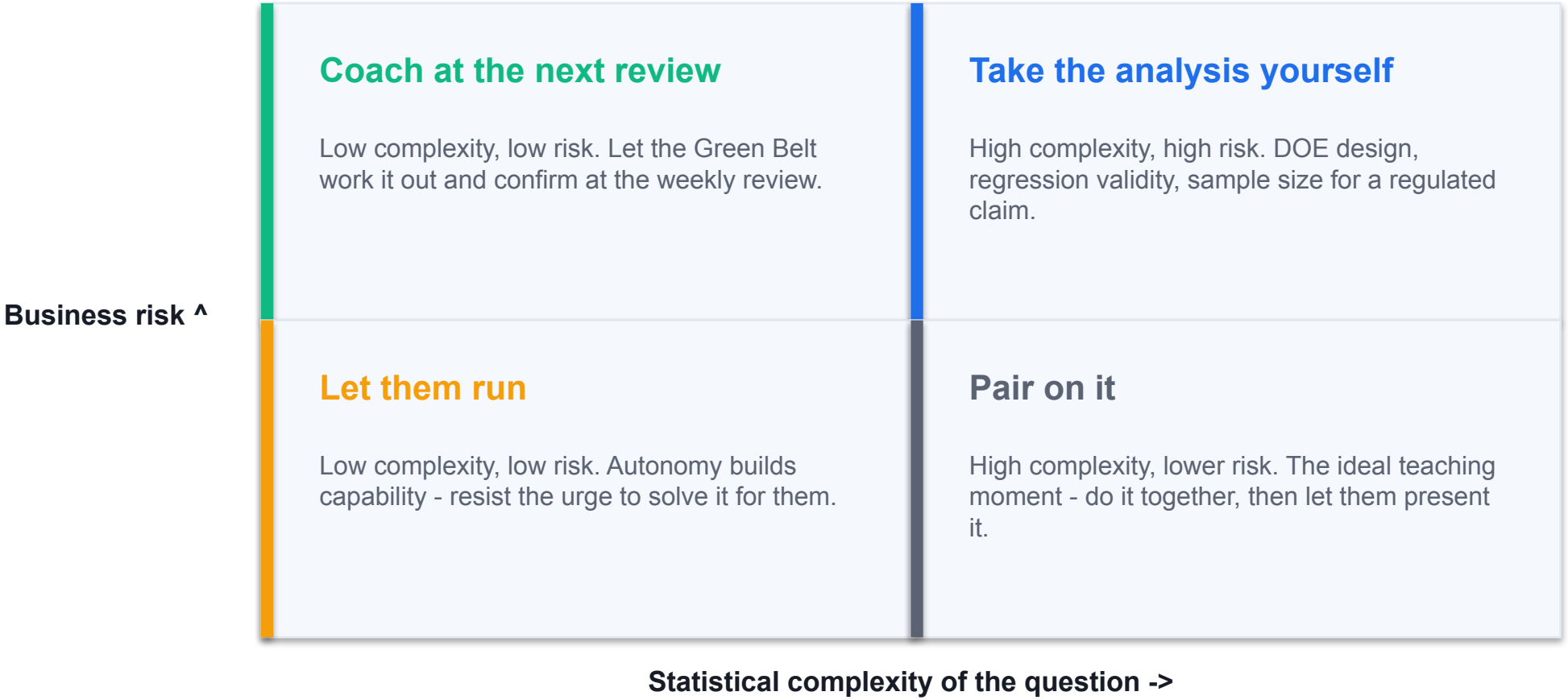
Deployment leader

Owns the belt pipeline, the project hopper and the programme-level scorecard.

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MENTORING JUDGEMENT

Escalation Rule - When a Green Belt Should Call You



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PROJECT PORTFOLIO SELECTION

Choosing the wrong project is the most expensive mistake a Black Belt can make.

A brilliantly executed project on a problem nobody cares about still ends with a cancelled programme. Selection is a Black Belt skill; a Green Belt is usually handed a project.

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SIX STEPS

The Portfolio Selection Cycle

1. GENERATE - build a hopper of candidates from COPQ, VOC, audits and strategy
2. SCREEN - remove anything already solved, out of scope or a known solution
3. SCORE - rank on weighted criteria with the sponsor in the room
4. SEQUENCE - schedule against belt capacity and site readiness
5. CHARTER - convert the top candidates into signed charters
6. REVIEW - re-score the hopper every quarter as strategy moves

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FILLING THE HOPPER

Where Good Candidate Projects Come From

1

The COPQ analysis

Follow the money. The biggest failure-cost buckets are usually the biggest opportunities.

2

Voice of the customer

Complaint themes, lost bids and churn interviews name problems customers actually feel.

3

Process performance data

Any metric off target, drifting, or with visibly wide variation is a candidate.

4

Audit and regulatory findings

Repeat non-conformances have a sponsor and a deadline already attached.

5

Strategy deployment

Hoshin breakthrough objectives cascade directly into project candidates.

6

Value stream maps

Bottlenecks, long queues and rework loops identified on the map are ready-made scopes.

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SIX CRITERIA

The Selection Criteria - What Actually Predicts Success

1

Strategic alignment

Traces to a stated business objective. Without it the sponsor disengages by Analyze.

2

Financial benefit

Hard, quantifiable and Finance-validatable. Soft benefits do not fund a programme.

3

Data availability

If the Y cannot be measured today, Measure phase becomes a six-month data project.

4

Scope and duration

3 to 6 months. Beyond that sponsors change, priorities move and teams disband.

5

Solution unknown

If everyone already knows the fix, it is a just-do-it, not a DMAIC project.

6

Process ownership

A named, willing process owner exists. No owner means no sustained control plan.

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SCORING WORKED THROUGH

The Weighted Scoring Matrix

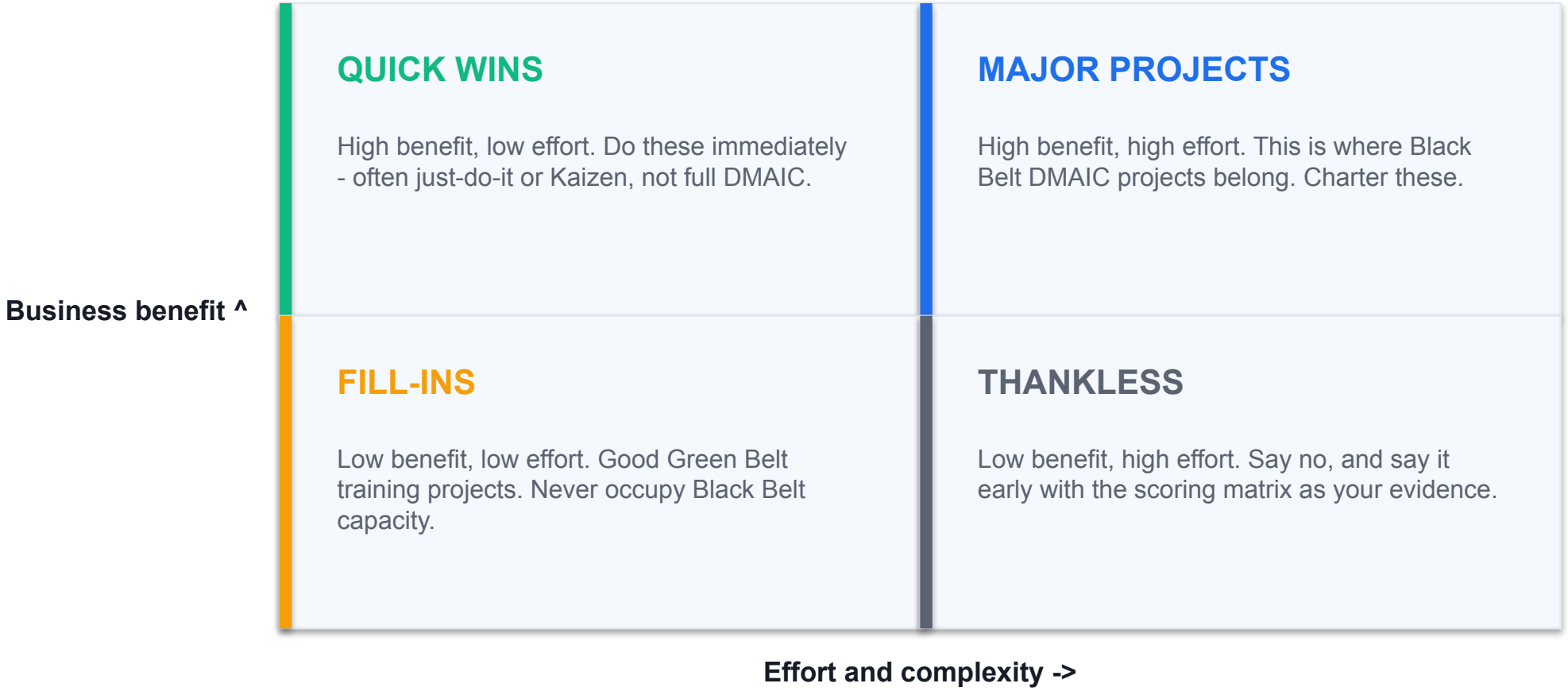
Weighted score	**Score = sum of (criterion weight x criterion rating)**	Rate each criterion 1 to 5, weights summing to 100 pct. Rank the hopper by total.
Meridian project A	**Suzhou solder yield improvement**	Align 5 x .30, Benefit 5 x .25, Data 4 x .20, Scope 4 x .15, Unknown 5 x .10 = 4.65.
Meridian project B	**Warehouse label reprint reduction**	Align 2 x .30, Benefit 2 x .25, Data 5 x .20, Scope 5 x .15, Unknown 2 x .10 = 2.85.
The decision	**Charter A, defer B, revisit at the next quarterly hopper review**	A is worth SGD 1.4m and is on strategy. B is worth SGD 40k and is a just-do-it.

Do the scoring WITH the sponsor. A matrix scored alone in your office convinces nobody.

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TRIAGE

The Benefit Versus Effort Screen



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HOSHIN KANRI

Strategy Alignment - Hoshin Kanri Cascade

1

Level 1 - Breakthrough objective

Meridian: reduce cost of poor quality by 30 pct within 12 months.

2

Level 2 - Annual objective

Lift group first-pass yield from 92.1 pct to 96.0 pct this financial year.

3

Level 3 - Site target

Suzhou final-test FPY from 88.7 pct to 95.0 pct by Q3.

4

Level 4 - Project Y

Reduce solder-joint defects at Suzhou reflow from 113,000 DPMO to 20,000 DPMO.

5

Catchball

Targets negotiated up and down until the site genuinely owns them - not imposed.

6

The test

Every project must trace up all four levels. If it cannot, it does not get chartered.

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PORTFOLIO DISCIPLINE

Saying No - A Core Black Belt Skill

1

No, it is a just-do-it

The cause and fix are already known. Hand it to the process owner today.

2

No, there is no measurable Y

Without a measurable output there is no baseline, no proof and no control.

3

No, the scope is a whole function

Boil-the-ocean scopes fail. Offer a scoped slice of the same problem instead.

4

No, there is no sponsor

A project without a sponsor has no decision maker and no resource. Decline it.

5

Not now - it is queued

Sequenced behind higher-scoring work. Show the matrix; the ranking does the arguing.

6

How to say it

Never a flat refusal. Show the criteria, the score and what you WILL do instead.

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PORTFOLIO MANAGEMENT

Managing a Live Portfolio of Four to Six Projects

1

Stagger the phases

Never have three projects in Analyze at once - the statistical load lands on you alone.

2

One scorecard

A single portfolio view: phase, tollgate date, benefit forecast, RAG status, blocker.

3

Protect Green Belt time

The commonest cause of stalled projects is a Green Belt pulled back to day job.

4

Kill fast

A project that misses two consecutive tollgates should be killed, not nursed.

5

Bank the benefits

Track realised versus forecast benefit monthly for 12 months after Control.

6

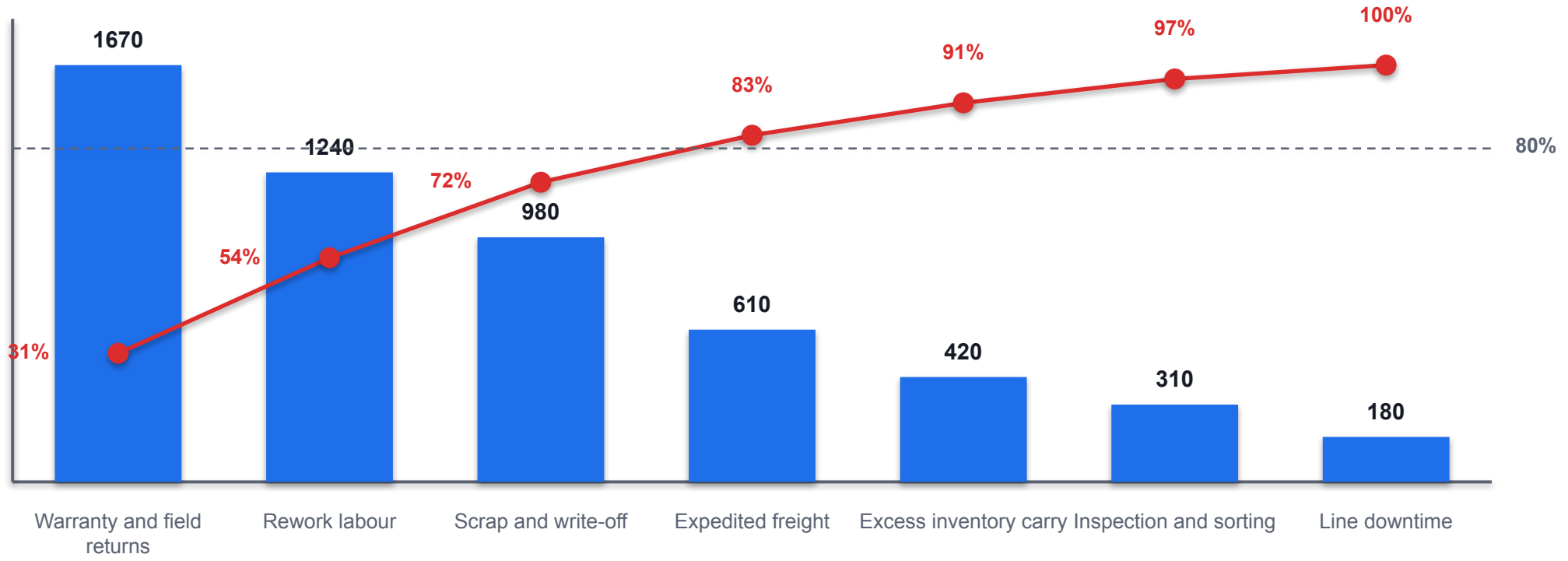
Refresh the hopper

Quarterly re-scoring keeps the portfolio aligned as strategy shifts under you.

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COPQ AT PORTFOLIO SCALE

Meridian COPQ by Category - Where the Portfolio Should Aim



SGD thousands. Three categories carry 71 pct of COPQ - that is where the first three charters go.

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COPQ AT SCALE

Rolling COPQ Up From Project to Programme

1

Project level

One charter, one Y, one benefit figure signed by one finance controller.

2

Site level

Sum the project benefits, subtract double-counting where two projects touch the same cost.

3

Programme level

Aggregate across sites, net of belt cost, training cost and any capital spend.

4

The double-count trap

Two projects both claiming the same scrap reduction is the fastest way to lose credibility.

5

The netting rule

Report gross benefit, programme cost and net benefit separately. Never only the gross.

6

Reporting cadence

Monthly to the deployment leader, quarterly to the steering committee, with Finance present.

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YOUR CAPSTONE PROJECT

Your capstone starts today.

Alongside the WSQ assessment, you build a capstone project on YOUR own workplace process - chosen today, developed in every lab, and presented to a simulated steering committee on Day 5.

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THE ASSESSMENT MODEL

How the Capstone Works

1

You choose it today

Pick a real process you own or influence at work. Real data, real sponsor, real benefit.

2

It grows in every lab

Each lab applies that session's tool to YOUR process, not to a textbook case.

3

It accumulates artefacts

Charter, SIPOC, MSA, capability study, hypothesis tests, DOE, control plan - all yours.

4

Day 5 consolidates

The final day assembles everything into one coherent project storyboard.

5

You present to a panel

A simulated steering committee tollgate review, with challenge questions.

6

Additional to the WSQ papers

You also sit the Written Assessment (SAQ) and the Case Study on Day 5. Both open book.

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THE FIVE-DAY ARC

Your Capstone Across the Five Days

1

DAY 1 - choose the process, draft the charter, define the Y and the business case

2

DAY 2 - SIPOC, VOC, CTQ flowdown, MSA and baseline capability on your data

3

DAY 3 - graphical analysis, hypothesis tests and regression to find your vital few Xs

4

DAY 4 - DOE, solution design, pilot plan, FMEA and your control plan

5

DAY 5 - consolidate the storyboard and present to the steering committee panel

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Choosing a Good Capstone Process - Today

1

You can access the data

Existing measurements you can actually obtain this week. No data, no capstone.

2

The Y is measurable

A continuous or countable output you can baseline - cycle time, yield, error rate, cost.

3

The solution is not obvious

If you already know the fix, choose something else - you will learn nothing.

4

Scope is a process, not a function

One process with a clear start and end. Not 'improve the supply chain'.

5

You have standing

A process you own, support or can influence. Not someone else's department.

6

It matters to someone

A named person would be pleased if it improved. That person is your sponsor.

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CHARTER ON DAY 1

Your Day 1 Capstone Charter - The Six Boxes

1

Problem statement

What, where, when, how big - in measurable terms. No cause, no blame, no solution.

2

Goal statement

From X to Y by when. Specific, measurable and agreed with the sponsor.

3

Business case

Why now, and what it is worth. Link to COPQ or to a strategic objective.

4

Scope

In scope and explicitly out of scope. The out-of-scope list prevents most scope creep.

5

Team and sponsor

Named sponsor, named process owner, named team members with time committed.

6

Milestones

The five tollgate dates. A charter without dates is an intention, not a plan.

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DAY 5 - THE FINAL TOLLGATE

The Day 5 Steering Committee Presentation

1

Format

15 minutes of storyboard plus 10 minutes of challenge questions from the panel.

2

Structure

One slide per DMAIC phase, showing the evidence that closed each tollgate.

3

The data must be yours

Real numbers from your own process. Invented data is obvious and scores nothing.

4

Defend the statistics

Expect to be asked why that test, that sample size, that confidence level.

5

State the benefit

A quantified benefit with the assumptions visible, as Finance would want to see it.

6

Name the risks

A credible presentation names what could still go wrong and how control handles it.

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ASSESSMENT CRITERIA

How Your Capstone Is Judged

1

Correct tool selection

Did you choose the right tool for the data type and the question asked?

2

Sound execution

Assumptions checked, sample sizes justified, conclusions supported by the output.

3

A clear causal chain

The Xs you improved are demonstrably the Xs your analysis proved mattered.

4

A workable control plan

Something the process owner could genuinely run after you leave.

5

A defensible benefit

A calculation a finance controller would sign, with assumptions declared.

6

Presentation under challenge

Held your position on evidence, and conceded gracefully where evidence was thin.

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DMAIC AT ALTITUDE

You know DMAIC. Now you have to be able to teach it.

The next slides recap all five phases at the depth a Black Belt needs to mentor Green Belts through them, run their tollgates, and recognise each phase's characteristic failure mode.

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THE FIVE PHASES

The DMAIC Roadmap



Define

- Charter and business case
- Voice of the customer
- CTQ flowdown
- SIPOC and scope
- Team and sponsor

Measure

- Operational definitions
- Data collection plan
- Measurement system analysis
- Baseline capability
- Process mapping

Analyze

- Graphical analysis
- Root cause hypotheses
- Hypothesis testing
- Regression and ANOVA
- Vital few Xs proven

Improve

- Solution generation
- Design of experiments
- Pilot and validate
- FMEA on the solution
- Implementation plan

Control

- Control plan
- Statistical process control
- Standard work and training
- Handover to process owner
- Benefit realisation

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DEFINE - Purpose, Tools, Tollgate, Failure Mode

1

Purpose

Establish what problem is being solved, for whom, worth how much, and within what boundary.

2

Core tools

Project charter, VOC capture, Kano, CTQ tree, SIPOC, stakeholder analysis, RACI.

3

Tollgate deliverable

A signed charter with a measurable Y, an agreed scope and a Finance-acknowledged business case.

4

Failure mode

A vague problem statement that contains a cause or a solution - guarantees a wandering project.

5

Mentoring cue

Green Belts write goals like 'improve quality'. Push until it reads 'from 88.7 pct to 95.0 pct by Q3'.

6

Time share

Roughly 15 pct of project effort. Rushing Define costs three times as much in Analyze.

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MEASURE - Purpose, Tools, Tollgate, Failure Mode

1

Purpose

Prove the measurement system is trustworthy, then establish an honest baseline of the Y.

2

Core tools

Operational definitions, data collection plan, MSA and Gage R and R, capability study, detailed process map.

3

Tollgate deliverable

A validated measurement system plus a baseline capability figure with its confidence interval.

4

Failure mode

Skipping MSA. If the gauge is the source of variation, every later conclusion is wrong.

5

Mentoring cue

Ask 'how do you know the data is right?' before you ever discuss what the data shows.

6

Time share

Roughly 25 pct of effort. This is where Green Belts most often need Black Belt rescue.

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ANALYZE - Purpose, Tools, Tollgate, Failure Mode

1

Purpose

Move from a long list of suspected Xs to a short list proven by data to drive the Y.

2

Core tools

Graphical analysis, fishbone, 5 Whys, hypothesis tests, ANOVA, regression, multi-vari studies.

3

Tollgate deliverable

Statistically validated root causes, with the p-values and effect sizes that justify them.

4

Failure mode

Confirmation bias - testing only the cause the team already believed, and stopping there.

5

Mentoring cue

Insist on practical significance alongside statistical significance. A tiny proven effect is still tiny.

6

Time share

Roughly 25 pct. This is the phase where the Black Belt's statistical ownership is most visible.

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IMPROVE - Purpose, Tools, Tollgate, Failure Mode

1

Purpose

Design, test and prove solutions that move the vital few Xs, and therefore move the Y.

2

Core tools

Design of experiments, response surface, solution selection matrix, pilot design, FMEA, poka-yoke.

3

Tollgate deliverable

Piloted solution with before-and-after data proving the Y improved, plus a full implementation plan.

4

Failure mode

Implementing without a pilot, then discovering the side effects in production at full scale.

5

Mentoring cue

Always ask what else this change touches. Optimising one Y at the cost of another is common.

6

Time share

Roughly 20 pct. Tempting to start here on day one - resist it, and stop your Green Belts too.

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CONTROL - Purpose, Tools, Tollgate, Failure Mode

1

Purpose

Lock the gain in so the process cannot drift back after the project team disperses.

2

Core tools

Control plan, SPC charts, standard work, visual management, training plan, response plan, audit schedule.

3

Tollgate deliverable

Signed control plan, trained operators, live charts and a formal handover to the process owner.

4

Failure mode

Handing over a control plan nobody was trained on - performance decays within one quarter.

5

Mentoring cue

Return at 3, 6 and 12 months. A gain that is not still there at 12 months was never a gain.

6

Time share

Roughly 15 pct. The most skipped phase and the reason most improvements silently reverse.

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When DMAIC Is the Wrong Roadmap

1. **The process does not exist**
   Use DMADV or DFSS. You cannot baseline a process that has never run.

2. **The cause and fix are known**
   Just do it. DMAIC on a known solution wastes months and burns sponsor goodwill.

3. **The problem is pure waste and flow**
   A Kaizen event or value stream mapping workshop moves faster than full DMAIC.

4. **It is a capital or IT project**
   Use project management. DMAIC solves variation problems, not installation problems.

5. **The scope is a whole business**
   Too big. Decompose into chartered projects with individual measurable Ys.

6. **There is no measurable output**
   Fix the measurement system first as a project in its own right, then run DMAIC.

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TOLLGATE GOVERNANCE

The Black Belt runs the tollgate. The sponsor decides.

A tollgate is not a status update. It is a formal decision point where evidence is presented and the sponsor chooses: proceed, adjust the charter, or kill the project.

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FIVE DECISION POINTS

The Five Tollgates

1. GATE D - is the problem worth solving and properly bounded?
2. GATE M - can we trust the data, and what is the true baseline?
3. GATE A - are the root causes statistically proven?
4. GATE I - does the piloted solution actually move the Y?
5. GATE C - is the gain locked in and formally handed over?

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GATE EVIDENCE

What Evidence Closes Each Gate

Gates D and M

Problem and baseline

- D: signed charter, measurable Y, agreed scope
- D: business case acknowledged by Finance
- D: named sponsor, process owner and team
- M: MSA result showing an acceptable Gage R and R
- M: baseline capability with confidence interval

Gates A and I

Causes and solution

- A: prioritised cause list with test results
- A: p-values, effect sizes and practical significance
- A: the vital few Xs explicitly named
- I: pilot data showing the Y moved, with statistics
- I: FMEA on the new process and an implementation plan

Gate C

Sustainment

- C: signed control plan with reaction rules
- C: live SPC charts owned by the process owner
- C: standard work updated and operators trained
- C: benefit validated and signed by Finance
- C: formal handover minute and 3-6-12 month review dates

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SIGN-OFF AUTHORITY

Who Signs Off What

1

Executive sponsor

The gate decision itself: proceed, adjust or kill. Signs the charter and the closure.

2

Process owner

Accepts the control plan and the standard work at Gate C. Owns it thereafter.

3

Finance controller

Validates the business case at Gate D and the realised benefit at Gate C.

4

Master Black Belt

Assures the methodology - particularly the MSA at Gate M and the statistics at Gate A.

5

Black Belt

Assembles the evidence, runs the meeting, frames the decision. Does not vote on it.

6

Quality or regulatory

Signs where a change touches a validated or regulated process - common at Meridian.

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GATE DISCIPLINE

Running a Tollgate That Actually Decides Something

1

Circulate evidence early

Pre-read 48 hours ahead. A gate is not the first time the sponsor sees the data.

2

Open with the decision

State the decision being asked for in the first minute, not the last.

3

Show the gaps honestly

Name what is incomplete. A sponsor who is surprised later stops trusting you.

4

Allow the kill option

If kill is never available, the gate is theatre and everybody knows it.

5

Minute the decision

Decision, owner, date and conditions recorded. Verbal approval evaporates.

6

Re-baseline the charter

Scope, timing or benefit changes get written into the charter at the gate.

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THE DECISION

Three Legitimate Gate Outcomes

1

Proceed

Evidence is sufficient. Resource confirmed for the next phase and the next gate date set.

2

Adjust

Scope, target, timing or resource is re-baselined into the charter, then the project continues.

3

Kill

The problem is smaller than believed, the data does not exist, or priorities have moved on.

4

Killing is not failure

A project killed at Gate M has cost weeks. The same project killed at Gate C has cost months.

5

The recycle option

Sometimes the answer is 'go back to Measure' - an adjust decision, not a proceed.

6

Track kill rate

A portfolio with a zero pct kill rate is not being screened honestly at the gates.

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MODULE CHECKPOINT

Foundations - What You Should Now Be Able to Do

1. **Explain Six Sigma three ways**
   As a metric, as a method and as a management philosophy, to three different audiences.

2. **Compute a sigma level**
   Yield to DPMO to Z to sigma level, declaring the 1.5 shift assumption every time.

3. **State your own role boundary**
   The five things you now own that a Green Belt does not.

4. **Score a project hopper**
   Build and defend a weighted scoring matrix with a sponsor in the room.

5. **Quantify COPQ at scale**
   Roll benefit up from project to programme without double counting.

6. **Run a tollgate**
   Assemble the evidence, frame the decision and let the sponsor decide.

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YOUR NEXT ACTION

Choose your capstone process before we break.

One real process, one measurable Y, one named sponsor. Everything you learn for the rest of the week gets applied to it. Bring what data you can access.

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FOUNDATIONS · HANDS-ON

Select and Charter Your Capstone Project

LAB 1

This is the most important lab of the course — everything you build for the next five days attaches to the project you select here. Screen candidate projects the way a Black Belt does: against strategic fit, financial benefit, data availability and risk. Then write the charter that will govern the work. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A scored project selection matrix and a signed-off capstone project charter.

Tools & techniques: Portfolio screening, strategic alignment, COPQ estimate, project charter, sponsor agreement

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Select and Charter Your Capstone Project

STEPS 1-5 OF 10

1. Choose your capstone process. Strongly preferred: a real process from your own workplace where you can obtain data. Fallback: the Meridian Medical Devices scenario.
2. List at least five candidate improvement projects within that process or business area. Write each as a PROBLEM, never as a solution.
3. Build a project selection matrix. Score each candidate 1-5 on: strategic fit, annualised financial benefit, data availability, scope manageability, sponsor commitment and technical risk.
4. Weight the criteria — strategic fit and financial benefit usually carry the highest weight — and compute a weighted total for each candidate.
5. Reject any candidate that is a solution in disguise ('deploy a new MES'), has no accessible data, or has no named sponsor. Record WHY each was rejected.

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Select and Charter Your Capstone Project

STEPS 6-10 OF 10

6	Select your capstone project — the highest weighted score that survives the rejection filters.
7	Estimate the Cost of Poor Quality for the selected project: scrap, rework, warranty, expedite, lost margin and capacity consumed. Annualise it.
8	Write the charter: business case, problem statement, goal statement (SMART), scope (in/out), preliminary Y, team, sponsor, milestones and expected benefit.
9	Write your problem statement to the four-part test: the process, the time period, the measurable gap and the business impact — with no cause and no solution in it.
10	Identify your sponsor and record what you need from them. Where possible, obtain real sign-off before Day 2.

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LAB 1 · VERIFY

Select and Charter Your Capstone Project

✅ Check your work

You have a weighted selection matrix with at least five scored candidates, a documented rejection rationale, an annualised COPQ figure, and a complete charter whose problem statement contains no cause and no solution.

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FOUNDATIONS · HANDS-ON

Portfolio Selection, Strategy Alignment and the Black Belt Role

LAB 2

A Green Belt is handed a project; a Black Belt decides which projects exist. Practise the portfolio view — trace every candidate up to a business objective, balance the portfolio across DMAIC maturity and risk, and articulate what you as Black Belt own that a Green Belt does not.

You'll build

A strategy-to-project traceability map and a Black Belt accountability charter.

Tools & techniques: Hoshin alignment, portfolio balance, belt role matrix, mentoring model, tollgate governance

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Portfolio Selection, Strategy Alignment and the Black Belt Role

STEPS 1-4 OF 8

1. Write down your organisation's top three business objectives for the current year (or Meridian's: OTIF to 95%, warranty cost down 30%, yield parity across plants).

2. Map each of your five candidate projects from Lab 1 to the objective it serves. Any project that maps to nothing is a candidate for cancellation.

3. Build the belt responsibility matrix: White, Yellow, Green, Black, Master Black Belt and Champion. For each, record who leads, who analyses, who mentors and who sponsors.

4. List the five things a Black Belt does that a Green Belt does not — lead a portfolio, own advanced statistics, mentor Green Belts, run tollgates, validate benefits with Finance.

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Portfolio Selection, Strategy Alignment and the Black Belt Role

STEPS 5-8 OF 8

5. Assess portfolio balance: are all your projects in one phase, one plant or one function? A healthy portfolio is spread across risk and time-to-benefit.

6. Identify which of your candidate projects should be delegated to a Green Belt under your mentorship rather than led by you. Record the reasoning.

7. Define the tollgate cadence for your capstone: what evidence closes Define, Measure, Analyze, Improve and Control, and who signs each one off.

8. Write your Black Belt accountability charter — what you own, what you delegate, what you escalate.

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LAB 2 · VERIFY

Portfolio Selection, Strategy Alignment and the Black Belt Role

✅ Check your work

Every candidate project traces to a named business objective, your belt matrix distinguishes Black from Green Belt accountability, and your capstone has a defined five-tollgate governance plan.

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FOUNDATIONS · HANDS-ON

DMAIC at Depth — Recap, Y = f(X) and Baseline Sigma for Your Project

LAB 3

A thorough recap of the DMAIC roadmap — not a skim, because you must be able to mentor Green Belts through it. Then apply it: express your capstone problem as Y = f(X), and convert your baseline defect data into DPMO and a sigma level so improvement can be proven numerically at the end. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A DMAIC tollgate map, a Y = f(X) model and a baseline DPMO / sigma level for your capstone.

Tools & techniques: DMAIC roadmap, tollgate deliverables, Y = f(X), DPU, DPO, DPMO, sigma level, sigma shift

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DMAIC at Depth — Recap, Y = f(X) and Baseline Sigma for Your Project

STEPS 1-5 OF 9

1. Recap each DMAIC phase in full: its purpose, its core tools, its tollgate deliverable and the most common way teams fail in it.
2. For each phase, write the one question that phase answers — Define: what problem? Measure: how big? Analyze: why? Improve: what fix? Control: how do we hold it?
3. Write your capstone project Y — the single output metric your customer actually feels. It must be measurable and it must already have data.
4. Brainstorm at least twelve candidate Xs that could drive that Y. Classify each as controllable, noise or SOP-governed.
5. Write the relationship formally as Y = f(X1, X2, ... Xn), and mark which Xs you can experiment on later in the DOE labs.

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DMAIC at Depth — Recap, Y = f(X) and Baseline Sigma for Your Project

STEPS 6-9 OF 9

6. Collect or estimate your baseline: units processed, defects observed and opportunities per unit over a defined recent period.

7. Calculate DPU, DPO and DPMO. Show the formulas: DPU = defects/units; DPO = defects/(units x opportunities); DPMO = DPO x 1,000,000.

8. Convert DPMO to a baseline sigma level using the conversion table, and state whether the 1.5-sigma long-term shift is included.

9. Record this baseline in your capstone pack — every improvement claim on Day 5 is measured against this number.

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LAB 3 · VERIFY

DMAIC at Depth — Recap, Y = f(X) and Baseline Sigma for Your Project

✅ Check your work

You can explain all five DMAIC tollgates without notes, your Y is a measurable customer-facing output with at least twelve classified Xs, and your baseline DPMO and sigma level are calculated from real counts.

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D

DMAIC · DEFINE

Define at Black Belt Depth — Charter, Stakeholders & Change

VOC recap · Advanced CTQ flowdown · Charter governance · Stakeholder & resistance analysis · Change leadership

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Key Concepts — Define (1)

1. **VOC, affinity and Kano — in depth**
   Re-derive VOC capture, affinity clustering and Kano classification in full — you must be able to teach these, not just use them.

2. **CTQ trees — in depth**
   Build the CTQ tree from need to driver to measurable requirement, with targets, specification limits and defect definitions.

3. **CTQ flowdown**
   Cascade a business-level Y down through sub-Ys to the process Xs a team can actually control.

4. **Charter governance**
   The charter is a living contract — re-baselined at each tollgate, not written once and filed.

5. **Stakeholder analysis**
   Map influence against support, then plan a specific move for every high-influence, low-support stakeholder.

6. **Resistance management**
   Resistance is data. Diagnose whether it is technical, political or cultural before responding.

7. **Change leadership**
   ADKAR and Kotter — Awareness, Desire, Knowledge, Ability, Reinforcement across the project life.

8. **Communication planning**
   Different audiences need different messages at different cadences — plan it, do not improvise.

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Key Concepts — Define (2)

1

Mentoring Green Belts

The Black Belt coaches Green Belt project definition rather than doing it for them.

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DEFINE AT BLACK BELT DEPTH

Define decides whether the project is worth doing at all

A Green Belt writes a charter. A Black Belt negotiates the charter, cascades the business Y to controllable Xs, reads the political terrain and leads the change. Same tools, higher altitude.

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What this phase delivers at Black Belt level

1

A charter that survives contact

Business case, problem, goal, scope and team, negotiated with the sponsor and re-baselined at every tollgate.

2

A validated CTQ flowdown

Business Y cascaded through sub-Ys to the process Xs a team can actually change.

3

A stakeholder and influence map

Influence versus support, with a named move for every high-influence blocker.

4

A resistance diagnosis

Technical, political or cultural - each demands a different response.

5

A change leadership plan

ADKAR at the individual level, Kotter at the organisation level.

6

A mentoring plan for the Green Belts

You will coach two to four GB charters through Define while running your own.

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Green Belt Define versus Black Belt Define

GREEN BELT

Runs one project inside a process

- Uses the charter template as given
- Collects VOC for one product or service
- Builds one CTQ tree
- Names the obvious stakeholders
- Escalates resistance to the sponsor

BLACK BELT

Runs a portfolio and coaches the belts

- Negotiates scope and benefit with the sponsor
- Segments customers and reconciles conflicting VOC
- Flows the business Y down to controllable Xs
- Maps influence, support and coalition paths
- Diagnoses and works the resistance directly

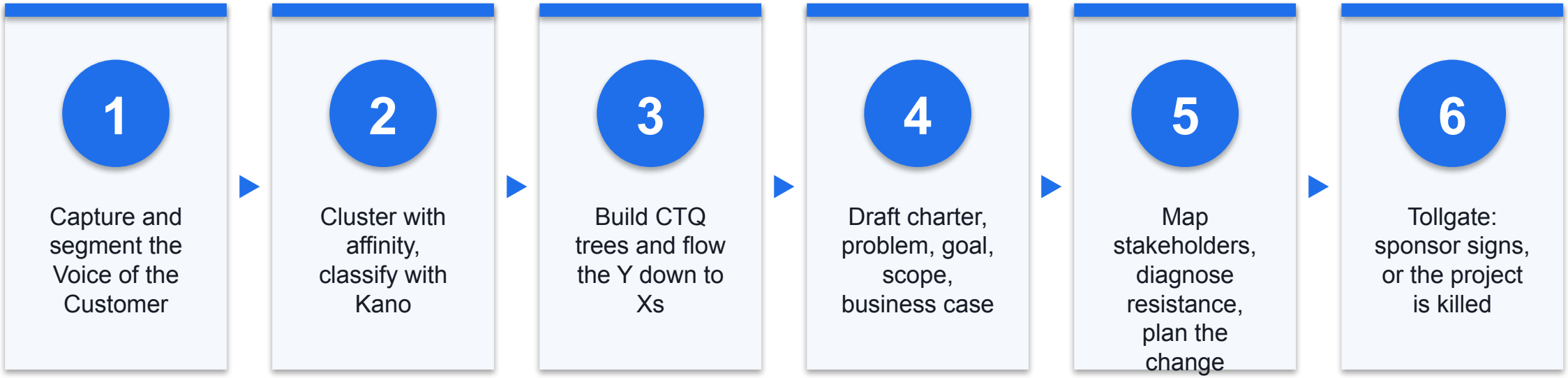
MASTER BLACK BELT

Owns the deployment

- Sets portfolio selection criteria
- Certifies charters before they start
- Audits benefit claims with Finance
- Coaches the Black Belts
- Owns the tollgate standard itself

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The Define sequence you will run and teach



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Meridian Medical Devices - the running case for this course

- Three plants: Singapore (HQ and pilot line), Penang (high volume), Suzhou (China domestic).
- Six assembly lines building infusion pump assemblies for hospital and homecare customers.
- Business pain: on-time in-full (OTIF) delivery is 87.4 percent against a customer commitment of 96 percent.
- First-pass yield sits at 91.2 percent group-wide, but Suzhou is 86.1 percent and Singapore is 95.0 percent.
- Warranty returns cost SGD 3.4 million last year, of which 61 percent trace to two failure modes.
- You will carry Meridian through every phase, and mirror it on your own capstone process.

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RECAP IN DEPTH - VOC

Voice of the Customer: you must be able to teach it, not just use it

Green Belts routinely collect VOC from whoever is easiest to reach. Your job is to make them segment, sample and capture verbatim before a single CTQ is written.

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Two families of VOC sources

REACTIVE

Arrives whether you ask or not

- Complaints and service desk tickets
- Warranty claims and field returns
- Credit notes, refunds, order cancellations
- Audit findings and regulatory observations
- Cheap and continuous, but biased to the angry

PROACTIVE

You must deliberately go and get it

- Structured surveys with a real sampling plan
- One-to-one interviews with buyers and users
- Focus groups of six to ten customers
- Gemba: watch a nurse actually set up the pump
- Costlier, but reaches the silent majority

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Customer segmentation - VOC is meaningless unsegmented

1

Segment by customer type

Hospital procurement, ward nursing staff, homecare patients and distributors want different things.

2

Segment by geography

Singapore private hospitals, Malaysian public tenders and China distributors buy on different criteria.

3

Segment by volume and value

The top 8 accounts are 62 percent of Meridian revenue. Weight their voice accordingly.

4

Segment by lifecycle stage

A first-time buyer values setup simplicity; a five-year user values spare parts availability.

5

Watch for conflicting voices

Procurement wants lowest unit price; nursing wants fastest changeover. Both are real.

6

Record the segment with the verbatim

An unattributed quote cannot be reconciled later when voices conflict.

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Verbatim capture - the discipline that Green Belts skip

CAPTURE VERBATIM

Their words, not yours

- Write exactly what was said, in quotes
- Do not paraphrase into company language
- Record who said it, when, and in what role
- Keep the emotion - it signals severity
- One statement per card or row

WHY IT MATTERS

Paraphrase destroys evidence

- 'Delivery is bad' loses which part is bad
- The team's bias enters at paraphrase time
- You cannot re-cluster paraphrased data
- Sponsors challenge summaries, not quotes
- Regulators want the source record

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Meridian - VOC verbatims from four segments

PROCUREMENT AND DISTRIBUTOR VOICE

- 'Two of the twelve pumps arrived, the rest slipped three weeks.'
- 'Nobody told me it was late - my ward planned around it.'
- 'Your Suzhou units fail calibration more often than Penang.'

– Source: 148 service tickets and 22 procurement interviews

CLINICAL AND PATIENT VOICE

- 'Setting up the infusion set takes too many steps at 3am.'
- 'The alarm tone is indistinguishable from the next bed.'
- 'I need a spare battery door within 48 hours, not 3 weeks.'

– Source: 9 ward gemba visits and 31 homecare surveys

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RECAP IN DEPTH - AFFINITY

The affinity diagram turns 200 verbatims into 6 themes

It is a silent, physical, team method. Run it wrong and you get the loudest person's categories. Run it right and the data speaks.

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The affinity method, done properly

1. Write one verbatim per card - no editing, no merging
2. Spread every card on the wall so all are visible
3. Cluster in SILENCE - anyone may move any card
4. Stop when movement settles; discuss only then
5. Name each cluster with a full sentence, not a noun
6. Check orphan cards - they are often the insight

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Naming clusters - weak versus strong

WEAK NAMES

Nouns that hide the meaning

- 'Delivery'
- 'Quality'
- 'Communication'
- 'Product issues'
- These could contain anything at all

STRONG NAMES

Full sentences that carry the finding

- 'Orders arrive late and we are not told'
- 'Suzhou units fail calibration more often'
- 'Setup has too many steps for night shift'
- 'Spare parts take weeks to reach the ward'
- A stranger can act on these

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Meridian - the six affinity themes from 214 verbatims

1

Orders arrive late and unannounced

68 cards. Late delivery plus no proactive notification. Ties directly to OTIF.

2

Partial shipments break ward planning

41 cards. In-full failure, distinct from the on-time failure.

3

Suzhou-built units fail calibration

37 cards. Plant-specific quality signal, not a design issue.

4

Setup takes too many steps

29 cards. Usability, mostly from night-shift nursing.

5

Alarm tones are not distinguishable

22 cards. Safety-adjacent; escalate to the design authority.

6

Spare parts lead time is weeks

17 cards. Aftermarket supply chain, likely a separate project.

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RECAP IN DEPTH - KANO

Kano tells you which requirements actually buy satisfaction

Not all CTQs are equal. Meeting a Must-Be earns you nothing; missing one loses the account. A Black Belt must classify before prioritising.

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The five Kano categories in full

MUST-BE

Expected, unspoken

- Absence causes severe dissatisfaction
- Presence causes no satisfaction at all
- Pump must not over-deliver dose
- Sterile packaging must be intact
- You cannot win here - only lose

ONE-DIMENSIONAL

More is linearly better

- Satisfaction rises with performance
- Customers will state these openly
- OTIF percentage, delivery lead time
- Battery run time in hours
- Where most CTQs and projects sit

DELIGHTER

Unexpected, unasked

- Presence delights; absence is unnoticed
- Customers cannot ask for these
- Auto-detection of the infusion set type
- Proactive delay alert before the due date
- Source of competitive differentiation

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The two categories teams forget

INDIFFERENT

Customer does not care either way

- Neither presence nor absence moves satisfaction
- Often internally-invented features
- Meridian example: 14 configurable report layouts
- Nursing staff use exactly one of them
- Stop investing here - this is pure waste

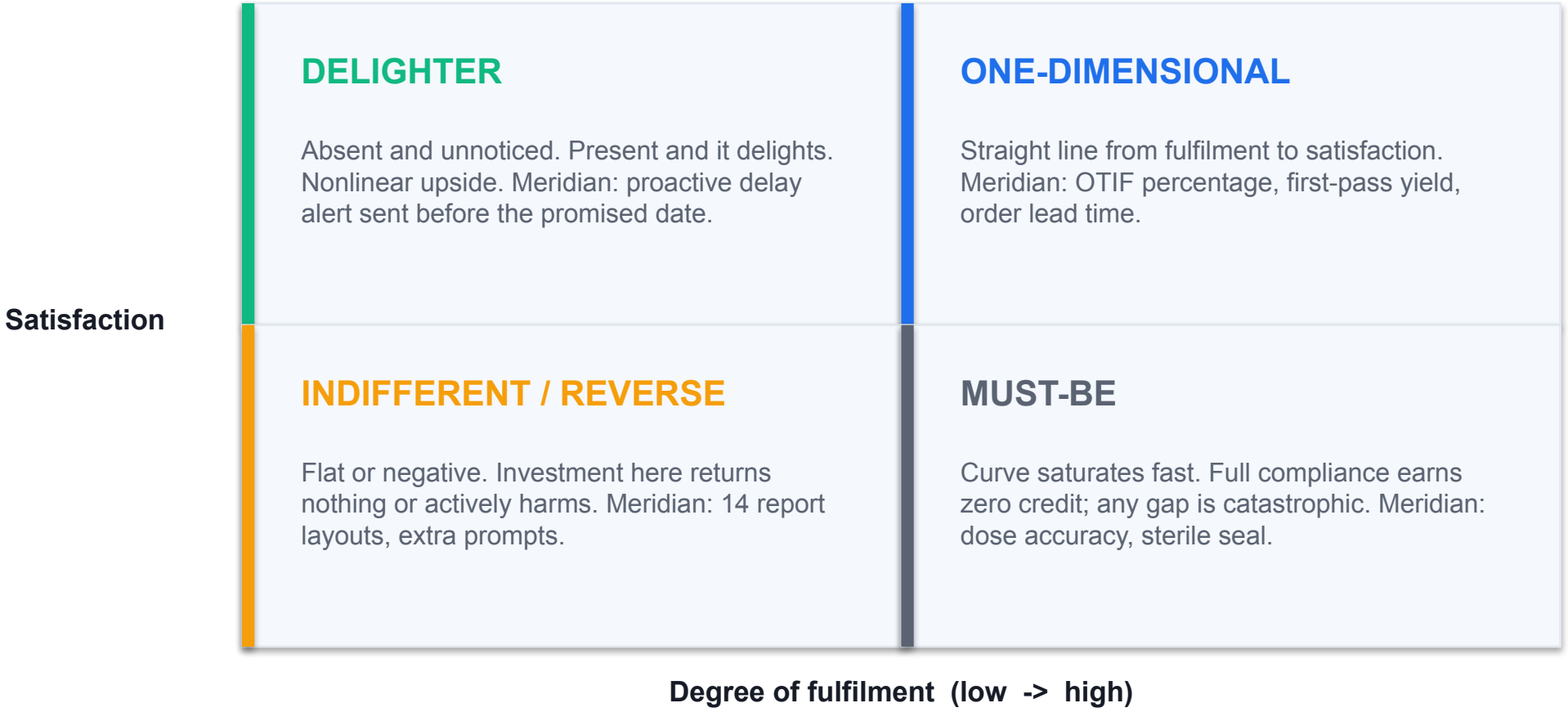
REVERSE

More of it makes it worse

- The feature actively harms satisfaction
- Meridian example: extra confirmation prompts
- Added for safety, hated at 3am on a ward
- Also: verbose alarm logs no one reads
- Removing these is a free improvement

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Kano positioned - satisfaction against fulfilment



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Category migration - Kano decays over time

1. A feature enters the market as a DELIGHTER
2. Competitors copy it; it becomes ONE-DIMENSIONAL
3. Everyone has it; it becomes a MUST-BE
4. Regulators or standards codify it as mandatory
5. Your differentiation is gone - find the next delighter

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How to run a Kano survey without corrupting it

1

Ask the functional question

'If the pump alerts you before a delivery is late, how do you feel?'

2

Ask the dysfunctional question

'If the pump does NOT alert you before a delivery is late, how do you feel?'

3

Use the same five-point scale

Like it / Expect it / Neutral / Live with it / Dislike it, on both questions.

4

Cross-tabulate to classify

The pair of answers maps to Must-Be, One-Dimensional, Delighter, Indifferent or Reverse.

5

Segment the results

Procurement and nursing will classify the same feature differently. Report both.

6

Re-run every 18 to 24 months

Categories migrate. A two-year-old Kano study is misleading, not merely stale.

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RECAP IN DEPTH - CTQ TREES

The CTQ tree converts a feeling into a number with limits

Need, then driver, then measurable requirement. Most weak Green Belt projects fail at the third level - they stop at a driver and call it a CTQ.

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The three levels of the CTQ tree

1

NEED - the general requirement behind the verbatim

2

DRIVER - the specific dimension that satisfies the need

3

MEASURABLE REQUIREMENT - metric, target and limits

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Meridian - one need drilled to three CTQs

NEED

What the customer requires

- 'My order arrives complete',
- 'on the date I was promised'
- Not measurable as written
- One need spawns several drivers

DRIVERS

The dimensions that satisfy it

- Arrives on the promised date
- Arrives complete in quantity
- I am told early if it will slip
- Each driver becomes its own CTQ

MEASURABLE REQUIREMENTS

What we will actually measure

- OTIF pct, target 96, LSL 96
- Line-fill rate pct, target 99
- Delay notice hours before due date,
- target at least 48 hours

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Every CTQ needs all five of these

1. **A metric with a unit**
   'OTIF percentage of order lines', not 'delivery performance'.

2. **An operational definition**
   Exactly how it is measured, by whom, from which system, and when the clock starts and stops.

3. **A target**
   The value we are aiming for. Meridian OTIF target is 96.0 percent.

4. **Specification limits**
   The customer's boundary of acceptability. LSL 96.0 percent; there is no upper limit here.

5. **A defect definition**
   The unambiguous statement of what counts as one defect. Here: one order line delivered late or short.

6. **A data source**
   Which system, which field, which report - so the Measure phase is not a rediscovery exercise.

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Operational definitions - vague versus operational

VAGUE

Two people will measure differently

- 'Delivered on time'
- 'Pump works correctly'
- 'Fast changeover'
- 'Good yield'
- Every one of these is arguable

OPERATIONAL

Two people get the same number

- 'Goods receipt timestamp in SAP is on or before the confirmed delivery date on the order line'
- 'Calibration test passes on first attempt, per procedure QC-114 revision 6'
- 'Minutes from last good unit of run A to first good unit of run B'
- 'Units passing all inline tests without rework, divided by units started'

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Turning a CTQ into a defect rate

Defect rate	**defects / units**	Meridian: 512 late lines out of 4,060 shipped lines = 12.6 pct defective
DPMO	**(defects / (units x opportunities)) x 1,000,000**	Meridian pump build: 38 defects, 1,200 units, 14 opportunities = 2,262 DPMO
First-pass yield	**units passing first time / units started**	Suzhou line 4: 861 of 1,000 pass without rework = 86.1 pct FPY

If a CTQ cannot be reduced to a countable defect, it is not yet a CTQ.

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BLACK BELT ONLY - CTQ FLOWDOWN

CTQ flowdown is the Black Belt's defining Define skill

A sponsor hands you a business Y such as 'OTIF must reach 96 percent'. Nobody can work on a Y. Your job is to cascade it until you reach Xs a team can actually turn.

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The flowdown cascade, level by level

1. BUSINESS Y - the number the sponsor is judged on
2. SUB-Y - the operational components that sum to Y
3. PROCESS Y - the output of one process step
4. CONTROLLABLE X - a setting, method or input a team owns
5. Verify: Y = f(X). If you cannot state the function, keep going

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Meridian - flowing OTIF down to controllable Xs

1

BUSINESS Y: OTIF = 87.4 pct

Target 96.0 pct. The sponsor's committed number to the top 8 hospital accounts.

2

SUB-Y 1: On-time = 91.2 pct

Order lines with goods receipt on or before the confirmed date.

3

SUB-Y 2: In-full = 95.8 pct

Order lines shipped at full ordered quantity, no partials.

4

PROCESS Y: Build cycle time

Days from works-order release to finished-goods putaway, per line.

5

PROCESS Y: First-pass yield

Units passing all inline tests without rework. Suzhou 86.1, Penang 92.4, Singapore 95.0.

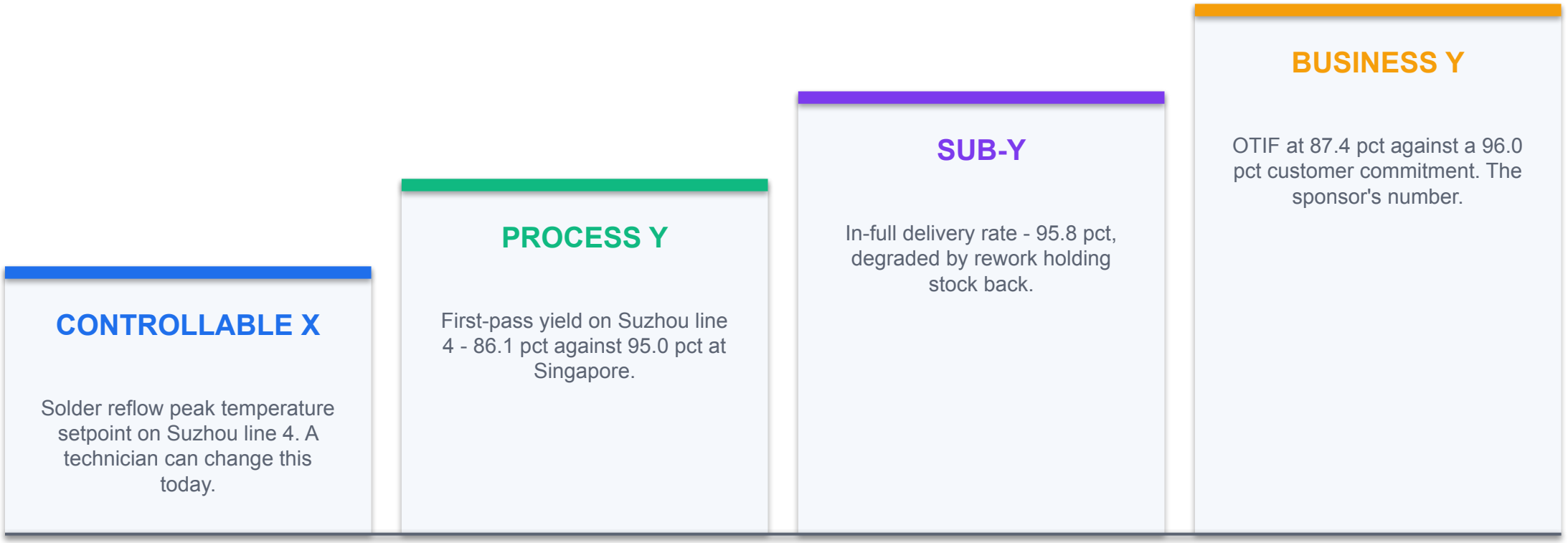
6

CONTROLLABLE Xs

Calibration interval, operator certification level, solder profile setpoint, kit-shortage rule.

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The flowdown as a ladder you climb DOWN



Read it top down to find the lever; read it bottom up to prove the benefit to Finance.

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Proving each level of the cascade arithmetically

OTIF composition	**OTIF = on-time pct x in-full pct**	Meridian: 0.912 x 0.958 = 0.874 = 87.4 pct. The cascade reconciles.
Yield rollup	**RTY = FPY1 x FPY2 x ... x FPYn**	Four Suzhou steps at 0.97, 0.96, 0.95, 0.975 gives RTY = 0.862
Benefit trace	**Y gain = sum of sub-Y gains x contribution**	Lifting Suzhou FPY 86.1 to 93.0 pct adds an estimated 2.9 pts of OTIF

If the arithmetic does not reconcile, the flowdown is wrong - not the data.

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Flowdown mistakes that kill projects

1

Stopping at a sub-Y

'Improve first-pass yield' is still a Y. Nobody can turn it. Keep cascading.

2

Jumping straight to a solution

'Buy a new reflow oven' is an answer, not an X. You have skipped Analyze.

3

Picking uncontrollable Xs

Customer order pattern and exchange rates are noise, not levers, for this team.

4

Losing the arithmetic link

If a 10 pct gain in X cannot be traced to a number in Y, Finance will reject the benefit.

5

Cascading only one branch

OTIF failed on both on-time AND in-full. Chase one and you cap your result.

6

Not testing the assumed function

Y = f(X) is a hypothesis in Define. Analyze must confirm it statistically.

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RECAP IN DEPTH - THE CHARTER

The charter is the contract between you and the sponsor

Six elements: business case, problem statement, goal statement, scope, team and milestones. Every one of them is negotiated, not merely filled in.

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The six elements of a Six Sigma project charter

1. **Business case**
   Why this project, why now, and what it is worth. Must be Finance-validated, not team-estimated.

2. **Problem statement**
   What is wrong, where, since when, how big - with no cause and no solution in it.

3. **Goal statement**
   SMART. From baseline to target by a date, expressed in the same metric as the problem.

4. **Scope in and out**
   Process start and stop, sites, product families, shifts - and an explicit out-of-scope list.

5. **Team and roles**
   Sponsor, Black Belt, process owner, Green Belts, SMEs, Finance representative.

6. **Milestones and tollgates**
   A date for each of D-M-A-I-C, and who chairs each tollgate review.

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The four-part test for a problem statement

1

1. What is wrong?

The specific undesirable outcome, in the customer's terms. 'Order lines arrive late or short.'

2

2. Where and when?

Which process, site, product and period. 'Across all three plants, Jan to Dec 2025.'

3

3. How big is the gap?

Baseline versus requirement, with the unit. 'OTIF 87.4 pct against a 96.0 pct commitment.'

4

4. What does it cost?

Business consequence in money or risk. 'SGD 1.9 million in penalties and expedite freight.'

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Problem statements - bad versus good

BAD

Fails the four-part test

- 'Delivery performance is poor because Suzhou is slow.'
- Contains a CAUSE - that is Analyze's job
- 'We need a new planning system.'
- Contains a SOLUTION in disguise
- 'Customers are unhappy.'
- No metric, no baseline, no gap, no cost

GOOD

Passes all four parts

- 'Between January and December 2025, OTIF delivery
- of infusion pump order lines from all three Meridian
- plants averaged 87.4 pct against a contracted 96.0 pct.
- The 8.6-point gap generated SGD 1.9 million in
- contractual penalties and expedite freight and placed
- two top-eight hospital accounts under review.'

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Goal statements - bad versus good

BAD

Unmeasurable or unbounded

- 'Significantly improve delivery.'
- 'Achieve world-class OTIF.'
- 'Reduce lateness as much as possible.'
- 'Fix the Suzhou plant by Q3.'
- None of these can be verified at Control

GOOD

Verifiable at the Control tollgate

- 'Raise OTIF for infusion pump order lines from
- 87.4 pct to at least 96.0 pct across all three
- plants by 31 December 2026, sustained for three
- consecutive months, delivering SGD 1.4 million
- in validated annualised benefit.'

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Meridian charter - scope in and out

WHAT THIS PROJECT OWNS

- IN SCOPE
  - Infusion pump assemblies - all six lines
  - All three plants: Singapore, Penang, Suzhou
  - Process from works-order release to goods receipt at customer
  - Top eight hospital accounts plus two distributors
  - All shifts, including night shift at Penang

WHAT IT EXPLICITLY DOES NOT

- OUT OF SCOPE
  - Spare parts and aftermarket supply - separate project
  - Product design changes - owned by the design authority
  - Syringe pump product family
  - Supplier selection and contract renegotiation
  - Anything upstream of the customer purchase order

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Building the business case with Finance, not around them

1. **Hard benefit**
   Cash that leaves or stops leaving the P&L. Penalties avoided, freight avoided, scrap avoided.

2. **Soft benefit**
   Capacity, time or risk released. Real, but do not claim it as savings without Finance agreeing.

3. **Cost avoidance**
   Spend that would have happened but now will not. Must be a committed, documented plan.

4. **Get the CFO's rate card**
   Use Finance's own labour, freight and scrap rates or your number will be disputed.

5. **Name a Finance signatory**
   One named person validates baseline and final benefit. Without this, benefits evaporate.

6. **Meridian case**
   SGD 1.9 million exposure; validated addressable benefit SGD 1.4 million annualised.

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BLACK BELT ONLY - CHARTER GOVERNANCE

The charter is a living contract, not a form you file

It is re-baselined at every tollgate. A charter that has not changed by the Analyze gate usually means nobody has looked at it.

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What happens to the charter at each tollgate

1. DEFINE gate - sponsor approves scope, baseline and benefit
2. MEASURE gate - baseline REVISED to the validated measurement
3. ANALYZE gate - scope narrowed to the proven vital few Xs
4. IMPROVE gate - benefit re-forecast from piloted results
5. CONTROL gate - benefit signed by Finance; charter closed

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Legitimate re-baselining versus scope creep

1

LEGITIMATE: measurement corrected

MSA shows the OTIF report double-counted partials. Baseline moves 87.4 to 89.1. Re-baseline.

2

LEGITIMATE: scope narrowed by data

Analyze proves 78 pct of the gap sits at Suzhou. Narrow the project. Re-baseline.

3

LEGITIMATE: external change

A top-eight account is lost. The benefit basis changed. Re-baseline or kill.

4

CREEP: new problem added

'While you are there, also fix the syringe pump line.' That is a second charter.

5

CREEP: target quietly lowered

Target slides 96 to 93 with no data reason. This is failure being renamed success.

6

CREEP: date extended without cause

Every extension must be traded against a named constraint at the tollgate.

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How a Black Belt runs a tollgate review

1. **Circulate the pack 48 hours ahead**
   Nobody should first see your data in the room. Objections land before, not during.

2. **Open with the charter as it stands**
   One slide: baseline, target, scope, benefit, and what has changed since last gate.

3. **Show the evidence, not the effort**
   Sponsors do not fund activity. Show what the data now proves that it did not before.

4. **State the decision you are asking for**
   Go, adjust or kill. Never let a tollgate end without an explicit sponsor decision.

5. **Record and publish the decision**
   Written, circulated same day. Verbal approvals evaporate under pressure.

6. **Be willing to recommend a kill**
   A Black Belt who has never killed a project is not testing hard enough.

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RECAP IN DEPTH - SIPOC

SIPOC fixes the boundary before anyone argues about the middle

Build it right to left, starting from the customer. Teams that start at Suppliers map the process they wish they had.

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Build SIPOC right to left

1. C - Who receives the output? Name them.
2. O - What exactly do they receive? Tie to CTQs.
3. P - The 4 to 7 high-level steps that produce it.
4. I - What the process consumes to run those steps.
5. S - Who provides each input, internal or external.

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Meridian - SIPOC for the pump fulfilment process

S — Suppliers	I — Inputs	P — Process	O — Outputs	C — Customers
• Component suppliers • Contract PCB assembler • Sterile packaging vendor • Planning (works orders) • Freight forwarders	• Purchase orders • PCB sub-assemblies • Pump housings and sets • Works order schedule • Calibration standards	• Release works order • Build and assemble • Calibrate and test • Pack and label • Ship and confirm	• Calibrated pump units • Certificate of conformity • Packing list and ASN • Delivery confirmation • Goods receipt record	• Hospital biomed teams • Ward nursing staff • Homecare patients • Distributors (MY, CN) • Meridian service desk

Read left to right: suppliers provide the inputs, the process transforms them, and the outputs go to the customers.

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SIPOC discipline points a Black Belt must enforce

1. **Fix the START and STOP explicitly**
   Meridian: starts at works-order release, stops at customer goods receipt. Written on the charter.

2. **Keep the process to 4 to 7 steps**
   More than seven and you have drifted into a process map, which belongs in Measure.

3. **Verb-noun for every step**
   'Calibrate and test', not 'Calibration'. Nouns hide who does what.

4. **Outputs must tie to the CTQs**
   If an output has no CTQ, ask whether the customer values it at all.

5. **Name the customers as people**
   'Ward nursing staff', not 'the hospital'. You cannot interview an institution.

6. **Do it in the room in 45 minutes**
   SIPOC is an alignment device. Perfecting it offline defeats its purpose.

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INTERACTIVE TOOL - USED IN LAB 6

Build your SIPOC online - alfredang.github.io/sipoc

1. Open the SIPOC tool in your browser - no login or licence needed
2. Name the process, then fix the START and STOP steps first
3. Work right to left: Customers, Outputs, Process, Inputs, Suppliers
4. Add 4 to 7 verb-noun process steps in the room with the team
5. Export the SIPOC into your capstone project pack

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BLACK BELT ONLY - STAKEHOLDERS

Every project you lead will be decided by people, not by data

You will map influence against support, and you will have a specific, named move for every person in the top-left quadrant.

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Stakeholder grid - influence against support



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The stakeholder analysis method

1. List everyone the change touches or who can stop it
2. Rate influence: can they stop or accelerate this?
3. Rate current support: from -2 opposed to +2 advocating
4. Rate REQUIRED support - it is rarely +2 for everyone
5. Write a named action and owner for every gap
6. Re-score at every tollgate - positions move

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Meridian - the stakeholder moves that mattered

1

Suzhou plant director

High influence, opposed. Move: private pre-brief, framed as capacity release, not defect hunting.

2

VP Operations (sponsor)

High influence, advocating. Move: ask them to open the kickoff and mandate data access.

3

Group Finance controller

High influence, neutral. Move: co-author the benefit method before any number is published.

4

Penang production manager

Medium influence, supportive. Move: make him the benchmark story the others learn from.

5

Ward nursing leads

Low influence, strongly supportive. Move: use their verbatims in every steering pack.

6

IT / MES data owner

Low influence, neutral, but blocking. Move: get data access written into the charter itself.

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RACI - who does what on the project

R - RESPONSIBLE

Does the work

- Black Belt: analysis and facilitation
- Green Belts: data collection at each plant
- Process owner: implements the changes
- There can be several R's per task

A - ACCOUNTABLE

Owns the outcome

- Exactly ONE per task, always
- Sponsor is A for benefit realisation
- Process owner is A for sustained control
- If two people are A, nobody is

C AND I

Consulted and Informed

- C: consulted BEFORE the decision
- Finance, Quality, Regulatory are usually C
- I: told AFTER the decision is made
- Confusing C with I creates most escalations

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BLACK BELT ONLY - RESISTANCE

Resistance is data. Diagnose it before you respond to it

The single most common Black Belt failure is answering a political objection with a technical argument. It never works, and it hardens the position.

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Three kinds of resistance, three different responses

TECHNICAL

'I do not believe your data'

- Sounds like: sample too small, wrong period
- Root: genuine doubt about method or evidence
- RESPONSE: show the MSA, widen the sample,
- invite them to audit the data themselves
- This is the EASY one - and the rarest

POLITICAL

'This costs me something'

- Sounds like: 'wrong time', 'no resources'
- Root: budget, headcount, status or blame at risk
- RESPONSE: negotiate. Find what they gain.
- Trade scope, credit or timing. Use the sponsor.
- Never argue data at a political objection

CULTURAL

'That is not how we work here'

- Sounds like: 'we tried that in 2019'
- Root: identity, history, professional pride
- RESPONSE: slow down. Pilot small, show a local
- win, let a respected insider carry the story
- The slowest to shift, and the most durable

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Diagnosing which one you are actually facing

1

Listen to what they ask for

A request for more data is technical. A request for delay is usually political.

2

Test with a hypothetical

'If the data were perfect, would you support it?' A no means it was never technical.

3

Ask who else feels this way

A widely shared objection is cultural. A lone strong one is usually political.

4

Check what they lose

Follow headcount, budget, autonomy and reputation. Political resistance always has a price tag.

5

Notice the history

'We tried that in 2019' is a cultural scar. Find out what actually happened then.

6

Escalate last, not first

Escalation converts a technical or cultural objection into a permanent political one.

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Resistance response grid - how hard to push

Strength of resistance

NEGOTIATE OR PAUSE High influence, strong resistance. Do not force this. Meet privately, find the trade, or delay the branch of scope that threatens them.	**CO-OPT AND CONVERT** High influence, mild resistance. Involve them in designing the pilot. Ownership converts more reliably than argument.
PROCEED AND MONITOR Low influence, strong resistance. Do not spend your capital here. Proceed, document, and revisit after the first visible win.	**INFORM AND MOVE ON** Low influence, mild resistance. Standard communication cadence. Most of it dissolves once results appear.

Their influence (low -> high)

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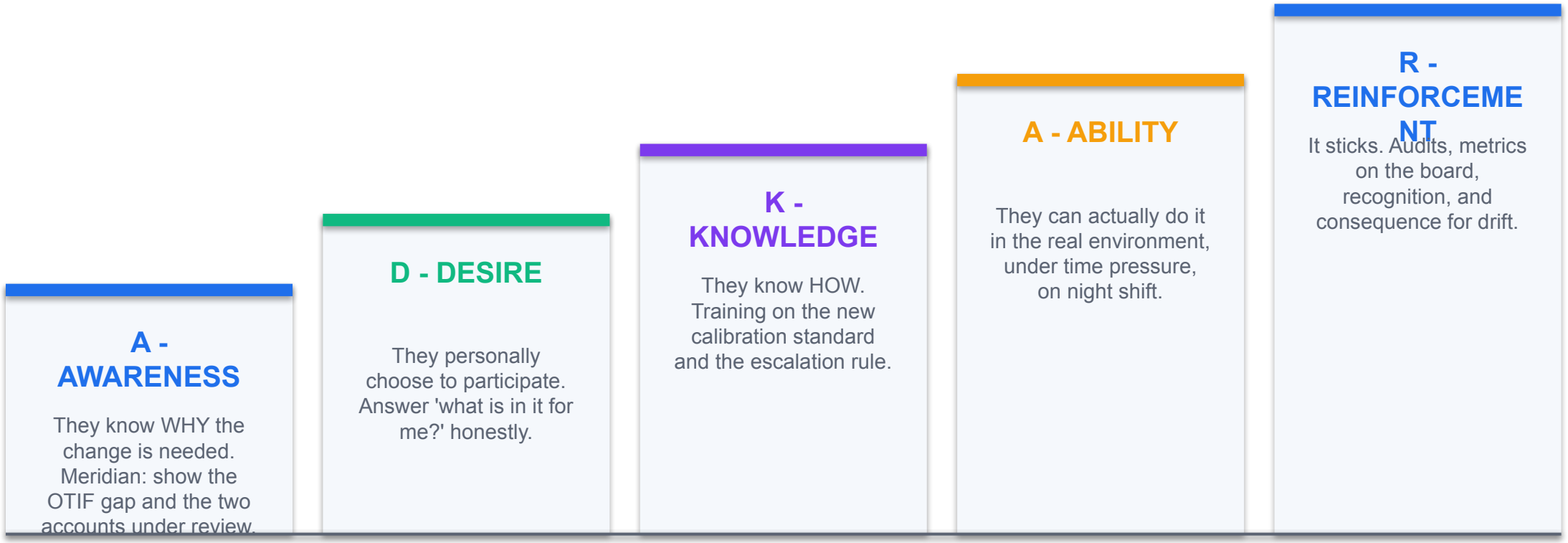
BLACK BELT ONLY - CHANGE LEADERSHIP

Six Sigma delivers a solution. Change leadership delivers the result

Two frameworks: ADKAR for the individual, Kotter for the organisation. You need both, and they operate at different levels.

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ADKAR - the individual change sequence



ADKAR is sequential. Diagnose which letter a person is stuck on - and fix THAT one only.

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Using ADKAR as a diagnostic, not a checklist

1

Stuck at Awareness

They say 'what problem?' Fix: share the customer verbatims and the OTIF trend, not the solution.

2

Stuck at Desire

They understand and still will not move. Fix: this is political. Negotiate, do not re-explain.

3

Stuck at Knowledge

Willing but unsure how. Fix: training, work instructions, a job aid at the station.

4

Stuck at Ability

Trained but cannot perform in practice. Fix: coaching at the line, tooling, time, or workload relief.

5

Stuck at Reinforcement

Did it for six weeks, then drifted. Fix: the Control plan and the response plan, not more training.

6

Wrong fix, wasted effort

Training a person stuck on Desire is the classic and expensive Black Belt error.

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Kotter's 8 steps - the organisational level

1

1. Create urgency

Make the status quo feel more dangerous than the change. Two accounts under review does this.

2

2. Build a guiding coalition

Sponsor plus the three plant directors plus Finance. Authority AND credibility.

3

3. Form a strategic vision

'96 percent OTIF, every account, every month' - repeatable in one sentence.

4

4. Enlist a volunteer army

Green Belts and line champions at each plant carry it further than you can.

5

5. Enable action, remove barriers

Get the MES data access and the pilot line time authorised, in writing.

6

6. Generate short-term wins

Penang line 2 pilot: FPY 92.4 to 96.1 in six weeks. Publish it everywhere.

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Kotter steps 7 and 8 - where projects actually fail

7. SUSTAIN ACCELERATION

Do not declare victory early

- The pilot win is evidence, not completion
- Roll to Suzhou and Singapore lines next
- Keep the coalition meeting after the pilot
- Most programmes stop here and quietly decay

8. INSTITUTE CHANGE

Make it the new normal

- Rewrite the SOP and the training pack
- Put OTIF on the plant daily management board
- Build it into new-hire onboarding
- Change the metric people are measured on

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BLACK BELT ONLY - COMMUNICATION

Different audiences need different messages at different cadences

A communication plan is a table with five columns, and it is written in Define - not improvised in Improve when the resistance arrives.

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The five columns of a communication plan

1. AUDIENCE - who exactly, by name or role
2. MESSAGE - what they need to know and to do
3. CHANNEL - the medium that reaches them
4. FREQUENCY - the cadence, fixed in advance
5. OWNER - one named person, not 'the team'

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Meridian - the communication plan in practice

1

Steering committee

Benefit, risk, decisions needed. One-page pack plus 30-min review. Monthly. Owner: Black Belt.

2

Plant directors

Site-level data, resource asks, what changes for them. Site visit. Fortnightly. Owner: sponsor.

3

Line supervisors

What changes at the station and when. Shift huddle. Weekly. Owner: process owner.

4

Operators (3 shifts)

The new standard, and why it matters to the patient. Board plus toolbox talk. Weekly.

5

Finance

Baseline method, benefit calculation, validation status. Working session. At each tollgate.

6

Customers (top 8)

Recovery plan and improvement trend, no internal detail. Account review. Quarterly.

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BLACK BELT ONLY - MENTORING

You will coach two to four Green Belt charters while running your own

The temptation is to rewrite their charter for them. Do that once and you own their project forever. Coach with questions instead.

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The Green Belt Define mistakes to watch for

1

Solution in disguise

'Implement a new scheduling system' is an answer, not a problem. Send them back to the data.

2

Scope creep at birth

A charter covering three processes and two sites will finish none of them. Cut it on day one.

3

Unmeasurable CTQs

'Improve communication' has no metric, no target, no defect definition. It is not a CTQ.

4

Cause in the problem statement

'Late because Suzhou is slow' pre-decides Analyze and biases every subsequent step.

5

Baseline taken on faith

The system report has never been validated. Measure will move the baseline - warn them now.

6

No sponsor, or the wrong one

If the sponsor cannot authorise the change to the process, they are not the sponsor.

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Coaching by question, not by correction

INSTEAD OF TELLING

You own it the moment you write it

- 'Your problem statement is wrong.'
- 'Change the target to 96 percent.'
- 'Take Suzhou out of scope.'
- 'Here, I have rewritten it for you.'

ASK

They own it, and they learn

- 'How would you prove that gap to Finance?'
- 'What does one defect look like, exactly?'
- 'Which part of this could you finish in 90 days?'
- 'Who has to agree, and have you asked them?'

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The Define tollgate checklist you will apply

1. **VOC evidence exists**
   Real verbatims, segmented, from more than one channel. Not the team's opinion.

2. **CTQs are measurable**
   Metric, operational definition, target, spec limits, defect definition, data source.

3. **Flowdown reconciles**
   The business Y decomposes arithmetically to the Xs the team proposes to work on.

4. **Charter passes all four tests**
   Problem is testable, goal is SMART, scope is bounded, business case is Finance-validated.

5. **Stakeholders are mapped**
   Every high-influence, low-support person has a named owner and a dated action.

6. **Sponsor has decided**
   An explicit, recorded go, adjust or kill. Not 'looks good, carry on'.

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DEFINE - CLOSE

If Define is wrong, everything downstream is precisely wrong

You can run flawless statistics on the wrong Y and deliver nothing. Spend the time here, negotiate hard, and be willing to recommend that the project should not run at all.

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WHAT YOU'LL DO

Hands-On Labs — Define

1

Lab 4 — Voice of the Customer, Affinity and Kano — In Depth

A VOC log, an affinity diagram and a Kano classification for your capstone… • Data: lab-04-voc-affinity-kano.xlsx

2

Lab 5 — CTQ Trees and CTQ Flowdown to Controllable Xs

A CTQ tree with targets and specification limits, plus a flowdown to controllable… • Data: lab-05-ctq-tree-flowdown.xlsx

3

Lab 6 — SIPOC, Process Mapping and Scope Governance

A SIPOC, a detailed swimlane process map and a documented scope boundary. • Data: lab-06-sipoc-process-map.xlsx

4

Lab 7 — Stakeholder Analysis, Resistance Management and Change Leadership

A stakeholder influence/support map, a resistance diagnosis and an ADKAR change… • Data: lab-07-stakeholder-analysis.xlsx

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DMAIC · DEFINE · HANDS-ON

Voice of the Customer, Affinity and Kano — In Depth

LAB 4

A full re-derivation of VOC, affinity clustering and Kano classification — taught to the depth at which you could run this session yourself for a Green Belt team. Then apply all three to your own capstone process. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A VOC log, an affinity diagram and a Kano classification for your capstone customers.

Tools & techniques: Reactive & proactive VOC, verbatim capture, affinity clustering, Kano model, customer segmentation

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Voice of the Customer, Affinity and Kano — In Depth

STEPS 1-5 OF 9

1. Recap the VOC sources in full: reactive (complaints, returns, warranty, escalations, social) and proactive (interviews, surveys, gemba, observation).
2. Segment your capstone customers — internal and external. Different segments will give contradictory VOC, and that contradiction is itself data.
3. Collect at least fifteen verbatim statements about your capstone process. Use the customer's own words; do not summarise them into your own.
4. Recap affinity diagram method: silent clustering first to avoid anchoring on the loudest voice, then name the clusters by consensus.
5. Build the affinity diagram for your verbatims. Aim for five to eight named themes.

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Voice of the Customer, Affinity and Kano — In Depth

STEPS 6-9 OF 9

6 Recap the Kano model in full: Must-Be, One-Dimensional, Delighter, Indifferent and Reverse — and how categories migrate over time as delighters become expectations.

7 Classify each affinity theme against Kano. Note explicitly that Must-Be requirements can only ever cause dissatisfaction, never satisfaction.

8 Identify the one or two themes that most drive your project Y, and record them as your CTQ candidates for the next lab.

9 Note how you would coach a Green Belt through this exercise — what mistake would you expect them to make?

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LAB 4 · VERIFY

Voice of the Customer, Affinity and Kano — In Depth

✅ Check your work

You have fifteen or more verbatims, every verbatim sits in a named affinity cluster, every cluster carries a Kano classification, and your CTQ candidates trace to specific verbatims.

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CTQ Trees and CTQ Flowdown to Controllable Xs

LAB 5

A customer need is not measurable; a CTQ is. Re-derive the CTQ tree in full, then go beyond Green Belt with CTQ FLOWDOWN — cascading a business-level Y through sub-Ys until you reach process Xs a team can actually control. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A CTQ tree with targets and specification limits, plus a flowdown to controllable Xs.

Tools & techniques: CTQ tree, need-driver-requirement, operational definitions, specification limits, Y-to-X cascade

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CTQ Trees and CTQ Flowdown to Controllable Xs

STEPS 1-5 OF 9

1. Recap the CTQ tree structure in full: Need -> Drivers -> Measurable Requirements, with each level more specific than the last.
2. Take your top VOC theme and drive it down: what is the need, what drives it, and what specifically must be measured?
3. For each CTQ, write a complete operational definition — what is measured, by whom, with what instrument, at what point, and what counts as a defect.
4. Set the target, the upper and lower specification limits, and the defect definition for each CTQ. A CTQ without limits cannot generate a defect count.
5. Verify each CTQ against the customer: would they agree this number represents what they asked for?

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CTQ Trees and CTQ Flowdown to Controllable Xs

STEPS 6-9 OF 9

6. Now flow down. Write your business-level Y (e.g. OTIF percentage) at the top of a cascade diagram.

7. Break that Y into sub-Ys — the process-level outputs that roll up into it (pick accuracy, line yield, changeover time, carrier dwell).

8. Break each sub-Y into the Xs that drive it. Continue until you reach Xs a team can directly control or experiment on.

9. Mark on the cascade which Xs you will measure in Day 2, test statistically in Day 3, and experiment on in Day 4.

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LAB 5 · VERIFY

CTQ Trees and CTQ Flowdown to Controllable Xs

✅ Check your work

Every CTQ has an operational definition, a target and specification limits, and your flowdown reaches at least one layer of Xs that are directly controllable and testable.

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SIPOC, Process Mapping and Scope Governance

LAB 6

Re-derive SIPOC and detailed process mapping in full, then apply them to bound your capstone. Scope creep kills more Black Belt projects than bad statistics — this lab builds the map that makes scope arguments objective. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A SIPOC, a detailed swimlane process map and a documented scope boundary.

Tools & techniques: SIPOC, process boundaries, swimlane mapping, handoff analysis, rework loops, scope control

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SIPOC, Process Mapping and Scope Governance

STEPS 1-5 OF 9

1. Recap SIPOC in full: Suppliers, Inputs, Process, Outputs, Customers — and why it is built right-to-left, starting from the customer.
2. Open the online SIPOC tool and build the SIPOC for your capstone process at five to seven high-level steps. Agree the first and last step explicitly — that IS your scope.

   https://alfredang.github.io/sipoc/
3. Recap process mapping symbols and conventions, and the difference between as-is and to-be maps.
4. Build a detailed swimlane map of your capstone process, with one lane per role or function.
5. Mark every handoff between lanes. Handoffs are where defects, delays and ownership gaps concentrate.

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SIPOC, Process Mapping and Scope Governance

STEPS 6-9 OF 9

6 Mark every decision point, inspection, rework loop and queue. Count the rework loops — each is a hidden factory.

7 Walk the process physically or through the system (gemba) and correct the map. The map in your head is never the map in reality.

8 Compare your map against the charter scope. Anything on the map outside the scope boundary must be explicitly marked out-of-scope.

9 Record the three questions you will ask when someone proposes expanding the scope mid-project.

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LAB 6 · VERIFY

SIPOC, Process Mapping and Scope Governance

✅ Check your work

Your SIPOC's first and last steps match your charter scope exactly, your swimlane map marks every handoff and rework loop, and the map has been validated by walking the actual process.

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DMAIC · DEFINE · HANDS-ON

Stakeholder Analysis, Resistance Management and Change Leadership

LAB 7

Black Belt projects fail on people far more often than on statistics. Map influence against support, diagnose resistance by type, and build an ADKAR-based change plan. This is Black-Belt-only material — the Green Belt course stops at RACI. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A stakeholder influence/support map, a resistance diagnosis and an ADKAR change plan.

Tools & techniques: Stakeholder mapping, RACI, influence-support grid, resistance types, ADKAR, Kotter, communication plan

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Stakeholder Analysis, Resistance Management and Change Leadership

STEPS 1-5 OF 9

1. List every stakeholder in your capstone: sponsor, process owner, operators, support functions, customers, finance, and anyone who can veto.
2. Build the RACI for your project — Responsible, Accountable, Consulted, Informed — and check there is exactly one Accountable per deliverable.
3. Plot each stakeholder on an influence (high/low) versus support (supportive/neutral/resistant) grid.
4. For every high-influence, low-support stakeholder, write a specific named action. 'Communicate more' is not an action.
5. Diagnose resistance type for each resistant stakeholder: technical (they doubt the solution), political (they lose something), or cultural (it conflicts with how things are done).

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Stakeholder Analysis, Resistance Management and Change Leadership

STEPS 6-9 OF 9

6. Match the response to the type — technical resistance needs evidence, political needs negotiation, cultural needs leadership modelling.

7. Build an ADKAR plan: how will you create Awareness, Desire, Knowledge, Ability and Reinforcement across the project life?

8. Write the communication plan: audience, message, channel, frequency and owner. Sponsors, operators and customers need different messages.

9. Identify the single stakeholder most likely to derail your capstone, and write your specific mitigation.

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LAB 7 · VERIFY

Stakeholder Analysis, Resistance Management and Change Leadership

✅ Check your work

Every stakeholder is plotted on the influence/support grid, every high-influence resistor has a named action matched to a diagnosed resistance type, and your ADKAR plan covers all five elements.

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PHASE RECAP

Recap — Define

- VOC, affinity and Kano — in depth — Re-derive VOC capture, affinity clustering and Kano classification in full — you must be able to teach these, not just use them.
- CTQ trees — in depth — Build the CTQ tree from need to driver to measurable requirement, with targets, specification limits and defect definitions.
- CTQ flowdown — Cascade a business-level Y down through sub-Ys to the process Xs a team can actually control.
- Charter governance — The charter is a living contract — re-baselined at each tollgate, not written once and filed.
- Stakeholder analysis — Map influence against support, then plan a specific move for every high-influence, low-support stakeholder.
- Resistance management — Resistance is data. Diagnose whether it is technical, political or cultural before responding.

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M

DMAIC · MEASURE

Measure at Black Belt Depth — Advanced Measurement Systems

MSA recap · Attribute MSA & Kappa · Nested & destructive gauge studies · Capability for non-normal data · Sampling strategy

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Key Concepts — Measure (1)

1

Process mapping & VSM — in depth

Re-derive SIPOC, swimlane maps, value stream maps, takt time and the eight wastes in full.

2

Data, sampling and sample size — in depth

Data types, operational definitions, sampling schemes and sample size formulas re-derived from first principles.

3

Continuous Gage R&R — in depth

Full treatment of accuracy, repeatability, reproducibility, linearity, stability and %study variation.

4

Yield, DPMO and capability — in depth

Classic yield, FPY, RTY, DPU, DPO, DPMO, sigma level and baseline Cp/Cpk, worked end to end.

5

Attribute MSA

When the measurement is a judgement (pass/fail, grade), Gage R&R does not apply — attribute agreement does.

6

Kappa analysis

Cohen's and Fleiss' Kappa quantify inspector agreement beyond chance. Kappa >= 0.75 is acceptable.

7

Nested gauge studies

When parts cannot be measured twice by every operator, the crossed study is invalid — nest it.

8

Destructive testing MSA

If measuring destroys the part, use near-identical part pairs and a nested design.

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Key Concepts — Measure (2)

1

Non-normal capability

Cp/Cpk assume normality. For non-normal data, transform (Box-Cox) or fit the correct distribution.

2

Ppk vs Cpk

Cpk uses within-subgroup variation; Ppk uses total variation. Reporting the wrong one flatters the process.

3

Sampling strategy at scale

Stratified and rational subgrouping decide what the control chart can even detect later.

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MEASURE - BLACK BELT DEPTH

A Black Belt owns the integrity of the number

Green Belts learn to measure. Black Belts learn when the measurement itself is lying, and how to prove it before a single improvement is funded.

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What Measure delivers at Black Belt level

1. **A mentor-grade recap**
   You must be able to teach mapping, VSM, sampling and Gage R&R to your Green Belts.

2. **Attribute MSA and Kappa**
   Judgement data needs agreement analysis, not Gage R&R. Most inspection is judgement.

3. **Nested and destructive studies**
   Crossed Gage R&R is invalid when measuring destroys the part.

4. **Non-normal capability**
   Anderson-Darling, Box-Cox, Johnson and distribution fitting for skewed data.

5. **Cp vs Cpk vs Pp vs Ppk**
   Short-term against long-term variation, and which one you may honestly report.

6. **Rational subgrouping**
   The subgrouping decision made here fixes what your control chart can ever detect.

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Green Belt Measure vs Black Belt Measure

Green Belt scope

Apply the standard toolkit

- Map the process and compute a baseline
- Run a continuous Gage R&R on a gauge
- Assume the data is roughly normal
- Report Cpk from the software output
- Sample size from the standard formula

Black Belt scope

Judge, adapt and defend the method

- Choose the RIGHT MSA for the data type
- Design nested studies for destructive tests
- Test normality and transform or fit correctly
- Separate short-term from long-term capability
- Design the subgrouping strategy deliberately

Mentoring scope

Teach and audit others

- Spot a Green Belt's invalid Gage R&R design
- Challenge an inflated opportunity count
- Catch Cpk reported on an unstable process
- Review data collection plans before launch
- Sign off the tollgate measurement evidence

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The Black Belt Measure sequence

1. Map the process and the value stream in full detail
2. Define the data: type, operational definition, collection plan
3. Design the sampling and subgrouping strategy
4. Validate the measurement system with the CORRECT MSA method
5. Test the distribution before computing any capability index
6. Report baseline yield, DPMO, sigma and honest capability

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Recap in depth: the process map

You will mentor Green Belts through this. That means knowing not just how to draw it, but every way a map goes wrong and how to challenge one.

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Standard process mapping symbols

SYMBOL	DESCRIPTION & USAGE
 Connector	• Connects any two steps and shows the path or direction of the process
 Delay	• Identifies when the process comes to a temporary halt • Also identifies what has to happen before the process resumes
 Document	• Indicates that a written document is prepared or used at that step of the process; the name of the document appears in the symbol

Oval = start/stop. Rectangle = activity. Diamond = decision. D-shape = delay. Document shape = paperwork produced.

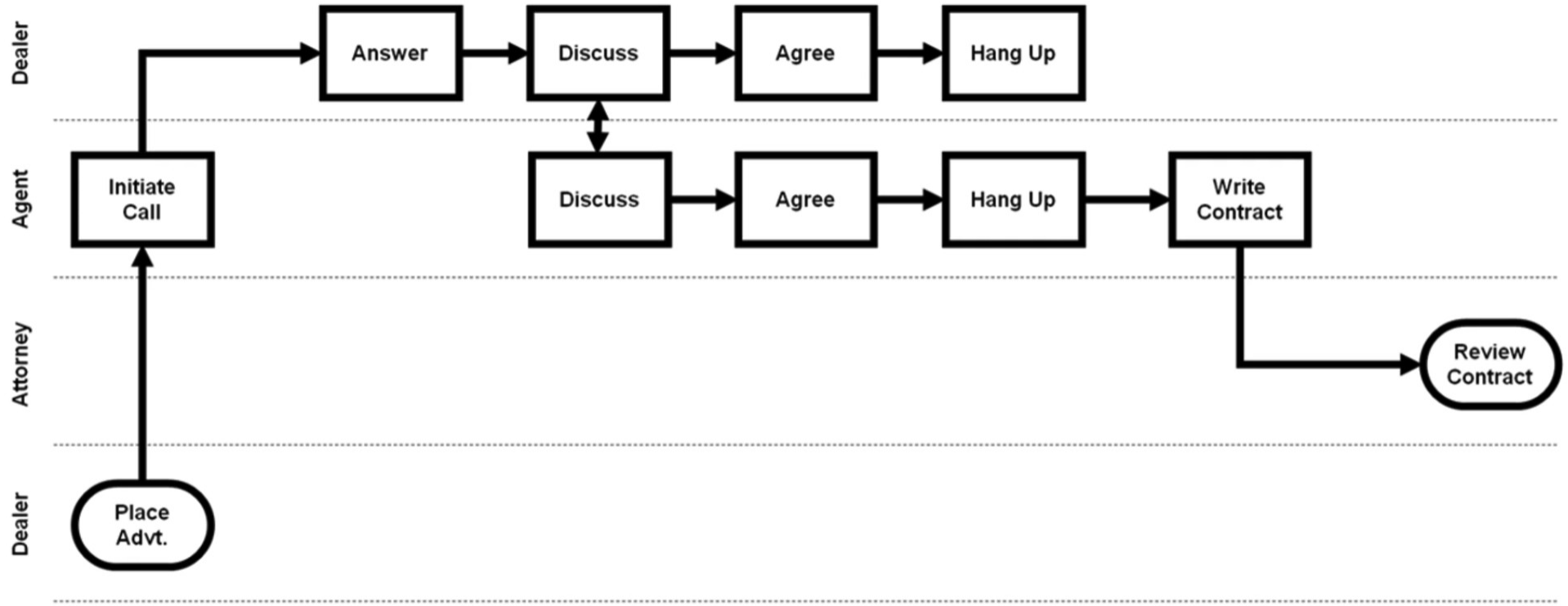
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How to build a defensible process map

1. Agree the start and stop boundary in writing with the process owner
2. Walk the process physically on every shift that runs it
3. Capture each step with the people who actually do the work
4. Sequence, then add every decision, delay and rework loop
5. Validate against a second shift and a second site
6. Load the data: touch time, queue time, defect rate, headcount

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Swimlane (deployment) process map



Each horizontal lane is one role or system. Every lane crossing is a handoff: a queue, a re-entry and a chance to lose context.

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Meridian Suzhou: swimlanes in pump final assembly

THE FLOW

- Planning releases the work order to the line
  - Handoff 1 to Kitting for component pick
- Kitting stages the assembly kit at the line head
  - Handoff 2 to Assembly cell 3
- Assembly builds, then calls Test
  - Handoff 3 to Functional Test, then to QA release

WHAT THE MAP REVEALED

- Eleven lane crossings per work order
  - Average queue at each crossing: 25 to 190 minutes
- Touch time across all steps: about 71 minutes
  - Everything else is waiting for the next lane
- First-pass yield 88.4 percent on the pump line
  - Most failures are found at Test, three lanes late

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Value-added analysis - the three categories

Value-Added

The customer would pay for it

- Physically transforms the product
- Done right the first time
- Meridian: assembly, calibration, sterile pack
- Typically 5 to 20 percent of steps
- Protect and speed these up

Business-Value-Added

Required but not customer value

- Design History File records, lot traceability
- IEC 60601 test evidence, regulatory retention
- In medical devices BVA is large and legitimate
- Meridian: DHR completion, sterilisation records
- Streamline it - you may not delete it

Non-Value-Added

Pure waste, delete it

- Inspection, rework, transport, waiting
- Meridian: triple inspection before Test
- Typically 60 to 90 percent of lead time
- This is your improvement opportunity
- Attack it first - it is free capacity

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MEASURE - BLACK BELT DEPTH

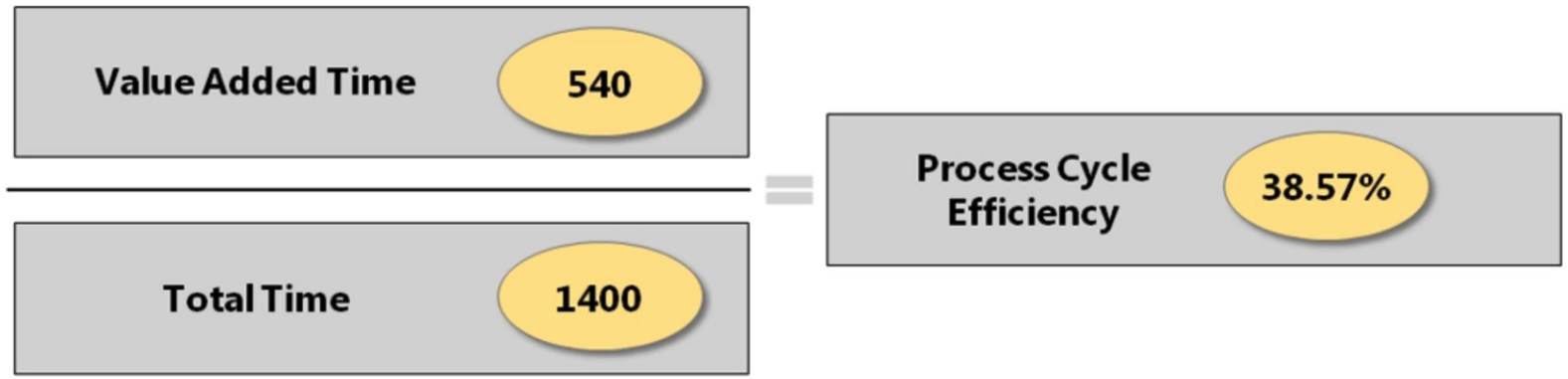
The value stream map is the Black Belt's system view

A process map shows the work. A value stream map shows the work, the signals that trigger it, and the time the customer actually waits.

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A complete value stream map

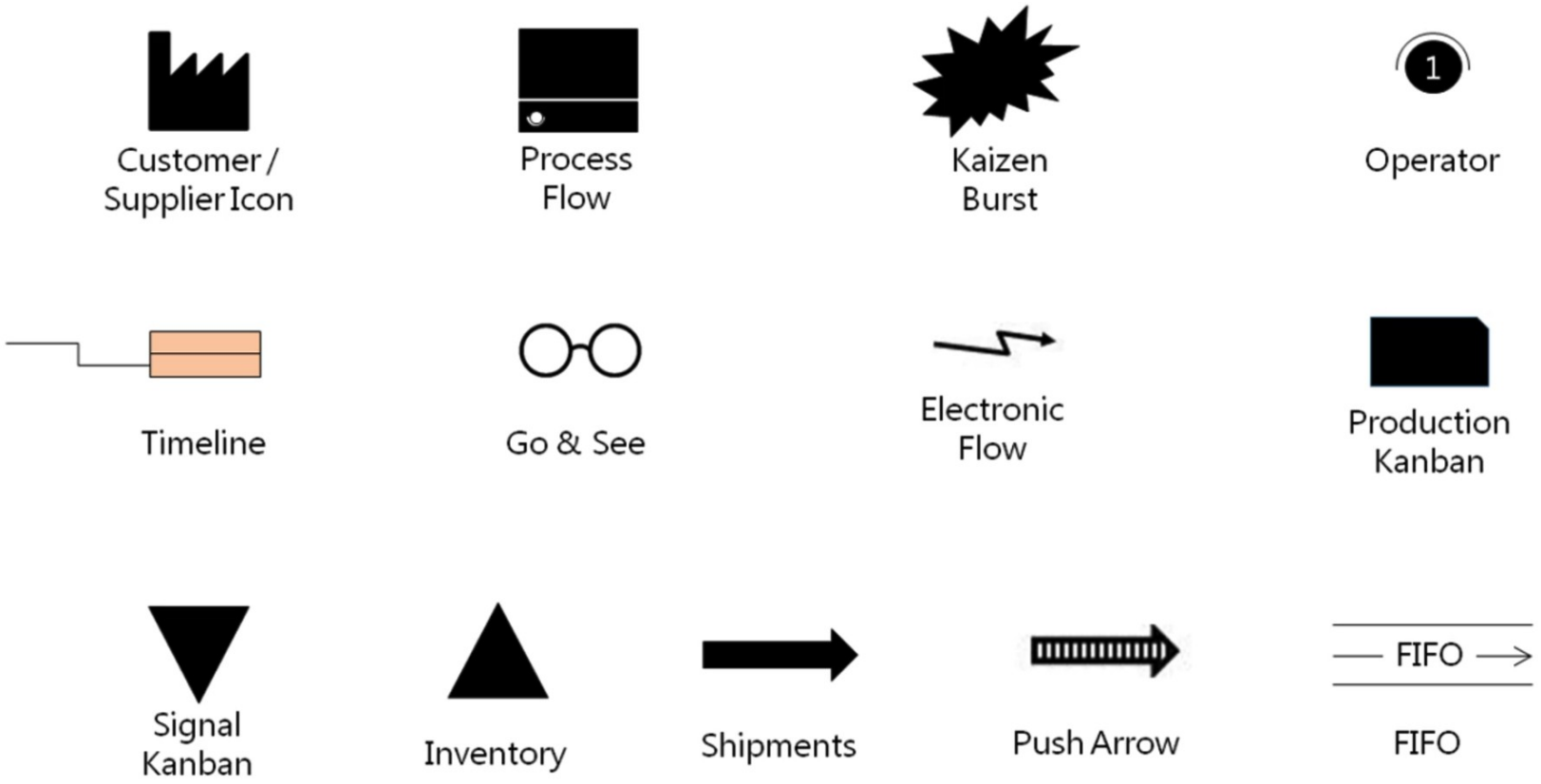
Non-Value Added Work	Essential Non-Value Added Work	Value Added Work
Actions that should be eliminated	Wasteful actions but essential under current operating procedures	Actions that add value to the customer



Information flow right to left across the top, material flow left to right along the bottom, and the ladder carrying the timeline.

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Standard VSM icons



Use the standard icon set so any Lean practitioner in Singapore, Penang or Suzhou can read your map without a legend.

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Building the Meridian pump value stream map

1. Choose one product family: the 3-channel infusion pump, Suzhou line 4
2. Walk the stream backwards from finished goods to the component supplier
3. Draw material flow with a data box under every process step
4. Add information flow: MRP release, kanban pull, expedite calls
5. Draw the ladder: process time below, wait time above
6. Compute lead time, VA time and process cycle efficiency

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The VSM timeline ladder - Meridian Suzhou line 4

Value-added time	**Sum of the lower rungs**	Kitting 8 min + Assembly 44 min + Test 14 min + Pack 5 min = 71 minutes.
Lead time	**Sum of the upper rungs plus VA**	Queues of 190 + 145 + 60 + 25 min plus 71 VA = 491 min, about 8.2 hours.
Process cycle efficiency	**PCE = VA time / Lead time**	71 / 491 = 0.145, so 14.5 percent. Above the 10 percent typical but far from world class.

World class discrete assembly runs above 25 percent PCE. The queues own the lead time, not the touch time.

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Takt time explained

$$T = \frac{T_a}{T_d}$$

where, T stands for Takt Time
T_a stands for Net Available Time
T_d is the total customer demand (units produced per day)

German word for "Drum beat"

Establishes the pace of the process needed to meet customer demand

Shows where the flow is uneven

Takt = available working time divided by customer demand for the same period. It is set by the customer, never by the process.

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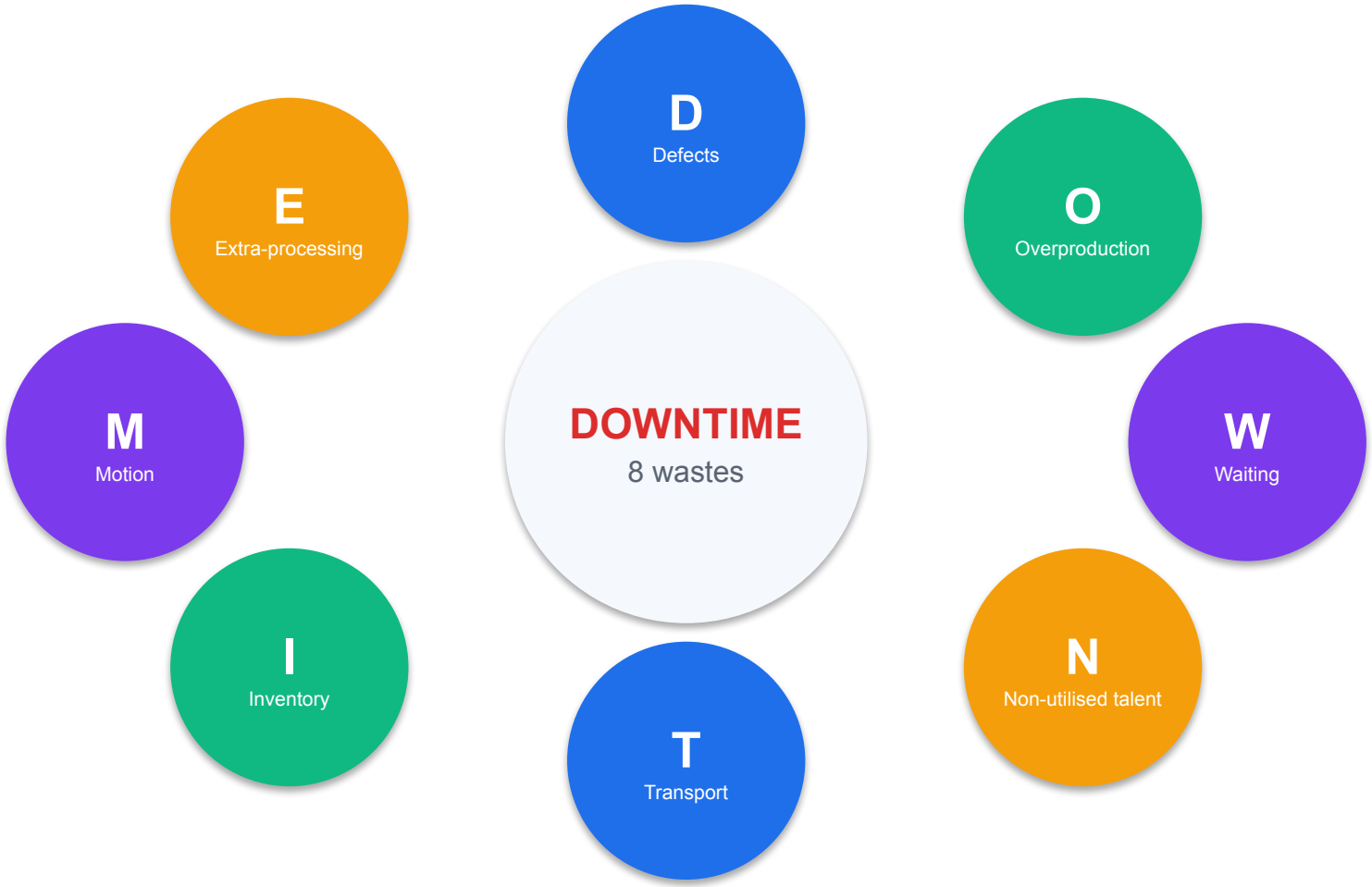
Takt time - Meridian worked example

Available time	**2 shifts x 7.5 h x 3600 s**	Suzhou line 4 runs two productive shifts of 7.5 hours = 54,000 seconds per day.
Takt time	**Takt = Available time / Demand**	Demand is 180 pumps per day. Takt = 54,000 / 180 = 300 seconds per pump.
Interpretation	**One pump every 300 seconds**	Assembly cycle time is 44 min = 2,640 s across 9 stations = 293 s per station. Just inside takt.

Use productive time only - exclude breaks, shift handover briefings and planned maintenance.

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The eight wastes of Lean - DOWNTIME



Waste consumes capacity, cash and lead time while adding nothing the customer values.

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DOWNTIME at Meridian Medical Devices

1

D - Defects

Pump occlusion sensors failing functional test and returning to assembly for rework.

2

O - Overproduction

Kitting a full day of assembly kits at 06:00 when only half will be consumed.

3

W - Waiting

Built pumps queuing 145 minutes for a free functional test rig.

4

N - Non-utilised talent

Station 6 operators know the fixture drifts, and were never asked.

5

T - Transport

Sub-assemblies trucked between the Penang and Suzhou plants for calibration.

6

I - Inventory

Six weeks of pressure sensors held against a single-source supply risk.

7

M - Motion

Assemblers reaching behind the bench for the torque driver 400 times a shift.

8

E - Extra-processing

Three manual inspections of the same solder joint before functional test.

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MEASURE - BLACK BELT DEPTH

Data type decides the entire analysis path

Choose the wrong data type and every tool downstream - MSA, capability, hypothesis test, control chart - is the wrong tool. This decision is made once, in Measure.

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The four measurement scales

1

Nominal

Named categories, no order. Defect type, carrier, plant. Only counts and modes are valid.

2

Ordinal

Ordered categories, unequal gaps. Severity 1-5, cosmetic grade A-D. Median, not mean.

3

Interval

Equal gaps, arbitrary zero. Degrees Celsius, calendar date. Differences valid, ratios not.

4

Ratio

Equal gaps, true zero. Time, weight, pressure, cost. Every statistic is valid.

5

Why Black Belts care

The scale caps which statistics are meaningful, whatever the software will compute.

6

The classic error

Averaging an ordinal severity score. 'Mean severity 2.7' has no physical meaning.

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Continuous vs discrete data

Continuous (variable)

Any value on a real scale

- Infinitely divisible: 4.37 hours, 12.63 mL/h
- Far more information per data point
- Needs 20 to 30 points for a decent estimate
- Enables capability, t-tests, regression
- Meridian: flow rate, torque, test duration

Discrete (attribute)

Counts and categories

- Whole counts or categories: 3 defects, pass/fail
- Very little information per data point
- May need hundreds of points for the same precision
- Proportions, chi-square, attribute charts
- Meridian: pump passes or fails functional test

The Black Belt move

Convert up wherever possible

- Behind every pass/fail sits a measured value
- 'Fail' hides the actual leak rate in mL/min
- Record the value, not just the verdict
- Sample size drops by an order of magnitude
- You also gain an early-warning signal

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Operational definitions - two people, one number

1

The test

Two trained people measure the same unit independently and record the same number.

2

Define the measure

'Occlusion alarm delay' means seconds from full occlusion to audible alarm.

3

Define the method

Which rig, which firmware version, which ambient temperature, which rounding.

4

Define the boundary

What is excluded: engineering builds, retest units, validation lots.

5

Define the decision

The exact criterion separating conforming from non-conforming.

6

Pilot it first

Run it with two appraisers. If they disagree, the definition is not finished.

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The data collection plan - the columns that matter

1

What

The exact measure and its units, tied to a written operational definition.

2

Who

The named person or automated system responsible for each observation.

3

When

Frequency, timing and duration - and whether it spans all shifts and all lines.

4

How

Instrument, fixture, system field, check sheet, exact recording method.

5

Sample size

Calculated from a target precision, never chosen for convenience.

6

Stratification

Factors captured alongside each point: shift, line, operator, lot, plant.

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Stratification factors you will wish you had recorded

1. **Plant and line**
   Singapore, Penang and Suzhou rarely behave as one population.

2. **Shift and day of week**
   Night shift and Monday start-up are classic hidden strata.

3. **Operator and appraiser**
   Needed later to separate reproducibility from process variation.

4. **Machine, fixture and tool ID**
   The only way to find a single drifting station.

5. **Component lot and supplier**
   Incoming variation often explains the output variation.

6. **Variant and firmware revision**
   Mixing variants creates artificial bi-modality in the histogram.

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MEASURE - BLACK BELT DEPTH

Sampling is a design decision, not an afterthought

The sampling scheme fixes what questions your data can ever answer. Get it wrong and no amount of clever analysis later can rescue it.

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The four probability sampling schemes

1. **Simple random**
   Every unit has an equal chance. Use a generator, never operator intuition.

2. **Stratified random**
   Split into homogeneous strata, sample within each. Best when strata differ.

3. **Systematic**
   Every Nth unit after a random start. Fails if a process cycle matches your interval.

4. **Cluster**
   Randomly select whole groups - a shift, a lot, a line - and measure all within.

5. **Meridian choice**
   Stratify by plant then by shift: the three plants have different equipment ages.

6. **Never**
   Convenience or judgement sampling. Selection probability is unknown, so bias is unmeasurable.

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Sampling techniques



Only probability-based methods support statistical inference. Convenience and judgement sampling do not.

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Sample size - CONTINUOUS data

Formula	**n = (1.96 s / d)^2**	s = standard deviation, d = acceptable margin of error, 1.96 = 95 percent confidence z-value.
Meridian example	**n = (1.96 x 0.42 / 0.15)^2**	Estimate mean pump flow-rate error within 0.15 mL/h; pilot of 25 units gave s = 0.42 mL/h.
Result	**(5.488)^2 = 30.1 -> n = 31**	Thirty-one pumps. Always round the calculated sample size UP, never down.

Halving the margin of error d quadruples the required sample size. Precision is bought, not wished for.

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Sample size - DISCRETE data

Formula	**n = (1.96/d)^2 x p(1-p)**	p = estimated proportion defective, d = margin of error on that proportion.
Meridian example	**n = (1.96/0.02)^2 x 0.116 x 0.884**	Functional-test failure rate is about 11.6 percent; estimate it within +/- 2 percentage points.
Result	**9604 x 0.1025 = 984.8 -> n = 985**	985 pumps against 31 for the continuous measure. Attribute data is thirty times more expensive.

This single comparison is the strongest argument for converting attribute measures to continuous ones.

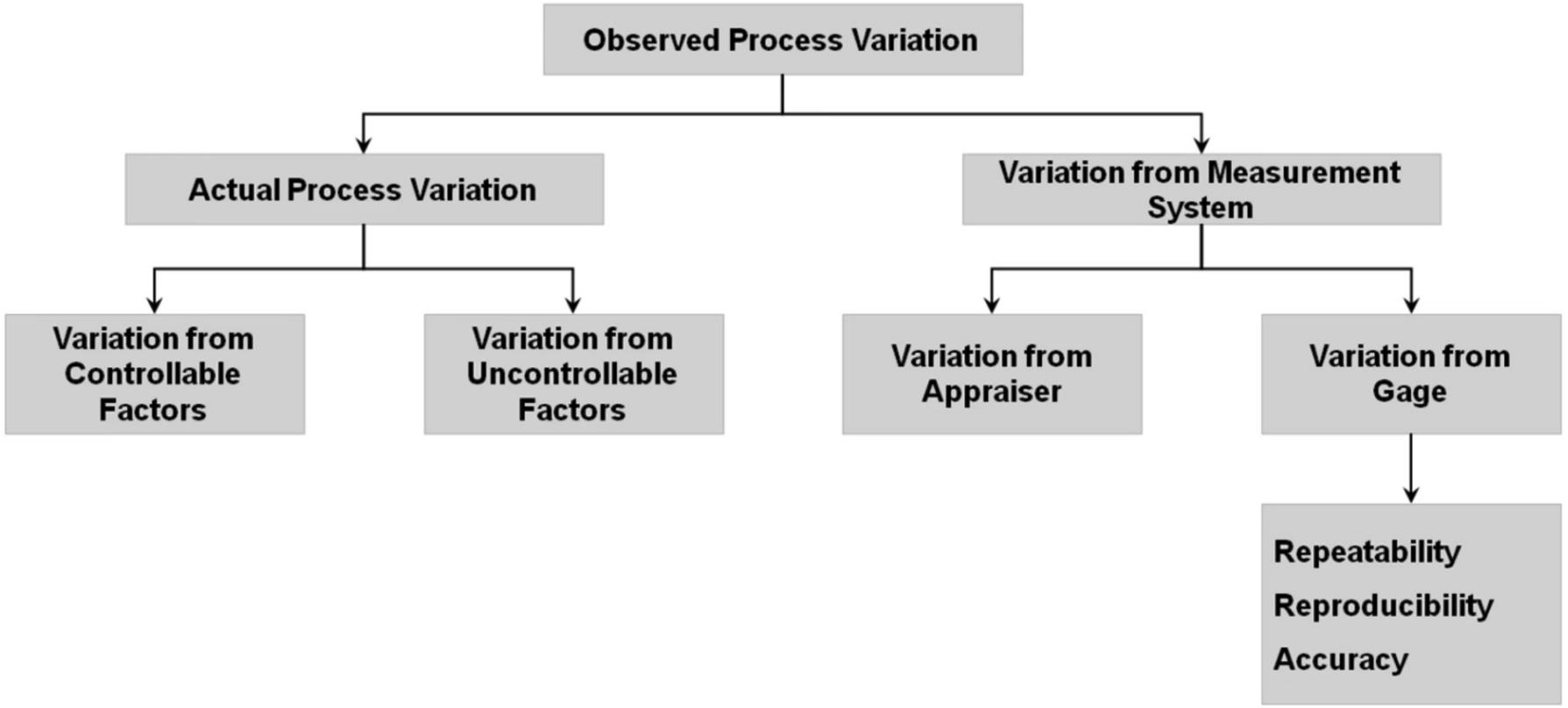
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Recap in depth: continuous Gage R&R

Total observed variation = true process variation + measurement system variation. Until you have split those two, every number you have is a guess with decimal places.

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Sources of variation - the MSA tree



Observed variation splits into true process variation and measurement system variation, which splits again into repeatability and reproducibility.

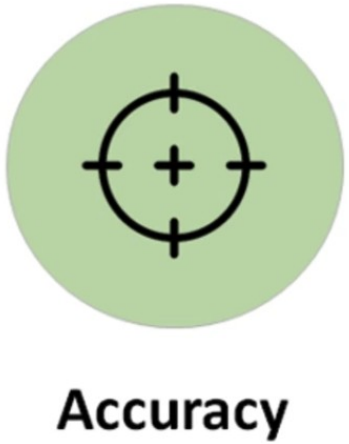
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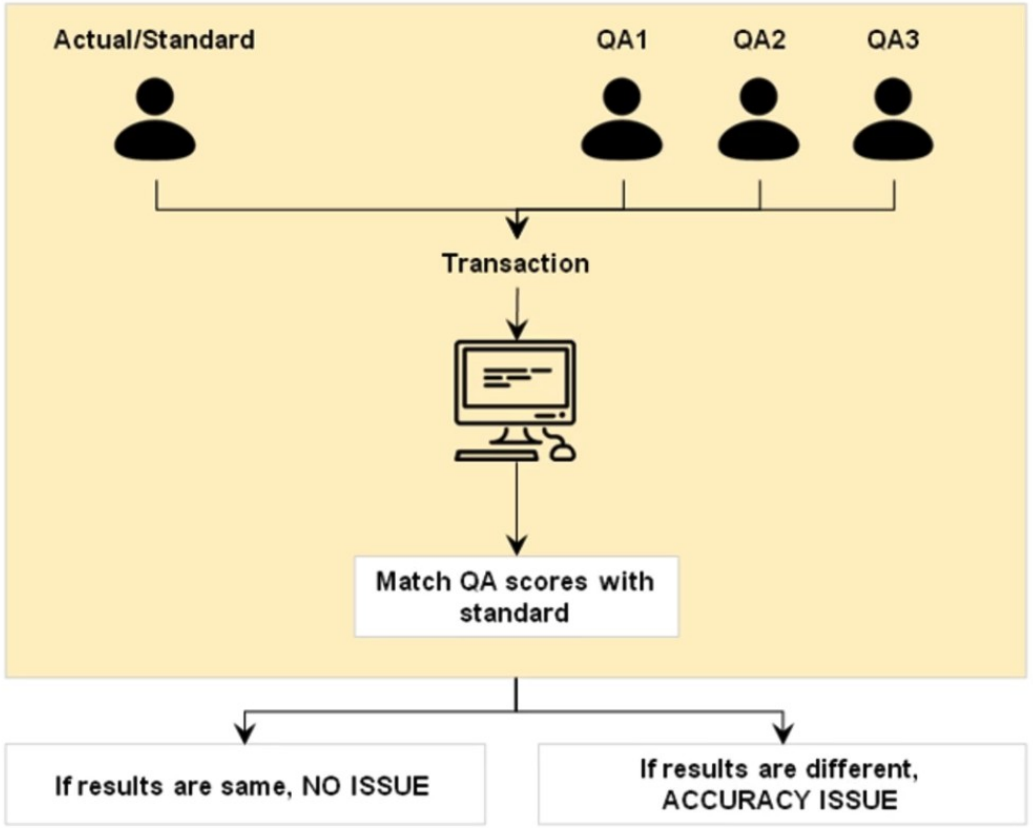
The variation equation

Total observed	**Var(total) = Var(process) + Var(MS)**	Variances add; standard deviations do not. Always work in variance terms.
Measurement system	**Var(MS) = Var(repeat) + Var(reprod)**	Repeatability is the equipment. Reproducibility is the appraiser.
Meridian flow rig	**Var(MS) = 0.0064 + 0.0021**	sigma(MS) = sqrt(0.0085) = 0.092 mL/h against a total sigma of 0.42 mL/h.

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Accuracy, linearity and stability



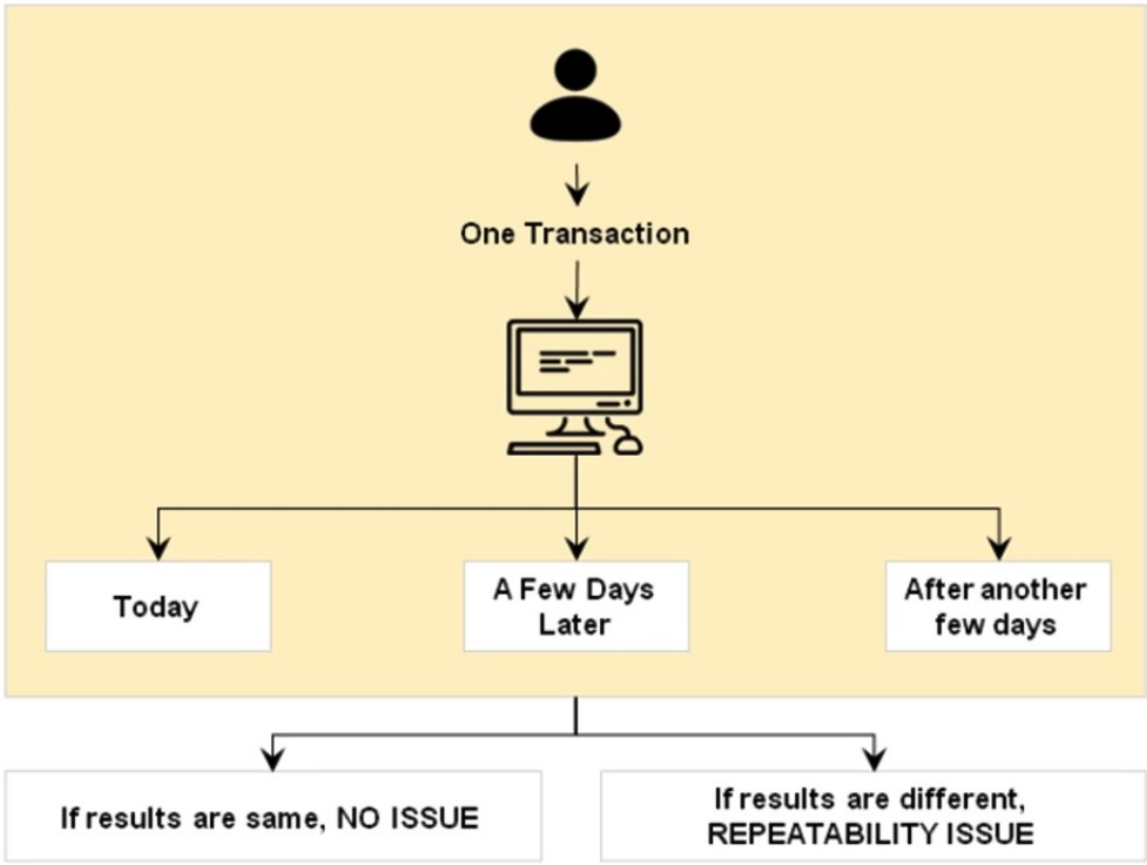


Accuracy is the offset from truth. Linearity is whether the offset changes across the range. Stability is whether it drifts over time.

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Repeatability and reproducibility





Gage R&R quantifies both components, so you know whether to fix the equipment or to retrain and re-standardise the people.

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Repeatability vs Reproducibility

Repeatability

Equipment variation - EV

- SAME appraiser, SAME part, repeated trials
- Variation inherent in the device or fixture
- A poor result means the instrument is the problem
- Fix: better gauge, better fixture, finer resolution
- Meridian: one technician, one pump, three reads

Reproducibility

Appraiser variation - AV

- DIFFERENT appraisers, SAME part, same conditions
- Variation caused by who is doing the measuring
- A poor result means definition or training failed
- Fix: operational definitions, training, standard work
- Meridian: three technicians on the same 10 pumps

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Designing a valid crossed Gage R&R

1. Select 10 parts spanning the FULL range of real process variation
2. Use 3 appraisers who normally perform this measurement
3. Each appraiser measures every part 3 times: 90 readings total
4. Randomise the order and blind the appraisers to prior readings
5. Run ANOVA to partition part, appraiser and interaction variance
6. Report %study variation, %tolerance, %contribution and ndc

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Design rules a Black Belt must enforce

1. **Parts span the real range**
   Ten near-identical parts make any gauge look excellent. This is the top error.

2. **Parts are NOT selected to spec**
   Sample from actual production output, not from a bin of known good units.

3. **Appraisers are the real users**
   Studying engineers proves nothing about how night shift measures.

4. **Randomise and blind**
   Otherwise trial 2 measures memory, not the gauge.

5. **Ten by three by three**
   Fewer parts or trials collapses the degrees of freedom for the interaction term.

6. **ANOVA, not X-bar/R**
   The ANOVA method estimates the appraiser-by-part interaction. Range method cannot.

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Gage R&R metrics - the calculations

% Study Variation	**%SV = 100 x sigma(MS) / sigma(total)**	Meridian: 100 x 0.092 / 0.42 = 21.9 percent. Marginal - usable with caution.
% Tolerance	**%Tol = 100 x 6 sigma(MS) / (USL-LSL)**	Spec is +/- 1.0 mL/h so tolerance = 2.0. 100 x 0.552 / 2.0 = 27.6 percent. Marginal.
Number of distinct categories	**ndc = 1.41 x sigma(part) / sigma(MS)**	sigma(part) = sqrt(0.42^2 - 0.092^2) = 0.410. ndc = 1.41 x 0.410/0.092 = 6.3 -> 6.

%Contribution is variance-based, so its thresholds are the SQUARES of the %SV thresholds: 1 and 9 percent.

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Gage R&R acceptance rules

1

Under 10 percent

GOOD. Accept the measurement system for process improvement work.

2

10 to 30 percent

MARGINAL. Acceptable only by cost, criticality and application judgement.

3

Over 30 percent

UNACCEPTABLE. Do not collect a baseline. Fix the system as a mini-project first.

4

ndc 5 or more

The system resolves at least 5 distinct levels of real part variation.

5

ndc under 5

The gauge cannot distinguish good parts from bad. Effectively a go/no-go device.

6

Meridian verdict

21.9 percent SV with ndc 6: usable for improvement, not for lot release.

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Acting on a failing Gage R&R

Repeatability dominates

The equipment is the problem

- Improve gauge resolution or condition
- Add a fixture to standardise part positioning
- Service, recalibrate or replace the device
- Automate the reading where feasible

Reproducibility dominates

People or method are the problem

- Rewrite the operational definition first
- Retrain every appraiser to one written standard
- Add master reference samples and visual aids
- Create standard work for the measurement itself

Interaction is significant

Appraisers differ by part type

- One appraiser struggles only with small parts
- Usually a fixture or a visibility problem
- Look at the appraiser-by-part interaction plot
- Fix the physical setup, not the person

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MEASURE - BLACK BELT DEPTH

BLACK BELT ONLY: when the measurement is a judgement

Most inspection in a medical device plant is a human deciding pass or fail. Gage R&R computes variances - and a pass/fail verdict has no variance to compute.

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Why continuous Gage R&R fails on judgement data

1

No continuous scale

Pass/fail has no mean and no standard deviation to partition.

2

ANOVA needs variance

The entire R&R method rests on decomposing variance. There is none here.

3

Repeat reads are identical

An appraiser almost always repeats their own verdict - falsely reassuring.

4

The real risk is different

Not noise around a value, but systematically calling bad units good.

5

Right tool: agreement

Attribute agreement analysis counts matches, then corrects for chance.

6

Meridian relevance

Cosmetic grading, solder-joint judgement and label legibility are all judgement calls.

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Continuous MSA vs Attribute MSA

Continuous Gage R&R

For measured values

- Data: a number on a real scale
- Method: ANOVA variance decomposition
- Outputs: %SV, %Tolerance, %Contribution, ndc
- Design: 10 parts x 3 appraisers x 3 trials
- Meridian: pump flow rate in mL/h

Attribute agreement

For judgements and grades

- Data: a category - pass/fail, grade A to D
- Method: agreement counting plus Kappa
- Outputs: % agreement and Kappa statistics
- Design: 30 to 50 parts x 3 appraisers x 2-3 trials
- Meridian: solder joint accept or reject

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Running an attribute agreement analysis

1. Select 30 to 50 parts, deliberately including borderline cases
2. Establish the true standard for every part via an expert panel
3. Label parts with opaque random codes, no grouping or sequence
4. Each of 3 appraisers assesses every part 2 to 3 times
5. Fully randomise the order between every trial
6. Compute within, between and versus-standard agreement plus Kappa

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The three agreement questions

1

Within-appraiser

Does one appraiser agree with THEMSELVES across repeated trials? Repeatability.

2

Between-appraiser

Do the appraisers agree with EACH OTHER? Reproducibility.

3

Versus the standard

Does each appraiser agree with the KNOWN TRUTH? Accuracy - the one that matters.

4

All appraisers vs standard

Does the whole inspection team, collectively, get it right?

5

The dangerous case

Perfect within and between agreement, poor versus standard: consistently wrong.

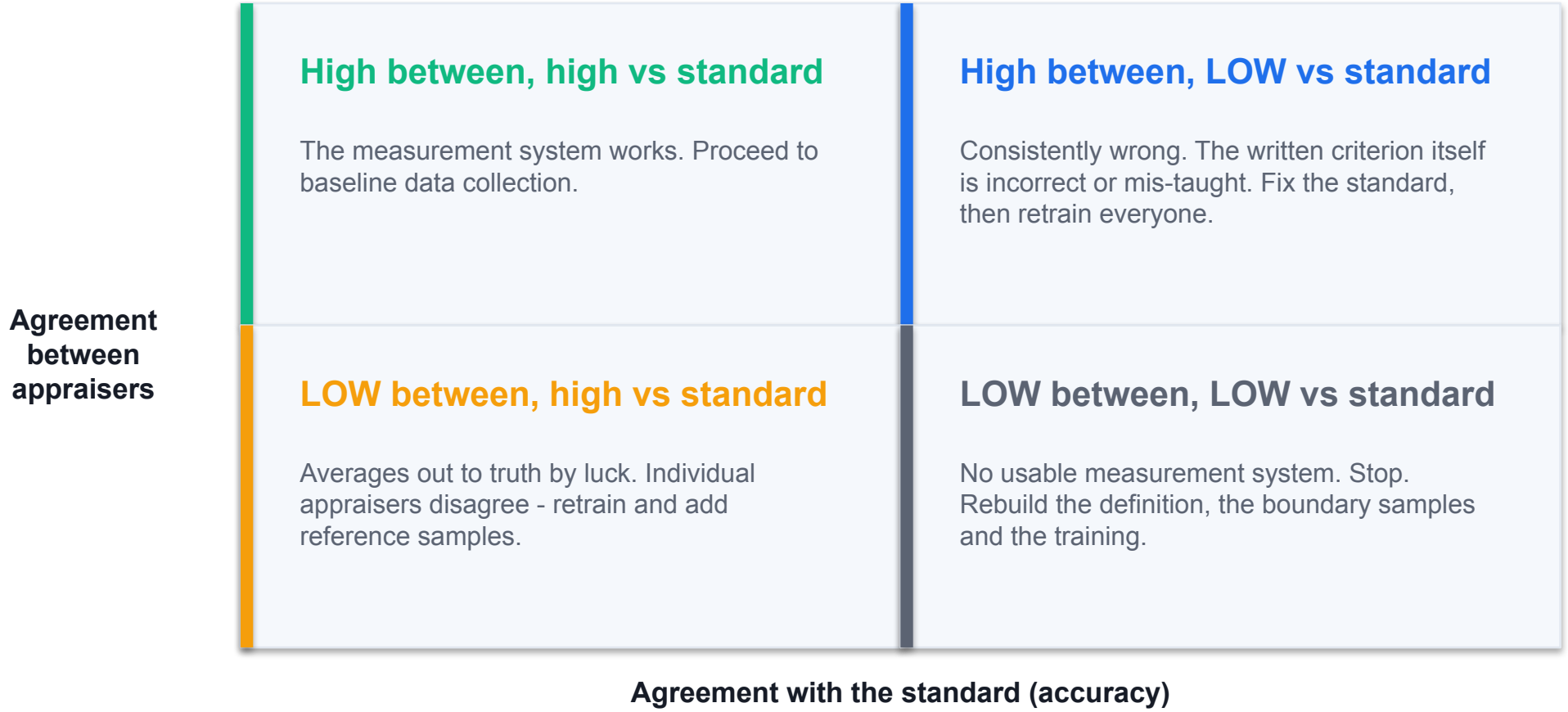
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Meridian example

Three solder inspectors agreed 96 percent with each other, 78 percent with the standard.

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Diagnosing an attribute MSA result



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Kappa: agreement after chance is removed

Two inspectors flipping coins on a pass/fail decision agree half the time. Raw agreement rewards them for that. Kappa does not.

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Cohen's Kappa - the chance correction

Kappa	**K = (Po - Pe) / (1 - Pe)**	Po = observed proportion agreeing. Pe = proportion expected to agree by chance alone.
Numerator	**Po - Pe**	Agreement actually achieved ABOVE what random guessing would deliver.
Denominator	**1 - Pe**	The maximum agreement available above chance. So Kappa is the fraction of that maximum earned.

K = 1 is perfect agreement. K = 0 is exactly chance. K below 0 means worse than random guessing.

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Meridian worked example - why 90 percent can be a poor Kappa

The data	**100 solder joints, 2 inspectors**	Both said PASS on 88. Both said FAIL on 2. They disagreed on 10 joints.
Observed agreement	**Po = (88 + 2)/100 = 0.90**	Ninety percent raw agreement. This is the number a manager will quote in a meeting.
Expected by chance	**Pe = (0.93x0.95) + (0.07x0.05)**	Inspector A passed 93, B passed 95. Pe = 0.8835 + 0.0035 = 0.887.

Now compute Kappa: K = (0.90 - 0.887) / (1 - 0.887) = 0.013 / 0.113 = 0.115. Effectively no real agreement.

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Why the Kappa collapsed at 90 percent agreement

1

The base rate is extreme

93 to 95 percent of joints pass, so chance agreement alone is already 88.7 percent.

2

The 90 percent is nearly free

Two inspectors who ALWAYS say pass would score 88 percent agreement.

3

Kappa exposes it

0.115 says the inspectors add almost nothing beyond a default 'pass' reflex.

4

Where they disagree matters

All 10 disagreements sit on the defective joints - the only ones that count.

5

The lesson

Never report raw agreement alone on a rare-event judgement. Always report Kappa.

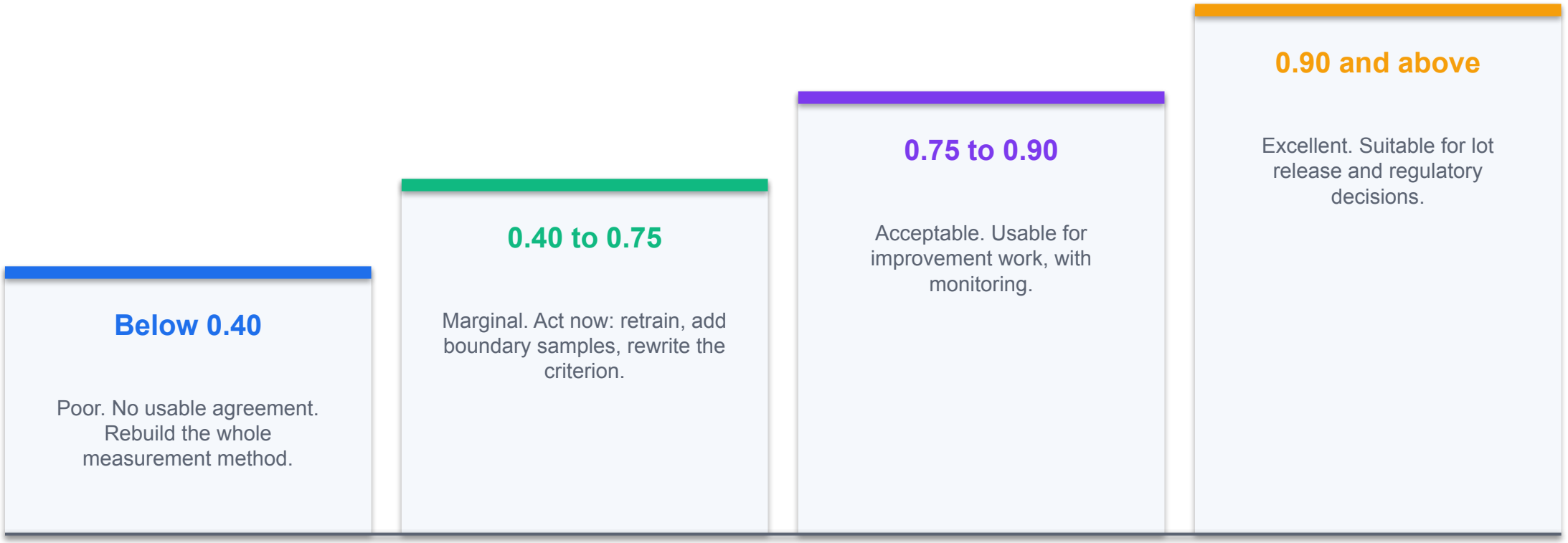
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The fix

Rebalance the study sample and retrain against physical boundary specimens.

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Kappa acceptance criteria



Six Sigma practice: below 0.75 you must act. At or above 0.90 the inspection system is genuinely reliable.

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Cohen's Kappa vs Fleiss' Kappa

Cohen's Kappa

Exactly TWO raters

- Pairwise agreement between two appraisers
- Or one appraiser against the known standard
- Or one appraiser against themselves, trial 1 vs 2
- With 3 appraisers you get 3 pairwise Kappas
- Meridian: inspector A vs inspector B

Fleiss' Kappa

THREE OR MORE raters

- One overall statistic across the whole panel
- Handles a different rater set per item if needed
- Same 0 to 1 scale and same acceptance thresholds
- Reports team agreement, not pairwise agreement
- Meridian: all 3 solder inspectors together

Weighted Kappa

For ORDERED categories

- Used when categories have a natural order
- Grade A vs B disagreement penalised less than A vs D
- Essential for cosmetic grading scales
- Unweighted Kappa treats all errors as equal
- Meridian: enclosure cosmetic grade A to D

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Attribute MSA traps that a Black Belt must catch

- Reporting raw percent agreement with no Kappa - the single most common and most misleading error.
- A study sample with 95 percent good units, which lets a lazy 'always pass' score above 90 percent.
- Omitting the known standard, so you can prove consistency but never prove correctness.
- Using only 20 parts, giving a Kappa confidence interval so wide the point estimate is meaningless.
- Testing the day-shift engineers when the real inspection happens on night shift with different lighting.
- Accepting Kappa 0.72 as 'close enough to 0.75' when the decision releases implanted-device components.

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MEASURE - BLACK BELT DEPTH

BLACK BELT ONLY: when measuring destroys the part

Leak testing to burst, peel strength, sterility, weld pull force. The part does not survive, so no appraiser can ever measure the same part twice. The crossed design is mathematically impossible.

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Crossed vs Nested gauge study designs

Crossed design

The standard Gage R&R

- EVERY appraiser measures EVERY part
- Requires the part to survive measurement
- Estimates part, appraiser AND their interaction
- 10 parts x 3 appraisers x 3 trials
- Meridian: pump flow rate, non-destructive

Nested design

For destructive measurement

- Each part is measured by ONE appraiser, ONCE
- Parts are NESTED within appraiser or within batch
- Cannot estimate the appraiser-by-part interaction
- Relies on homogeneous batches as pseudo-repeats
- Meridian: seal peel strength, destructive

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The homogeneous batch assumption

1

The core idea

If parts within a batch are truly identical, different parts act as repeat measures.

2

How to build one

Consecutive units, one lot, one machine, one operator, one short time window.

3

The critical risk

Any real within-batch variation is counted as measurement repeatability.

4

Consequence

Repeatability is CONFOUNDED with within-batch part variation and always overstated.

5

So the result is conservative

A nested study that passes is trustworthy. One that fails may be a false alarm.

6

Meridian batches

Ten seal pouches from one sealer, one reel, one 5-minute window, per batch.

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Designing a nested destructive study at Meridian

1. Form 30 homogeneous batches of 3 sterile-barrier pouches each
2. Confirm homogeneity: same sealer, reel, settings, 5-minute window
3. Assign each batch to ONE appraiser at random
4. That appraiser peel-tests all 3 pouches from their batches
5. Run a NESTED ANOVA: part within batch, batch within appraiser
6. Report %study variation, remembering repeatability is inflated

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Nested study - Meridian seal peel strength

The design	**30 batches x 3 pouches, nested**	Ten batches per appraiser, three appraisers, 90 destructive peel tests in total.
Repeatability estimate	**sigma(rep) = 0.31 N**	This is equipment noise PLUS true within-batch pouch variation. It cannot be separated.
% Study variation	**100 x 0.34 / 1.08 = 31.5%**	sigma(MS) = 0.34 N against total 1.08 N. Above 30 percent - but the estimate is inflated.

A nested result above 30 percent needs investigation of batch homogeneity BEFORE condemning the gauge.

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Other situations that force a nested design

1. **Consumed chemical assays**
   The aliquot is gone once it has been analysed. No second read exists.

2. **Sterility and bioburden**
   The test article is destroyed by the test procedure itself.

3. **Pull-force testing**
   Weld, crimp and solder joints are broken in order to obtain the reading.

4. **Life and burst testing**
   The unit is deliberately taken to failure to get the number.

5. **Altering measurements**
   Any test that meaningfully changes the part, even if it survives.

6. **The rule of thumb**
   If measuring twice would not give the same part, you need a nested design.

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Recap in depth: yield, defects and sigma level

Four different yield numbers can be computed from the same process. Three of them flatter you. A Black Belt insists on the fourth.

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DUDO - agree these four before computing anything

1

D - Defect

Any single failure to meet a customer requirement - one CTQ unmet on one unit.

2

U - Unit

The deliverable the customer receives. At Meridian: one infusion pump.

3

D - Defect opportunity

Every distinct way one unit could fail. Count only realistic, inspected modes.

4

O - Observed defects

The actual defect count found in the sample you inspected.

5

Why opportunities matter

DPMO normalises complexity so a simple product compares fairly with a complex one.

6

Be conservative

Inflating the opportunity count flatters your sigma level. Auditors always check this.

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Four ways to measure yield

Classic yield

Units out / units in

- Counts anything that eventually shipped
- Rework and retest are completely invisible
- Always the most flattering number available
- Meridian pump line: 98.2 percent

First pass yield

Right first time at ONE step

- Units through with no rework or repair
- Computed step by step across the line
- Exposes rework at each individual station
- Meridian assembly step: 88.4 percent

Rolled throughput yield

Right first time ACROSS all steps

- RTY = FPY1 x FPY2 x ... x FPYn
- Probability a unit passes every step cleanly
- Falls fast as the step count rises
- Meridian pump line: 71.6 percent

Normalised yield

The per-step average

- Ynorm = RTY^(1/n) for n steps
- The typical yield of an average step
- Useful for comparing lines of different lengths
- Meridian: 0.716^(1/6) = 94.6 percent

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Rolled throughput yield - Meridian Suzhou line 4

RTY formula	**RTY = FPY1 x FPY2 x ... x FPYn**	Multiply the first pass yield of every step in the chain.
Six steps	**0.982 x 0.951 x 0.884 x 0.974 x 0.962 x 0.991**	Kitting, sub-assembly, main assembly, calibration, functional test, final pack.
Result	**RTY = 0.716**	71.6 percent, against a classic yield of 98.2 percent. The 26.6 point gap is the hidden factory.

Each step looks respectable. Only the product of them reveals what the customer experience really is.

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DPU, DPO and DPMO

DPU	**DPU = Defects / Units**	Average number of defects carried by each unit produced.
DPO	**DPO = Defects / (Units x Opportunities)**	Defects per single opportunity, normalised for product complexity.
DPMO	**DPMO = DPO x 1,000,000**	Scaled to a million opportunities so any two processes can be compared directly.

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DPMO - Meridian pump baseline

Units and defects	**3,600 pumps, 486 defects**	One month of Suzhou line 4 output, with all defect records from every station.
DPU and DPO	**DPU = 486/3600 = 0.135**	With 14 inspected opportunities per pump: DPO = 486 / (3600 x 14) = 0.00964.
DPMO and sigma	**DPMO = 9,643**	Between 4 sigma (6,210) and 3.5 sigma (22,750). Meridian sits at about 3.84 sigma.

The 1.5 sigma shift convention is built into these DPMO tables. State that assumption whenever you report sigma level.

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The sigma scale - DPMO at each level



Each level is roughly a ten-fold defect reduction. Meridian at 3.84 sigma has two full levels to climb.

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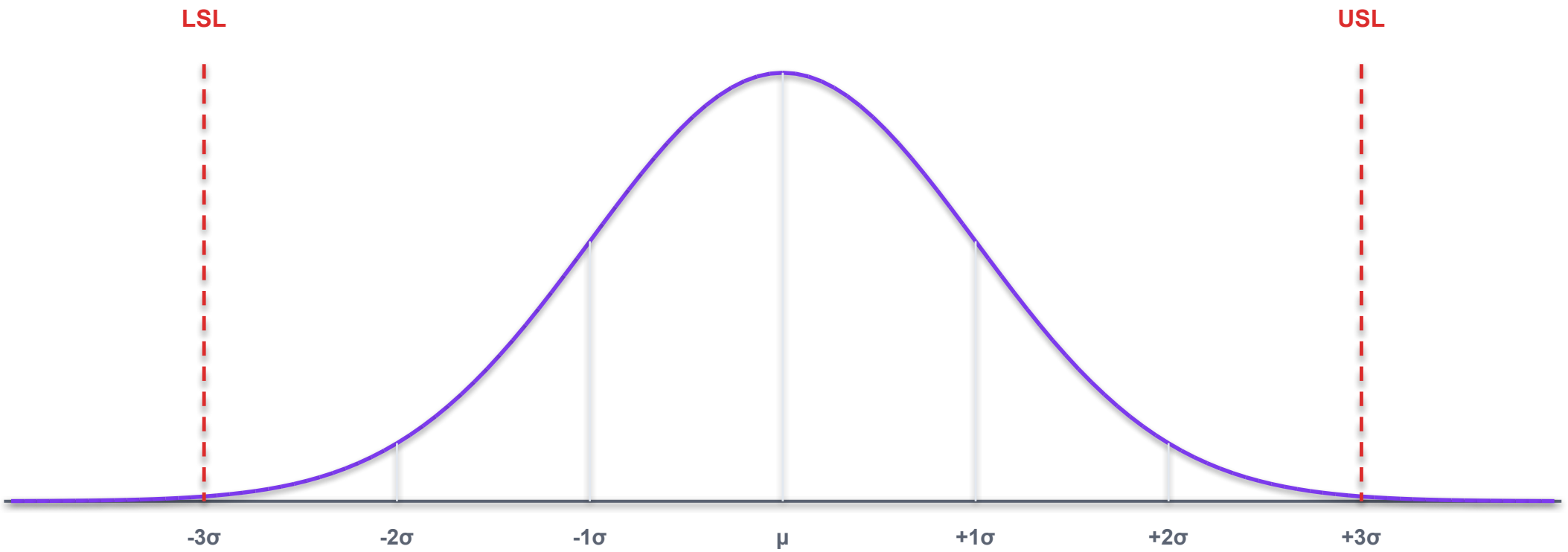
MEASURE - BLACK BELT DEPTH

BLACK BELT ONLY: capability when the data is not normal

Cp, Cpk, Pp and Ppk all assume a normal distribution. Cycle times, leak rates and impurity levels almost never are. Running the index anyway produces a confident, wrong number.

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The normal distribution and the empirical rule



68.26 percent within +/-1 sigma, 95.46 percent within +/-2 sigma, 99.73 percent within +/-3 sigma. Capability indices depend entirely on this.

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Why non-normality wrecks a capability index

1

Cpk is a tail probability

It converts a distance in sigma into a defect rate using normal tails.

2

Skewed data has fatter tails

A right-skewed process produces far more high-side defects than Cpk predicts.

3

The error is not small

A Cpk of 1.33 on lognormal data can hide a real defect rate ten times higher.

4

Bounded data is never normal

Leak rate, impurity and cycle time are bounded at zero and skew right.

5

Software will not warn you

It computes Cpk on any column of numbers you give it, silently.

6

Black Belt discipline

Test normality BEFORE computing any capability index. Every single time.

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The non-normal capability decision path

1. Plot the data: histogram, probability plot, and a time-ordered run chart
2. Confirm stability first - capability on an unstable process is meaningless
3. Run Anderson-Darling: p below 0.05 rejects normality
4. Try a transformation : Box-Cox first, then Johnson if Box-Cox fails
5. If no transform works, fit the correct distribution: Weibull, lognormal, exponential
6. Compute capability in that distribution and report in ORIGINAL units

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Anderson-Darling normality test

The hypotheses	**H0: data is normal**	H1: data is not normal. AD is preferred because it weights the TAILS, which drive capability.
The decision rule	**p < 0.05 -> reject normality**	p at or above 0.05 means you have no evidence against normal. Not proof that it IS normal.
Meridian example	**AD = 1.84, p = 0.002**	Pump leak-rate data on 120 units. Normality firmly rejected - do not compute Cpk on this raw data.

With very large n, AD rejects trivial departures. With small n it lacks power. Always look at the probability plot too.

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Three routes for non-normal capability

Box-Cox transform

A power transformation family

- Y' = (Y^lambda - 1)/lambda, lambda estimated from data
- lambda = 0 is the natural log; 0.5 is the square root
- Requires strictly positive data
- Simple, widely accepted, easy to explain
- Meridian leak rate: lambda = 0 works well

Johnson transform

A flexible three-family system

- Johnson SB, SL and SU cover bounded and unbounded
- Succeeds on many datasets where Box-Cox fails
- Handles negative values and bounded ranges
- Harder to explain to a steering committee
- Use when Box-Cox residuals still fail AD

Fit the distribution

Model the physics instead

- Weibull for life, strength and time-to-failure
- Lognormal for multiplicative, right-skewed data
- Exponential for constant-hazard arrival times
- No transformation, so no back-transform needed
- Preferred when theory suggests the distribution

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Box-Cox on Meridian pump leak rate

Raw data	**AD p = 0.002, right skewed**	120 pumps, leak rate in mL/min, USL 0.50, bounded below at zero. Clearly non-normal.
Transform	**lambda = 0, so Y' = ln(Y)**	Box-Cox estimates lambda near zero, giving the natural log. AD on ln(Y) now gives p = 0.41.
Capability	**Ppk computed on ln(Y)**	USL transformed to ln(0.50) = -0.693. Ppk = 0.94 - back-report as 0.94 in original terms.

Always transform the SPEC LIMITS too, and always present the final result in the customer's original units.

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The percentile method - capability without a transform

- Normal capability compares spec limits against the 0.135 and 99.865 percentiles of the fitted normal.
- The percentile method uses the same two percentiles taken from the FITTED non-normal distribution.
- Ppu = (USL - median) / (P99.865 - median) and Ppl = (median - LSL) / (median - P0.135).
- Ppk is then the smaller of Ppu and Ppl, exactly as in the normal case.
- The advantage: the result is already in original units, with no back-transformation needed.
- Minitab calls this the ISO / Z-score method; state clearly which one you used in the report.

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MEASURE - BLACK BELT DEPTH

BLACK BELT ONLY: Cp, Cpk, Pp and Ppk are four different claims

The four indices differ in one thing only: which sigma goes in the denominator. That single choice decides whether you are describing a good day or a real year.

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The capability index family

Which sigma is used

Cp - ignores centring

Uses WITHIN-subgroup sigma (short-term). Potential capability: what the process could do if perfectly centred.

Cpk - accounts for centring

Uses WITHIN-subgroup sigma (short-term). Actual short-term capability, the smaller of the two one-sided indices.

Pp - ignores centring

Uses TOTAL sigma (long-term). Overall potential performance across everything the process actually did.

Ppk - accounts for centring

Uses TOTAL sigma (long-term). The honest number: what the customer actually received over the period.

Does the index account for centring?

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The four indices

Cp and Pp	**(USL - LSL) / 6 sigma**	Cp uses sigma-within (from R-bar/d2). Pp uses sigma-total (the ordinary sample sigma).
Cpk	**min[(USL-Xbar), (Xbar-LSL)] / 3 s(within)**	The worse of the two sides, so an off-centre process is correctly penalised.
Ppk	**min[(USL-Xbar), (Xbar-LSL)] / 3 s(total)**	Same structure, but the denominator includes ALL the between-subgroup variation.

sigma-within is estimated from within-subgroup ranges. sigma-total is the plain standard deviation of all data.

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Meridian worked example - the Cpk / Ppk gap

The process	**Pump flow accuracy, USL +1.0, LSL -1.0**	60 subgroups of 5 units over 12 weeks. Overall mean = +0.14 mL/h.
Short term	**s(within) = 0.19 -> Cpk = 1.51**	Cpk = min(1.0-0.14, 0.14+1.0) / (3 x 0.19) = 0.86 / 0.57 = 1.51. Looks comfortably capable.
Long term	**s(total) = 0.42 -> Ppk = 0.68**	Ppk = 0.86 / (3 x 0.42) = 0.68. The same process, honestly reported, is NOT capable.

A Cpk of 1.51 with a Ppk of 0.68 does not mean the gauge is wrong. It means the process is drifting between subgroups.

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Reading the Cpk to Ppk gap

1

Cpk approximately Ppk

The process is stable. Within-subgroup and total variation agree. Trust both.

2

Cpk much greater than Ppk

Significant between-subgroup drift, shift or special cause. Cpk is flattering you.

3

Meridian: 1.51 vs 0.68

A large gap: shift-to-shift and lot-to-lot drift the control chart must catch.

4

The ratio is a diagnostic

Cpk/Ppk is roughly a stability index. Far above 1 means unstable.

5

Never report Cpk alone

On an unstable process Cpk describes a good hour, not the customer's year.

6

What the customer got

Ppk. Always Ppk. That is the number that maps to real received defect rates.

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When to report which index

Report Cp / Cpk

Short-term, stable, controlled

- The process is demonstrably in statistical control
- Subgroups are rationally formed
- Machine capability studies over a short run
- Process validation runs under fixed conditions
- State that it is short-term capability

Report Pp / Ppk

Long-term, real performance

- Baseline capability in the Measure phase
- Any process not yet proven stable
- Customer or regulator performance reporting
- Post-improvement verification in Control
- This is the default at Meridian

Report BOTH

The Black Belt standard

- The gap between them is itself the finding
- It quantifies the opportunity from stabilising alone
- It prevents the accusation of cherry-picking
- It tells the sponsor where to invest first
- Always with the sample period stated

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MEASURE - BLACK BELT DEPTH

BLACK BELT ONLY: rational subgrouping

How you group the data decides what variation lands inside a subgroup and what lands between them. That single choice determines what your control chart is capable of ever detecting.

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The rational subgrouping principle

1

Within-subgroup

Should contain ONLY common-cause noise. This becomes the yardstick for the limits.

2

Between-subgroup

Is what the chart tests against that yardstick. This is what you want to detect.

3

The design question

What source of variation do I want this chart to catch? Put it BETWEEN subgroups.

4

Consecutive units

The classic rational subgroup: 5 consecutive pumps, minimising within-subgroup spread.

5

Control limits follow

Limits are computed from within-subgroup variation, so wide subgroups mean wide limits.

6

Get it wrong

Wide limits that flag nothing, or tight limits that flag everything. Both useless.

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Two subgrouping strategies, same data

Consecutive sampling

5 units back to back

- Within-subgroup captures only machine noise
- Shift and lot differences land BETWEEN subgroups
- Control limits are tight
- The chart DETECTS shift and lot changes
- This is what you almost always want

Spread sampling

1 unit per hour over 5 hours

- Within-subgroup absorbs the drift across the shift
- Drift is now inside, not between, subgroups
- Control limits become wide
- The chart CANNOT detect the drift it swallowed
- A stable-looking chart on a drifting process

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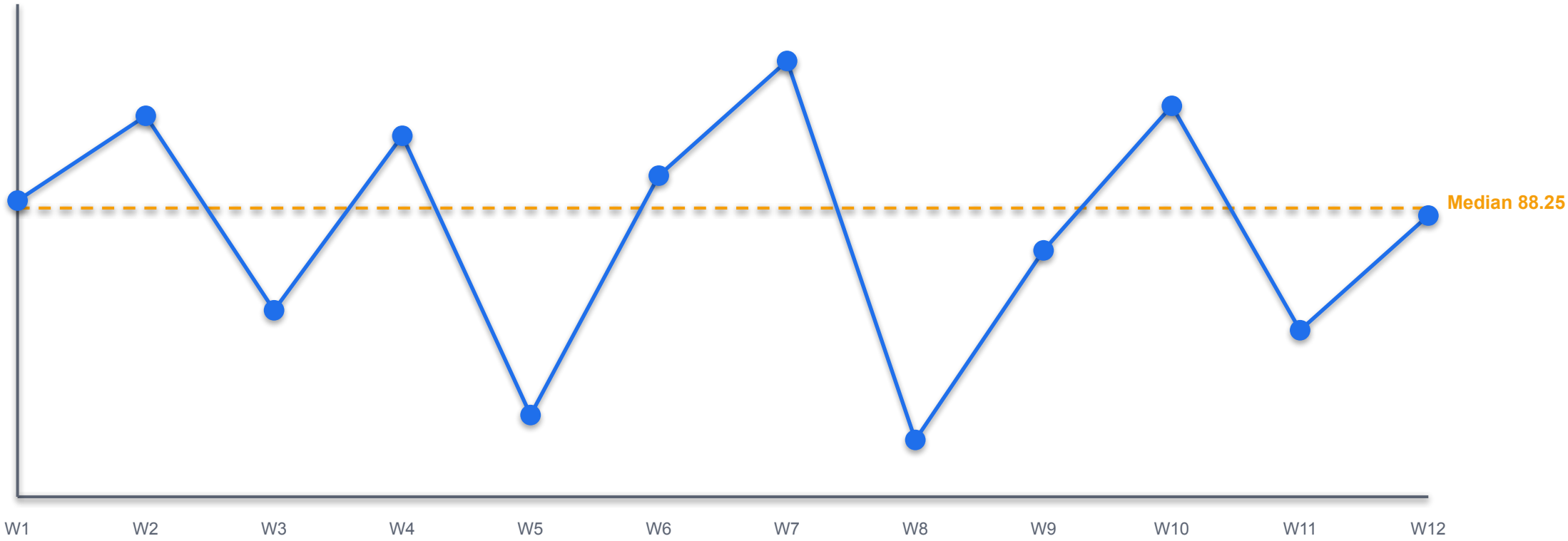
Meridian subgrouping - the same 300 pumps, two ways

Consecutive design	**5 consecutive pumps, hourly**	s(within) = 0.19. Control limits Xbar +/- 0.25. Shift changes flag clearly on the chart.
Spread design	**1 pump every 12 min for an hour**	s(within) = 0.38. Control limits Xbar +/- 0.51. The same shift changes now sit inside the limits.
The consequence	**Same process, opposite verdict**	Design one says 14 out-of-control signals in 12 weeks. Design two says the process is perfectly stable.

Nothing about the process changed. Only the subgrouping. This is why the Black Belt owns this decision.

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Meridian Suzhou line 4 - weekly first pass yield



Common-cause variation around a stable median. The process is consistently at 88 percent, not occasionally. That means a system fix, not a people fix.

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The Measure tollgate evidence pack

1. **Validated process map and VSM**
   Walked, verified on a second shift, with the timeline ladder loaded.

2. **Data collection plan**
   Operational definitions, sampling scheme, stratification factors, sample sizes.

3. **MSA evidence**
   The RIGHT study for the data type, with acceptance criteria met or a fix plan attached.

4. **Distribution evidence**
   Normality test result and the transformation or fitted distribution used.

5. **Baseline performance**
   RTY, DPMO, sigma level, and Ppk with Cpk alongside for the stability gap.

6. **Refined problem statement**
   The Define scope narrowed to where the data proves the pain sits.

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Black Belt Measure failure modes

1. **Gage R&R on judgement data**
   Running a variance study on pass/fail data. Use attribute agreement and Kappa.

2. **Raw agreement with no Kappa**
   90 percent agreement on a rare-event call can hide a Kappa near zero.

3. **Crossed design on destructive tests**
   Mathematically impossible. It means the study was never really run.

4. **Cpk on non-normal data**
   A confident index computed on an assumption that was never tested.

5. **Cpk on an unstable process**
   Reporting a good hour as if it were the customer's year. Report Ppk.

6. **Default subgrouping**
   Accepting whatever the software offers, then wondering why nothing ever flags.

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Measure phase - Black Belt checkpoint

- Two inspectors agree on 90 of 100 parts. Why might their Kappa still be near zero?
- Your peel test destroys the pouch. Design a valid MSA and state what you cannot estimate.
- Anderson-Darling gives p = 0.002 on your leak-rate data. State your next three actions.
- Cpk is 1.51 and Ppk is 0.68 on the same data. What does that gap tell you, and what do you report?
- You want a chart that detects shift-to-shift differences. How must you form the subgroups, and why?
- Compute Meridian's DPMO from 3,600 pumps, 486 defects and 14 opportunities, and give the sigma level.

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WHAT YOU'LL DO

Hands-On Labs — Measure (1)

1. **Lab 8 — Value Stream Mapping, Takt Time and the Eight Wastes — In Depth**
   A current-state VSM with a value-added ratio, takt time and a quantified waste log. • Data: lab-08-value-stream-map.xlsx

2. **Lab 9 — Data Types, Operational Definitions and the Data Collection Plan**
   A data collection plan with operational definitions for every metric in your… • Data: lab-09-data-collection-raw-extract.xlsx

3. **Lab 10 — Sampling Strategy, Sample Size and Rational Subgrouping**
   Calculated sample sizes for your capstone metrics and a rational subgrouping scheme. • Data: lab-10-sampling-frame.xlsx

4. **Lab 11 — Continuous Gage R&R — In Depth**
   A completed crossed Gage R&R study with %study variation and ndc for your capstone… • Data: lab-11-gage-rr-continuous.xlsx

5. **Lab 12 — Attribute MSA and Kappa Analysis (Black Belt Only)**
   An attribute agreement study with within-, between- and versus-standard Kappa… • Data: lab-12-attribute-msa-kappa.xlsx

6. **Lab 13 — Nested and Destructive Gauge Studies (Black Belt Only)**
   A nested gauge study design and analysis for a destructive or single-measurement… • Data: lab-13-nested-destructive-gauge.xlsx

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WHAT YOU'LL DO

Hands-On Labs — Measure (2)

1

Lab 14 — Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

A baseline capability study for your capstone Y with a documented normality… • Data: lab-14-baseline-capability.xlsx

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DMAIC · MEASURE · HANDS-ON

Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

LAB 8

A full re-derivation of value stream mapping — material AND information flow, value-added ratio, takt time and the eight wastes — then applied to your own capstone process to expose where lead time is actually consumed. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A current-state VSM with a value-added ratio, takt time and a quantified waste log.

Tools & techniques: VSM icons, material & information flow, lead time ladder, takt time, VA/NVA ratio, eight wastes

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Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

STEPS 1-5 OF 10

1. Recap the VSM in full: what it shows that a process map does not — inventory, information flow, lead time and the push/pull signal.
2. Recap the standard VSM icons: process box, data box, inventory triangle, push arrow, supermarket, truck, information flow and the timeline ladder.
3. Map the current state of your capstone value stream from customer order to delivery. Include the information flow that triggers each step.
4. Populate each data box: cycle time, changeover time, uptime, batch size, number of operators and first-pass yield.
5. Draw the timeline ladder underneath. Put value-added time on the lower rung and waiting/queue time on the upper rung.

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Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

STEPS 6-10 OF 10

6. Calculate total lead time, total value-added time, and the value-added ratio (VA time / lead time). Most processes land under 5 percent.

7. Calculate takt time = available production time / customer demand for the period. Compare takt against each step's cycle time to find the bottleneck.

8. Recap the eight wastes in full (DOWNTIME: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra processing).

9. Walk your value stream and log every instance of each waste. Quantify each in time or money — an unquantified waste will not win funding.

10. Mark the three points on the VSM where the largest waste concentrates. These become improvement candidates for Day 4.

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LAB 8 · VERIFY

Value Stream Mapping, Takt Time and the Eight Wastes — In Depth

✅ Check your work

Your VSM shows both material and information flow, you have calculated the value-added ratio and takt time from real numbers, and every logged waste carries a time or dollar quantification.

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Data Types, Operational Definitions and the Data Collection Plan

LAB 9

Re-derive data classification and operational definitions in full, then build the data collection plan that will feed every statistical test in Days 3 and 4. Bad data collected now cannot be rescued by clever statistics later. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A data collection plan with operational definitions for every metric in your capstone.

Tools & techniques: Continuous vs discrete, nominal/ordinal/interval/ratio, operational definitions, data collection plan, stratification factors

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Data Types, Operational Definitions and the Data Collection Plan

STEPS 1-5 OF 10

1. Recap the data hierarchy in full: nominal, ordinal, interval and ratio — and why the type determines which statistical test is legal.
2. Recap continuous versus discrete (attribute) data, and why continuous data needs far smaller samples to reach the same confidence.
3. List every metric your capstone requires: the project Y, each sub-Y from your flowdown, and each candidate X you intend to test.
4. Classify each metric by data type. Flag any metric currently collected as attribute that could be captured as continuous — convert it if you can.
5. Write a complete operational definition for each metric: what is measured, the instrument, the unit, the sampling point, the collector and the defect criterion.

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Data Types, Operational Definitions and the Data Collection Plan

STEPS 6-10 OF 10

6. Test each operational definition by having two people apply it independently to the same item. If they disagree, the definition is not yet operational.

7. Identify your stratification factors — plant, line, shift, operator, product family, supplier. You must capture these at collection time or you can never stratify later.

8. Build the data collection plan table: metric, definition, type, source, sample size, frequency, collector, stratification fields and start date.

9. Design the check sheet or data form. Make the easy path the correct path, or collectors will improvise.

10. Dry-run the plan on a small batch and fix whatever breaks before committing to full collection.

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LAB 9 · VERIFY

Data Types, Operational Definitions and the Data Collection Plan

✅ Check your work

Every capstone metric has a written operational definition that two people apply identically, stratification fields are captured at source, and the plan has survived a dry run.

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DMAIC · MEASURE · HANDS-ON

Sampling Strategy, Sample Size and Rational Subgrouping

LAB 10

Re-derive sampling schemes and sample size formulas in full, then go beyond Green Belt with RATIONAL SUBGROUPING — the decision that silently determines what your control charts in Day 4 will be capable of detecting at all. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

Calculated sample sizes for your capstone metrics and a rational subgrouping scheme.

Tools & techniques: Random/stratified/systematic/cluster sampling, sample size for continuous & discrete data, rational subgrouping, sampling bias

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Sampling Strategy, Sample Size and Rational Subgrouping

STEPS 1-5 OF 10

1. Recap the four sampling schemes in full: simple random, stratified, systematic and cluster — with the failure mode of each.
2. Identify the sampling bias risks in your capstone: convenience sampling, day-shift-only sampling, and sampling only when the process is running well.
3. Recap the continuous-data sample size formula n = (1.96 s / d)^2, where s is the estimated standard deviation and d is the precision you require.
4. Apply it to your project Y. Estimate s from historical data or a pilot sample, choose your required precision d, and calculate n.
5. Recap the discrete-data formula n = (1.96/d)^2 x p(1-p), where p is the estimated proportion defective.

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Sampling Strategy, Sample Size and Rational Subgrouping

STEPS 6-10 OF 10

6. Apply it to your attribute metrics. Note how much larger n becomes — this is the cost of measuring in pass/fail.

7. Now design rational subgroups. A subgroup must contain only common-cause variation, so that between-subgroup variation shows up as a signal.

8. Decide what varies WITHIN your subgroup and what varies BETWEEN subgroups. Getting this backwards makes control limits so wide that nothing ever signals.

9. Set subgroup size and sampling frequency for your capstone Y, and state what shift size you would be able to detect.

10. Document the sampling plan and check it against the data collection plan from Lab 9 for consistency.

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LAB 10 · VERIFY

Sampling Strategy, Sample Size and Rational Subgrouping

✅ Check your work

Your sample sizes are calculated from a real estimate of s or p, your subgrouping scheme places common-cause variation within subgroups, and you can state the shift size your plan can detect.

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Continuous Gage R&R — In Depth

LAB 11

A full re-derivation of measurement system analysis for continuous data. Before any of the statistics in Days 3 and 4 can be trusted, the measurement system must be proven — an invalid gauge invalidates every conclusion that follows. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A completed crossed Gage R&R study with %study variation and ndc for your capstone metric.

Tools & techniques: Accuracy, bias, linearity, stability, repeatability, reproducibility, %study variation, ndc, %tolerance

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Continuous Gage R&R — In Depth

STEPS 1-5 OF 10

1. Recap the components of measurement variation in full: observed variation = actual process variation + measurement system variation.
2. Recap the five MSA properties: accuracy (bias), linearity, stability, repeatability (equipment) and reproducibility (operator).
3. Distinguish repeatability from reproducibility precisely — same operator repeating, versus different operators on the same part.
4. Design a crossed Gage R&R for your capstone metric: 10 parts spanning the full process range, 3 operators, 3 repeat measurements each.
5. Select parts that span the real range of production. Parts clustered near the mean will flatter the gauge and hide its weakness.

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Continuous Gage R&R — In Depth

STEPS 6-10 OF 10

6	Randomise the measurement order and blind the operators to previous readings, or you will measure memory rather than the gauge.
7	Run the study and record all 90 measurements in the standard crossed layout.
8	Calculate %study variation for repeatability, reproducibility and total Gage R&R. Apply the acceptance rule: under 10% acceptable, 10-30% marginal, over 30% unacceptable.
9	Calculate the number of distinct categories (ndc). ndc must be 5 or more for the gauge to distinguish parts usefully.
10	If the gauge fails, decide the fix: recalibrate, retrain operators, tighten the operational definition or replace the instrument. Then re-run.

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LAB 11 · VERIFY

Continuous Gage R&R — In Depth

✅ Check your work

Your study uses parts spanning the full process range with randomised blinded order, and you have computed %study variation and ndc with a stated accept/reject decision.

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DMAIC · MEASURE · HANDS-ON

Attribute MSA and Kappa Analysis (Black Belt Only)

LAB 12

BLACK BELT ONLY — the Green Belt course does not teach this. When the measurement is a human judgement (pass/fail, cosmetic grade, claim approve/reject), Gage R&R does not apply at all. Attribute agreement analysis and Cohen's/Fleiss' Kappa are the correct tools, and most organisations discover their inspectors agree far less than assumed. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack,

You'll build

An attribute agreement study with within-, between- and versus-standard Kappa values.

Tools & techniques: Attribute agreement analysis, Cohen's Kappa, Fleiss' Kappa, within/between appraiser agreement, agreement vs standard

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Attribute MSA and Kappa Analysis (Black Belt Only)

STEPS 1-6 OF 11

1. Establish why Gage R&R fails for attribute data: there is no continuous scale on which to compute variance components.
2. Identify a judgement-based measurement in your capstone — visual inspection, cosmetic grading, document approval, claim adjudication or defect classification.
3. Select 30 to 50 sample items spanning clear-pass, clear-fail and deliberately borderline cases. Borderline items are where agreement actually breaks down.
4. Have an expert establish the KNOWN STANDARD (true) classification for every item. Without a standard you can measure consistency but never correctness.
5. Have 3 appraisers each classify every item twice, in randomised order, blinded to their own previous answer and to each other.
6. Calculate WITHIN-appraiser agreement — does each inspector agree with themselves? Self-disagreement is the most common and most ignored failure.

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Attribute MSA and Kappa Analysis (Black Belt Only)

STEPS 7-11 OF 11

7	Calculate BETWEEN-appraiser agreement — do the inspectors agree with each other?
8	Calculate agreement VERSUS THE STANDARD — are they collectively correct, or consistently wrong together?
9	Compute Cohen's Kappa for appraiser pairs and Fleiss' Kappa across all appraisers. Kappa corrects raw agreement for agreement expected by chance.
10	Apply the acceptance rule: Kappa >= 0.90 excellent, 0.75-0.90 acceptable, below 0.75 requires action. Note that 90% raw agreement can still yield a poor Kappa.
11	Where agreement fails, prescribe the fix: boundary samples, photographic standards, tightened operational definitions or appraiser retraining. Then re-run.

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LAB 12 · VERIFY

Attribute MSA and Kappa Analysis (Black Belt Only)

✅ Check your work

Your study includes borderline items and a known standard, you have computed within-appraiser, between-appraiser and versus-standard agreement plus Kappa, and you have an action plan for any Kappa below 0.75.

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Nested and Destructive Gauge Studies (Black Belt Only)

LAB 13

BLACK BELT ONLY. The crossed Gage R&R of Lab 11 assumes every operator can measure every part repeatedly. Destructive tests (tensile, burst, seal strength, chemical assay) break that assumption entirely — measuring consumes the part. The nested design is the correct answer, and using a crossed study here produces confident nonsense. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone

You'll build

A nested gauge study design and analysis for a destructive or single-measurement test.

Tools & techniques: Nested vs crossed designs, destructive testing MSA, homogeneous batches, part-to-part confounding, variance components

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Nested and Destructive Gauge Studies (Black Belt Only)

STEPS 1-5 OF 10

1. State precisely why a crossed design is invalid for destructive tests: the part no longer exists to be measured a second time.
2. Identify a destructive or non-repeatable measurement in your capstone — or use the Meridian seal-strength test on infusion pump tubing.
3. Understand the nested design: each operator measures DIFFERENT parts, and parts are nested within batch rather than crossed with operator.
4. Establish homogeneous batches. The design assumes parts within a batch are effectively identical — if they are not, part variation contaminates repeatability.
5. Justify batch homogeneity: same lot, same machine, same setup, consecutive production, minimal elapsed time.

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Nested and Destructive Gauge Studies (Black Belt Only)

STEPS 6-10 OF 10

6. Design the study: 3 operators, 10 batches, 3 parts per operator per batch, all parts within a batch treated as replicates.

7. Run the study and record results in the nested layout — operator within batch, not operator crossed with part.

8. Compute variance components for repeatability and reproducibility using the nested model. Note that repeatability is now confounded with within-batch part variation.

9. State that confounding explicitly in your report. A nested study OVERSTATES repeatability variation, so it is a conservative test.

10. Decide accept/reject, and record the design choice and its rationale for the Day 5 presentation — expect this to be challenged.

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LAB 13 · VERIFY

Nested and Destructive Gauge Studies (Black Belt Only)

✅ Check your work

You can explain why a crossed study is invalid for your measurement, your batches are justified as homogeneous, and your report states the repeatability/part-variation confounding explicitly.

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DMAIC · MEASURE · HANDS-ON

Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

LAB 14

Re-derive capability analysis in full, then go beyond Green Belt: the distinction between Cpk and Ppk that most practitioners get wrong, and what to do when the data is not normal — which is most of the time in cycle-time and defect data. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A baseline capability study for your capstone Y with a documented normality decision.

Tools & techniques: Cp, Cpk, Pp, Ppk, within vs overall variation, normality testing, Box-Cox transformation, non-normal capability

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Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

STEPS 1-6 OF 11

1. Recap capability in full: Cp = (USL - LSL) / 6 sigma measures spread only; Cpk also accounts for centring and is always <= Cp.
2. Recap the acceptance benchmarks: Cpk >= 1.33 capable, >= 1.67 for critical characteristics, and the relationship between Cpk and sigma level.
3. Learn the distinction most practitioners get wrong: Cp/Cpk use WITHIN-subgroup (short-term) variation; Pp/Ppk use TOTAL (long-term) variation.
4. Understand the consequence: reporting Cpk on an unstable process flatters it, because between-subgroup drift is excluded from the estimate.
5. Establish the rule — the process must be STABLE before capability means anything. Check stability with a control chart first.
6. Test your capstone Y for normality using Anderson-Darling or Shapiro-Wilk. Read the p-value: p < 0.05 rejects normality.

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Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

STEPS 7-11 OF 11

7. Recognise that cycle time, waiting time and defect counts are routinely non-normal and usually right-skewed — this is expected, not an error.

8. For non-normal data, choose your route: apply a Box-Cox or Johnson transformation, or fit the correct distribution (Weibull, lognormal, exponential) directly.

9. Apply the transformation, re-test normality, then compute capability on the transformed scale — and report results in the ORIGINAL units.

10. Calculate both Cpk and Ppk for your capstone Y and explain the gap between them. A large gap means the process drifts between subgroups.

11. Record the baseline capability in your capstone pack alongside the Lab 3 sigma level.

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LAB 14 · VERIFY

Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

✅ Check your work

You have confirmed stability before computing capability, tested and documented normality with a p-value, handled non-normality by transformation or distribution fitting, and reported both Cpk and Ppk with an explanation of the gap.

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PHASE RECAP

Recap — Measure

- Process mapping & VSM — in depth — Re-derive SIPOC, swimlane maps, value stream maps, takt time and the eight wastes in full.
- Data, sampling and sample size — in depth — Data types, operational definitions, sampling schemes and sample size formulas re-derived from first principles.
- Continuous Gage R&R — in depth — Full treatment of accuracy, repeatability, reproducibility, linearity, stability and %study variation.
- Yield, DPMO and capability — in depth — Classic yield, FPY, RTY, DPU, DPO, DPMO, sigma level and baseline Cp/Cpk, worked end to end.
- Attribute MSA — When the measurement is a judgement (pass/fail, grade), Gage R&R does not apply — attribute agreement does.
- Kappa analysis — Cohen's and Fleiss' Kappa quantify inspector agreement beyond chance. Kappa >= 0.75 is acceptable.

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A

DMAIC · ANALYZE

Analyze at Black Belt Depth — Advanced Statistical Analysis

Hypothesis recap · Multiple regression · ANOVA · Non-parametric tests · Multi-vari studies · Model diagnostics

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ANALYZE · 22% OF THE COURSE

Key Concepts — Analyze (1)

1

Variation, run charts and Pareto — in depth

Common vs special cause, run chart patterns, stratification and Pareto analysis re-derived in full.

2

Fishbone and 5 Whys — in depth

Cause-and-effect analysis by 5M+E, 5 Whys discipline and multi-voting — the tools you will coach Green Belts through.

3

Hypothesis testing — in depth

H0/Ha formulation, alpha, p-values, confidence levels, test selection and Type I/II errors, worked thoroughly.

4

Simple regression & correlation — in depth

Correlation coefficients, the regression line and the causation trap — the foundation multiple regression builds on.

5

Multiple regression

Model Y against several Xs at once — the Green Belt's simple linear regression will not do here.

6

Model diagnostics

R-squared adjusted, residual plots, multicollinearity (VIF) — a model that fits is not yet a model that is valid.

7

ANOVA

Compare three or more group means in one test instead of inflating alpha with repeated t-tests.

8

Two-way ANOVA & interactions

Two factors at once reveals interaction effects that one-factor-at-a-time analysis hides entirely.

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Key Concepts — Analyze (2)

1

Non-parametric tests

Mann-Whitney, Kruskal-Wallis and Mood's median when the data will not meet normality assumptions.

2

Multi-vari studies

Separate positional, cyclical and temporal variation before spending money hunting the wrong X.

3

Chi-square & contingency

Test association between categorical variables — defect type against shift, line or supplier.

4

From statistics to story

Translate a coefficient and a p-value into money, risk and a decision the sponsor can make.

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ANALYZE - BLACK BELT DEPTH

Analyze is where the Black Belt earns the title

Measure told us Meridian runs 8.9 percent OTIF misses and 91.4 percent first-pass yield. Analyze proves WHICH Xs drive that - with statistics a sponsor will fund and an auditor will accept.

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What Analyze delivers at Black Belt level

1

A validated, ranked list of Xs

Two or three proven drivers, each with a p-value and an effect size in units

2

A fitted predictive model

Y = f(X) with coefficients, adjusted R-squared and validated residuals

3

Interaction knowledge

Which fixes work everywhere and which only work at one plant

4

A variation family diagnosis

Positional, cyclical or temporal - so Improve spends money in the right place

5

Assumption-clean evidence

Every test with its normality, independence and variance check documented

6

A translated business case

Coefficients converted into dollars, risk and a decision the sponsor can sign

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Green Belt Analyze vs Black Belt Analyze

GREEN BELT

One X at a time

- 2-sample t-test, chi-square, simple regression
- Tests one factor against Y in isolation
- Reports r and R-squared, stops there
- Assumes normality without checking it
- Answers: is this X related to Y?

BLACK BELT

Many Xs, with diagnostics

- Multiple regression, ANOVA, multi-vari, non-parametrics
- Models several Xs simultaneously and their interactions
- Reports adjusted R-squared, VIF and residual plots
- Tests every assumption and switches test when violated
- Answers: how much of Y does each X explain, and is it valid?

THE MENTOR ROLE

You coach both

- You must teach a Green Belt the simple tools cleanly
- You must know when their simple tool is wrong
- You review their test selection before they run it
- You catch the assumption they did not check
- You translate their output for the sponsor

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The Analyze route map for this module

1. Recap variation, run charts, Pareto and cause tools in full depth
2. Recap hypothesis testing and build the test-selection decision tree
3. Advance into multiple regression and model diagnostics
4. Advance into ANOVA, interactions and non-parametric alternatives
5. Run a multi-vari study to isolate the variation family
6. Translate every statistic into money and a sponsor decision

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The Meridian Medical Devices case we will use throughout

1

Three plants

Singapore (lines 1-2), Penang (lines 3-4), Suzhou (lines 5-6)

2

The product

Infusion pump assemblies - Class IIb regulated medical devices

3

Primary Y

OTIF delivery to hospital distributors, currently 91.1 percent

4

Secondary Y

First-pass yield at final functional test, currently 91.4 percent

5

The pain

Warranty returns at 2.8 percent, costing SGD 4.1 million per year

6

The mandate

Steering committee wants OTIF to 97 percent within two quarters

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ANALYZE - RECAP IN DEPTH

Recap 1 - Variation, and the discipline not to react

A Black Belt is asked about a moved number every single day. Everything you do in Analyze rests on telling ordinary variation from a real signal.

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Common cause vs special cause variation

COMMON CAUSE

Inherent, random, always present

- Built into the process design itself
- Many small sources acting together at once
- Predictable within a range - the process is stable
- Fix requires a SYSTEM change owned by management
- Meridian: line 3 yield varying between 89 and 94 percent

SPECIAL CAUSE

Assignable to one identifiable event

- Something changed - it is not part of the system
- One traceable source you can name and date
- Unpredictable - appears as a signal on the chart
- Fix is LOCAL: find it, remove it, prevent recurrence
- Meridian: Suzhou yield fell to 71 percent the week the solder pot was replaced

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Why the distinction decides your entire strategy

1

Wrong diagnosis, wrong owner

Special cause is a local fix. Common cause needs management to change the system.

2

Blame follows the error

Treating common cause as special sends you hunting an operator who is not at fault.

3

Roughly 85 percent is systemic

Deming: most defects come from the process, not the person running it.

4

It picks the tool

Special cause: investigate the event. Common cause: run the full DMAIC.

5

Stability comes first

You cannot judge capability or test causes on an unstable process.

6

It licenses inaction

Knowing variation is common cause is what permits you to leave the process alone.

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ANALYZE - RECAP IN DEPTH

Tampering - the most expensive habit in manufacturing

Adjusting a stable process in response to common cause variation does not reduce variation. It roughly DOUBLES it. Deming proved this with a funnel and a marble.

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The Deming funnel experiment

1. Fix a funnel above a target, drop 50 marbles, mark where each lands
2. Rule 1: never move the funnel. Marbles scatter in a stable circle around the target
3. Rule 2: move the funnel to correct each miss. Scatter grows about 40 percent wider
4. Rule 3: move the funnel from the target to the opposite of the last drop. Scatter explodes
5. Rule 4: aim the funnel at the last marble. The pattern drifts away without limit
6. Lesson: adjusting a stable process on one data point always makes it worse

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Tampering at Meridian - a real Tuesday

THE TRIGGER

One ordinary point moves

- Penang line 4 first-pass yield drops from 92 to 89 percent
- Both values sit inside normal common cause range
- Nothing in the process actually changed
- But the number appears in red on the daily dashboard

THE REACTION

The supervisor adjusts

- Reflow oven zone 3 setpoint raised by 6 degrees C
- Two operators moved from line 3 to line 4 for the shift
- The adjustment is well intentioned and quick
- No data supports it - only the single point

THE RESULT

Variation roughly doubles

- Line 3 now understaffed and starts missing takt
- Higher reflow temperature warps two connector housings
- Next swing is bigger, prompting a bigger correction
- Rule: never adjust a stable process on one data point

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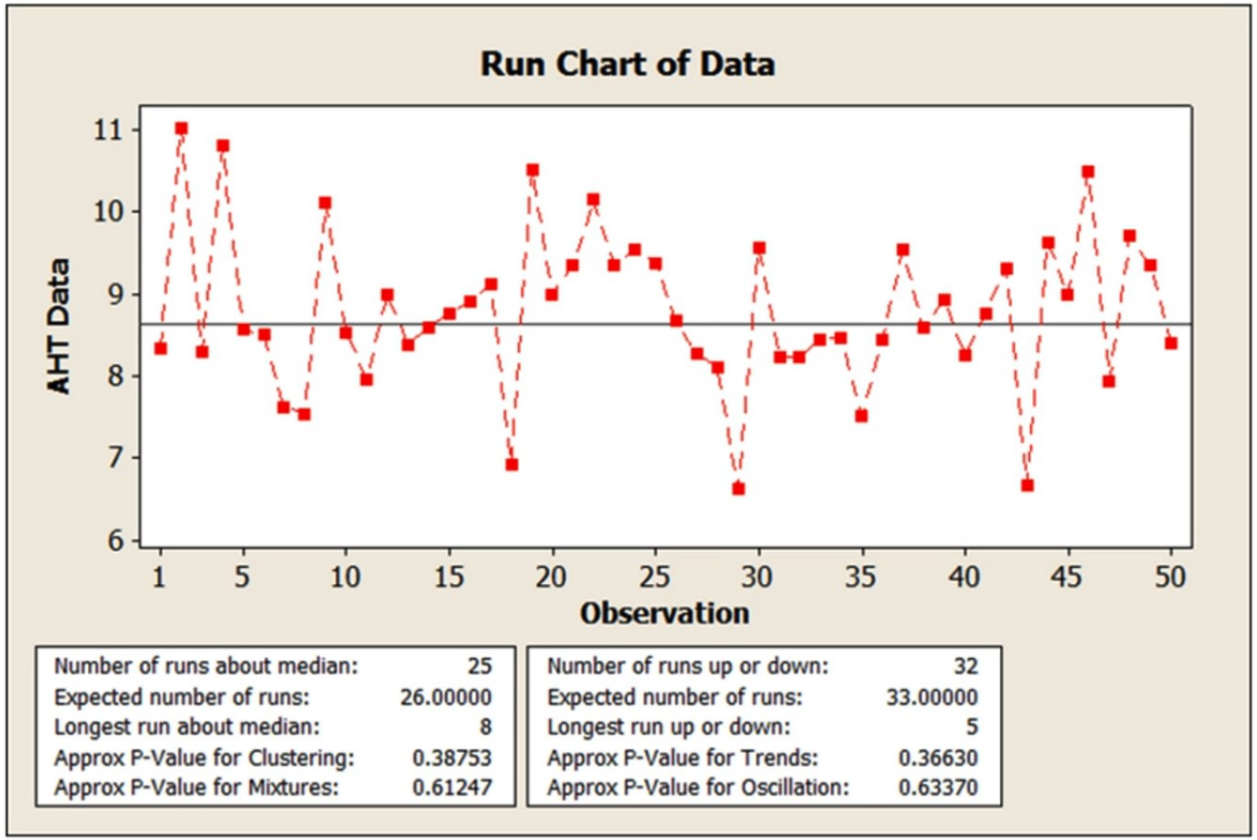
ANALYZE - RECAP IN DEPTH

Run charts - six patterns, six run rules

The run chart is the cheapest signal detector in the toolkit. A Black Belt reads all six patterns and knows the exact rule that fires each one.

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Anatomy of a run chart



Time on the horizontal axis, the measure on the vertical axis, and a MEDIAN centre line - not a mean, because the median resists outliers.

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How to build a run chart correctly

1

Plot in TIME ORDER

Sequence is the whole point - sorting the data destroys the information

2

Use the MEDIAN centre line

The median is not dragged by the extreme points you are hunting

3

At least 15 to 20 points

Fewer points cannot reliably reveal a trend, shift or oscillation

4

Keep the interval constant

Daily or weekly - do not mix, and do not skip periods

5

Annotate known events

Mark the oven rebuild or the supplier change so signals can be explained

6

Read patterns, not points

One high point is noise. A PATTERN is a signal worth investigating.

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The six run chart patterns

1

TREND

6 or more consecutive points all rising or all falling

2

SHIFT

8 or more consecutive points all on one side of the median

3

CLUSTER

Points bunched in one region - grouping by an untracked variable

4

MIXTURE

Too few crossings - points avoid the median, two populations combined

5

OSCILLATION

Rapid regular up-down sawtooth - usually the fingerprint of tampering

6

BIAS

Astronomical point far outside the rest - a one-off assignable event

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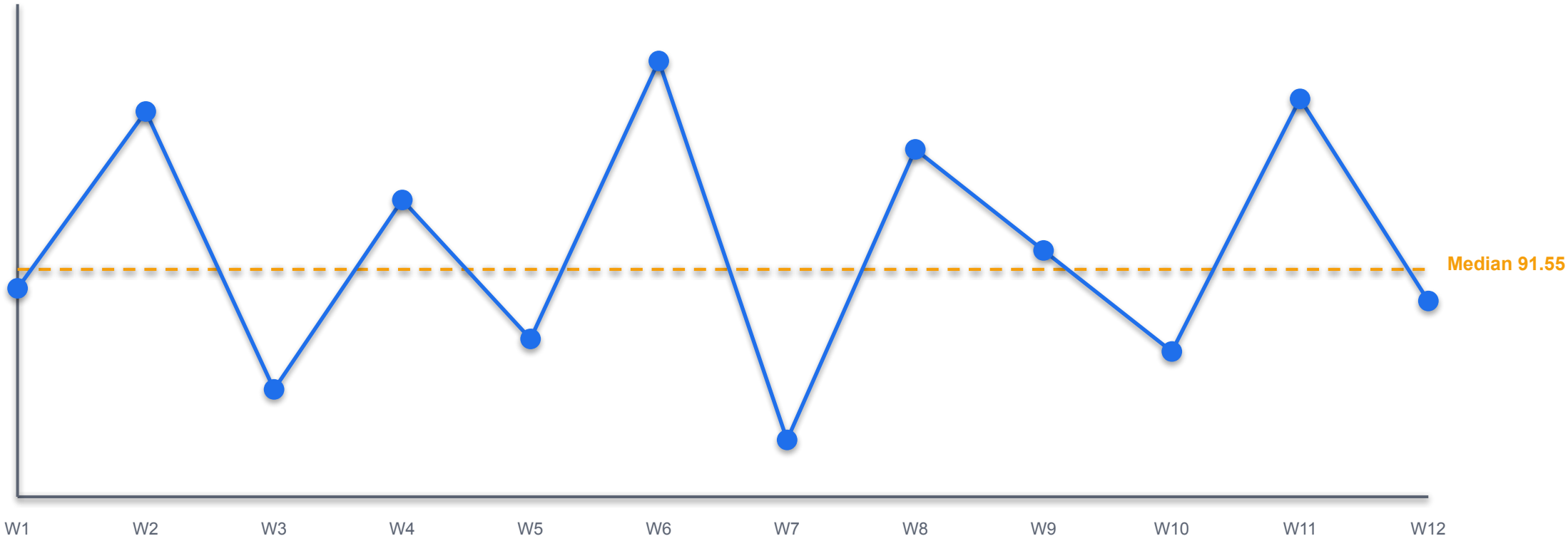
Run rules - the arithmetic behind the patterns

Too few runs (mixture)	**runs < expected** **expected = n/2 + 1**	24 points, expect 13 runs. Meridian line 5 gave 6 runs - two populations mixed.
Too many runs (oscillation)	**runs > n/2 + 1** **with no long runs**	24 points gave 20 runs - a sawtooth. Operators were re-tuning the tester every shift.
Shift	**8 consecutive points** **one side of median**	Suzhou line 6 yield sat below median for 11 straight days after the stencil change.
Trend	**6 consecutive points** **all increasing or decreasing**	Penang line 3 cycle time rose 7 shifts in a row as the feeder wore.

A run is an unbroken sequence of points on the same side of the median. Points sitting exactly on the median are ignored.

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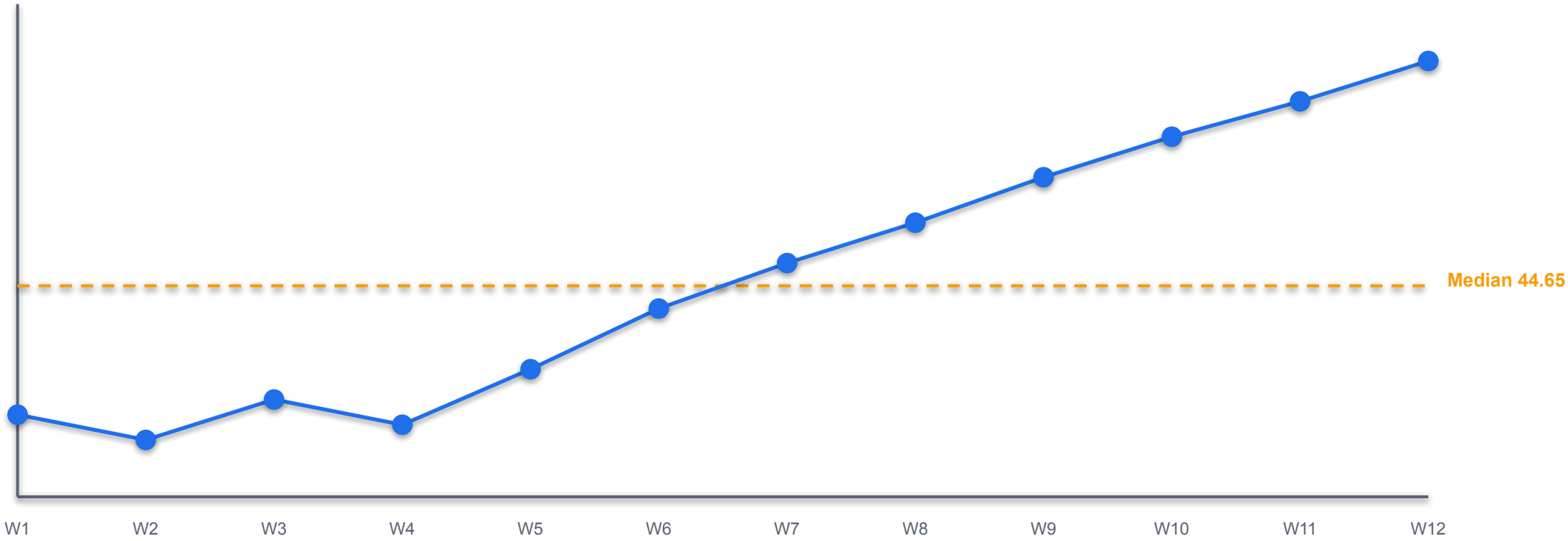
Meridian line 1 first-pass yield - 12 weeks (percent)



Points scatter randomly either side of the median with no run, trend or oscillation - a STABLE process showing only common cause variation.

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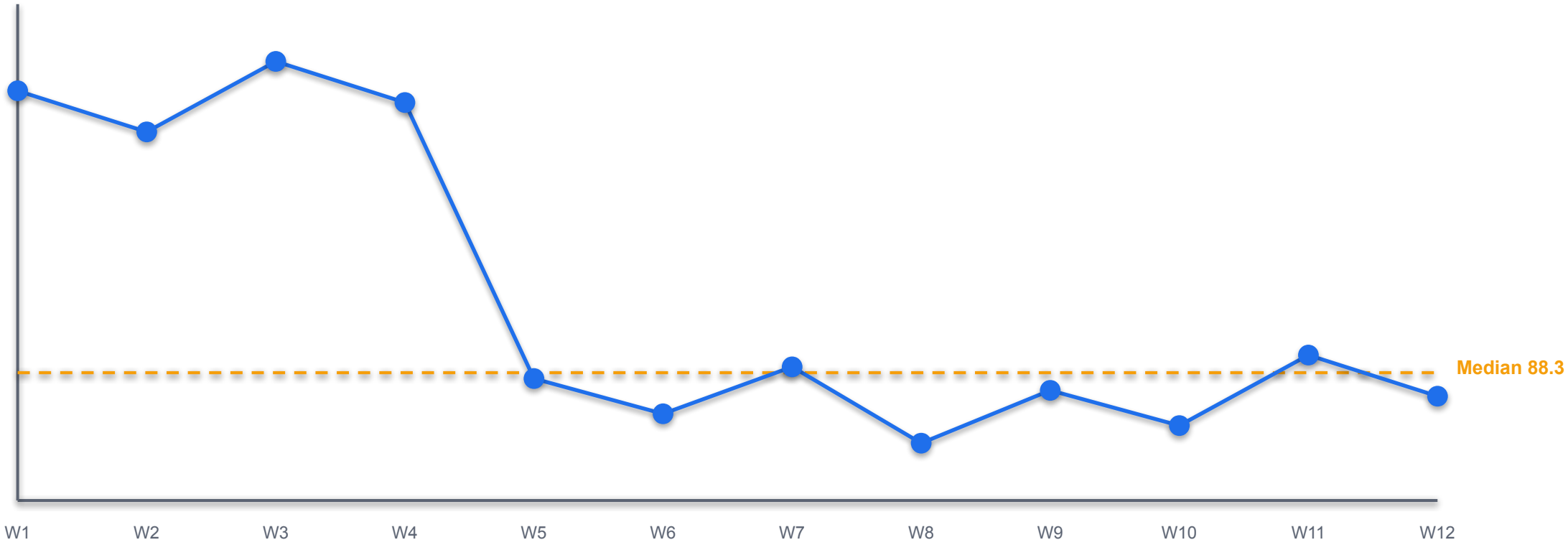
Meridian line 3 - a TREND appears (cycle time, minutes)



Seven consecutive rising points from W6 - the TREND rule fires. Investigation found progressive wear on the component feeder cam.

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Meridian line 6 - a SHIFT appears (yield, percent)



Eight consecutive points below the median from W5 - the SHIFT rule fires. Traced to the new solder paste stencil installed that weekend.

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ANALYZE - RECAP IN DEPTH

Recap 2 - Stratification and Pareto, weighted by COST

Counting defects tells you what happens most. Costing defects tells you what to fix first. These are frequently not the same list.

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Stratification - the questions that split the data

1. **WHO**
   Operator, team, shift, inspector, supervisor

2. **WHAT**
   Product family, model variant, defect type, component supplier

3. **WHERE**
   Plant, line, station, geographic customer region

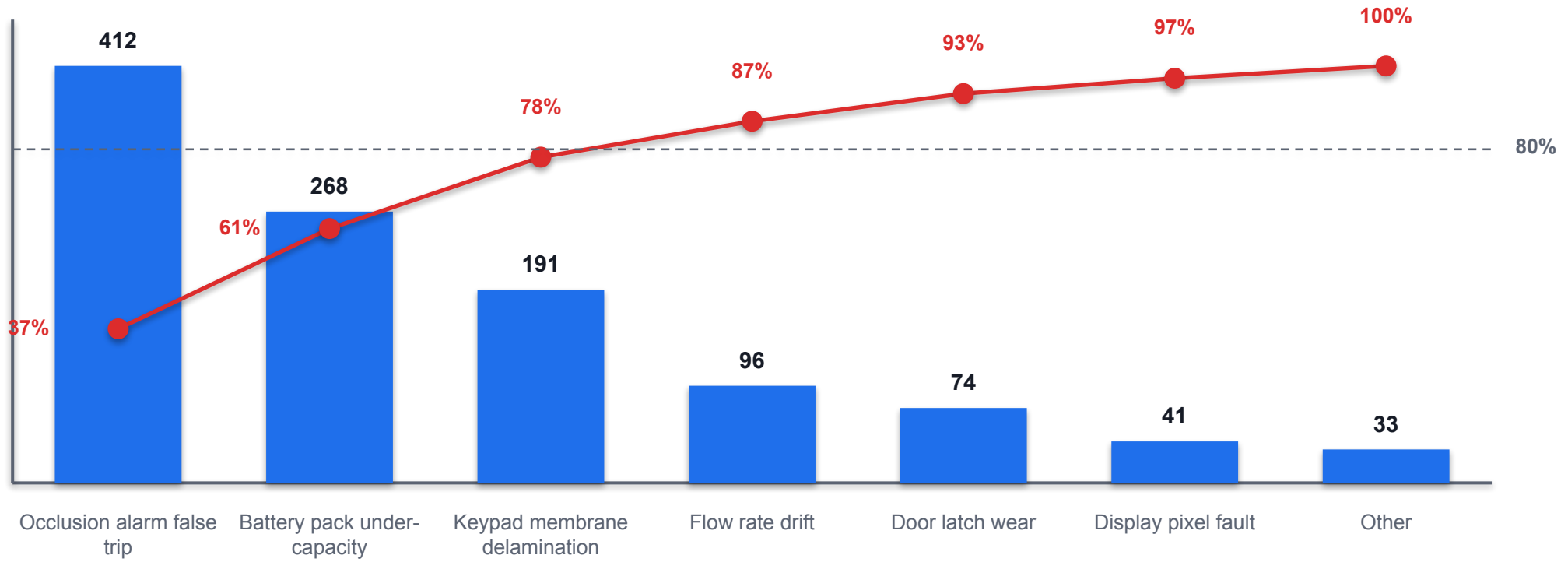
4. **WHEN**
   Shift, day of week, hour of shift, month, before or after a change

5. **HOW**
   Method version, machine, tooling set, changeover type

6. **HOW MUCH**
   Batch size, run length, order value, order line count

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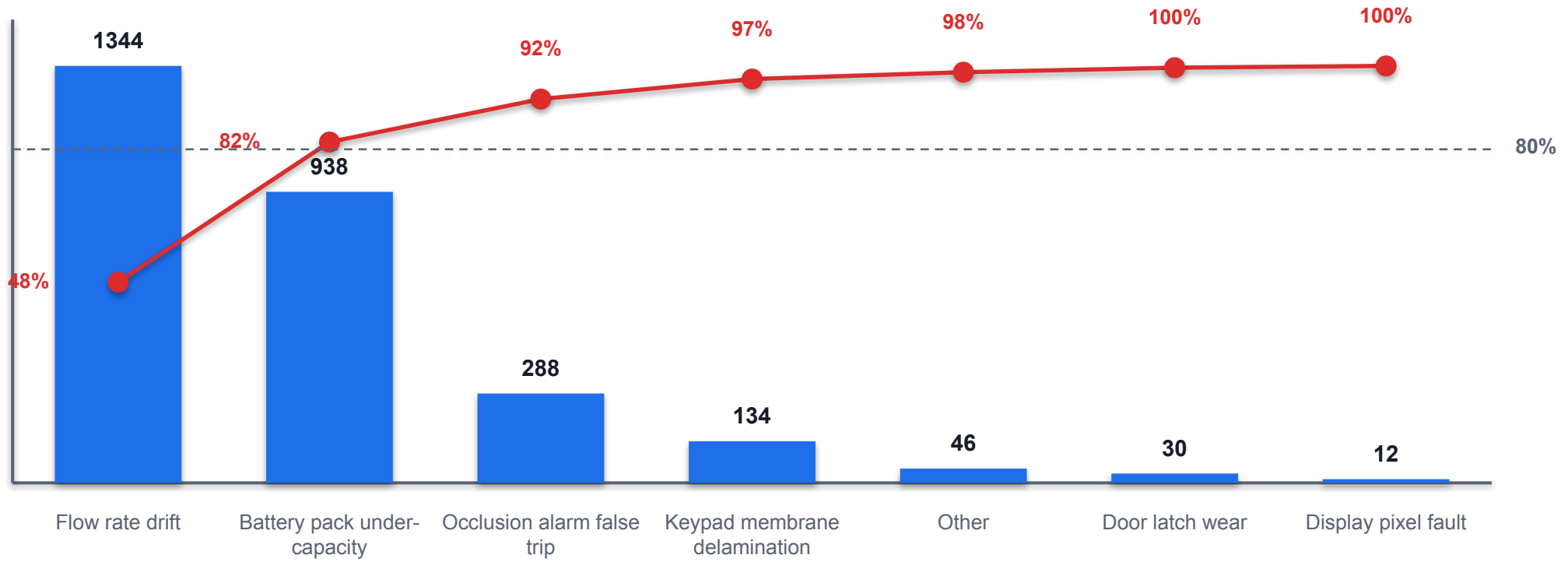
Meridian warranty returns by defect type - BY COUNT



By COUNT, occlusion alarm false trips dominate at 38 percent of all returns. This is where a Green Belt would stop.

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Meridian warranty returns by defect type - BY COST (SGD 000)



By COST the order inverts. Flow rate drift is only 96 units but each one triggers a field recall investigation at SGD 14,000. Fix cost, not count.

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Weighting a Pareto correctly

Cost weight	**cost = count x unit cost of one occurrence**	Flow rate drift: 96 x SGD 14,000 = SGD 1,344,000 - the true top category.
Cumulative percent	**cum% = running total / grand total x 100**	Flow drift 49%, battery 84%. Two categories carry 84 percent of the money.
Vital few test	**Do 20 percent of categories carry 80 percent**	2 of 7 categories (29 percent) carry 84 percent of cost - a valid Pareto.

If every category is roughly equal, there is no vital few - stratify on a different dimension before proceeding.

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Boxplots - what they show that a Pareto cannot

THE FIVE NUMBERS

What the box is made of

- Minimum (excluding outliers) at the lower whisker
- Q1 at the bottom of the box - 25th percentile
- Median as the line inside the box
- Q3 at the top of the box - 75th percentile
- IQR = Q3 - Q1, the width of the middle half

WHAT TO READ

Shape, spread, outliers

- Box width shows spread - narrower is more consistent
- Median off-centre in the box means skew
- Points beyond 1.5 x IQR are flagged outliers
- Side-by-side boxes compare groups instantly
- Overlapping boxes warn you the difference may not be real

MERIDIAN EXAMPLE

Yield by plant

- Singapore: median 93.1, IQR 1.8 - tight and high
- Penang: median 92.4, IQR 2.1 - similar and stable
- Suzhou: median 88.6, IQR 6.4 - lower AND far more variable
- The spread difference matters as much as the mean difference
- This is what sends us to a variance test, not just a t-test

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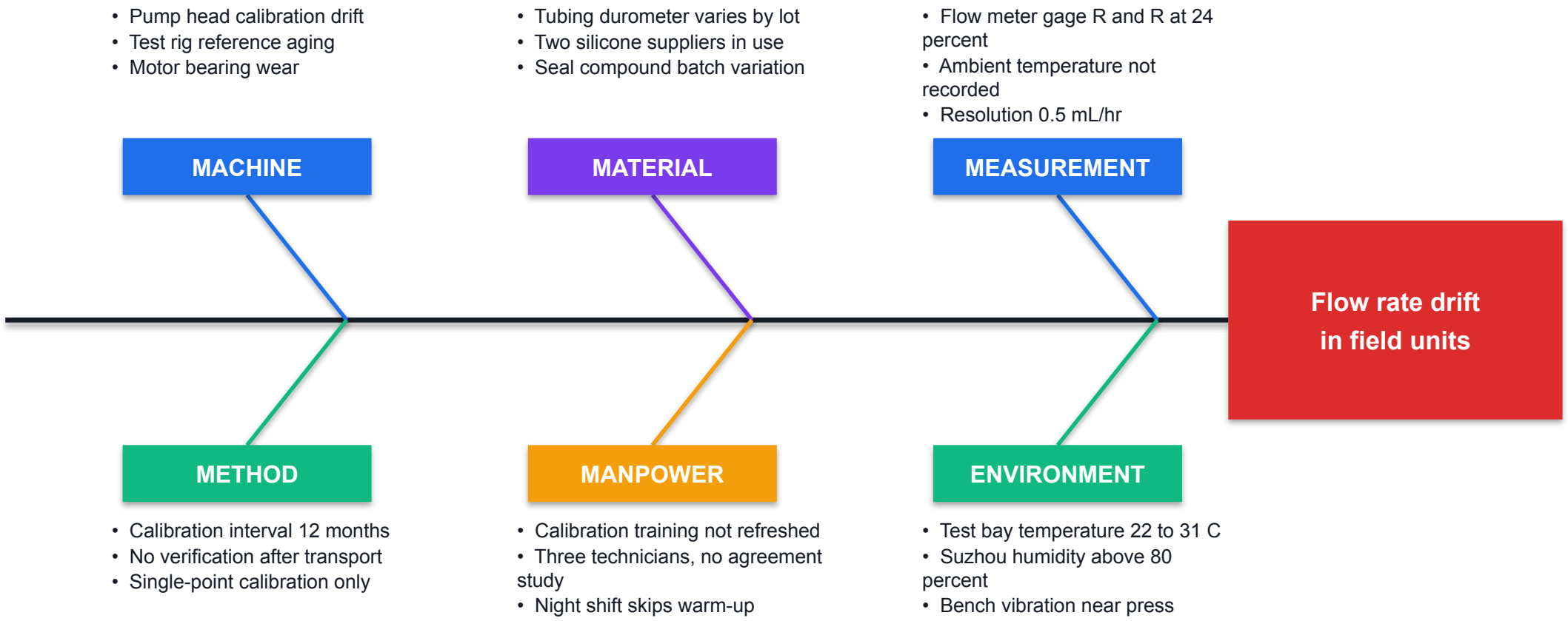
ANALYZE - RECAP IN DEPTH

Recap 3 - Generating causes you can actually test

The fishbone does not find the root cause. It produces a candidate list good enough that the statistics have something worth testing.

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Fishbone 5M plus E - Meridian flow rate drift



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The 5M plus E categories - what belongs in each

1

MACHINE

Equipment, tooling, fixtures, software, calibration state

2

METHOD

Procedures, work instructions, sequence, settings, sampling plan

3

MATERIAL

Raw material, components, consumables, supplier and lot variation

4

MANPOWER

Skill, training, experience, staffing level, fatigue, handover

5

MEASUREMENT

Gage capability, resolution, inspector agreement, data capture

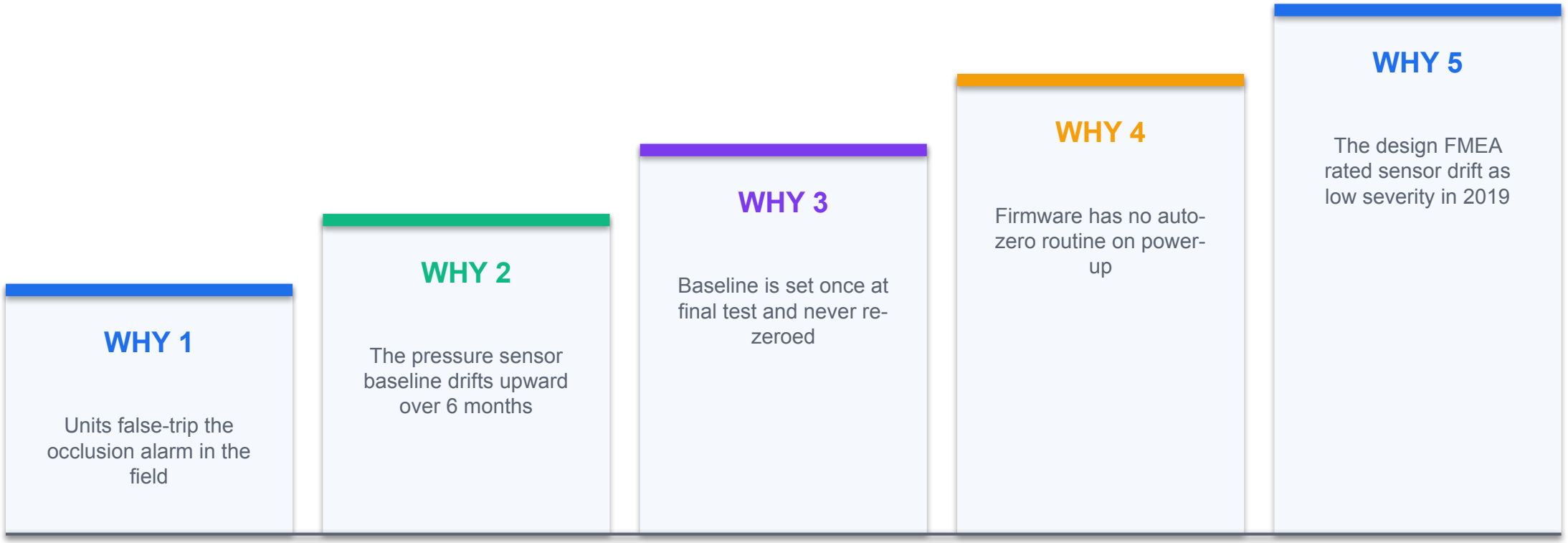
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ENVIRONMENT

Temperature, humidity, vibration, cleanliness, layout, lighting

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5 Whys on Meridian occlusion false trips



Root cause lands on a PROCESS decision (the design FMEA rating), not on a person. That is the discipline.

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5 Whys discipline - the rules that keep it honest

1

Stop at a process, never a person

'The technician forgot' is not a root cause - ask why the process allowed it

2

Each why must be verifiable

If you cannot get data to confirm a link, you are guessing not analysing

3

Five is a guide, not a quota

Sometimes three, sometimes seven - stop when the answer is actionable

4

Branch when there are two answers

A why with two valid causes becomes two chains, not a forced single line

5

Test the chain backwards

Read it as 'therefore' from the bottom up - it must still make sense

6

Confirm the fix breaks the chain

If removing the root cause would not stop the effect, you have the wrong root

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Multi-voting a 40-cause fishbone down to 5

1. Cluster duplicate and near-duplicate causes and re-word them clearly
2. Give each person N/3 votes where N is the number of causes
3. Vote silently and simultaneously - no discussion during voting
4. Tally, keep the top third, discard the rest into the parked log
5. Repeat with fewer votes until 5 to 7 candidates remain
6. Rank the survivors on the cause-and-effect matrix before testing

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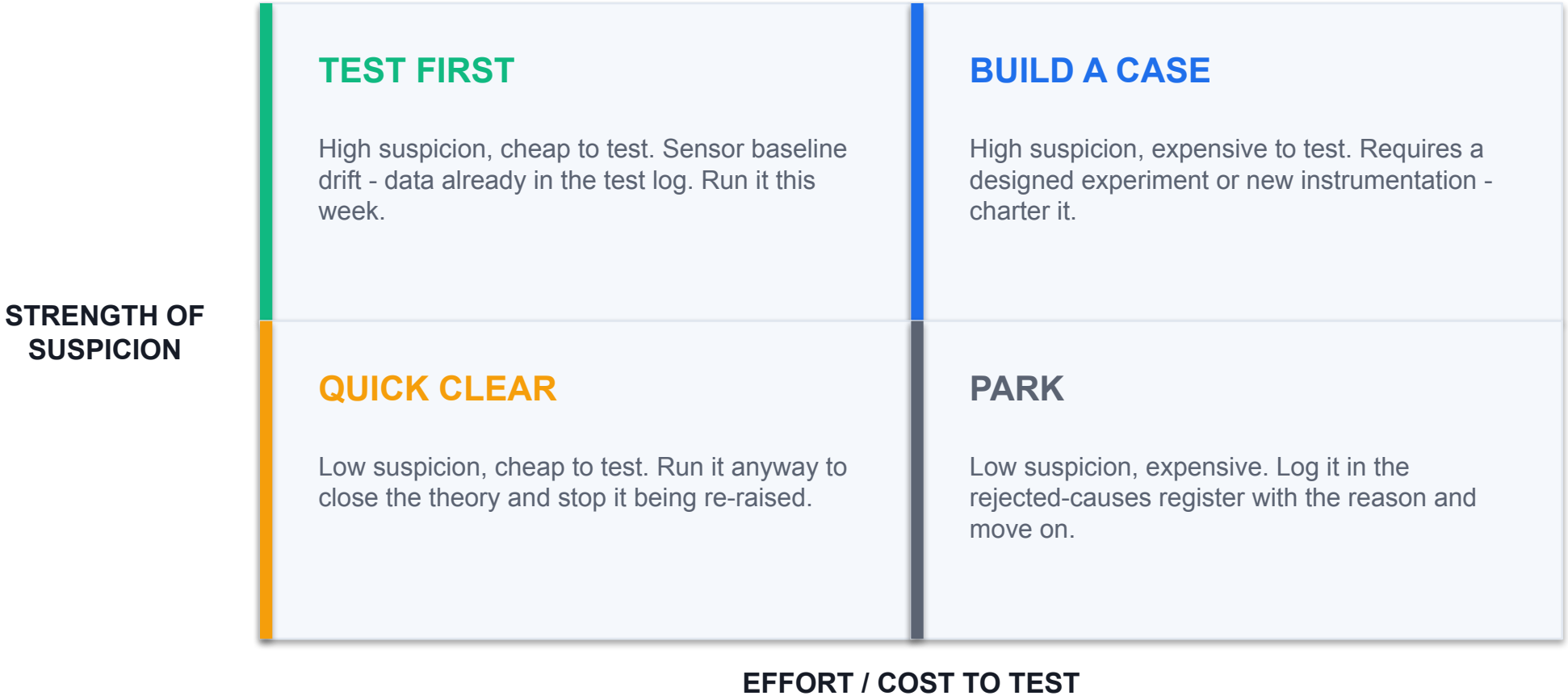
The cause-and-effect matrix - scoring what to test first

Output weight	**Each Y rated 1-10 by customer importance**	Meridian: OTIF weight 9, first-pass yield 8, warranty cost 10.
Relationship score	**Each X vs each Y: 0, 1, 3 or 9**	Sensor baseline drift vs warranty cost = 9 (strong direct relationship).
Total score	**sum of (weight x relationship)**	Sensor drift: 9x1 + 8x3 + 10x9 = 123 - the highest scoring X, test it first.

The matrix converts opinion into a ranked test order. It does NOT prove anything - the hypothesis test does.

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Choosing which causes to test with statistics



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ANALYZE - RECAP IN DEPTH

Recap 4 - Hypothesis testing, in full

This is the single most-taught and most-misused topic in Six Sigma. As a Black Belt you must be able to derive it, defend it and correct a Green Belt who has it wrong.

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The hypothesis testing procedure

1. State the practical question in plain business language
2. Formulate H0 and Ha, and decide one-tail or two-tail
3. Choose alpha and the required power, then compute the sample size
4. Select the correct test from the Y and X data types
5. Check the assumptions: normality, independence, equal variance
6. Run the test, read the p-value, then translate back to the business

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H0 and Ha - formulating them correctly

NULL - H0

The status quo, the boring answer

- Always contains equality: =, <= or >=
- 'There is no difference' / 'the change did nothing'
- It is what we ASSUME true until the data forces us to abandon it
- We never 'accept' H0 - we fail to reject it
- Meridian: mean yield Singapore = mean yield Suzhou

ALTERNATIVE - Ha

The claim you must prove

- Never contains equality: not equal, > or <
- 'There IS a difference' / 'the change helped'
- It carries the burden of proof
- It is what you write the project benefit against
- Meridian: mean yield Singapore is not equal to mean yield Suzhou

COMMON ERRORS

What a Black Belt corrects

- Putting the interesting claim in H0
- Writing Ha after seeing the data
- Using a one-tail test to rescue a marginal p-value
- Reporting 'we proved there is no difference'
- Testing a hypothesis nobody could act on either way

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Type I and Type II error - the decision table

YOUR DECISION \ THE TRUTH ABOUT THE PROCESS	
CORRECT H0 is true and you fail to reject it. You correctly concluded nothing changed. Probability = 1 - alpha.	**TYPE I ERROR - alpha** H0 is true but you reject it. A FALSE ALARM. You claim a difference that is not there. PRODUCER'S RISK.
TYPE II ERROR - beta H0 is false but you fail to reject it. A MISS. A real difference goes undetected. CONSUMER'S RISK.	**CORRECT - POWER** H0 is false and you reject it. You detected a real difference. Probability = 1 - beta = POWER.

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Producer's risk vs consumer's risk

TYPE I - PRODUCER'S RISK

alpha, typically 0.05

- You reject a good batch, or claim a fix that does nothing
- Meridian: scrapping a conforming lot of pump housings
- Cost falls on the PRODUCER - wasted material and rework
- You spend money implementing an improvement with no effect
- Controlled directly by your choice of alpha

TYPE II - CONSUMER'S RISK

beta, often 0.10 to 0.20

- You pass a bad batch, or miss a real improvement
- Meridian: shipping drifting pumps to a hospital
- Cost falls on the CUSTOMER - and on your warranty and reputation
- In a regulated medical device this is the risk that ends careers
- Controlled by sample size and effect size, via power

THE TRADE-OFF

You cannot minimise both

- Lowering alpha to 0.01 raises beta at fixed n
- The only way to reduce both is a LARGER SAMPLE
- Choose alpha by consequence, not by habit
- Medical device safety claim: alpha 0.01 is defensible
- Exploratory screening: alpha 0.10 is defensible

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Alpha, beta, power and confidence

Alpha	**alpha = P(reject H0 | H0 true)**	alpha = 0.05 means a 5 percent chance of a false alarm on this test.
Confidence level	**confidence = 1 - alpha**	alpha = 0.05 gives a 95 percent confidence level and 95 percent CI.
Beta	**beta = P(fail to reject H0 | H0 false)**	beta = 0.20 means a 20 percent chance of missing a real difference.
Power	**power = 1 - beta**	power = 0.80: an 80 percent chance of detecting the shift you care about.

Standard practice: alpha = 0.05, power >= 0.80. Raise power to 0.90 or 0.95 for safety-critical medical device decisions.

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Sample size - worked for Meridian

2-sample t sample size	**n per group = 2 x (Z_a/2 + Z_b)^2 x s^2 / d^2**	Detect a 1.5 pp yield gap, s = 2.2, alpha .05, power .80: n = 2 x 7.85 x 4.84 / 2.25 = 34 per line.
1-proportion sample size	**n = (Z_a/2)^2 x p(1-p) / d^2**	Confirm OTIF at 91 percent within 2 pp: n = 3.84 x 0.0819 / 0.0004 = 787 orders.
Effect of halving delta	**n scales as 1 / d^2**	Halving the detectable gap from 1.5 pp to 0.75 pp needs FOUR times the sample: 136 per line.

Z for alpha 0.05 two-tail = 1.96; Z for power 0.80 = 0.84. Always compute n BEFORE collecting data.

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ANALYZE - RECAP IN DEPTH

The p-value - what it actually means

The p-value is the probability of seeing data this extreme or more extreme IF the null hypothesis were true. It is not the probability that H0 is true.

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What a p-value is and is not

IT IS

A conditional probability

- P(data this extreme | H0 is true)
- A measure of how surprising your data is under H0
- p < alpha means reject H0
- Meridian p = 0.003: data this extreme would be rare if plants matched
- 'If it ain't low, let it go' - the field mnemonic

IT IS NOT

Four common misreadings

- NOT the probability that H0 is true
- NOT the probability your conclusion is wrong
- NOT a measure of how big or important the effect is
- NOT proof of no difference when p is large
- p = 0.06 and p = 0.04 are not meaningfully different evidence

ALWAYS REPORT WITH

The context that makes it useful

- The effect size in real units, not just significance
- The confidence interval around that effect
- The sample size the test ran on
- Which assumptions were checked and how
- The business consequence of the effect

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The test selection decision tree

Almost every hypothesis testing mistake is a test SELECTION mistake. Start from the data type of Y, then the data type and number of levels of X.

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How to select the right test in four questions

1. What is the data type of Y - continuous or discrete/attribute
2. What is the data type of X - and how many levels or groups does it have
3. Are the samples independent, or paired on the same units
4. Do the assumptions hold - normal, independent, equal variance

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CONTINUOUS Y - which test

1

1-sample t-test

One group vs a target value. Is Suzhou mean yield equal to 93 percent?

2

2-sample t-test

Two independent groups. Singapore line 1 vs Penang line 3 cycle time.

3

Paired t-test

Same units measured twice. Yield before and after the stencil change, same line.

4

One-way ANOVA

Three or more groups, one factor. Yield across all 6 Meridian lines.

5

Two-way ANOVA

Two factors at once plus their interaction. Plant x shift on yield.

6

F-test / Levene

Comparing VARIANCES, not means. Is Suzhou more variable than Singapore?

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DISCRETE / ATTRIBUTE Y - which test

1

1-proportion test

One proportion vs a target. Is OTIF equal to the 97 percent goal?

2

2-proportion test

Two proportions compared. Penang scrap rate vs Suzhou scrap rate.

3

Chi-square test

Association between two categorical variables. Defect type vs shift.

4

Chi-square goodness of fit

Does the observed distribution match an expected one?

5

Fisher's exact test

Same as chi-square but for small expected counts below 5.

6

Logistic regression

Predict a pass/fail Y from continuous Xs - beyond this course's core

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The decision tree in three branches

Y CONTINUOUS, X DISCRETE

Comparing means or variances

- 1 group vs target -> 1-sample t
- 2 independent groups -> 2-sample t
- 2 paired measurements -> paired t
- 3+ groups, 1 factor -> one-way ANOVA
- 2 factors -> two-way ANOVA with interaction

Y CONTINUOUS, X CONTINUOUS

Quantifying relationship

- One X -> correlation then simple regression
- Several Xs -> multiple regression
- Curved relationship -> add a quadratic term
- Always follow with residual diagnostics
- Never report r without a scatterplot

Y DISCRETE

Comparing proportions

- 1 proportion vs target -> 1-proportion test
- 2 proportions -> 2-proportion test
- Two categorical variables -> chi-square contingency
- Expected cell count below 5 -> Fisher's exact
- Discrete Y from continuous X -> logistic regression

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Worked 2-sample t-test - Singapore vs Suzhou yield

The hypotheses	**H0: mu_SG = mu_SZ** **Ha: mu_SG not= mu_SZ**	Two-tailed, alpha = 0.05. n = 40 lots per plant.
The data	**SG: xbar 93.1, s 1.9** **SZ: xbar 88.6, s 4.7**	Levene p = 0.002 -> variances are NOT equal, so use Welch's t-test.
The statistic	**t = (93.1 - 88.6) /** **sqrt(1.9^2/40 + 4.7^2/40)**	t = 4.5 / 0.801 = 5.62, df approx 51.4, p < 0.001. Reject H0.
The translation	**4.5 pp yield gap,** **95% CI [2.9, 6.1]**	4.5 pp on 210,000 Suzhou units = 9,450 fewer good units, SGD 2.6M at SGD 275 margin.

Note the sequence: hypotheses BEFORE data, assumption check BEFORE the test, business translation AFTER the p-value.

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Where alpha lives on the distribution



With alpha = 0.05 two-tailed, the rejection regions are the outer 2.5 percent in each tail. A test statistic landing there is judged too extreme to be chance alone.

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ANALYZE - RECAP IN DEPTH

Recap 5 - Correlation and simple regression

Hypothesis tests tell you IF an X matters. Regression tells you HOW MUCH. This is the foundation everything in the Black Belt half of Analyze is built on.

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Correlation and simple linear regression

Correlation r	**r = cov(X,Y) / (s_x x s_y)**	Meridian oven zone-3 temp vs yield: r = -0.72. Strong negative linear relationship.
Coefficient of determination	**R-squared = r^2**	r = -0.72 gives R-squared = 0.52. Oven temperature explains 52 percent of yield variation.
Regression line	**Y = b0 + b1 X**	Yield = 118.4 - 0.104 x (oven temp C). Each extra degree costs 0.104 pp of yield.
Slope meaning	**b1 = change in Y per one unit change in X**	Running 6 C hotter costs 0.62 pp yield = about 1,300 units a year at Penang volume.

Always plot the scatter before trusting r. A curved or clustered relationship can produce a misleading r.

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Correlation is not causation - the four explanations

X CAUSES Y

The one you hope for

- The mechanism is physically plausible
- X precedes Y in time
- Changing X in a trial changes Y
- Meridian: reflow temperature -> solder joint quality
- Only a designed experiment can confirm this

Y CAUSES X

Reverse causation

- The arrow runs the other way
- Meridian: high scrap correlates with more inspectors
- The scrap did not come from the inspectors
- The inspectors were added BECAUSE of the scrap
- Check which came first before writing the story

A THIRD VARIABLE

Confounding or coincidence

- Z drives both X and Y
- Meridian: humidity drives both scrap and absenteeism
- Or pure coincidence in a small sample
- Or you tested 40 Xs and 2 came up 'significant' by chance
- This is why a Black Belt confirms in Improve with a DOE

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Multiple regression

A Green Belt models Y against one X. Real processes have several Xs acting at once, and they are correlated with each other. From here on, this is beyond Green Belt.

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The multiple regression model

The model	**Y = b0 + b1X1 + b2X2 + ... + bkXk + e**	Y = first-pass yield; X1 oven temp, X2 humidity, X3 paste age, X4 line speed.
Intercept b0	**Predicted Y when every X equals zero**	b0 = 118.4. Often has no physical meaning - do not over-interpret it.
Coefficient bi	**Change in Y per unit of Xi, HOLDING all other Xs CONSTANT**	b1 = -0.081: one extra degree C costs 0.081 pp yield at fixed humidity, paste age and speed.
Error term e	**Everything the model does not explain**	Assumed independent, normal, mean zero, constant variance. The residuals test this.

'Holding other Xs constant' is the whole point of multiple regression, and the phrase Green Belts most often omit.

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ANALYZE - BLACK BELT ONLY

Meridian fitted model - first-pass yield

Fitted equation	**Yield = 118.4 - 0.081(Temp) - 0.147(Humidity) - 0.043(PasteAge) + 0.019(Speed)**	Fitted on 180 lots across all 6 lines over 9 months of production data.
Coefficient p-values	**Temp .000, Humidity .001 PasteAge .014, Speed .382**	Line speed is not significant at alpha .05 - it is a candidate for removal.
Model fit	**R-squared = 0.741 adj R-squared = 0.735**	The four Xs together explain about 74 percent of yield variation between lots.

Read the coefficient WITH its p-value. A large coefficient with p = 0.38 is noise dressed as insight.

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Interpreting a coefficient the right way

SAY THIS

Correct and defensible

- Each 1 C rise in oven temp is associated with 0.081 pp lower yield
- ...holding humidity, paste age and line speed constant
- ...within the observed range of 232 to 251 C
- ...with 95 percent CI [-0.104, -0.058]
- Association, quantified, bounded, with uncertainty

NOT THIS

Overclaims a Black Belt must catch

- 'Temperature causes 0.081 pp of yield loss' - regression is not a DOE
- 'Cutting temp 20 C gains 1.6 pp' - extrapolating outside the data range
- 'Temp is the biggest driver' - coefficients in different units are not comparable
- 'The model explains yield' - it explains 74 percent of the VARIATION observed
- Any causal verb without an experiment behind it

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Backward elimination - building a parsimonious model

1. Fit the full model with every candidate X you have reason to include
2. Find the term with the highest p-value above your cutoff, typically 0.10
3. Remove that single term and refit the entire model
4. Check adjusted R-squared - it should hold or rise, never drop materially
5. Repeat until every remaining term is significant
6. Validate the final model on held-out data before you present it

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Backward elimination on the Meridian model

Step 0 - full model	**4 terms, adj R2 = 0.735** **Speed p = 0.382**	Speed has the highest p-value and is well above the 0.10 cutoff. Remove it.
Step 1 - refit	**3 terms, adj R2 = 0.737** **PasteAge p = 0.011**	Adjusted R-squared actually ROSE. Speed was costing more than it contributed.
Step 2 - check	**All p-values below 0.05** **adj R2 = 0.737**	Final model: Yield = 119.1 - 0.079(Temp) - 0.151(Humidity) - 0.045(PasteAge).

Never remove more than one term at a time - removing an X changes every other coefficient in the model.

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Overfitting - the trap that ends careers

WHAT IT LOOKS LIKE

Suspiciously good

- R-squared of 0.97 on 24 data points with 9 terms
- Every coefficient significant, none physically explainable
- Interaction and quadratic terms nobody can justify
- The model predicts the training data almost perfectly
- It fails completely on next month's production

WHY IT HAPPENS

The model learns noise

- Too many terms relative to the number of observations
- Terms chosen by p-value hunting, not by process knowledge
- The same data used to both build and judge the model
- Rule of thumb: at least 10 to 15 observations per term
- Meridian: 180 lots supports about 12 terms, not 40

HOW TO PREVENT IT

Discipline, not software

- Hold out 20 to 30 percent of the data before you start modelling
- Fit on the training set only, never touch the holdout
- Compare R-squared on training vs holdout - a big drop means overfit
- Prefer terms you can explain physically
- Judge on adjusted R-squared, and keep the model small

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ANALYZE - BLACK BELT ONLY

Validating the Meridian model on held-out data

Split	**180 lots ->** **135 train / 45 holdout**	Split by TIME: months 1-7 train, months 8-9 holdout. Mimics real deployment.
Training fit	**adj R2 = 0.737** **RMSE = 1.12 pp**	Fitted on 135 lots. Three terms, all significant at alpha 0.05.
Holdout performance	**R2 on holdout = 0.708** **RMSE = 1.24 pp**	Only a small drop - the model generalises. A drop to R2 0.31 would mean overfit.

If holdout performance collapses, the model is describing last quarter, not the process. Do not present it.

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Model diagnostics

A model that FITS is not yet a model that is VALID. Adjusted R-squared says nothing about whether the assumptions hold. The residuals do.

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The four residual plots you must always produce

1

Residuals vs FITTED

Should be a formless cloud. Any pattern means the model is wrong.

2

Normal probability plot

Residuals should follow the straight line - confirms the normality assumption.

3

Residuals vs ORDER

Plotted in time order. Any drift or cycling means the errors are not independent.

4

Histogram of residuals

Should look roughly bell shaped and centred on zero

5

Residuals vs each X

Curvature against one X shows exactly which term needs a quadratic

6

The rule

If any plot shows structure, the model is not finished - fix it before presenting

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Residuals vs fitted - diagnosing the shape

FUNNEL SHAPE

Non-constant variance

- Spread widens as fitted values rise (heteroscedasticity)
- Prediction intervals are wrong at the wide end
- Meridian: yield errors are larger at low-yield lots
- Fix: transform Y (log or Box-Cox), or use weighted regression
- Do NOT just report it and move on

CURVATURE

A missing quadratic term

- Residuals arc: negative, then positive, then negative
- The relationship is real but not linear
- Meridian: yield falls off at BOTH low and high oven temperature
- Fix: add a squared term (Temp^2) and refit
- Adjusted R-squared usually jumps sharply when you do

DRIFT vs ORDER

Autocorrelation

- Residuals plotted in time order trend or cycle
- The independence assumption is violated - p-values are wrong
- Meridian: paste age drifts up through the week and resets Monday
- Fix: add the time-varying X to the model, or use a time-series method
- Never ignore - this makes every p-value optimistic

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What to do when a residual plot shows structure

1. Identify which plot failed and what shape it showed
2. Funnel: transform Y with log or Box-Cox and refit
3. Curvature: add the squared term for the offending X and refit
4. Drift: add the time-varying X, or block by time period
5. Outliers: investigate physically before deciding to keep or exclude
6. Refit and re-plot - never present a model whose residuals you have not seen

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ANALYZE - BLACK BELT ONLY

Multicollinearity - when your Xs explain each other

If two Xs move together, the model cannot tell which one drives Y. Coefficients become unstable, signs flip, and standard errors inflate.

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Variance Inflation Factor

VIF	**VIF_i = 1 / (1 - R2_i)**	R2_i is from regressing Xi against all the OTHER Xs, not against Y.
VIF = 1	**Xi is uncorrelated with the other Xs**	Ideal. The coefficient estimate is as precise as the data allows.
VIF > 5	**R2_i > 0.80 CONCERN**	Meridian: oven temp VIF = 6.2, conveyor speed VIF = 5.9 - they move together.
VIF > 10	**R2_i > 0.90 SERIOUS**	Standard error inflated over 3x. The coefficient is effectively uninterpretable.

VIF above 5 warrants action; above 10 the coefficient must not be quoted to a sponsor as a standalone effect.

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Meridian collinearity worked through

THE PROBLEM

Two Xs, one signal

- Oven temperature and conveyor speed correlate at r = 0.91
- Operators raise temperature whenever they raise speed
- Temp coefficient -0.081, speed coefficient +0.019 with p = 0.38
- VIF: temp 6.2, speed 5.9 - both above the concern threshold
- The model cannot separate their effects from this data

THE FIX

Engineer one variable

- Create thermal exposure index = time above 217 C liquidus
- That single variable captures what both Xs jointly control
- Refit: index coefficient -0.196, p < 0.001, VIF = 1.3
- Adjusted R-squared holds at 0.734 with one fewer term
- The coefficient is now stable and physically meaningful

THE LESSON

Design beats correction

- Observational data cannot break a correlation the process created
- Only a designed experiment can vary temp and speed independently
- This is exactly why Analyze hands over to DOE in Improve
- Report VIF alongside every multiple regression, always
- Never present a coefficient with VIF above 10

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Analysis of Variance

Six lines, one question: are their mean yields the same? Running fifteen t-tests would guarantee a false alarm. ANOVA answers it in one test.

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ANALYZE - BLACK BELT ONLY

Alpha inflation - why repeated t-tests fail

Number of pairs	**pairs = k(k-1)/2**	4 groups gives 6 pairwise t-tests. Meridian's 6 lines gives 15.
Family-wise error	**FWE = 1 - (1 - alpha)^m**	4 groups, 6 tests, alpha .05: FWE = 1 - 0.95^6 = 0.265 - about 26 percent.
At 6 groups	**1 - 0.95^15**	FWE = 0.537. You are MORE likely than not to raise at least one false alarm.
ANOVA's answer	**One test, one alpha**	One F-test at alpha = 0.05 keeps the overall false alarm risk at 5 percent.

This is the single clearest reason a Black Belt reaches for ANOVA rather than a stack of t-tests.

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The F-statistic - between over within

The logic	**F = variance BETWEEN groups / variance WITHIN groups**	If the groups are really the same, both estimate the same sigma^2 and F is near 1.
Between (MS treatment)	**SS_between / (k-1)**	Meridian 6 lines: SS_between = 214.7, df = 5, MS = 42.9.
Within (MS error)	**SS_within / (N-k)**	SS_within = 361.2, df = 174, MS = 2.08. This is the pooled noise.
The test	**F = MS_between / MS_within**	F = 42.9 / 2.08 = 20.6, df 5 and 174, p < 0.001. At least one line differs.

A large F means the gaps between group means are large relative to the noise inside the groups.

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ANALYZE - BLACK BELT ONLY

Post-hoc testing - which lines actually differ

The F-test says 'somebody is different'. Tukey's HSD tells you who, while holding the family-wise error rate at your chosen alpha.

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ANALYZE - BLACK BELT ONLY

Tukey's Honestly Significant Difference

The statistic	**HSD = q_(alpha,k,df) x sqrt(MS_error / n)**	q from the studentised range table: k=6, df=174, alpha .05 gives q = 4.03.
Meridian HSD	**4.03 x sqrt(2.08 / 30)**	HSD = 4.03 x 0.263 = 1.06 pp. Any two line means differing by more than 1.06 pp differ significantly.
The comparison	**|mean_i - mean_j| > HSD**	Line 1 93.2 vs line 5 88.4: gap 4.8 pp > 1.06 -> significantly different.
The non-difference	**Line 1 93.2 vs line 2 92.9**	Gap 0.3 pp < 1.06 -> NOT significantly different. Do not chase this gap.

Tukey controls family-wise error across ALL pairwise comparisons - which a stack of t-tests does not.

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Post-hoc results across the six Meridian lines

GROUP A - HIGH

Not different from each other

- Line 1 Singapore: mean 93.2 pp
- Line 2 Singapore: mean 92.9 pp
- Line 3 Penang: mean 92.4 pp
- All pairwise gaps below HSD of 1.06
- Treat these three as one performance population

GROUP B - MIDDLE

Distinct from both others

- Line 4 Penang: mean 91.1 pp
- Differs from line 1 (2.1 pp) and from line 5 (2.7 pp)
- Sits alone between the two clusters
- Worth a separate targeted investigation
- Recent stencil change is the leading hypothesis

GROUP C - LOW

The project target

- Line 5 Suzhou: mean 88.4 pp
- Line 6 Suzhou: mean 88.9 pp
- Not different from each other - a PLANT effect, not a line effect
- 4.8 pp below the best lines, worth SGD 2.6M a year
- Analyze focus moves to what is common to Suzhou

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Two-way ANOVA and interaction effects

One factor at a time hides the most important finding in a multi-site business: that a fix which works at one plant can backfire at another.

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The two-way ANOVA model and its three tests

The model	$Y = \mu + A_i + B_j + (AB)_{ij} + e$	A = plant main effect, B = shift main effect, AB = the interaction term.
Main effect A	$F_A = MS_A / MS_{error}$	Plant: $F = 27.4$, $p < 0.001$. Plant means genuinely differ.
Main effect B	$F_B = MS_B / MS_{error}$	Shift: $F = 1.58$, $p = 0.211$. Overall, shift does NOT move yield on average.
Interaction AB	$F_{AB} = MS_{AB} / MS_{error}$	Plant x shift: $F = 4.12$, $p = 0.004$. The shift effect DEPENDS on which plant.

Read the INTERACTION first. If it is significant, the main effects cannot be interpreted on their own.

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The Meridian plant x shift interaction

SINGAPORE

Night shift is BETTER

- Day shift yield 92.6 pp
- Night shift yield 93.8 pp - 1.2 pp better
- Air conditioning load is lower overnight
- Bay temperature holds within 2 C at night
- Night shift looks like the model to copy

PENANG

Almost no shift effect

- Day shift yield 91.9 pp
- Night shift yield 91.6 pp - a 0.3 pp gap, not significant
- Climate control is independent of external load
- Shift makes essentially no difference here
- A shift-based intervention would waste money

SUZHOU

Night shift is WORSE

- Day shift yield 89.7 pp
- Night shift yield 87.6 pp - 2.1 pp WORSE
- Overnight humidity rises above 80 percent, no dehumidification
- Solder paste absorbs moisture; voiding rises
- Copying Singapore's night-shift practice would HURT Suzhou

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ANALYZE - BLACK BELT ONLY

This is why interactions matter for Improve

The main effect said shift does not matter (p = 0.21). The interaction said it matters in opposite directions at two plants. Roll out the wrong fix and you destroy value.

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Non-parametric tests

Cycle times, repair times and warranty ages are almost never normal. When the assumption fails and transformation does not rescue it, switch tests - do not proceed.

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Parametric vs non-parametric

PARAMETRIC

Assumes a distribution

- Assumes the data or residuals are normal
- Tests MEANS, using the actual data values
- More powerful WHEN the assumption holds
- Sensitive to outliers and skew
- t-tests, ANOVA, regression

NON-PARAMETRIC

Distribution-free

- Makes no distributional assumption
- Tests MEDIANS, using the RANKS of the data
- Slightly less powerful when normality does hold
- Robust to outliers and to heavy skew
- Mann-Whitney, Kruskal-Wallis, Wilcoxon, Mood's median

WHEN TO SWITCH

The decision rule

- Anderson-Darling $p < 0.05$ and Box-Cox does not fix it
- Small sample where normality cannot be assessed at all
- Ordinal data - satisfaction scores, severity grades
- Heavy outliers that are real and cannot be excluded
- When the MEDIAN is the more meaningful business measure

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ANALYZE - BLACK BELT ONLY

Mann-Whitney - Penang vs Suzhou repair time

The situation	**Repair time is right-skewed** **AD p = 0.001**	Box-Cox did not normalise it. A 2-sample t-test is not defensible here.
The hypotheses	**H0: median_PN = median_SZ** **Ha: medians differ**	n = 45 repairs per plant, alpha = 0.05, two-tailed.
The method	**Rank all 90 values together, sum by group**	W = 1642 for Penang. Compare against the Mann-Whitney distribution.
The result	**p = 0.008** **Reject H0**	Median 4.2 hours Penang vs 6.8 hours Suzhou. A real 2.6-hour gap.

Report the MEDIANS, never the means, when you have used a rank-based test.

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Kruskal-Wallis across all six lines

The situation	**Yield residuals non-normal after an oven fault**	Two lines carry genuine outliers that engineering confirmed are real.
The hypotheses	**H0: all 6 medians equal Ha: at least one differs**	N = 180 lots, alpha = 0.05.
The statistic	**H approximately chi-square with k-1 df**	H = 41.7, df = 5, p < 0.001. At least one line median differs.
Follow-up	**Pairwise Mann-Whitney with Bonferroni**	alpha adjusted to 0.05/15 = 0.0033. Suzhou lines separate from the rest, as ANOVA found.

Kruskal-Wallis is the ANOVA of ranks. It answers the same question without the normality assumption.

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Wilcoxon signed-rank and Mood's median

WILCOXON SIGNED-RANK

The paired test

- Same units measured before and after a change
- Ranks the ABSOLUTE differences, then applies the signs
- Meridian: calibration time before vs after the new fixture, 30 techs
- Median saving 11 minutes, p = 0.002
- The non-parametric replacement for the paired t-test

MOOD'S MEDIAN

The robust test

- Counts how many observations per group are above the grand median
- Then runs a chi-square on that contingency table
- Extremely robust to outliers and to unequal spread
- The least powerful of the non-parametric family
- Use when the data is badly contaminated but must still be tested

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ANALYZE - BLACK BELT ONLY

BLACK BELT ONLY - Multi-vari studies

Before spending money hunting one X, find out WHERE the variation lives. Most teams chase machine settings when the variation is actually within a single part.

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The three families of variation

POSITIONAL

Within-piece / within-run

- Variation WITHIN a single part or across positions
- Left cavity vs right cavity in the same mould
- Position 1 to 12 across the panel on the SMT line
- Meridian: connector pin 1 vs pin 20 solder volume
- Points to: fixturing, tooling, thermal gradient, layout

CYCLICAL

Piece-to-piece / batch-to-batch

- Variation between consecutive pieces or batches
- Unit 1 vs unit 2 vs unit 3 in the same run
- Lot A vs lot B of the same material
- Meridian: paste lot changes between production batches
- Points to: material lots, setup, operator, changeover

TEMPORAL

Time-to-time

- Variation across hours, shifts, days, weeks
- Morning vs afternoon; Monday vs Friday
- Before vs after a maintenance event
- Meridian: humidity rising through the Suzhou night shift
- Points to: environment, wear, drift, fatigue, seasonality

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Nested sampling design at Meridian - line 5 Suzhou

1

Y measured

Solder paste deposit volume in cubic mm at final inspection

2

POSITIONAL level

3 positions on each panel: left, centre, right

3

CYCLICAL level

3 consecutive panels sampled back-to-back at each time point

4

TEMPORAL level

5 time points across the shift: 07:00, 10:00, 13:00, 16:00, 19:00

5

Total observations

3 x 3 x 5 = 45 measurements over one full shift

6

Rule during study

Change nothing - no setting adjustment, no lot change, no operator swap

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ANALYZE - BLACK BELT ONLY

Variance components from the Meridian study

Total variation	**s2_total = s2_pos + s2_cyc + s2_temp**	Total observed variance in deposit volume: 0.0412 mm^6.
Positional	**s2_pos / s2_total**	0.0289 / 0.0412 = 70.1 percent. Left-to-right across the panel dominates.
Cyclical	**s2_cyc / s2_total**	0.0071 / 0.0412 = 17.2 percent. Panel-to-panel variation is modest.
Temporal	**s2_temp / s2_total**	0.0052 / 0.0412 = 12.7 percent. Drift across the shift is the smallest family.

70 percent positional redirects the whole project: the answer is stencil and squeegee, not shift scheduling.

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Reading the multi-vari chart

WHAT YOU SEE

Three visual signatures

- Tall vertical spread within each small group = POSITIONAL
- Groups sitting at different heights close together = CYCLICAL
- The whole cluster drifting left to right = TEMPORAL
- One glance tells you which family dominates
- No statistics needed to see the answer

MERIDIAN RESULT

Positional dominates

- Within each panel, left to right spans 0.34 mm^3
- Between consecutive panels, means differ by only 0.09 mm^3
- Across the shift, the cluster drifts just 0.07 mm^3
- The vertical spread visibly dwarfs everything else
- Confirmed by variance components at 70 percent positional

WHAT IT KILLS

Theories now closed

- 'It is the night shift' - temporal is only 12.7 percent
- 'It is the paste lot' - cyclical is only 17.2 percent
- 'It is operator technique' - would show as cyclical, and does not
- Three expensive investigations avoided in one shift of data
- Log them in the rejected-causes register with the numbers

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ANALYZE - BLACK BELT ONLY

Chi-square and contingency tables

When both Y and X are categorical - defect type against shift, line or supplier - chi-square is the test that decides whether the association is real.

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The chi-square test of independence

Hypotheses	**H0: the two variables are INDEPENDENT**	Ha: there is an association between defect type and plant.
Expected count	**E = (row total x col total) / grand total**	Row 'solder void' 240, col 'Suzhou' 300, grand 900: E = 240x300/900 = 80.
The statistic	**chi2 = sum of (O - E)^2 / E**	Suzhou solder voids: observed 118 vs expected 80 -> (38)^2/80 = 18.05 for that cell.
Degrees of freedom	**df = (rows-1) x (cols-1)**	4 defect types x 3 plants: df = 3 x 2 = 6. chi2 = 47.3, p < 0.001. Reject H0.

A significant chi-square proves association, not causation, and does not say WHICH cells drive it.

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Meridian contingency table - defect type by plant

1

Solder void

Singapore 54, Penang 68, Suzhou 118. Expected 80 each - Suzhou is far over.

2

Sensor drift

Singapore 41, Penang 47, Suzhou 62. Mildly over-represented at Suzhou.

3

Keypad delamination

Singapore 62, Penang 58, Suzhou 60. Almost exactly as expected everywhere.

4

Battery capacity

Singapore 43, Penang 87, Suzhou 60. Penang is the outlier here.

5

Column totals

Singapore 200, Penang 260, Suzhou 300, grand total 760 defects

6

Conclusion

chi2 = 47.3, p < 0.001. Defect PROFILE differs by plant, not just defect volume.

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ANALYZE - BLACK BELT DEPTH

From statistics to a decision the sponsor can make

No steering committee has ever approved funding because of a p-value. Your last job in Analyze is translation - into money, into risk, into a decision.

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Translating three statistics into business language

A COEFFICIENT

b = -0.151 for humidity

- Statistical: each 1 percent RH costs 0.151 pp of yield
- Business: Suzhou runs 22 percent RH above Singapore
- That is 3.3 pp of yield, or 6,900 units a year
- At SGD 275 contribution margin: SGD 1.9M a year
- Decision: fund the SGD 340k dehumidification retrofit

A P-VALUE

p = 0.004 on the interaction

- Statistical: the shift effect depends on the plant
- Business: the Singapore night-shift practice must NOT be rolled out globally
- Copying it to Suzhou would cost about 2.1 pp of yield
- Risk avoided: roughly SGD 1.2M a year of value destruction
- Decision: approve plant-specific solutions, not one standard

AN R-SQUARED

adj R-squared = 0.737

- Statistical: three Xs explain 74 percent of yield variation
- Business: control these three and most of the problem is controllable
- The remaining 26 percent needs a designed experiment to find
- Confidence: we know where to aim the Improve phase budget
- Decision: approve a DOE on the three confirmed Xs

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Analyze tollgate checklist

1

Root causes validated

Each surviving X has a test, a p-value and an effect size in real units

2

Assumptions documented

Normality, independence and variance checks shown for every test

3

Model diagnostics clean

Residual plots produced, VIF reported, holdout validation done

4

Rejected causes logged

Every cleared theory recorded with the evidence that cleared it

5

Benefit quantified

Each confirmed X carries a dollar value the finance partner accepts

6

Improve scope defined

The DOE factors and the pilot sites are named and agreed

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ANALYZE - BLACK BELT DEPTH

Analyze is complete when the causes are PROVEN

Not brainstormed, not suspected, not obvious. Proven - with a test, an effect size, a validated model and a dollar figure attached to each one.

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Apply it to YOUR capstone this afternoon

1. **Chart your Y over time**
   Run chart with median and run rules - is your process even stable?

2. **Pareto your defects by COST**
   Weight by cost of one occurrence, not by count

3. **Run one fishbone and 5 Whys**
   Multi-vote to five testable candidate Xs

4. **Select and run two tests**
   Use the decision tree; document the assumption checks

5. **Fit a multiple regression**
   Report adjusted R-squared, VIF and all four residual plots

6. **Write two sponsor sentences**
   The decision you want, and the dollar value behind it

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WHAT YOU'LL DO

Hands-On Labs — Analyze (1)

1. **Lab 15 — Variation, Run Charts, Pareto and Stratification — In Depth**
   A run chart with pattern analysis, a Pareto chart and a stratified defect analysis. • Data: lab-15-run-chart-pareto-stratification.xlsx

2. **Lab 16 — Fishbone, 5 Whys and Cause Prioritisation — In Depth**
   A fishbone diagram, 5 Whys chains and a prioritised list of testable candidate… • Data: lab-16-cause-prioritisation.xlsx

3. **Lab 17 — Hypothesis Testing — In Depth, With Test Selection**
   Completed hypothesis tests on your capstone's candidate causes with documented… • Data: lab-17-hypothesis-testing.xlsx

4. **Lab 18 — Multiple Regression and Model Diagnostics (Black Belt Only)**
   A validated multiple regression model with full residual and multicollinearity… • Data: lab-18-multiple-regression.xlsx

5. **Lab 19 — ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)**
   One-way and two-way ANOVA on your capstone data plus a non-parametric confirmation. • Data: lab-19-anova-interactions-nonparametric.xlsx

6. **Lab 20 — Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)**
   A multi-vari chart decomposing your capstone variation plus a chi-square… • Data: lab-20-multivari-chisquare.xlsx

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WHAT YOU'LL DO

Hands-On Labs — Analyze (2)

1

Lab 21 — Analyze Tollgate — Proving Root Cause and Telling the Data Story

An Analyze tollgate pack with proven root causes, evidence and a business-language… • Data: lab-21-analyze-tollgate-evidence.xlsx

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DMAIC · ANALYZE · HANDS-ON

Variation, Run Charts, Pareto and Stratification — In Depth

LAB 15

A full re-derivation of the foundation of all Six Sigma analysis: the distinction between common and special cause variation. Acting on common cause as if it were special (tampering) makes processes worse — this is the single most expensive mistake in process management. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A run chart with pattern analysis, a Pareto chart and a stratified defect analysis.

Tools & techniques: Common vs special cause, run chart patterns, tampering, Pareto principle, stratification, boxplots

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Variation, Run Charts, Pareto and Stratification — In Depth

STEPS 1-6 OF 11

1. Recap common versus special cause variation in full, with the management response each demands — process change versus local investigation.
2. Understand tampering (Deming's funnel): adjusting a stable process in response to common-cause noise increases variation. Demonstrate it with the bead or funnel experiment.
3. Plot your capstone Y as a run chart over time, with the median as the centre line.
4. Recap and apply the six run chart patterns in full: trend, shift, cluster, mixture, oscillation and bias — with the run rules that detect each.
5. Mark any special-cause signals and investigate what changed at that point in time. Special causes must be explained, not averaged away.
6. Recap the Pareto principle and build a Pareto chart of defect categories for your capstone, with the cumulative percentage line.

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Variation, Run Charts, Pareto and Stratification — In Depth

STEPS 7-11 OF 11

7. Identify the vital few categories accounting for roughly 80 percent of the impact. Weight by cost, not count, where defect costs differ materially.

8. Now stratify. Re-draw the Pareto separately by plant, line, shift, operator, product family and supplier.

9. Look for a stratification factor where the Pareto shape changes dramatically — that factor is a strong candidate X and tells you where to focus.

10. Build boxplots of your Y across each stratification factor to compare medians, spread and outliers visually before testing statistically.

11. Record your top three candidate Xs from stratification — these feed directly into the hypothesis tests in Lab 17.

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LAB 15 · VERIFY

Variation, Run Charts, Pareto and Stratification — In Depth

✅ Check your work

Your run chart is assessed against all six patterns, your Pareto is weighted appropriately, and stratification has produced at least three named candidate Xs for statistical testing.

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DMAIC · ANALYZE · HANDS-ON

Fishbone, 5 Whys and Cause Prioritisation — In Depth

LAB 16

Re-derive cause-and-effect analysis in full — to the depth at which you can facilitate it for a Green Belt team, which is a core Black Belt accountability. The output is a prioritised, testable cause list, not an unmanageable wall of sticky notes. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A fishbone diagram, 5 Whys chains and a prioritised list of testable candidate causes.

Tools & techniques: Ishikawa 5M+E, brainstorming facilitation, 5 Whys, multi-voting, cause-and-effect matrix, hypothesis formulation

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Fishbone, 5 Whys and Cause Prioritisation — In Depth

STEPS 1-6 OF 11

1. Recap brainstorming facilitation in full: divergence before convergence, no criticism during generation, and how to stop the highest-paid opinion from dominating.
2. Recap the fishbone structure: the effect at the head, and 5M+E bones — Manpower, Method, Machine, Material, Measurement, Environment.
3. Write your capstone problem statement precisely at the head of the fishbone. A vague effect produces vague causes.
4. Facilitate the fishbone with your team, generating at least 25 candidate causes across all six bones. Thin bones signal blind spots, not absence of causes.
5. Recap the 5 Whys and its discipline: continue until the answer is a process or system failure, never a person to blame.
6. Apply 5 Whys to your top five fishbone causes. Note where a single chain branches — real causal chains are often trees, not lines.

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Fishbone, 5 Whys and Cause Prioritisation — In Depth

STEPS 7-11 OF 11

7. Recap multi-voting and apply it to reduce your cause list to the ten most plausible candidates.

8. Build a cause-and-effect matrix: score each candidate cause against each CTQ, weighted by CTQ importance, to produce a ranked cause list.

9. Filter the ranked list for TESTABILITY — for each cause, can you state a hypothesis and obtain data? An untestable cause cannot be proven in Analyze.

10. Convert your top five testable causes into formal hypothesis statements ready for Lab 17.

11. Note how you would coach a Green Belt team through this session, and the two facilitation mistakes you would watch for.

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LAB 16 · VERIFY

Fishbone, 5 Whys and Cause Prioritisation — In Depth

✅ Check your work

Your fishbone carries 25+ causes across all six bones, 5 Whys chains terminate at process rather than person, and you have five testable causes written as formal hypotheses.

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DMAIC · ANALYZE · HANDS-ON

Hypothesis Testing — In Depth, With Test Selection

LAB 17

A full re-derivation of hypothesis testing, including the test selection logic that Green Belts most often get wrong. You must be able to choose the right test, defend the choice, and explain the result in business language. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

Completed hypothesis tests on your capstone's candidate causes with documented conclusions.

Tools & techniques: H0/Ha, alpha, beta, power, p-values, t-tests, paired t, F-test, proportion tests, test selection logic

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Hypothesis Testing — In Depth, With Test Selection

STEPS 1-6 OF 12

1. Recap the logic of hypothesis testing in full: assume H0 (no difference), then ask how likely the observed data would be if H0 were true.
2. Recap H0 and Ha formulation. H0 always contains the equality; Ha carries the claim you are trying to establish.
3. Recap alpha, beta and power. Alpha is producer's risk (Type I, false alarm); beta is consumer's risk (Type II, missed signal); power = 1 - beta.
4. Understand the practical consequence: an underpowered test that fails to reject H0 has proven nothing. Check power before concluding 'no difference'.
5. Learn the test selection logic in full. Continuous Y: 1 sample vs target = 1-sample t; 2 independent groups = 2-sample t; paired = paired t; 3+ groups = ANOVA; variances = F-test or Levene.
6. Learn the discrete-Y branch: 1 proportion = 1-proportion test; 2 proportions = 2-proportion test; categorical association = chi-square.

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Hypothesis Testing — In Depth, With Test Selection

STEPS 7-12 OF 12

7	Check the assumptions before every test: normality, independence and equal variance. A violated assumption invalidates the p-value entirely.
8	Take your five candidate causes from Lab 16 and select the correct test for each. Write down the justification for each selection.
9	State H0 and Ha for each test, set alpha (normally 0.05), and confirm you have adequate sample size and power.
10	Run each test and record the p-value. Apply the decision rule: p < alpha rejects H0.
11	Distinguish statistical from PRACTICAL significance. A p-value of 0.001 on a 0.2 percent improvement is real but worthless — always report effect size alongside p.
12	Build your rejected-causes log so nobody re-litigates cleared causes in Improve, and record confirmed causes in your capstone pack.

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LAB 17 · VERIFY

Hypothesis Testing — In Depth, With Test Selection

✅ Check your work

Every test has a documented selection justification and assumption check, results report effect size alongside the p-value, and you maintain both a confirmed-cause and a rejected-cause log.

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Multiple Regression and Model Diagnostics (Black Belt Only)

LAB 18

BLACK BELT ONLY — the Green Belt course teaches simple linear regression with one X. Real processes have several interacting Xs. Multiple regression models them together, but a model that fits is not yet a model that is valid, so the diagnostics matter as much as the coefficients. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A validated multiple regression model with full residual and multicollinearity diagnostics.

Tools & techniques: Multiple regression, coefficients, adjusted R-squared, residual analysis, multicollinearity/VIF, model selection, overfitting

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Multiple Regression and Model Diagnostics (Black Belt Only)

STEPS 1-5 OF 13

1. Recap simple linear regression: the fitted line, slope interpretation, R-squared and the correlation-is-not-causation trap.
2. Extend to multiple regression: Y = b0 + b1X1 + b2X2 + ... + e, where each coefficient is the effect of that X holding all other Xs constant.
3. Understand why that 'holding constant' clause matters — it is what makes multiple regression fundamentally different from running several simple regressions.
4. Assemble your capstone dataset with the Y and at least four candidate Xs, with enough rows (a common rule of thumb is 10-15 observations per X).
5. Fit the full model and record the coefficients, their p-values, R-squared and adjusted R-squared.

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Multiple Regression and Model Diagnostics (Black Belt Only)

STEPS 6-9 OF 13

6. Use ADJUSTED R-squared, not R-squared, to compare models. Plain R-squared always rises when you add an X, even a random one.

7. Check multicollinearity with Variance Inflation Factors. VIF > 5 is a concern, VIF > 10 is serious — correlated Xs make coefficients unstable and signs nonsensical.

8. Resolve multicollinearity by dropping or combining redundant Xs, then refit.

9. Run the residual diagnostics in full: residuals versus fitted values (should show no pattern), normal probability plot of residuals, and residuals versus order (independence).

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Multiple Regression and Model Diagnostics (Black Belt Only)

STEPS 10-13 OF 13

10 Interpret residual patterns: a funnel shape indicates non-constant variance, curvature indicates a missing quadratic term, and drift over order indicates autocorrelation.

11 Reduce to the final model using backward elimination — drop the least significant X, refit, and repeat until all remaining Xs are significant.

12 Guard against overfitting: hold back a validation subset, or use cross-validation, and confirm the model predicts data it has not seen.

13 Translate the final model into business language: for each significant X, state what a one-unit change does to Y in customer or dollar terms.

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LAB 18 · VERIFY

Multiple Regression and Model Diagnostics (Black Belt Only)

✅ Check your work

Your final model has all VIFs under 5, residual plots show no pattern, adjusted R-squared is reported, the model has been validated on held-out data, and each coefficient is stated in business terms.

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ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

LAB 19

BLACK BELT ONLY. Comparing three or more groups with repeated t-tests inflates the false-positive rate badly — ANOVA solves this. Two-way ANOVA then reveals interaction effects that one-factor analysis cannot see. And when normality fails, non-parametric tests are the correct answer rather than proceeding regardless. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the

You'll build

One-way and two-way ANOVA on your capstone data plus a non-parametric confirmation.

Tools & techniques: One-way ANOVA, F-statistic, post-hoc Tukey, two-way ANOVA, interaction effects, Mann-Whitney, Kruskal-Wallis, Mood's median

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ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

STEPS 1-5 OF 13

1. Understand alpha inflation: comparing 4 groups pairwise needs 6 t-tests, and at alpha = 0.05 the family-wise false-positive risk rises to about 26 percent.
2. Recap one-way ANOVA: it partitions total variation into between-group and within-group components, and the F-statistic is their ratio.
3. State the ANOVA hypotheses: H0 is that all group means are equal; Ha is that at least one differs — ANOVA does not tell you which.
4. Check the ANOVA assumptions: normality of residuals, independence, and equal variances (test with Levene's or Bartlett's).
5. Run one-way ANOVA on a capstone factor with 3+ levels — plant, line, shift, product family or supplier.

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ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

STEPS 6-9 OF 13

6. If ANOVA is significant, run a post-hoc Tukey HSD to identify WHICH pairs differ, with the family-wise error rate controlled.

7. Extend to two-way ANOVA with two factors at once — for example plant AND shift on your capstone Y.

8. Read the interaction term. A significant interaction means the effect of one factor DEPENDS on the level of the other — for instance night shift hurts Suzhou but not Singapore.

9. Plot the interaction. Non-parallel lines indicate interaction; crossing lines indicate a strong one that invalidates any single-factor conclusion.

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ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

STEPS 10-13 OF 13

10 Understand why this matters for Improve: with a significant interaction, a fix that works at one plant may fail or backfire at another.

11 Now the non-parametric branch. When normality fails and transformation does not fix it, switch tests rather than ignoring the violation.

12 Map each test to its non-parametric equivalent: 2-sample t -> Mann-Whitney; one-way ANOVA -> Kruskal-Wallis; paired t -> Wilcoxon signed-rank; and Mood's median for heavy outliers.

13 Re-run one of your significant findings using its non-parametric equivalent. Agreement across both strengthens your Day 5 defence considerably.

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LAB 19 · VERIFY

ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

✅ Check your work

Your ANOVA assumptions are tested and documented, significant results carry a post-hoc analysis, you have interpreted an interaction plot, and at least one finding is confirmed non-parametrically.

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DMAIC · ANALYZE · HANDS-ON

Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

LAB 20

BLACK BELT ONLY. Before spending money hunting a root cause, find out WHERE the variation actually lives. A multi-vari study separates positional, cyclical and temporal variation — and frequently reveals that the team has been investigating entirely the wrong family. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A multi-vari chart decomposing your capstone variation plus a chi-square association test.

Tools & techniques: Multi-vari studies, positional/cyclical/temporal families, variation decomposition, chi-square, contingency tables, expected frequencies

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Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

STEPS 1-5 OF 13

1. Understand the three variation families. POSITIONAL: within a single unit or across positions in a machine. CYCLICAL: unit to unit, batch to batch. TEMPORAL: shift to shift, day to day.
2. Understand the payoff: if 70 percent of variation is positional, then investigating shift patterns is wasted effort no matter how thorough the investigation.
3. Design a multi-vari sampling plan for your capstone: sample multiple positions within units, consecutive units, and repeat across time periods.
4. Ensure the sampling captures all three families simultaneously — this is a nested sampling design, not three separate studies.
5. Collect the data and plot the multi-vari chart, with positional variation innermost and temporal outermost.

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Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

STEPS 6-9 OF 13

6. Read the chart: the family showing the largest vertical spread dominates. That is where your root cause investigation should concentrate.

7. Quantify the contribution of each family as a percentage of total variation using variance components.

8. Re-prioritise your candidate cause list from Lab 16 against the dominant variation family, and drop causes that belong to a family contributing little.

9. Now chi-square. Build a contingency table of two categorical variables from your capstone — defect type against shift, supplier or line.

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Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

STEPS 10-13 OF 13

10 Calculate expected frequencies under independence: expected = (row total x column total) / grand total.

11 Check the validity condition: all expected frequencies should be 5 or more, or the chi-square approximation breaks down and categories must be combined.

12 Compute the chi-square statistic and its p-value, and conclude whether defect type is associated with your categorical factor.

13 Translate a significant association into a testable process hypothesis — association is not causation, but it does tell you where to look.

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LAB 20 · VERIFY

Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

✅ Check your work

Your multi-vari chart captures all three families with quantified variance contributions, your cause list is re-prioritised against the dominant family, and your chi-square satisfies the expected-frequency condition.

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DMAIC · ANALYZE · HANDS-ON

Analyze Tollgate — Proving Root Cause and Telling the Data Story

LAB 21

The Analyze tollgate. Consolidate everything from Labs 15-20 into a proven root cause set, translate the statistics into business language, and rehearse the challenges you will face. Sponsors do not fund p-values; they fund decisions. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

An Analyze tollgate pack with proven root causes, evidence and a business-language summary.

Tools & techniques: Evidence consolidation, root cause validation, statistical-to-business translation, tollgate review, challenge preparation

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Analyze Tollgate — Proving Root Cause and Telling the Data Story

STEPS 1-6 OF 11

1. Assemble every analytical result from Labs 15-20: stratification, hypothesis tests, regression model, ANOVA, multi-vari and chi-square.
2. For each candidate cause, record the verdict — PROVEN (statistically supported), REJECTED (tested and cleared) or UNTESTED (no data available).
3. Apply the triangulation rule: a root cause is strongest when supported by more than one independent method — for instance stratification plus a hypothesis test plus a regression coefficient.
4. Check each proven cause for practical significance. Quantify how much of the Y gap each explains, in percent and in dollars.
5. Rank your proven causes by the size of their contribution to the Y gap.
6. Confirm your causes are ACTIONABLE — a proven cause you cannot influence belongs in the risk register, not the improvement plan.

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Analyze Tollgate — Proving Root Cause and Telling the Data Story

STEPS 7-11 OF 11

7. Translate each finding into one business sentence: 'Night shift at Suzhou runs 2.3 percentage points lower first-pass yield, costing about $180k annually.'

8. Build the tollgate pack: baseline, analysis method, proven causes, rejected causes, quantified impact and the recommendation to proceed.

9. Rehearse the challenges you should expect: was the sample adequate, were assumptions checked, could a confounding variable explain this, is the measurement system valid?

10. Prepare your evidence for each challenge — this is exactly what the Day 5 steering committee will probe.

11. Present the tollgate to a peer group and capture their challenges, then strengthen any weak points before Day 4.

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LAB 21 · VERIFY

Analyze Tollgate — Proving Root Cause and Telling the Data Story

✅ Check your work

Every candidate cause carries a verdict, proven causes are triangulated across multiple methods and quantified in dollars, and you have rehearsed a prepared response to each expected challenge.

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PHASE RECAP

Recap — Analyze

- Variation, run charts and Pareto — in depth — Common vs special cause, run chart patterns, stratification and Pareto analysis re-derived in full.
- Fishbone and 5 Whys — in depth — Cause-and-effect analysis by 5M+E, 5 Whys discipline and multi-voting — the tools you will coach Green Belts through.
- Hypothesis testing — in depth — H0/Ha formulation, alpha, p-values, confidence levels, test selection and Type I/II errors, worked thoroughly.
- Simple regression & correlation — in depth — Correlation coefficients, the regression line and the causation trap — the foundation multiple regression builds on.
- Multiple regression — Model Y against several Xs at once — the Green Belt's simple linear regression will not do here.
- Model diagnostics — R-squared adjusted, residual plots, multicollinearity (VIF) — a model that fits is not yet a model that is valid.

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DMAIC · IMPROVE

Improve at Black Belt Depth — Design of Experiments

Solution selection · Lean countermeasures · FMEA · Full & fractional factorial DOE · Confounding · RSM · Robust design

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Key Concepts — Improve (1)

1. **Solution generation & selection — in depth**
   Brainstorming, brainwriting, benchmarking and weighted solution selection matrices, re-derived in full.

2. **Lean countermeasures — in depth**
   5S, poka-yoke, pull, JIT, Heijunka, Jidoka and standard work — the Lean toolkit taught thoroughly.

3. **FMEA and RPN — in depth**
   Severity, Occurrence and Detection scoring, RPN calculation and the re-scored post-mitigation FMEA.

4. **Why DOE beats OFAT**
   One-factor-at-a-time misses interactions and wastes runs. DOE varies factors together, by design.

5. **Full factorial design**
   2^k designs estimate every main effect and every interaction — the gold standard when runs are affordable.

6. **Main effects & interaction plots**
   Read the design: a non-parallel interaction plot means the effect of one X depends on another.

7. **Fractional factorial design**
   2^(k-p) designs screen many factors in few runs by deliberately trading away high-order interactions.

8. **Confounding & resolution**
   Resolution III, IV and V describe exactly what you gave up. Know it before you interpret the result.

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Key Concepts — Improve (2)

1

Response surface methodology

Once the vital few Xs are known, RSM finds the optimum settings, not just the significant factors.

2

Robust design (Taguchi)

Set the process so it performs well despite noise you cannot control — not just on average.

3

Piloting and scale-up

Confirm the DOE optimum holds in production before rolling it out across sites.

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IMPROVE

Green Belts pick a solution. Black Belts design the experiment that finds it.

Improve at Black Belt level is where Design of Experiments enters the toolkit. You stop guessing at settings and start estimating effects.

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What Improve delivers at Black Belt level

1. **A generated solution set**
   Many candidates, each traced to a statistically proven root cause

2. **A weighted selection decision**
   Criteria weighted WITH the sponsor before any solution is scored

3. **A designed experiment**
   Factors, levels and runs chosen deliberately, not one factor at a time

4. **Optimum settings with evidence**
   Prediction equation plus confirmation runs at the chosen setpoint

5. **A robust operating window**
   Settings that hold up against the noise you cannot control

6. **A risk-assessed, piloted change**
   FMEA re-scored after mitigation, then piloted before rollout

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Green Belt Improve vs Black Belt Improve

Green Belt

Choose and pilot a solution

- Brainstorm, then vote or use a simple matrix
- Apply known Lean countermeasures
- FMEA the new process, act on top RPNs
- Pilot, measure, roll out

Black Belt

Design experiments to find the optimum

- Everything the Green Belt does, taught well enough to mentor it
- Plus: factorial designs to estimate effects and interactions
- Plus: fractional screening when factors are many
- Plus: RSM and robust design to optimise and de-sensitise

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SOLUTION GENERATION

You cannot select a good solution from a shallow list.

Generation and selection are separate cognitive acts. Mixing them kills the good ideas before they are fully formed.

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Six generation techniques a Black Belt must run

1. **Classic brainstorming**
   Aloud, in group; quantity first, zero criticism, build on others

2. **Brainwriting 6-3-5**
   6 people write 3 ideas in 5 minutes, then pass the sheet on

3. **SCAMPER**
   Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse

4. **Anti-solution thinking**
   Ask how to make it WORSE, then invert every answer

5. **Benchmarking**
   Borrow proven designs from inside, from rivals, from other industries

6. **Analogy and TRIZ prompts**
   How does another industry solve the same physical problem?

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Brainwriting 6-3-5 mechanics

1. Seat 6 people; each gets a 3-column sheet
2. Round 1: write 3 ideas in 5 minutes, in silence
3. Pass the sheet to the left
4. Read what is there, then add 3 BUILDS not 3 new ideas
5. Repeat until each sheet has been round the table
6. 108 ideas in 30 minutes with no dominant voice

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Anti-solution thinking at Meridian Suzhou

Make it worse

- How could we make first-pass yield MUCH worse?
  - Let each operator set cure temperature by feel
  - Change adhesive lots mid-shift with no record
  - Run the test rig without daily calibration
  - Give no torque spec on the housing screws
  - Start the shift before the line reaches temperature

Invert it

- Now invert each answer into a countermeasure
  - Lock cure temperature to a recipe in the controller
  - One adhesive lot per shift, logged and traceable
  - Daily calibration check as a line-start gate
  - Torque-controlled driver with a verified setpoint
  - Warm-up interlock before the first unit is released

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Benchmarking as a repeating cycle



Plan, collect, analyse, adapt. The gap analysis is where the learning sits - copying the practice without understanding it fails.

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Three types of benchmarking at Meridian

Internal

Another Meridian plant or line

- Singapore Line 2 runs 96.4 percent first-pass yield
- Suzhou Line 5 runs 91.2 percent on the same product
- Cheapest, fastest, fully comparable data
- Risk: you only ever copy your own habits

Competitive

A rival infusion pump maker

- Most relevant comparison of performance
- Hardest data to obtain legitimately
- Use teardowns, public filings, industry surveys
- Risk: confidentiality and competition law

Functional

Best-in-class in ANY industry

- For leak testing, study automotive fuel systems
- For clean assembly, study semiconductor packaging
- Usually the source of the breakthrough
- Risk: needs the most adaptation effort

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LEAN COUNTERMEASURES

Before you design an experiment, exhaust the countermeasures that need no experiment.

Most defects are not an optimisation problem. They are an absence of standard work, mistake-proofing and flow.

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The Lean countermeasure toolkit

1. **5S**
   Workplace organisation so the abnormal condition becomes visible instantly

2. **Poka-yoke**
   Design so the error cannot be made, or cannot pass undetected

3. **Pull systems**
   Downstream consumption authorises upstream production - no overproduction

4. **Just-in-Time**
   The right part, the right quantity, exactly when needed

5. **Heijunka**
   Level the mix and volume so demand spikes do not propagate

6. **Jidoka**
   Automation with a human touch - the machine stops itself on abnormality

7. **Standard work**
   The current best-known method, documented, trained and followed

8. **Kaizen events**
   Short, focused, cross-functional bursts of improvement on one process

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5S: the foundation everything else stands on

SORT

- Distinguishes the work that needs to be done, and eliminated the unnecessary steps

STRAIGHTEN

- Follows the maxim, "*A place for everything and everything in its place.*"

SHINE

- Clean and remove the clutter from the workplace

STANDARDIZE

- Refers to efforts to develop and implement standardized procedures
- This will make the work environment more orderly

SUSTAIN

- Reinforces the prior four steps with repeated and continuous performance

Sort, Set in order, Shine, Standardise, Sustain. Without Sustain the other four decay within a quarter.

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The three levels of poka-yoke

Prevention

The error cannot occur

- Strongest level - always prefer this
- Asymmetric connector: the sensor only fits one way
- Fixture will not close on a mis-oriented housing
- Cost is in the design, not in the running

Detection

The error occurs but cannot pass

- Second best - catches it at the source
- Vision system rejects a missing O-ring before potting
- Torque driver blocks the next step if the count is short
- Requires a working sensor and a real interlock

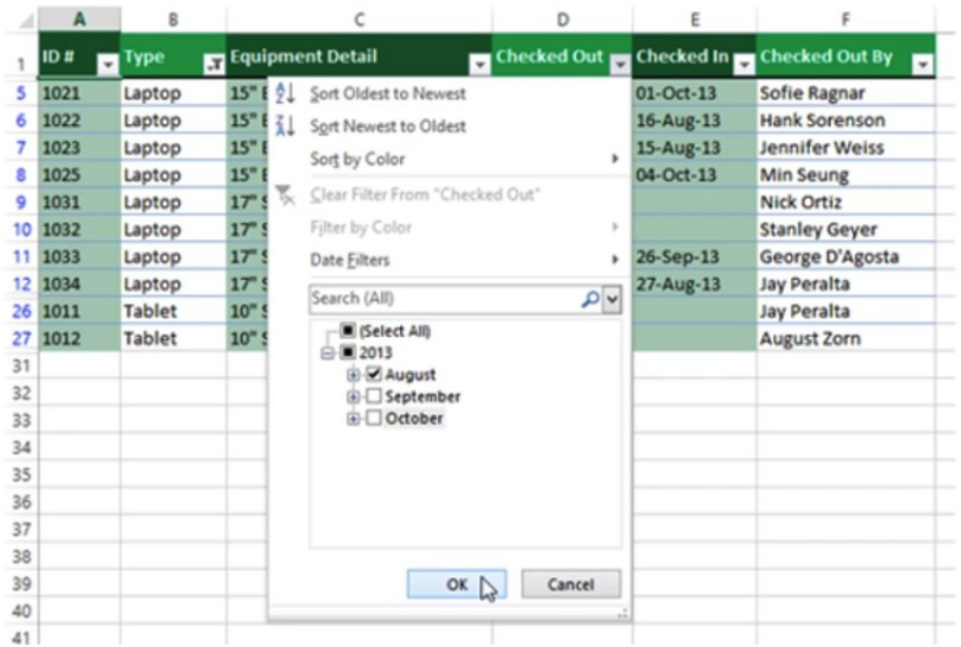
Warning

The operator is alerted

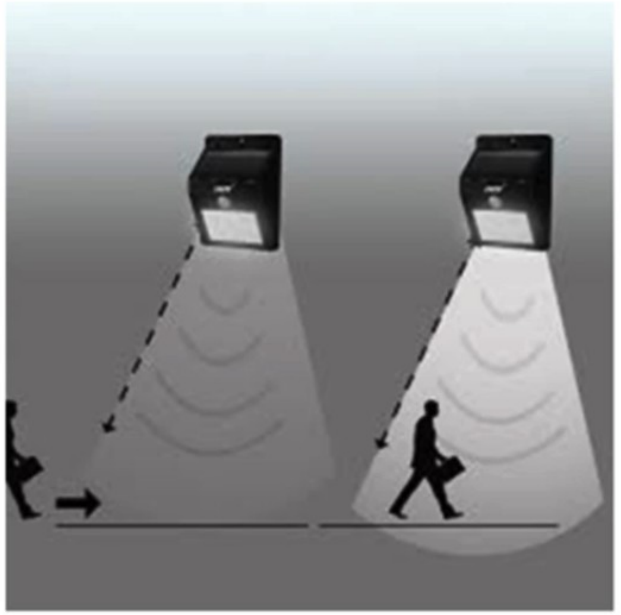
- Weakest level - relies on human response
- Andon light and buzzer on out-of-spec cure temperature
- Use only when prevention and detection are infeasible
- Degrades fast once alarms become routine

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Poka-yoke in practice



EXCEL FILTER

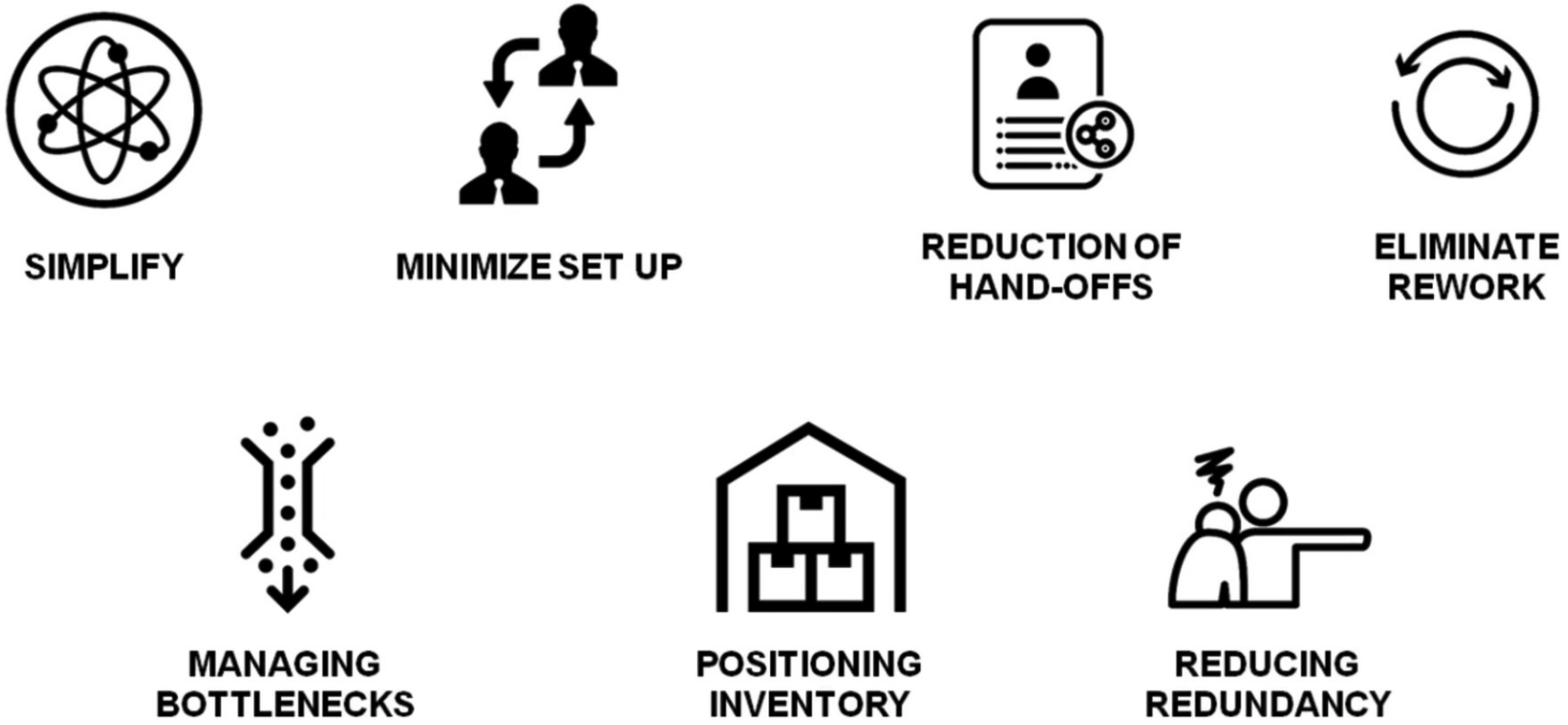


SENSOR/MOTION LIGHTING

Every good poka-yoke is cheap, obvious and impossible to defeat casually. If operators bypass it, the design is wrong.

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Push vs pull production



Push builds to a forecast and accumulates inventory. Pull builds only what the next process has consumed.

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JIT, Heijunka and Jidoka: what each one actually does

1. **JIT - the goal**
   Right part, right quantity, right time; inventory is the enemy

2. **JIT - the enabler**
   Short changeover, reliable equipment, capable suppliers

3. **Heijunka - the goal**
   Level the mix so every model is built every day

4. **Heijunka - the effect**
   A demand spike stops amplifying upstream (no bullwhip)

5. **Jidoka - the goal**
   Never pass a defect to the next process

6. **Jidoka - the effect**
   The line stops, the andon pulls help, the cause gets fixed now

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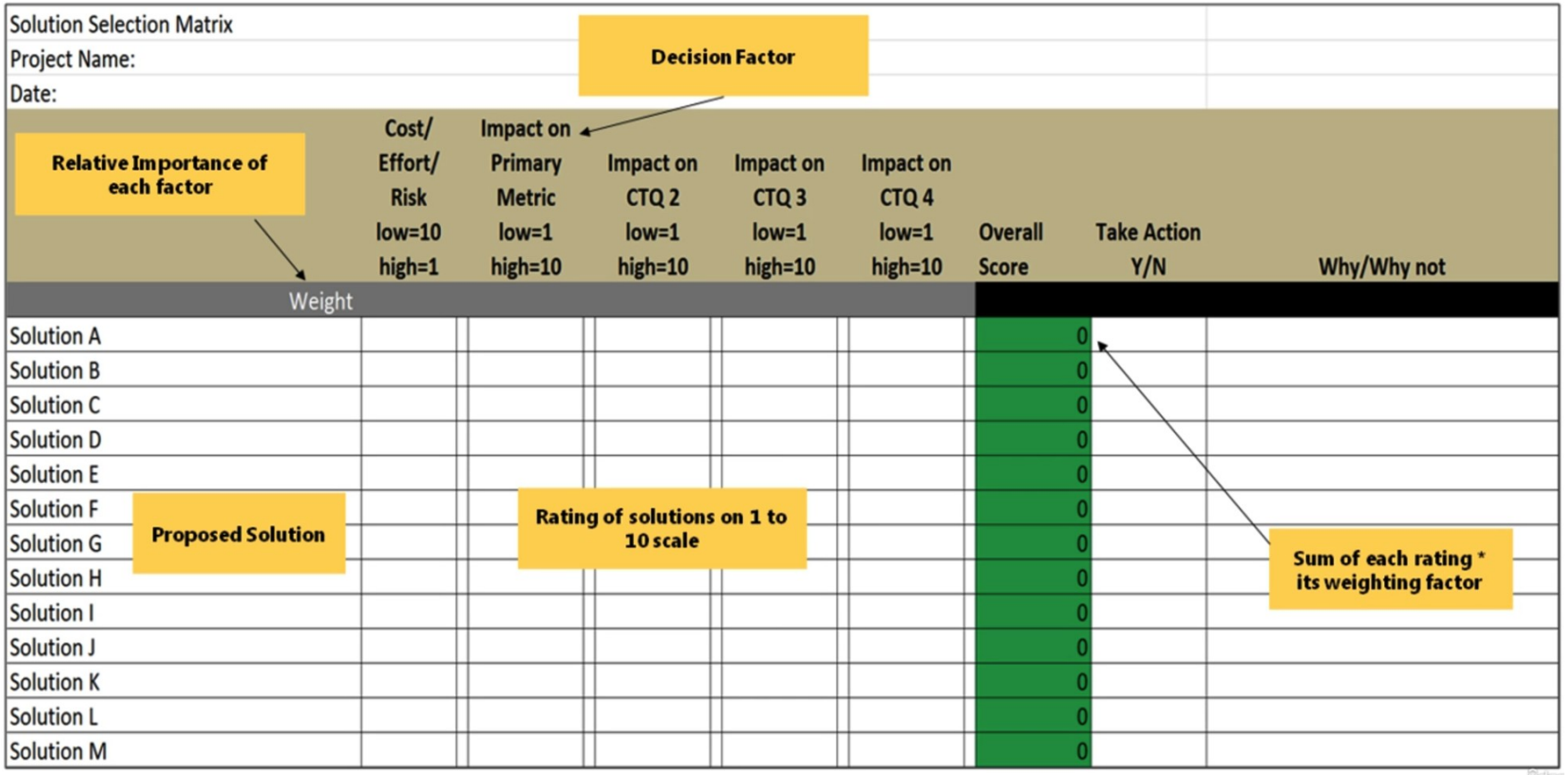
SOLUTION SELECTION

Weight the criteria with the sponsor BEFORE anyone scores a solution.

Weight the criteria after you see the scores and you have not made a decision. You have rationalised one.

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The weighted solution selection matrix



Criteria across the top with agreed weights; solutions down the side; weighted total drives the shortlist, not the decision.

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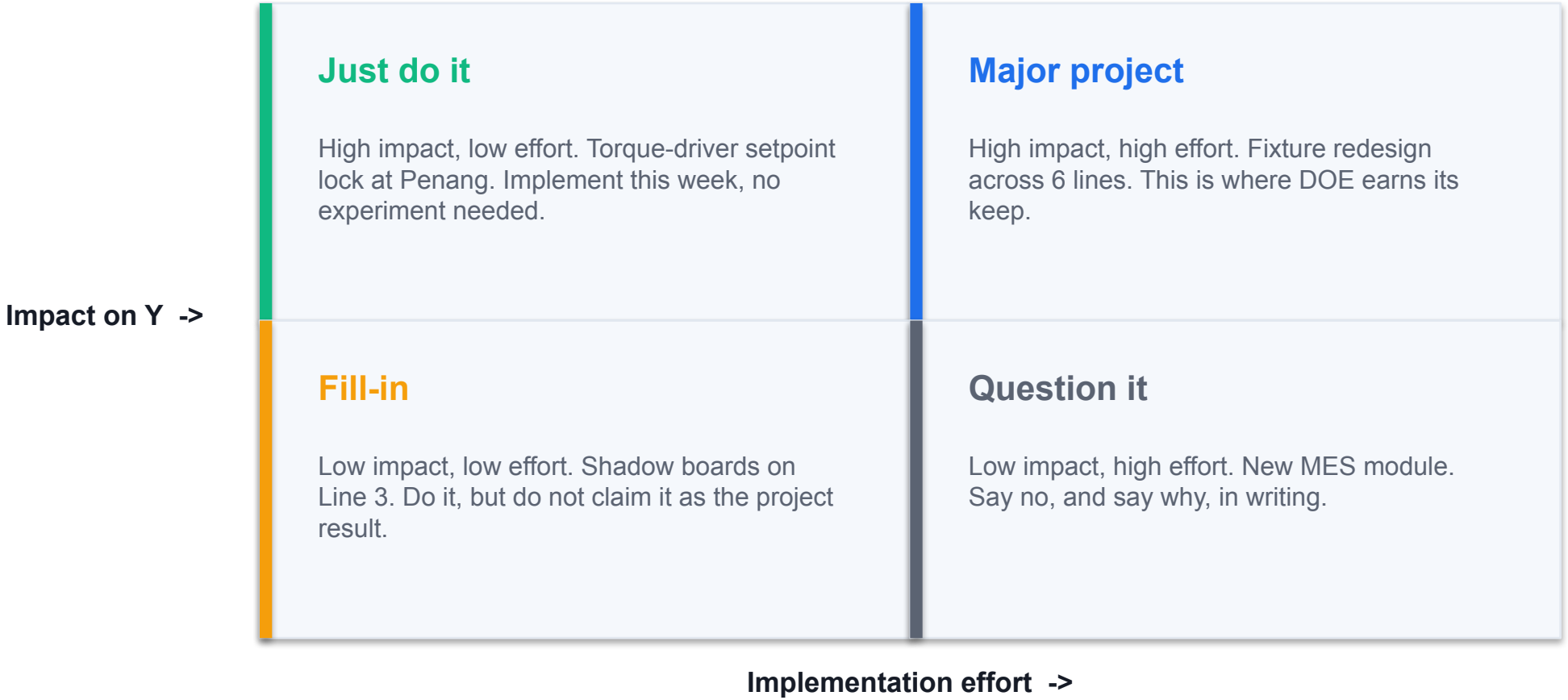
Meridian: scoring three candidate solutions

Criteria + weights	**Impact 40, Cost 25, Speed 20, Risk 15**	Weights agreed with the VP Ops before scoring began
Fixture redesign	**(5x40)+(2x25)+(3x20)+(4x15) = 370**	Highest impact, slowest to implement, chosen
Extra inspector	**(2x40)+(1x25)+(5x20)+(3x15) = 250**	Fast and cheap but adds cost forever and detects only

A solution that scores well only because it is cheap and fast is usually a containment, not a fix.

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Impact vs effort: where the shortlist really sits



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FMEA

FMEA is a design tool, not a paperwork exercise.

Done before implementation it prevents failures. Done after, it documents them.

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The FMEA worksheet

Failure modes and effects analysis (FMEA)

Project:	FMEA - TECHNOLGY	Date:	Update the date
FMEA Team:	Names of Team Members	Prepared by:	Names of Individuals who prepare this worksheet

SEV = How severe is effect on the customer?
OCC = How frequent is the cause likely to occur?
DET = How probable is detection of cause?
RPN = Risk priority number in order to rank concerns; calculated as SEV x OCC x DET

Business Function	Process step	Potential failure mode	Potential failure effects	SEV	Potential causes	OCC	Current process controls	DET	RPN	Actions recommended	Responsibility (target date)	Actions taken	New SEV	New OCC	New DET	New RPN
Which Functional Department?	What is the step?	In what ways can the step go wrong?	What is the impact on the customer if the failure mode is not prevented or corrected?	10	What causes the step to go wrong? (i.e., How could the failure mode occur?)	10	What are the existing controls that either prevent the failure mode from occurring or detect it should it occur?	10	1000	What are the actions for reducing the occurrence of the cause or for improving its detection? You should provide actions on all high RPNs and on severity ratings of 9 or 10.	Who is responsible for the recommended action? What date should it be completed by?	What were the actions implemented? Include completion month/year (then recalculate resulting RPN).	10	10	10	1000
	Connectivity	No calls	• Loss of Business • Client Dissatisfaction • Impact on Repeat Business • Impact on Revenue • Communication Breakdown	8	• CMS link failure • Issues with PG (Peripheral Gateway) server for incoming calls • Issues with outbound calls which are routed through PG. • For outbound calls: Issues with the trunks, either from SP end or from local end.	7	• Continuous Monitoring of CMS link • Monitoring of session running on CMS for PG • Any issue with PG, Technology will escalate to GS telephony team for resolution	3	168	• Voice DR Infrastructure • Appropriate Notification procedure (Escalation Matrix) • Awareness training of Escalation Matrix to second rung of frontline supervisors	• DR Plan (Operations): Dec 2009 • Reviewed Escalation Matrix: 12th Feb					0
		Call Drops	• Loss of Business • Client Dissatisfaction • Impact on Repeat Business • Impact on Revenue • Communication Breakdown	8	• Fluctuation in trunks at local end or from Service Provider end	2	• Monitoring the status of the trunks & alarms on PBX	1	16	• Timely review of Escalation Matrix • Increase Awareness to follow Escalation Matrix	Responsibility: Shiv Target: 12th Feb					0
		Skills mismatch /Incorrect Mapping	• Customer effected as unable to connect to the agent with specific skills • Agents getting incorrect calls • Incorrect Call Routing • SLA Miss	4	• Improper skills assigned to the agent	2	• Process Mission team to take care of the skill mapping	1	8	• Prompt Notification to the Technology Team	• Team leaders & Mission team • As and when required					0
			• Customer/Agent dissatisfaction		• Faulty headsets • Faulty QD chord		• Supervisors to replace faulty			• Proper Asset Maintenance (Technology Team to follow Checklist)						

Process step, failure mode, effect, cause, current controls, then S, O, D and RPN. Actions, then a re-score.

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Severity, Occurrence, Detection: three different questions

Severity (S)

How bad is the EFFECT?

- Scored on the effect, never on the cause
- 1 = no discernible effect
- 10 = hazard without warning, patient harm
- Severity can only be changed by REDESIGN

Occurrence (O)

How often does the CAUSE happen?

- Scored on the cause, from real data if you have it
- 1 = failure is eliminated by design
- 10 = failure is almost inevitable
- Reduce by attacking the cause: poka-yoke, controls

Detection (D)

How likely are we to CATCH it?

- The scale is INVERTED - read it carefully
- 1 = control almost certainly detects it
- 10 = no control exists, or it cannot detect
- Reduce by better detection: sensors, gauges, tests

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FMEA - THE DETECTION TRAP

Detection is scored backwards. Low D means GOOD detection.

This is the single most common FMEA scoring error. A team that scores D=10 for 'we always catch it' inverts every priority in the table.

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RPN and the Meridian cure-station FMEA

RPN	**RPN = S x O x D**	Range 1 to 1000; a rank, not a measurement
Incomplete cure	**S=9, O=6, D=7 -> RPN = 378**	Field separation risk, cause frequent, dial check is weak
Wrong adhesive lot	**S=8, O=3, D=4 -> RPN = 96**	Serious effect, rarer cause, lot scan catches most

RPN is ordinal. An RPN of 378 is not 'four times worse' than 96 - it is simply higher priority.

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FMEA - THE SEVERITY OVERRIDE

Severity 9 or 10 gets an action regardless of RPN.

A failure that can harm a patient is not acceptable because it is rare and we usually catch it. The severity override outranks the arithmetic.

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Re-scoring the Meridian FMEA after mitigation

Before	**S=9, O=6, D=7 -> RPN = 378**	Dial check by operator; no interlock; cure verified by feel
Action taken	**Recipe lock + thermocouple interlock**	Controller refuses to release below 62 C; logged per unit
After	**S=9, O=2, D=2 -> RPN = 36**	S unchanged - the effect is still a fluid path breach

Severity almost never moves. If your re-scored FMEA shows S falling, challenge it.

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PILOTING

Define success before the pilot starts, or the pilot cannot fail.

A pilot without pre-agreed criteria always succeeds, because the criteria get written afterwards to match the result.

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The five things a pilot must define BEFORE it starts

1. **Scope**
   Which line, which product, which shift. Meridian: Penang Line 3, pump model M400

2. **Duration**
   Long enough for the effect to show. Meridian: 4 weeks, about 9600 units

3. **Success criteria**
   The number that decides. First-pass yield at or above 95.5 percent

4. **Control group**
   Line 4 runs unchanged so you can separate the change from the season

5. **Stop conditions**
   What ends the pilot early. Any scrap rate above 3 percent, any safety event

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Cost-benefit for the Meridian fixture redesign

Payback period	**Investment / annual net benefit**	SGD 285,000 / SGD 610,000 = 0.47 yr, about 5.6 months
Net Present Value	**NPV = sum of CFt / (1+r)^t - I**	3 yrs at 8 pct: 565+523+484 = 1572k, minus 285k = SGD 1.287m
Decision rule	**NPV > 0 and payback < 18 months**	Meridian finance gate; both satisfied, project funded

Get the benefit number signed by Finance, not by the project team. Credibility depends on it.

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DESIGN OF EXPERIMENTS

One factor at a time is the most expensive way to learn the least.

OFAT costs more runs, gives narrower conclusions, and is structurally blind to the one thing that usually matters most: interaction.

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OFAT vs designed experiment

OFAT

Change one thing, hold the rest

- Feels rigorous and is intuitive to managers
- CANNOT detect interactions - structurally impossible
- Conclusions valid only at the held levels of the others
- Needs more runs for less information
- Optimum found is local, not global

Designed experiment

Change several together, by design

- Every factor varied across every combination
- Estimates main effects AND interactions
- Each run contributes to every effect estimate (hidden replication)
- Fewer runs, tighter estimates, wider validity
- The optimum found is defensible across the region

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The run-count comparison: 3 factors, 2 levels each

OFAT	**Baseline + 2 runs per factor**	About 12-16 runs with replication, and zero interaction estimates
Full factorial	**2^3 = 8 runs**	8 runs give 3 main effects, 3 two-factor, 1 three-factor interaction
Precision	**Each effect uses ALL 8 runs**	Effect estimate has the precision of a 4-vs-4 comparison, not 1-vs-1

This is 'hidden replication': every run is used in every effect estimate, so DOE gets more from less.

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Why OFAT cannot see an interaction

1

The setup

Vary temperature at fixed pressure. Then vary pressure at fixed temperature

2

What you learn

The effect of temperature AT THAT ONE PRESSURE. Nothing more

3

What you miss

That temperature helps at high pressure and hurts at low pressure

4

The Meridian case

Cure temp raised yield at 4.2 bar clamp, lowered it at 2.8 bar

5

The consequence

OFAT recommended 62 C. The true optimum was 68 C at 4.2 bar

6

The cost

Six months and 41 confirmation runs before the interaction was found

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DOE vocabulary: the words you must use precisely

1 **Factor**
An input X you deliberately set. Cure temperature, clamp pressure, dwell time

2 **Level**
A setting of a factor. Low (-1) and high (+1) in a two-level design

3 **Response**
The output Y you measure. First-pass yield, bond strength, leak rate

4 **Run**
One execution of the process at one specific factor combination

5 **Effect**
The change in the mean response caused by moving a factor from low to high

6 **Interaction**
When the effect of one factor DEPENDS on the level of another

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Replication vs repetition: the distinction that matters

Replication

Re-run the whole run

- Set the machine up again from scratch
- Captures setup, material and operator variation
- This is what gives you PURE ERROR
- Meridian: 8 runs replicated twice = 16 trials

Repetition

Re-measure the same run

- Same setup, measure the part 3 times
- Captures only measurement variation
- Understates the true error, inflates significance
- Useful for MSA, useless as replication

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DOE - RANDOMISATION

Randomisation is not optional. It is what makes the inference valid.

Without randomisation, any drift in ambient temperature, tool wear or operator fatigue is silently added to whichever factor you happened to change last.

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Blocking: when you cannot randomise away a nuisance

The problem

A known nuisance factor

- Meridian must run 16 trials over 2 days
- Day-to-day variation is real and large
- Randomising alone leaves the day effect in the error

The solution

Block on the nuisance

- Run 8 trials on day 1, 8 on day 2 as two blocks
- Choose which 8 so the block is confounded with ABC only
- Day effect is estimated and removed from the error

The trade

You sacrifice one effect

- The blocked effect can no longer be estimated
- Sacrifice the highest-order interaction - ABC
- Randomise WITHIN each block, always

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FULL FACTORIAL

The 2^k full factorial: every factor at every combination of the others.

k factors at 2 levels means 2^k runs. It estimates every main effect and every interaction, cleanly. It is the gold standard when you can afford the runs.

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How big is a 2^k design?

2 factors	**2^2 = 4 runs**	2 main effects, 1 two-factor interaction
3 factors	**2^3 = 8 runs**	3 main, 3 two-factor, 1 three-factor interaction
5 factors	**2^5 = 32 runs**	5 main, 10 two-factor, 16 higher-order - runs grow fast

Effects estimated always equals runs minus one. 8 runs estimate 7 effects plus the grand mean.

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The Meridian 2^3 experiment: pump housing bond strength

1

Response Y

Bond pull strength in newtons; spec is 180 N minimum

2

Factor A

Cure temperature: 58 C (low, -1) or 68 C (high, +1)

3

Factor B

Clamp pressure: 2.8 bar (low, -1) or 4.2 bar (high, +1)

4

Factor C

Dwell time: 20 s (low, -1) or 35 s (high, +1)

5

Design

2^3 = 8 runs, replicated twice, 16 trials in randomised order

6

Held constant

Adhesive lot, operator, fixture, ambient humidity band

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The design matrix in standard (Yates) order

1 **Run 1**
A = -1, B = -1, C = -1 Y = 154 N

2 **Run 2**
A = +1, B = -1, C = -1 Y = 171 N

3 **Run 3**
A = -1, B = +1, C = -1 Y = 162 N

4 **Run 4**
A = +1, B = +1, C = -1 Y = 206 N

5 **Run 5**
A = -1, B = -1, C = +1 Y = 161 N

6 **Run 6**
A = +1, B = -1, C = +1 Y = 180 N

7 **Run 7**
A = -1, B = +1, C = +1 Y = 172 N

8 **Run 8**
A = +1, B = +1, C = +1 Y = 219 N

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Replication and pure error at Meridian

Replicate pairs	**Run 1: 154, 149 Run 8: 219, 213**	Whole runs re-set-up, not the same part re-measured
Pure error SD	**s from the within-pair ranges**	Average range 5.4 N -> s about 4.8 N across 8 pairs
Significance yardstick	**Effect significant if it exceeds ~2 x SE**	SE of an effect = 2s / sqrt(N) = 2(4.8)/4 = 2.4 -> flag > 4.8 N

Without replication you have no pure error, so you must rely on normal plots or effect sparsity instead.

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MAIN EFFECTS

A main effect is the average change in Y when a factor goes from low to high.

Average all the runs where A is high, average all the runs where A is low, subtract. Every run contributes to every effect.

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Calculating the main effects at Meridian

Effect of A	**avg(A high) - avg(A low)**	(171+206+180+219)/4 - (154+162+161+172)/4 = 194.0 - 162.25 = +31.75 N
Effect of B	**avg(B high) - avg(B low)**	(162+206+172+219)/4 - (154+171+161+180)/4 = 189.75 - 166.5 = +23.25 N
Effect of C	**avg(C high) - avg(C low)**	(161+180+172+219)/4 - (154+171+162+206)/4 = 183.0 - 173.25 = +9.75 N

Cure temperature (A) is the dominant factor; dwell time (C) is real but small.

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Calculating interaction effects

AB interaction	**avg(AB=+1) - avg(AB=-1)**	(154+206+161+219)/4 - (171+162+180+172)/4 = 185.0 - 171.25 = +13.75 N
AC interaction	**avg(AC=+1) - avg(AC=-1)**	(154+162+180+219)/4 - (171+206+161+172)/4 = 178.75 - 177.5 = +1.25 N
ABC interaction	**Sign column = A x B x C**	(171+162+161+219)/4 - (154+206+180+172)/4 = 178.25 - 178.0 = +0.25 N

Sign column for AB is just A multiplied by B, row by row. Same arithmetic, different column.

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Main effect plot vs interaction plot

Main effect plot

Y against one factor's two levels

- One line per factor, low on the left, high on the right
- STEEPER slope means a LARGER effect
- Flat line means the factor does nothing in this range
- Meridian: A is steepest, C is nearly flat

Interaction plot

Y against factor A, one line per level of B

- PARALLEL lines mean NO interaction
- NON-PARALLEL lines mean an interaction exists
- Crossing lines mean a strong reversal of effect
- Meridian AB: lines diverge sharply to the right

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Reading the Meridian AB interaction plot

1. **At low clamp pressure**
   Raising temperature 58 -> 68 C gains only 18 N

2. **At high clamp pressure**
   The same temperature rise gains 45 N

3. **The lines**
   Diverge sharply - the classic non-parallel signature

4. **The meaning**
   Temperature is worth far more when clamp pressure is high

5. **The decision**
   Set both high; never optimise them independently

6. **The OFAT trap**
   Testing temperature at 2.8 bar would have concluded 'temperature barely matters'

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DOE - INTERPRETING INTERACTIONS

When an interaction is significant, never interpret its main effects alone.

The statement 'raising cure temperature adds 32 N' is only true on average. It is 18 N in one condition and 45 N in another.

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Three ways to decide which effects are real

Pareto of effects

Bar chart of absolute effect size

- Bars sorted largest to smallest
- A reference line drawn at the critical value
- Bars beyond the line are significant
- Fastest visual read for a review meeting

Normal plot of effects

Effects on normal probability paper

- Inert effects fall on a straight line (they are pure noise)
- REAL effects fall OFF the line, at the extremes
- Works even with NO replication - the key advantage
- Half-normal plot is the same idea on absolute values

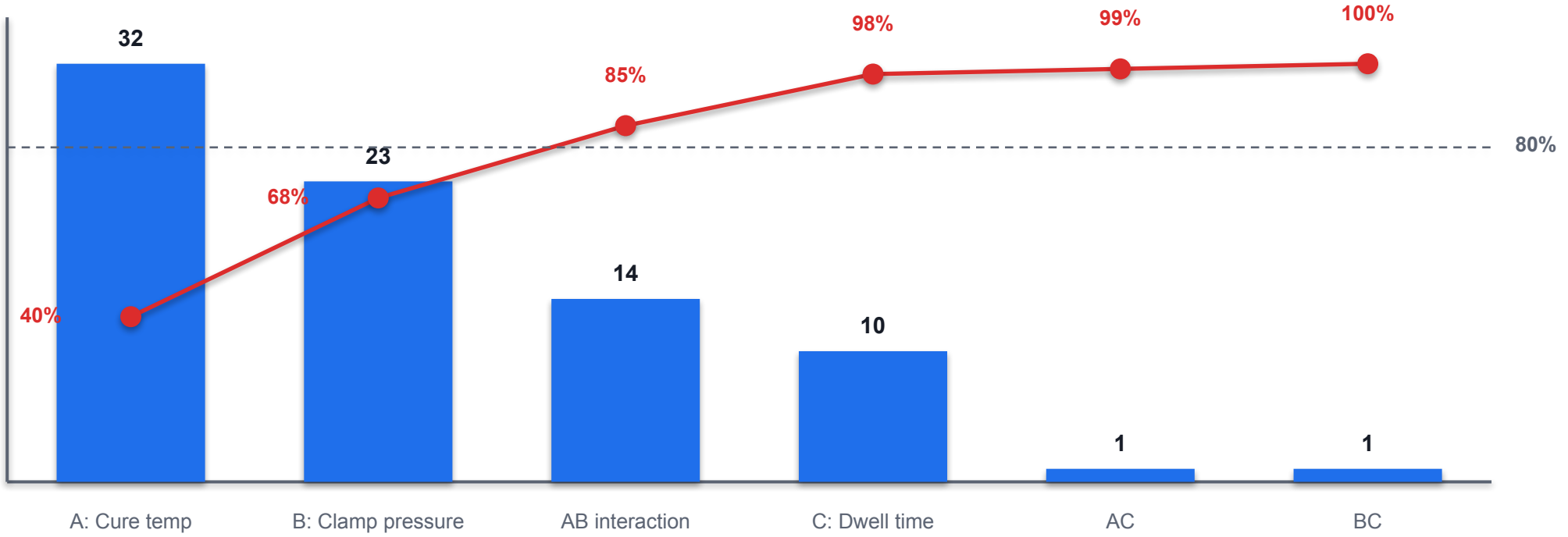
ANOVA / p-values

Formal test against pure error

- Requires replication to get pure error
- Effect significant at $p < 0.05$
- Gives R-squared and lack-of-fit as a bonus
- The number a sponsor will ask you for

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Pareto of effects: Meridian bond strength



Absolute effect size in newtons. Everything above the 4.8 N critical value is real: A, B, AB and C.

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Reading a normal probability plot of effects

1

The logic

If a factor does nothing, its estimated effect is pure noise, so it is normal

2

Noise effects

Plot as a straight line through the middle of the chart

3

Real effects

Fall clearly OFF that line, out at the tails

4

Positive effects

Sit above and to the right of the line

5

Negative effects

Sit below and to the left of the line

6

Why it matters

This works with an unreplicated design where you have no pure error

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Building the Meridian prediction equation

General form	**Y = b0 + b1(A) + b2(B) + b12(AB)**	Coefficients are HALF the effects, in coded -1/+1 units
Meridian model	**Y = 178.1 + 15.9A + 11.6B + 6.9AB**	Grand mean 178.1 N; A, B and AB retained; C dropped for cycle time
Predicted optimum	**A=+1, B=+1**	178.1 + 15.9 + 11.6 + 6.9 = 212.5 N predicted at 68 C, 4.2 bar

Coefficient = effect / 2 because the coded range from -1 to +1 spans 2 units.

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CONFIRMATION RUNS

If the confirmation runs fail, you do NOT roll out.

A prediction is a hypothesis about the future. Confirmation runs test it. Rolling out an unconfirmed optimum is gambling with the sponsor's money.

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Running the confirmation properly

1. Set the process at the predicted optimum: 68 C, 4.2 bar
2. Run 8-12 independent confirmation trials, randomised
3. Compare the observed mean to the predicted 212.5 N
4. Check it falls inside the prediction interval
5. Only then move to pilot and rollout

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Meridian confirmation result

Predicted	**212.5 N, PI 203 to 222 N**	From the model at A=+1, B=+1
Observed	**n=10, mean 209.4 N, s = 5.1 N**	Inside the prediction interval - the model is confirmed
Decision	**Confirmed -> pilot on Penang Line 3**	Bond strength margin over the 180 N spec rises from -12 to +29 N

If observed had come back at 186 N, the model would be rejected and you return to the design.

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FRACTIONAL FACTORIAL

With 7 factors, a full factorial is 128 runs. A quarter-day of screening is 16.

Fractional factorials buy that economy by deliberately trading away the high-order interactions you almost certainly do not need.

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The economics of 2^(k-p)

Full factorial	**2^7 = 128 runs**	At 45 min per run that is 96 hours of line time
Half fraction	**2^(7-1) = 64 runs**	One generator sacrificed; still very expensive
Sixteenth fraction	**2^(7-4) = 16 runs**	12 hours. Screens 7 factors down to the vital 2 or 3

p is the number of factors 'generated' from the others. Each unit of p halves the run count.

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How a fractional design is built

1

Start with a base design

For 2^(7-4): a full 2^3 in factors A, B, C - just 8 rows

2

Generators

Assign the extra factors to interaction columns: D=AB, E=AC, F=BC, G=ABC

3

Defining relation

I = ABD = ACE = BCF = ABCG and all their products

4

Alias structure

Every effect is aliased with several others; the relation tells you which

5

The consequence

You estimate a SUM of effects, never one effect on its own

6

The bet

That the high-order terms in each sum are negligible - effect sparsity

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Working out an alias: the arithmetic

Generator	**D = AB, so I = ABD**	Multiply both sides by D; D squared = I, the identity column
Alias of A	**A x I = A x ABD = BD**	So the A column also carries BD. You estimate A + BD
Alias of AB	**AB x ABD = D**	The AB column IS the D column. They cannot be separated

Rule: multiply any effect by the defining relation, and any letter squared drops out as I.

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CONFOUNDING

Confounding means two effects share one column. You cannot tell them apart.

The design does not fail. It simply reports a sum. Knowing WHICH sum before you interpret the result is the whole discipline.

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Resolution III, IV and V: what each one costs you

Resolution III

Shortest word has 3 letters

- MAIN effects aliased with 2-FACTOR interactions
- Cheapest design; 2^(7-4) screens 7 factors in 8 runs
- A big 'A' effect might really be a big BC interaction
- Use ONLY to screen many factors for the vital few
- Never use it to make a final settings decision

Resolution IV

Shortest word has 4 letters

- Main effects CLEAR of 2-factor interactions
- But 2-factor interactions are aliased WITH EACH OTHER
- 2^(7-3) in 16 runs; the usual screening workhorse
- AB + CE is one column: you know an interaction is there
- You cannot yet say WHICH interaction it is

Resolution V

Shortest word has 5 letters

- Main effects AND 2-factor interactions all clear
- 2FIs aliased only with 3-factor interactions
- 2^(5-1) in 16 runs; the characterisation standard
- Safe to build a prediction model from
- Costs more runs - buy it when the decision matters

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Meridian: screening 7 factors on the Suzhou potting line

1

The factors

A cure temp, B clamp, C dwell, D adhesive lot, E humidity, F fixture, G operator

2

The design

2^(7-3) Resolution IV, 16 runs, randomised over 2 blocked days

3

The result

A, C and F stand off the normal plot; B, D, E, G sit on the line

4

The alias caution

The AC + FG column was significant - which one is it?

5

The follow-up

Fold over to de-alias, or run a 2^3 full factorial on A, C, F

6

The outcome

7 factors reduced to 3 in 12 hours of line time

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Fold-over: de-aliasing without starting again

1. Run the original Res III fraction, 8 runs
2. Reverse ALL signs to build the mirror fraction, 8 more runs
3. Combine the two: 16 runs total
4. The combined design is Resolution IV - main effects now clear
5. Alternative: reverse ONE factor's signs to free that factor and its 2FIs

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RESPONSE SURFACE METHODOLOGY

Two-level designs fit planes. Real optima sit on curves.

A 2^k design has only two points per factor. Two points define a straight line. It is structurally incapable of seeing curvature.

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Centre points: the cheapest curvature test there is

What they are

Runs at the midpoint of every factor

- Coded (0, 0, 0) - all factors at their mid-level
- Meridian: 63 C, 3.5 bar, 27.5 s
- Add 3 to 5 of them to any 2-level design

What they test

Factorial mean vs centre mean

- If no curvature, the centre runs land on the fitted plane
- If they sit well above or below, curvature is present
- A formal t-test on the difference gives a p-value

Free bonus

Pure error and drift detection

- Repeated centre points give pure error with no extra design
- Run some early and some late to detect process drift
- Costs 3-5 runs; tells you whether RSM is even needed

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Meridian curvature test at the potting station

Factorial mean	**avg of the 8 corner runs**	178.1 N
Centre mean	**avg of 5 centre-point runs**	196.3 N - well above the plane, not on it
Curvature	**196.3 - 178.1 = +18.2 N, p = 0.003**	Significant curvature -> the linear model is inadequate, go to RSM

A positive curvature this large usually means the optimum lies INSIDE the region you tested.

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Steepest ascent: getting to the right neighbourhood first

1. Fit the first-order model in the current region
2. The gradient points in the direction of steepest improvement
3. Take steps along that path, proportional to each coefficient
4. Run single trials along the path until Y stops improving
5. Set up a new factorial with centre points at that best point
6. Test for curvature there; if present, run the RSM design

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The two standard response surface designs

Central Composite (CCD)

Factorial + centre + axial

- Factorial core: the 2^k corner points
- Centre points: 3 to 6 replicated runs at (0,0)
- Axial or 'star' points at (+/-alpha, 0) and (0, +/-alpha)
- 5 levels per factor; alpha = 1.682 for k=3 gives rotatability
- 2^3 CCD = 8 + 6 + 6 = 20 runs

Box-Behnken (BBD)

Midpoints of the edges

- No corner points at all - only edge midpoints and centres
- Only 3 levels per factor, easier to set on real equipment
- Never runs all factors at their extremes simultaneously
- Use when a corner combination is unsafe or impossible
- 3-factor BBD = 15 runs, slightly cheaper than the CCD

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The Meridian quadratic model

Model form	**Y = b0 + b1A + b2B + b11A^2 + b22B^2 + b12AB**	Six terms for two factors; needs at least 6 distinct design points
Fitted	**Y = 218.4 + 9.1A + 6.2B - 7.8A^2 - 4.1B^2 + 3.4AB**	R-squared 0.96; both quadratic terms strongly significant
Stationary point	**Solve dY/dA = 0 and dY/dB = 0**	A = +0.71, B = +1.05 -> 71.4 C and 4.4 bar, predicted 224.7 N

Both quadratic coefficients are negative, so the surface is a dome and the stationary point is a MAXIMUM.

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Classifying the stationary point: three outcomes

Maximum

Both quadratic terms negative

- The surface is a dome; contours are closed ellipses
- The stationary point is the optimum you want
- Meridian bond strength: -7.8 and -4.1, a clean maximum

Minimum

Both quadratic terms positive

- A bowl; contours are closed ellipses around the low point
- This is the optimum when smaller Y is better
- Leak rate and warranty returns behave this way

SADDLE POINT

Mixed signs

- A max in one direction and a min in the other
- Contours are hyperbolas, not closed ellipses
- The stationary point is NOT an optimum - do not set there
- Optimise along a ridge, at the edge of the safe region

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RSM - THE SADDLE

A saddle point is the trap. It satisfies the maths and fails in production.

The gradient is zero, so the software reports a stationary point. But move a little in one direction and Y falls off a cliff.

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Optimising several responses at once: desirability

The problem

Responses conflict

- Bond strength wants 71 C and long dwell
- Cycle time wants short dwell
- Adhesive cost wants low clamp pressure
- No single setting is best for all three

Desirability functions

Convert each Y to 0-1

- d = 0 means completely unacceptable
- d = 1 means fully at target
- Larger-is-better, smaller-is-better, target-is-best shapes
- Weight each response by its business importance

Overall D

Geometric mean of the d values

- D = (d1 x d2 x ... x dn)^(1/n)
- Geometric, so any d = 0 forces D = 0
- One unacceptable response kills the whole setting - correctly
- Maximise D across the design space to get the compromise

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ROBUST DESIGN

Do not fight the noise. Design so the noise stops mattering.

Taguchi's insight: you cannot control ambient humidity, but you can find the settings at which ambient humidity barely affects your output.

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Control factors vs noise factors

Control factors

You set them and they stay set

- Cure temperature setpoint, clamp pressure, dwell time
- Fixture design, adhesive grade, recipe parameters
- These are the dials you are allowed to turn
- Their levels go in the inner array

Noise factors

Real, influential, not controllable

- Ambient humidity, incoming adhesive viscosity variation
- Operator-to-operator differences, tool wear, line voltage
- Too expensive or impossible to control in production
- Deliberately VARIED in the outer array during the experiment

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THE TAGUCHI INSIGHT

A lower mean with far less variation usually beats a higher mean that swings.

Meridian setting A gives 224 N plus or minus 22. Setting B gives 214 N plus or minus 5. B never goes below spec. A does, one batch in twenty.

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Meridian: two settings, same spec, different risk

Setting A	**mean 224 N, s = 11 N, LSL 180**	Z = (224-180)/11 = 4.0 -> about 32 ppm below spec
Setting B	**mean 214 N, s = 2.6 N, LSL 180**	Z = (214-180)/2.6 = 13.1 -> effectively zero risk
Choice	**Choose B**	10 N of mean given up to remove all the tail risk. Correct trade

Six Sigma is about variation, not averages. The robust setting wins even with a lower mean.

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The three signal-to-noise ratios

Smaller-is-better

Defects, leak rate, cycle time

- S/N = -10 log10(mean of y-squared)
- Ideal target is zero; no negative values possible
- Meridian leak rate and warranty returns use this

Larger-is-better

Bond strength, yield, life

- S/N = -10 log10(mean of 1/y-squared)
- No upper bound; more is always better
- Meridian bond pull strength uses this

Nominal-is-best

Dimensions, fill volume, flow rate

- S/N = 10 log10(mean-squared / variance)
- Two-step: maximise S/N, then adjust the mean to target
- Meridian pump flow rate calibration uses this

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S/N ratio for the Meridian bond-strength study

Setting A	**y = 241, 228, 219, 208**	S/N (larger-better) = 27.0 dB; mean 224, wide spread
Setting B	**y = 217, 214, 212, 213**	S/N (larger-better) = 26.6 dB; mean 214, very tight
Robustness	**Compare the SPREAD across noise**	A ranges 33 N across the noise conditions; B ranges 5 N

S/N alone can mislead. Always inspect mean and variance separately as well as the ratio.

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The robust operating window

1

Definition

The range of settings over which Y stays acceptable despite noise

2

Where to find it

The flat plateau on the response surface, not the sharp peak

3

Meridian's window

68 to 73 C at 4.1 to 4.6 bar keeps bond strength above 205 N

4

Why it matters

Production drifts. Setpoints wander. Operators round numbers

5

What you hand over

A window in the control plan, not a single number nobody can hold

6

The Black Belt test

Would this setting survive a wet Tuesday in Suzhou? If not, move it

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DOE STRATEGY

Screen, characterise, optimise, robustify, confirm.

That sequence is the Black Belt's experimental strategy. Each stage answers a different question and buys the right amount of information.

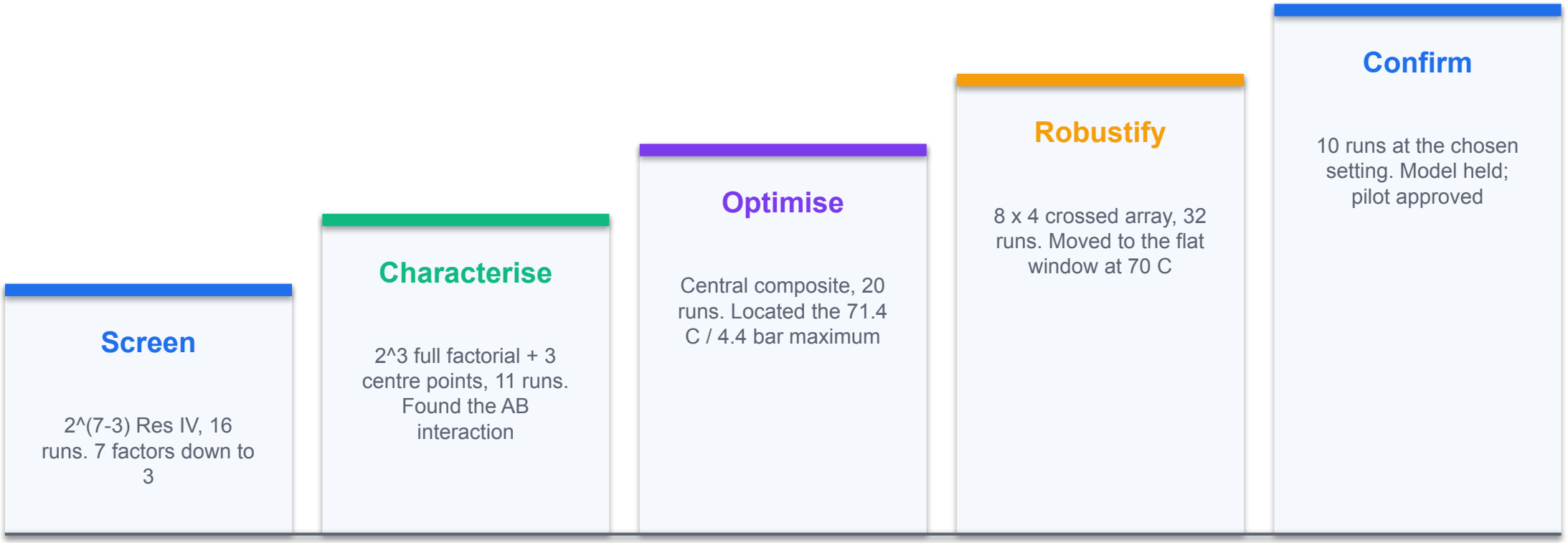
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The complete experimental strategy

1. SCREEN: Res III or IV fractional over many factors
2. CHARACTERISE: full or Res V factorial over the vital few
3. TEST CURVATURE: add centre points; move by steepest ascent if needed
4. OPTIMISE: CCD or Box-Behnken, fit the quadratic, find the stationary point
5. ROBUSTIFY: cross with noise factors, choose the flat window
6. CONFIRM: independent runs at the chosen setting before any rollout

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Runs invested at each stage of the Meridian programme



89 runs total across five stages. A single 2^7 full factorial would have been 128 runs and answered less.

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Meridian Improve phase: the result

1. **First-pass yield**
   Penang Line 3 moved from 91.2 to 96.8 percent over the 4-week pilot

2. **Bond strength margin**
   From 12 N below the 180 N spec floor to 29 N above it

3. **Warranty returns**
   Projected reduction of 61 percent on the M400 pump family

4. **FMEA**
   Highest RPN fell from 378 to 36 after the recipe lock and interlock

5. **Financial**
   SGD 610k annual benefit, 5.6 month payback, NPV SGD 1.29m

6. **Transferability**
   The robust window transferred to Suzhou with no re-optimisation

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NEXT: CONTROL

You have found the optimum. Now the hard part: making it stay.

Every improvement decays without a control system. Control is where the savings become permanent, or quietly evaporate.

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WHAT YOU'LL DO

Hands-On Labs — Improve

1. **Lab 22 — Solution Generation, Selection and Lean Countermeasures — In Depth**
   A scored solution selection matrix mapping each solution to a proven root cause. • Data: lab-22-solution-selection.xlsx

2. **Lab 23 — Full Factorial Design of Experiments (Black Belt Only)**
   A designed 2^k full factorial experiment with main effects and interaction analysis. • Data: lab-23-full-factorial-doe.xlsx

3. **Lab 24 — Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)**
   A fractional factorial screening design with a documented confounding structure. • Data: lab-24-fractional-factorial-doe.xlsx

4. **Lab 25 — Response Surface Methodology and Robust Design (Black Belt Only)**
   An RSM study locating optimal settings plus a robust design analysis against noise… • Data: lab-25-rsm-central-composite.xlsx

5. **Lab 26 — FMEA, Pilot Design and Implementation Planning — In Depth**
   A completed FMEA with re-scored RPNs and a full pilot plan with success criteria. • Data: lab-26-fmea-worksheet.xlsx

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DMAIC · IMPROVE · HANDS-ON

Solution Generation, Selection and Lean Countermeasures — In Depth

LAB 22

Re-derive solution generation and the full Lean countermeasure toolkit, then select rigorously with a weighted matrix. Every solution must attack a PROVEN root cause from Lab 21 — solutions that address unproven causes are how projects quietly fail. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A scored solution selection matrix mapping each solution to a proven root cause.

Tools & techniques: Brainstorming, brainwriting, SCAMPER, benchmarking, 5S, poka-yoke, pull/JIT, Heijunka, Jidoka, standard work, selection matrix

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Solution Generation, Selection and Lean Countermeasures — In Depth

STEPS 1-6 OF 12

1. Take your ranked proven root causes from Lab 21. Every solution generated in this lab must trace to one of them.
2. Recap divergent generation techniques in full: brainstorming, brainwriting (6-3-5), SCAMPER, anti-solution thinking and analogy from other industries.
3. Recap benchmarking and its four steps — plan, collect, analyse, adapt. Look outside your industry, where the best analogous processes usually are.
4. Generate at least three candidate solutions for each proven root cause. One solution per cause means you have not really explored the space.
5. Recap the Lean countermeasure toolkit in full: 5S, poka-yoke, pull systems, JIT, Heijunka (levelling), Jidoka (autonomation) and standard work.
6. Recap poka-yoke levels specifically: prevention (error impossible), detection (error caught immediately) and warning (error flagged). Always prefer prevention.

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Solution Generation, Selection and Lean Countermeasures — In Depth

STEPS 7-12 OF 12

7	Map each Lean tool to the root causes in your capstone it could plausibly address, and generate at least two poka-yoke ideas.
8	Build the weighted solution selection matrix. Criteria typically: impact on Y, implementation cost, time to benefit, technical risk, ease of adoption and sustainability.
9	Weight the criteria with your sponsor, not alone — the weighting reflects business priorities that are theirs to set.
10	Score every candidate solution and compute weighted totals.
11	Sanity-check the top-ranked solutions: do they collectively address the causes explaining most of the Y gap? If not, revisit generation.
12	Identify which solutions have controllable factors suitable for DOE optimisation in Labs 23-25.

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LAB 22 · VERIFY

Solution Generation, Selection and Lean Countermeasures — In Depth

✅ Check your work

Every solution traces to a proven root cause, criteria weights are agreed with the sponsor, and your selected set collectively addresses the majority of the quantified Y gap.

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Full Factorial Design of Experiments (Black Belt Only)

LAB 23

BLACK BELT ONLY — the Green Belt course mentions DOE in passing; the Black Belt owns it. One-factor-at-a-time experimentation is slower, less informative and structurally incapable of detecting interactions. Full factorial designs estimate every main effect AND every interaction. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A designed 2^k full factorial experiment with main effects and interaction analysis.

Tools & techniques: OFAT vs DOE, 2^k factorial, factors & levels, randomisation, replication, blocking, main effects, interaction plots, effect significance

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Full Factorial Design of Experiments (Black Belt Only)

STEPS 1-5 OF 15

1. Understand why OFAT fails: it cannot detect interactions at all, and it requires more runs for less information than a factorial design.
2. Recap the DOE vocabulary in full: factor, level, response, run, effect, interaction, replication, randomisation, blocking and experimental error.
3. Select 3 factors from your capstone that you can actually control and vary — temperature, speed, pressure, batch size, staffing level or method variant.
4. Set two levels for each factor — low (-1) and high (+1). Set them wide enough to produce a detectable effect but within safe operating limits.
5. Build the 2^3 design matrix: 8 runs covering every combination of the three factors, in standard order.

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Full Factorial Design of Experiments (Black Belt Only)

STEPS 6-10 OF 15

6. Decide replication. Replicates give an estimate of pure experimental error and let you test effect significance — 2 replicates gives 16 runs.

7. RANDOMISE the run order. Randomisation protects against unknown time-varying nuisance factors and is not optional.

8. Use blocking if runs must span shifts, days or material lots — block on the nuisance factor so it does not contaminate the effects.

9. Execute the experiment, holding all non-experimental factors constant and recording anything unexpected during each run.

10. Calculate main effects: the average response at the high level minus the average at the low level, for each factor.

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Full Factorial Design of Experiments (Black Belt Only)

STEPS 11-15 OF 15

11	Calculate interaction effects for AB, AC, BC and the three-way ABC.
12	Plot main effects (steeper slope means larger effect) and interaction plots (non-parallel lines mean interaction present).
13	Test effect significance using a Pareto of effects or a normal probability plot of effects — points off the line are real effects.
14	Build the prediction equation from the significant effects and identify the factor settings that optimise your Y.
15	Run confirmation trials at the predicted optimum. If the confirmation fails, the model is missing something — do not roll out.

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LAB 23 · VERIFY

Full Factorial Design of Experiments (Black Belt Only)

✅ Check your work

Your design is fully randomised with replication, you have computed and plotted both main effects and interactions, effect significance is assessed statistically, and confirmation runs validate the predicted optimum.

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Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

LAB 24

BLACK BELT ONLY. A full factorial on 7 factors needs 128 runs — usually unaffordable. Fractional factorials screen many factors in a fraction of the runs by deliberately trading away the ability to separate certain effects. The critical skill is knowing exactly WHAT you traded away, which is what resolution tells you. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering

You'll build

A fractional factorial screening design with a documented confounding structure.

Tools & techniques: 2^(k-p) designs, generators, defining relation, alias structure, confounding, Resolution III/IV/V, screening, fold-over designs

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Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

STEPS 1-5 OF 13

1. Understand the economics: 2^7 full factorial = 128 runs; a 2^(7-4) fractional = 16 runs. The saving is real but it is paid for in resolving power.
2. Understand confounding (aliasing): in a fractional design, some effects are mathematically indistinguishable from others.
3. Learn the generator and defining relation — the algebra that determines exactly which effects are aliased with which.
4. Learn the resolution scale precisely. RESOLUTION III: main effects aliased with two-factor interactions. RESOLUTION IV: main effects clear of two-factor interactions, but two-factor interactions aliased with each other. RESOLUTION V: main effects and two-factor interactions all clear.
5. Understand the selection rule: use Resolution III only for early screening of many factors; use IV or V when you intend to interpret interactions.

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Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

STEPS 6-9 OF 13

6. Select 5 or more candidate factors from your capstone for a screening experiment.

7. Choose your fraction based on your run budget, and write down the resulting resolution BEFORE running anything.

8. Generate the design matrix and write out the complete alias structure. You must be able to state what each estimated effect is confounded with.

9. Apply effect sparsity and hierarchy: usually few factors dominate, and low-order effects are more likely to be real than high-order ones.

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Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

STEPS 10-13 OF 13

10 Randomise and execute the screening experiment.

11 Analyse the effects and identify the vital few factors. Where an important effect is confounded, state the ambiguity explicitly rather than assuming the main effect is the real one.

12 Resolve critical ambiguity with a fold-over design — a second fraction that de-aliases the effects you care about.

13 Take the screened vital few factors forward to a full factorial or RSM study in Lab 25.

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LAB 24 · VERIFY

Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

✅ Check your work

You stated the design resolution before running, wrote out the complete alias structure, and any ambiguous conclusion is either declared explicitly or resolved by a fold-over.

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Response Surface Methodology and Robust Design (Black Belt Only)

LAB 25

BLACK BELT ONLY. Factorial designs find WHICH factors matter; RSM finds the OPTIMUM SETTINGS by modelling curvature that a two-level design cannot see. Taguchi robust design then goes further — setting the process so it performs consistently despite noise you cannot control. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

An RSM study locating optimal settings plus a robust design analysis against noise factors.

Tools & techniques: Curvature & centre points, steepest ascent, central composite design, Box-Behnken, contour & surface plots, Taguchi, signal-to-noise, control vs noise factors

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Response Surface Methodology and Robust Design (Black Belt Only)

STEPS 1-5 OF 15

1. Understand the limitation of 2-level designs: with only two levels per factor they fit a straight line and cannot detect curvature at all.

2. Add CENTRE POINTS to your factorial design to test for curvature. A significant centre-point effect means the true response is curved and a linear model will mislead you.

3. If no curvature is present and you are far from the optimum, use steepest ascent — move the factor settings in the direction of greatest improvement and re-experiment.

4. Once curvature appears, you are near the optimum and need a second-order design.

5. Learn the central composite design (CCD): a factorial core, plus centre points, plus axial (star) points that add the third level needed to estimate quadratic terms.

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Response Surface Methodology and Robust Design (Black Belt Only)

STEPS 6-10 OF 15

6. Learn the Box-Behnken alternative: fewer runs than CCD and it never requires extreme corner combinations, which matters when corners are unsafe or infeasible.

7. Design and run an RSM study on the 2-3 vital factors from Lab 24.

8. Fit the second-order model including linear, quadratic and interaction terms.

9. Generate contour and 3D surface plots to visualise the response surface and locate the optimum region.

10. Find the stationary point and classify it — maximum, minimum or saddle. A saddle point means the optimum lies at a boundary, not in the middle.

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Response Surface Methodology and Robust Design (Black Belt Only)

STEPS 11-15 OF 15

11	Use desirability functions if you must optimise several responses at once with competing optima.
12	Now robust design. Separate your factors into CONTROL factors (you set them) and NOISE factors (ambient conditions, material variation, operator, wear).
13	Understand the Taguchi insight: choose control factor settings at which the response is LEAST SENSITIVE to noise — a slightly lower mean with far less variation usually wins.
14	Compute the signal-to-noise ratio for candidate settings, using the correct form for your objective (smaller-is-better, larger-is-better or nominal-is-best).
15	Select the robust operating window and run confirmation trials under deliberately varied noise conditions.

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LAB 25 · VERIFY

Response Surface Methodology and Robust Design (Black Belt Only)

✅ Check your work

Curvature is tested with centre points, your second-order model is visualised as a contour or surface plot with a classified stationary point, and the robust settings are confirmed under varied noise conditions.

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DMAIC · IMPROVE · HANDS-ON

FMEA, Pilot Design and Implementation Planning — In Depth

LAB 26

Re-derive FMEA in full, then design the pilot that proves your improvement works before it is scaled. A Black Belt never rolls out an unpiloted change across sites — the cost of being wrong at scale is exactly what the pilot buys down. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A completed FMEA with re-scored RPNs and a full pilot plan with success criteria.

Tools & techniques: FMEA, severity/occurrence/detection, RPN, risk mitigation, pilot design, success criteria, implementation planning, cost-benefit

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FMEA, Pilot Design and Implementation Planning — In Depth

STEPS 1-5 OF 14

1. Recap FMEA in full: failure mode, effect, cause, current controls, and the three ratings that combine into the risk priority number.
2. Recap the scoring scales precisely: Severity (impact if it occurs), Occurrence (likelihood) and Detection (probability of catching it BEFORE the customer does — note the inverted scale).
3. Build the FMEA for your selected improvement. For each process step, ask what could fail, what happens if it does, and what would cause it.
4. Score S, O and D on 1-10 scales and calculate RPN = S x O x D.
5. Apply the severity override rule: any failure mode with Severity 9-10 requires action regardless of how low its RPN is.

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FMEA, Pilot Design and Implementation Planning — In Depth

STEPS 6-10 OF 14

6. Rank by RPN and define mitigation for every high-RPN mode, targeting the component you can most cheaply move — usually Detection, sometimes Occurrence.

7. RE-SCORE after mitigation and record the residual RPN. An FMEA without re-scoring has not demonstrated that risk was actually reduced.

8. Design the pilot: scope (which line, site or segment), duration, sample size and the exact data to be collected.

9. Define quantitative success criteria BEFORE the pilot starts, expressed against your Lab 3 baseline. Criteria defined afterwards get rationalised.

10. Plan the comparison: before/after, or better, a control group running the unchanged process concurrently to rule out external effects.

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FMEA, Pilot Design and Implementation Planning — In Depth

STEPS 11-14 OF 14

11 Define the stop conditions — what result would cause you to halt the pilot rather than push through.

12 Build the implementation plan: tasks, owners, dates, training, communication, systems changes and documentation updates.

13 Complete the cost-benefit analysis: implementation cost, annualised benefit, payback period and NPV where the investment is material.

14 Run the pilot (or design it fully for post-course execution) and test the results statistically against the baseline — a visual improvement is not proof.

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LAB 26 · VERIFY

FMEA, Pilot Design and Implementation Planning — In Depth

✅ Check your work

Your FMEA is re-scored post-mitigation with the severity override applied, pilot success criteria were defined before starting, and the improvement is tested statistically against the documented baseline.

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PHASE RECAP

Recap — Improve

- Solution generation & selection — in depth — Brainstorming, brainwriting, benchmarking and weighted solution selection matrices, re-derived in full.
- Lean countermeasures — in depth — 5S, poka-yoke, pull, JIT, Heijunka, Jidoka and standard work — the Lean toolkit taught thoroughly.
- FMEA and RPN — in depth — Severity, Occurrence and Detection scoring, RPN calculation and the re-scored post-mitigation FMEA.
- Why DOE beats OFAT — One-factor-at-a-time misses interactions and wastes runs. DOE varies factors together, by design.
- Full factorial design — 2^k designs estimate every main effect and every interaction — the gold standard when runs are affordable.
- Main effects & interaction plots — Read the design: a non-parallel interaction plot means the effect of one X depends on another.

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C

DMAIC · CONTROL

Control at Black Belt Depth — Advanced SPC & Sustaining

SPC recap · CUSUM · EWMA · Short-run charts · Multivariate SPC · Control plan governance · Financial validation · Handover

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Key Concepts — Control (1)

1. **SPC and control charts — in depth**
   Chart anatomy, control limits, chart selection (Xbar-R, Xbar-S, I-MR, p, np, c, u) and out-of-control rules, in full.

2. **Capability and control plans — in depth**
   Cp, Cpk, Pp, Ppk, control limits vs specification limits, control plans, SOPs and response plans re-derived.

3. **Limits of Shewhart charts**
   Shewhart charts are slow to detect small sustained shifts — which is exactly what a drifting process does.

4. **CUSUM charts**
   Cumulative sum charts accumulate small deviations, detecting a sustained 1-sigma shift far faster.

5. **EWMA charts**
   Exponentially weighted moving average — tunable memory, strong for small shifts and autocorrelated data.

6. **Short-run SPC**
   High-mix low-volume processes never gather 25 subgroups per part. Standardise and chart across parts.

7. **Multivariate SPC**
   When several correlated characteristics matter, Hotelling's T-squared beats several separate charts.

8. **Control plan governance**
   The Black Belt audits control plans across the portfolio — not just their own project.

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Key Concepts — Control (2)

1

Financial benefits validation

Benefits are not real until Finance signs them. Hard, soft and cost-avoidance are booked differently.

2

Handover and closure

Transfer to the process owner, verify sustained performance, then formally close and harvest the lessons.

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An improvement you cannot hold

is an improvement you did not make. Control is where the money is banked.

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What Control delivers at Black Belt level

1. **Recap taught to teaching depth**
   SPC, chart selection, run rules, capability, control plan - deep enough to coach

2. **Advanced detection schemes**
   CUSUM and EWMA find the small sustained shifts Shewhart charts miss for weeks

3. **Charts for awkward processes**
   Short-run SPC for high-mix, multivariate SPC for correlated characteristics

4. **Autocorrelation handled**
   Chemical and continuous-flow data breaks the Shewhart assumption outright

5. **Portfolio control plan governance**
   You audit control plans across every project, not just your own

6. **Finance-validated benefits**
   Hard, soft and cost-avoidance benefits signed off before closure

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The Control phase step sequence

DEFINE	MEASURE	ANALYZE	IMPROVE	CONTROL
What problem are your trying to solve?	What is the extent of the problem?	Why is the problem occurring?	What do you propose to do and why?	How will you ensure that the problem stays fixed?
1. Understand customer & business requirements	4. Identify what to measure	7. Identify performance gaps	10. Generate, prioritize and select solution(s)	13. Institutionalize the solution(s)
2. Complete the project charter	5. Plan and collect data	8. Ascertain critical root-causes	11. Pilot solution(s)	14. Replicate and share best practices
3. Complete high-level as is process map	6. Determine baseline performance	9. Validate root-causes	12. Validate impact of solution(s)	15. Celebrate and recognize success
Toll Gate – Define	Toll Gate - Measure	Toll Gate - Analyze	Toll Gate - Improve	Toll Gate - Control

Monitor the improved process, document it, then hand it over to a named owner with a written reaction plan.

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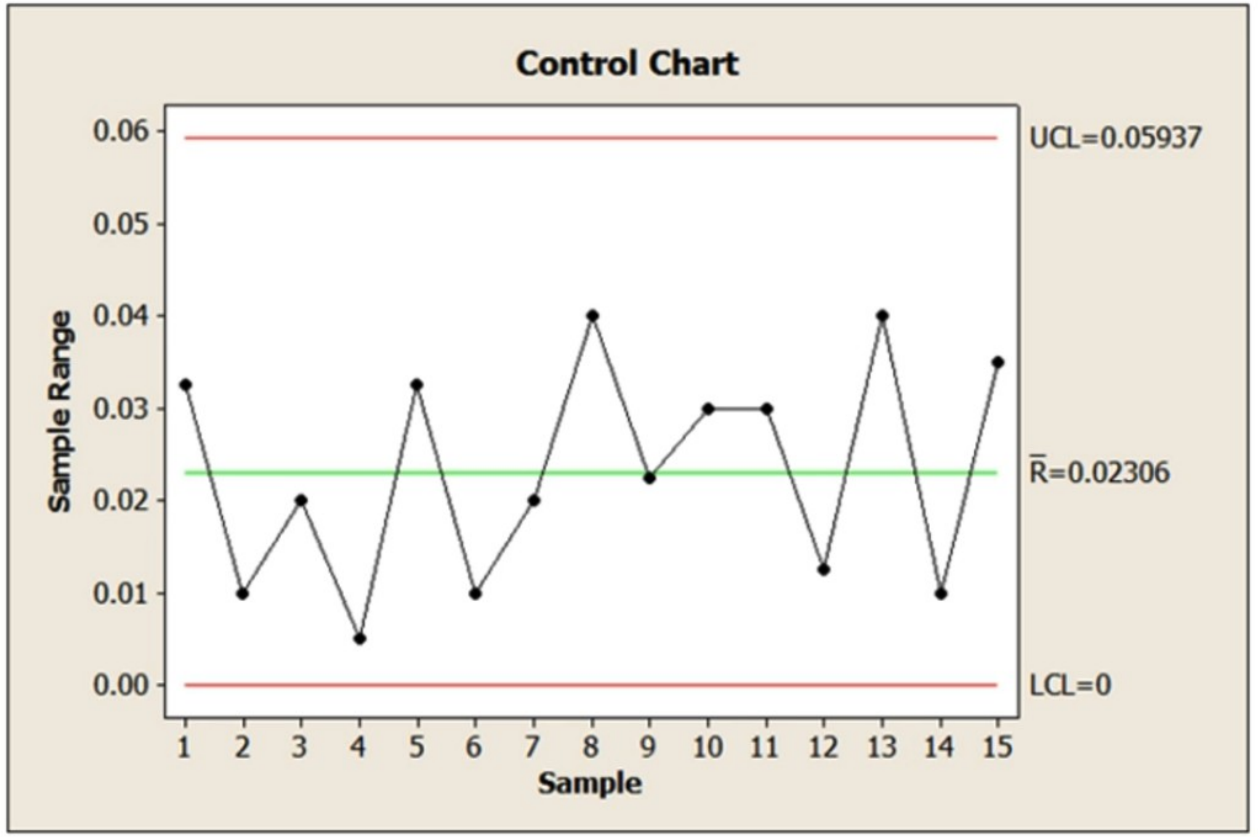
RECAP IN DEPTH - SPC

Statistical Process Control

Re-derived in full - because you must be able to teach it, not merely use it.

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Anatomy of a control chart



Centre line at the process average, upper and lower control limits at plus and minus 3 sigma of the plotted statistic.

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Chart anatomy - every element and what it means

1

Centre line (CL)

The process average of the plotted statistic - Xbar-bar, Rbar, pbar

2

Upper control limit (UCL)

CL + 3 sigma of the plotted statistic - the voice of the process

3

Lower control limit (LCL)

CL - 3 sigma; truncated at zero for ranges and attribute counts

4

Plotted points

Subgroup statistics IN TIME ORDER - order is the entire information content

5

Zones A, B, C

Each 1 sigma wide either side of CL - the run rules are defined on these

6

Sample and frequency

Rational subgroup size and interval, chosen deliberately, recorded on the chart

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Why 3 sigma - and not 2 or 4

At 3 sigma

Shewhart's economic balance

- False alarm rate 0.27 percent - about 1 in 370 points
- Roughly one false signal per year on a daily chart
- Cheap enough that people still trust the chart
- Wide enough to ignore ordinary common cause noise
- Not a statistical law - an economic compromise

At 2 sigma

Too many false alarms

- False alarm rate 4.6 percent - 1 in 22 points
- A daily chart cries wolf every month
- Operators stop investigating - the chart dies

At 4 sigma

Too many missed signals

- False alarms almost vanish - so do real signals
- Real shifts sit inside the limits for months
- The chart becomes decorative

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RECAP - THE CLASSIC ERROR

Control limits are NOT specification limits

They come from different sources, mean different things, and must never be plotted together.

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Control limits vs specification limits

Control limits

The VOICE OF THE PROCESS

- Calculated FROM the process data itself
- Answer: what is this process actually doing?
- Move when the process genuinely changes
- Used to detect special causes
- Live on the control chart

Specification limits

The VOICE OF THE CUSTOMER

- Set by the customer, design or regulator
- Answer: what does the customer require?
- Change only via engineering change control
- Used to judge conformance and capability
- Live on the drawing and in the capability study

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Four consequences of confusing the two

1. **Spec limits drawn on the chart**
   Every point inside spec looks fine - real special causes are invisible

2. **Limits widened to pass**
   Somebody sets UCL at the spec so the chart stops alarming. Fraud, not control

3. **In control read as capable**
   A stable process can be stably terrible. Control says nothing about the spec

4. **Capable read as in control**
   A capable process can be drifting. Capability says nothing about stability

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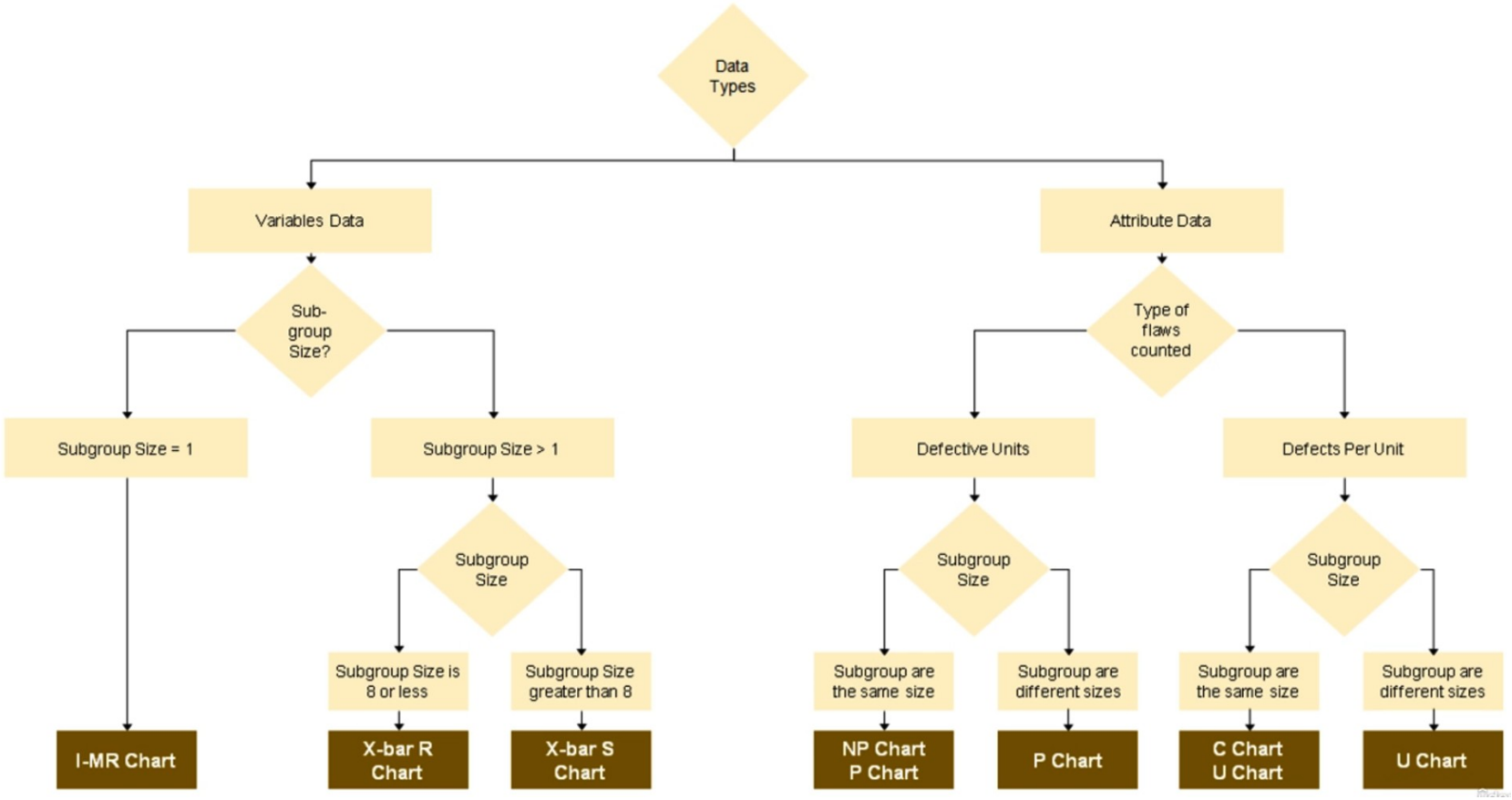
RECAP - CHART SELECTION

Chart selection is a decision tree

Data type, then subgroup size, then what you are counting. Get it wrong and the limits are wrong.

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The control chart selection map



Continuous data branches on subgroup size; attribute data branches on defectives versus defects and on constant versus varying sample.

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RECAP - CHART SELECTION

CONTINUOUS data - select on subgroup size n

1

n = 1 -> I-MR

Individuals and moving range. Slow processes, batch results, one reading per period

2

n = 2 to 8 -> Xbar-R

Average and range. The workhorse - range is easy for operators to compute

3

n >= 9 -> Xbar-S

Average and standard deviation. At large n the range wastes information

4

Always a PAIR

One chart for the centre, one for the spread. Read the spread chart FIRST

5

Why the pair matters

If the range chart is out of control, the Xbar limits are computed from noise

6

Meridian example

Fill volume n=5 every hour -> Xbar-R; monthly OTIF percentage n=1 -> I-MR

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RECAP - CHART SELECTION

ATTRIBUTE data - defectives versus defects

1

DEFECTIVE = the unit fails

One pump either passes or fails. Binomial. Count units, not faults

2

DEFECT = one fault found

One pump may carry three faults. Poisson. Count faults, not units

3

np chart

Number defective, CONSTANT sample size - e.g. exactly 200 pumps inspected daily

4

p chart

Proportion defective, VARYING sample size - limits step in and out with n

5

c chart

Count of defects, CONSTANT opportunity - faults per finished pump assembly

6

u chart

Defects per unit, VARYING opportunity - faults per batch of differing size

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RECAP - CHART SELECTION

Meridian - four attribute charts chosen correctly

np chart

Singapore final test

- Exactly 250 pumps tested every shift
- Count of pumps failing final test
- Constant n, defectives -> np

p chart

Penang incoming inspection

- Lot sizes vary from 80 to 600 units
- Proportion of lots rejected
- Varying n, defectives -> p

c chart

Suzhou cosmetic audit

- One finished pump audited per hour
- Number of cosmetic faults per pump
- Constant opportunity, defects -> c

u chart

Warranty returns

- Faults reported per 1000 units shipped
- Shipment volume varies each month
- Varying opportunity, defects -> u

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Computing control limits with the constants

- Upper Control Limit (UCL):

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

- Lower Control Limit (LCL):

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

  - If LCL is negative, set $LCL = 0$.

- Example:

  For $\bar{c} = 3.4$:

$$UCL = 3.4 + 3\sqrt{3.4} = 3.4 + 3(1.8439) = 3.4 + 5.5317 = 8.93$$

$$LCL = 3.4 - 3\sqrt{3.4} = 3.4 - 5.5317 = -2.13 \approx 0$$

  - **UCL = 8.93, LCL = 0**

The A2, D3 and D4 constants convert the average range into 3-sigma limits without ever computing a standard deviation.

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RECAP - LIMITS

Xbar-R control limits - the constants

Xbar chart	UCL/LCL = Xbar-bar +/- A2 x Rbar	A2 depends only on subgroup size n
R chart upper	UCL(R) = D4 x Rbar	D4 = 2.114 at n=5
R chart lower	LCL(R) = D3 x Rbar	D3 = 0 for n <= 6, so LCL(R) = 0

A2 = 0.577 at n=5, 0.729 at n=4, 1.023 at n=3. The constants replace a sigma estimate.

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RECAP - LIMITS

Meridian worked example - Singapore fill volume, n = 5

Data	**Xbar-bar = 10.02 mL, Rbar = 0.18 mL**	25 subgroups of 5, hourly, over 3 shifts
Xbar limits	**10.02 +/- 0.577 x 0.18**	UCL = 10.124, LCL = 9.916 mL
R chart limits	**UCL = 2.114 x 0.18 = 0.381**	LCL = 0 x 0.18 = 0 (n = 5)

Spec is 9.80 to 10.20 mL. The control limits sit well inside spec - process voice narrower than customer voice.

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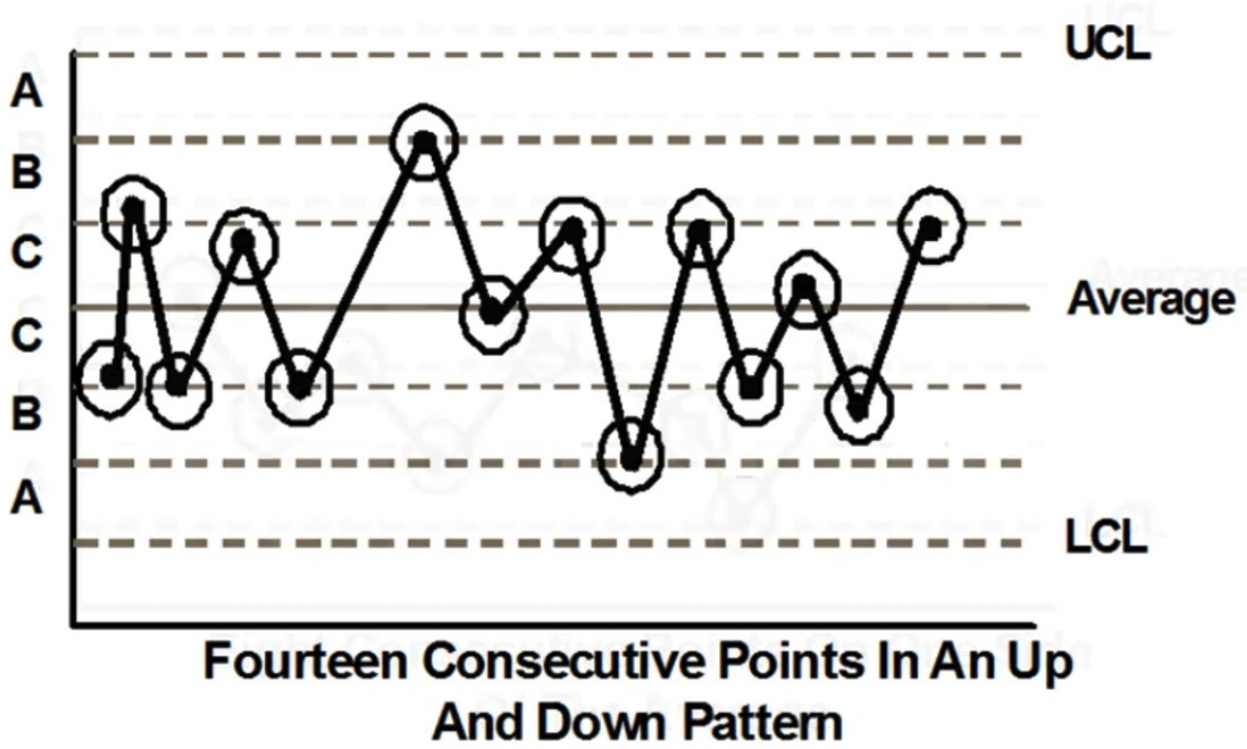
RECAP - RUN RULES

A point inside the limits can still be a signal

The Western Electric run rules read the PATTERN, not just the extremes.

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Out-of-control patterns on a chart



Beyond-limit points, runs on one side, trends and hugging - each pattern has a different physical cause.

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RECAP - RUN RULES

The Western Electric rules in full

1

Rule 1 - beyond 3 sigma

Any single point outside a control limit. The classic special cause

2

Rule 2 - 2 of 3 beyond 2 sigma

Two of three consecutive points in zone A on the SAME side

3

Rule 3 - 4 of 5 beyond 1 sigma

Four of five consecutive points beyond 1 sigma, SAME side

4

Rule 4 - 8 on one side

Eight consecutive points on one side of the centre line - a shift

5

Rule 5 - trend

Six or seven consecutive points steadily rising or falling - tool wear, fouling

6

Rule 6 - hugging the CL

15 consecutive points inside 1 sigma. Suspiciously good - check the data

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The Black Belt caveat on run rules

1

Rules multiply false alarms

Each rule has its own false-alarm rate - all six together is far above 0.27 percent

2

All six on a daily chart

Combined false alarm roughly 1 percent - about one nuisance signal per quarter

3

Choose rules deliberately

Rules 1 and 4 on a stable, well-understood process is often enough

4

Never let software decide

Default rule sets differ between packages. Specify yours in the control plan

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RECAP - CAPABILITY

Capability recap - Cp, Cpk, Pp, Ppk

Four indices, two questions: is the spread small enough, and is it centred?

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RECAP - CAPABILITY

The four capability indices

Cp	**Cp = (USL - LSL) / (6 x s-within)**	Potential - spread only, ignores centring
Cpk	**Cpk = min[(USL-Xbar), (Xbar-LSL)] / (3 x s-within)**	Actual - penalises off-centre
Pp / Ppk	**Same formulas with s-TOTAL**	Long-term performance including shift-to-shift drift

Cp and Cpk use WITHIN-subgroup variation; Pp and Ppk use TOTAL variation. Reporting the wrong pair flatters the process.

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Meridian worked example - Suzhou seal force

Given	**USL = 62 N, LSL = 48 N, Xbar = 56.4 N**	s-within = 1.55 N, s-total = 2.10 N
Cp and Cpk	**Cp = 14 / 9.30 = 1.51**	Cpk = min(5.6, 8.4) / 4.65 = 1.20
Pp and Ppk	**Pp = 14 / 12.60 = 1.11**	Ppk = 5.6 / 6.30 = 0.89

Cpk 1.20 but Ppk 0.89 - the process is good within a shift and drifts BETWEEN shifts. That gap is the project.

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Reading the gaps between the indices

1

Cp much greater than Cpk

Spread is fine, process is OFF-CENTRE. Re-centre - do not launch variation work

2

Cpk much greater than Ppk

Good within subgroup, drifting between them. Hunt the between-shift X

3

Cp roughly equals Cpk

Centred. Any shortfall is genuine spread - variation reduction is the right project

4

Cpk equals Ppk

Within and total variation agree - the process is genuinely stable over time

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RECAP - CONTROL PLAN

The control plan format

Process			Method						Action		
					Sample						
Process Step	What is being Controlled?	Input or Output?	Standard Applied?	How is it measured?	Size	Frequency	Responsible Party	Where Recorded?	Control Point	Response Plan?	Responsible Party?

One row per characteristic: what is measured, how, how often, by whom, against what limits, and what happens on a signal.

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RECAP - CONTROL PLAN

Every control plan row - the mandatory columns

1. **Process step and characteristic**
   Where in the flow, and exactly which X or Y is being controlled
2. **Specification and limits**
   Target with tolerance, plus the control limits if charted
3. **Measurement method and gauge**
   Which instrument, and the MSA result that qualified it
4. **Sample size and frequency**
   The rational subgroup and interval, chosen not inherited
5. **Control method**
   Chart type, checklist, poka-yoke, automated interlock
6. **Reaction plan and owner**
   The named person and the exact action on an out-of-control signal

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RECAP - CONTROL PLAN

The three layers that hold a gain

SOP

The method written down

- The new way of working, step by step
- Version controlled with an effective date
- Training record for every affected operator
- Reviewed when the process changes

Visual management

The state visible at a glance

- Chart at the machine, not in a monthly pack
- Red/green status visible from the aisle
- Shadow boards, floor markings, andon
- A stranger can see if the process is healthy

Response plan

The action on a signal

- IF the chart signals THEN do exactly this
- Contain, adjust, escalate - in that order
- Named owner and escalation path
- Rehearsed, not written and filed

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RECAP - CONTROL PLAN

Mistake-proofing beats monitoring

1

Level 1 - eliminate

Redesign so the error is physically impossible. The strongest control

2

Level 2 - prevent

Poka-yoke: the fixture only accepts the part one way round

3

Level 3 - detect at source

Automatic interlock stops the line the moment the error occurs

4

Level 4 - detect downstream

Inspection or a control chart finds it after the fact

5

Level 5 - depend on people

Training and vigilance. The weakest control - it decays

6

Design rule

Push every control as far UP this list as the budget allows

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BLACK BELT ONLY - ARL

Average Run Length - the honest measure of a chart

1

ARL definition

The average number of subgroups plotted before the chart signals

2

ARL-0 - in control

How long between FALSE alarms. For 3-sigma Shewhart, ARL-0 = 370

3

ARL-1 - out of control

How long to DETECT a real shift. Smaller is better

4

The number that matters

ARL-1 for the smallest shift you actually care about catching

5

Chart comparison rule

Compare charts at EQUAL ARL-0, then see which has the lower ARL-1

6

Why it is not taught at GB

Green Belts use the chart. Black Belts choose between charts

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BLACK BELT ONLY - ARL

Shewhart ARL - the uncomfortable arithmetic

In control	**ARL-0 = 1 / 0.0027 = 370**	One false alarm every 370 subgroups
3-sigma shift	**ARL-1 = about 2**	Detected almost immediately - the easy case
1-sigma shift	**ARL-1 = about 44**	Forty-four subgroups before the chart notices

Rule 1 alone, subgroup n=5. The 1-sigma case is the one that quietly destroys the project benefit.

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BLACK BELT ONLY - ARL

Meridian - what an ARL of 44 costs

The shift

Penang line 2 fill volume

- Fill mean drifts 1 sigma high after a pump reseal
- Still inside spec - no customer complaint yet
- Xbar-R chart, subgroups hourly, n = 5

The delay

44 subgroups = 44 hours

- Roughly 5.5 shifts before the chart signals
- About 26,000 units filled off target
- Overfill cost alone about SGD 19,000

The structural cause

The chart has no memory

- Each point judged ALONE against the limits
- Yesterday's evidence is discarded entirely
- A shift small relative to sigma never breaches

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BLACK BELT ONLY - CUSUM

BLACK BELT ONLY: CUSUM

Accumulate every small deviation from target until the total becomes undeniable.

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How the CUSUM works

1

Deviation from TARGET

Each point contributes its distance from target, not from the mean

2

Deviations accumulate

Small biased-high deviations add up; random ones cancel out

3

Reset at zero

The sum is floored at zero so an in-control process stays flat

4

Two one-sided sums

An UPPER CUSUM for high shifts and a LOWER CUSUM for low shifts

5

Signal on h

Signal when either sum exceeds the decision interval h

6

The payoff

A 1-sigma shift detected in about 8 subgroups instead of 44

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The CUSUM equations

Upper CUSUM	$C^+(i) = \max[0, x(i) - (T + k) + C^+(i-1)]$	Accumulates evidence of a HIGH shift
Lower CUSUM	$C^-(i) = \max[0, (T - k) - x(i) + C^-(i-1)]$	Accumulates evidence of a LOW shift
Signal rule	**Signal if $C^+ > h$ or $C^- > h$**	h is the decision interval, in sigma units

T = target. Both sums start at zero and are floored at zero - that floor is what keeps a stable process flat.

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Choosing k and h - the two design parameters

1. **k = the reference value**
   Set k to HALF the shift you want to detect, in sigma units

2. **k for a 1-sigma shift**
   k = 0.5 sigma. This is by far the most common setting

3. **h = the decision interval**
   Typically 4 to 5 sigma. h = 5 with k = 0.5 gives ARL-0 near 465

4. **Larger h**
   Fewer false alarms, slower detection. Trades ARL-1 for ARL-0

5. **Smaller k**
   Tuned for smaller shifts, but slower against large ones

6. **The trade you are making**
   CUSUM is fast on SMALL shifts and slightly SLOWER than Shewhart on huge ones

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Meridian worked CUSUM - Penang fill volume

Design	**T = 10.00 mL, sigma = 0.04, k = 0.02, h = 0.20**	k = 0.5 sigma, h = 5 sigma
Points 1-4	**10.01, 10.02, 10.03, 10.04**	C+ = 0, 0, 0.01, 0.03 - accumulating
Points 5-8	**10.04, 10.05, 10.04, 10.05**	C+ = 0.05, 0.08, 0.10, 0.13 - climbing

By point 11 C+ exceeds h = 0.20 and signals. No individual point came near the 3-sigma Shewhart limit of 10.12.

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CUSUM versus Shewhart on the same Penang data

Shewhart Xbar-R

Judges each point alone

- UCL = 10.124 mL - never breached
- Rule 4 (8 on one side) eventually fires at point 13
- No signal at all if the drift is under 8 points
- Detected the 1-sigma shift at subgroup 44 on average

CUSUM

Accumulates the evidence

- Signals at point 11 in this run
- Average detection about 8 subgroups
- Roughly 5 times faster on this shift size
- The plotted sum also estimates WHEN the shift began

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BLACK BELT ONLY: EWMA

Exponentially weighted moving average - tunable memory, and it survives autocorrelation.

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BLACK BELT ONLY - EWMA

The EWMA statistic

The recursion	**z(i) = lambda x(i) + (1 - lambda) z(i-1)**	z(0) = target; lambda between 0 and 1
Weight decay	**Weight on the point k steps back = lambda(1-lambda)^k**	Geometrically decaying memory
Control limits	**T +/- L x sigma x sqrt[lambda / (2 - lambda)]**	L is typically 2.7 to 3.0

Every past point still contributes - but its weight halves away geometrically. That is the memory.

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BLACK BELT ONLY - EWMA

Choosing lambda - the single design decision

1

lambda = 0.10 to 0.20

Long memory, narrow limits. Best for SMALL sustained shifts. The usual choice

2

lambda = 0.20

The most common default - detects a 1-sigma shift in about 10 subgroups

3

lambda = 0.40 and above

Short memory, wider limits - behaviour approaches a Shewhart chart

4

lambda = 1.00

No memory at all - the EWMA IS the individuals chart

5

Match lambda to shift size

Small shift to detect -> small lambda. Large shift -> larger lambda

6

Set L with lambda

L = 2.7 with lambda = 0.10 gives ARL-0 near 500. Tune the pair, not one

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BLACK BELT ONLY - EWMA

Meridian worked EWMA - Suzhou cure oven temperature

Design	**T = 180 C, sigma = 1.2, lambda = 0.20, L = 2.7**	Steady-state limits 180 +/- 1.08 C
Points 1-4	**180.6, 181.1, 180.9, 181.4**	z = 180.12, 180.32, 180.43, 180.63
Points 5-7	**181.2, 181.6, 181.5**	z = 180.75, 180.92, 181.03 - signals at point 8

Individual readings never exceed 183.6 (the 3-sigma individuals limit). The smoothed statistic crosses in 8 points.

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CUSUM or EWMA - choosing between them

CUSUM

Sharper on a known shift size

- Slightly faster when you know the shift to detect
- The sum estimates when the shift started
- Two statistics to plot and read
- Less intuitive for operators

EWMA

More forgiving and more general

- One smoothed line plotted like a normal chart
- Performs well across a RANGE of shift sizes
- Handles AUTOCORRELATED data - CUSUM does not
- Easier for operators to interpret

In practice

The Black Belt's default

- EWMA for most processes - readability wins
- CUSUM when the shift size is known and detection speed is critical
- Either one ALWAYS paired with a Shewhart chart for spikes

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BLACK BELT ONLY: Autocorrelation

If this reading predicts the next one, Shewhart limits are far too narrow - and the chart screams.

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Where autocorrelation comes from

1. **Chemical and batch processes**
   Reactor contents carry over - this hour's concentration sets next hour's

2. **Continuous flow**
   Tanks, ovens and dryers have thermal inertia measured in hours

3. **High-frequency sampling**
   Sample faster than the process can change and every reading resembles the last

4. **Feedback control loops**
   A PID controller deliberately makes the next reading depend on the last

5. **Meridian case**
   Suzhou adhesive viscosity sampled every 5 minutes from a 400 L heated vessel

6. **The tell-tale**
   Lag-1 autocorrelation above about 0.5 on the baseline data

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What autocorrelation does to a Shewhart chart

The mechanism

Sigma is underestimated

- Moving-range sigma assumes consecutive points are independent
- Correlated points sit close together, so MR is small
- Small MR gives a small sigma estimate
- Small sigma gives limits far too narrow

The symptom

Constant false alarms

- 20 to 40 percent of points outside the limits
- Long runs on one side firing Rule 4 continually
- The process is genuinely stable - the chart is wrong

The damage

The chart is abandoned

- Operators learn every signal is noise
- They stop investigating - including the real ones
- This is how SPC programmes quietly die

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Diagnosing and fixing autocorrelation

1. Plot the data in time order and look for long smooth waves
2. Compute the lag-1 autocorrelation on 50+ baseline points
3. If it exceeds about 0.5, the Shewhart assumption is broken
4. Fix 1: sample LESS often, until consecutive readings are independent
5. Fix 2: use an EWMA chart, which tolerates the correlation
6. Fix 3: fit a time-series model and chart the RESIDUALS

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BLACK BELT ONLY - SHORT RUN

BLACK BELT ONLY: Short-run SPC

High-mix low-volume never gets 25 subgroups per part number. Chart the deviation instead.

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BLACK BELT ONLY - SHORT RUN

The short-run problem

1

The rule you cannot meet

Control limits need 25+ subgroups. A 40-unit run gives you 8

2

High mix, low volume

Meridian Singapore runs 34 pump variants across one assembly cell

3

Per-part charts fail

34 charts, none with enough data, all with meaningless limits

4

The cell is the constant

The same fixture, operator and method make every variant

5

The insight

Chart the PROCESS, not the part. Variation belongs to the cell

6

The method

Transform every part onto one common scale, then chart them together

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BLACK BELT ONLY - SHORT RUN

The two short-run transformations

DNOM chart	**plotted = x - nominal(part)**	Deviation from nominal. Use when spread is similar across parts
Standardised chart	**plotted = (x - nominal) / sigma(part)**	Use when parts have DIFFERENT spreads
Limits	**DNOM: 0 +/- A2 x Rbar-pooled**	Standardised: 0 +/- 3, always

Both centre every part at zero, so one chart carries the whole product mix on a single scale.

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BLACK BELT ONLY - SHORT RUN

Meridian worked DNOM - Singapore assembly cell

Part A	**Nominal 24.0 mm, measured 24.03**	Plotted = +0.03
Part B	**Nominal 31.5 mm, measured 31.46**	Plotted = -0.04
Part C	**Nominal 18.2 mm, measured 18.25**	Plotted = +0.05

Three different parts, one chart, one set of limits. The cell now has 300 points instead of 34 charts of 8.

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BLACK BELT ONLY - MULTIVARIATE

BLACK BELT ONLY: Multivariate SPC

Five separate charts inflate your false alarms - and still miss the failure that matters most.

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BLACK BELT ONLY - MULTIVARIATE

Problem 1 - false alarm inflation

One chart	P(signal) = 0.0027	The familiar 3-sigma false alarm rate
Five charts	1 - (1 - 0.0027)^5 = 0.0134	1.34 percent - five times worse
Ten charts	1 - (1 - 0.0027)^10 = 0.0267	2.67 percent - a false alarm every 37 points

The same alpha-inflation logic as repeated t-tests. More charts means more crying wolf.

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BLACK BELT ONLY - MULTIVARIATE

Problem 2 - the failure separate charts CANNOT see

Two correlated CTQs

Meridian pump body

- Bore diameter and piston diameter
- Historically they track together - r = 0.86
- Both machined from the same billet, same thermal cycle

Both charts pass

Each looks perfectly normal

- Bore = 12.03 mm, well inside its limits
- Piston = 11.88 mm, well inside its limits
- Two green charts, no signal, no action taken

The product is scrap

The COMBINATION is impossible

- That bore-piston pair has never occurred before
- Clearance is 0.15 mm against a 0.06 mm design
- The correlation broke - only a joint chart sees it

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BLACK BELT ONLY - MULTIVARIATE

Hotelling's T-squared

The statistic	**T2 = n (x-bar - mu)' S-inverse (x-bar - mu)**	Squared distance in p dimensions
What S-inverse does	**The covariance matrix inverse**	It accounts for the CORRELATION between variables
Control limit	**UCL from the F-distribution**	Function of p variables, m subgroups, n subgroup size

T2 is a generalised squared distance from the process centroid, scaled by how the variables normally co-vary.

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BLACK BELT ONLY - MULTIVARIATE

Reading the T-squared chart

1

One chart, all variables

A single line replaces p separate charts - and one honest alpha

2

NO lower control limit

T2 is a squared distance, so it cannot be negative. Only an upper limit

3

The limit comes from F

UCL uses the F-distribution, not 3 sigma - the geometry is different

4

It sees correlation breaks

A point can signal with EVERY variable inside its own limits

5

It does not say which variable

The signal is a distance. Diagnosis needs decomposition

6

Pair it with a spread chart

T2 watches the centroid; a generalised variance chart watches the spread

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BLACK BELT ONLY - MULTIVARIATE

When multivariate SPC earns its keep

1

Use it when CTQs correlate

Correlation above about 0.5 - separate charts are then misleading

2

Use it when p is large

Five or more characteristics on one product or process

3

Use it for assemblies

Mating dimensions where the FIT, not each part, is the requirement

4

Do NOT use it if independent

Uncorrelated variables - separate charts are clearer and diagnostic

5

Do NOT use it without software

The matrix algebra is not a spreadsheet job for the shop floor

6

Do NOT use it without training

A T2 signal with no decomposition is a mystery, not a control

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BLACK BELT ONLY - GOVERNANCE

BLACK BELT ONLY: Governing control plans

You do not audit your own project. You audit the portfolio - and find the decay.

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BLACK BELT ONLY - GOVERNANCE

The audit cadence across a portfolio

1

30 days after handover

Is the chart being plotted? Does the owner know they own it?

2

60 days

Have any signals occurred? Was the reaction plan actually followed?

3

90 days

Is capability holding? Has the benefit rate held against the Finance model?

4

Then quarterly

Rolling sample of control plans across all closed projects

5

Trigger audits

Any customer complaint, scrap spike or process change forces an off-cycle audit

6

Report to the steering committee

Sustainment status by project - a standing agenda item

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BLACK BELT ONLY - GOVERNANCE

The common decay mode - and how to catch it

What auditors see

Everything looks compliant

- Charts printed, posted, up to date
- SOP present at the workstation
- Owner named on the control plan
- The tick-box audit passes cleanly

What is actually happening

Signals are never actioned

- Out-of-control points plotted and left uncircled
- No investigation record for any of them
- The chart has become a record, not a control
- The gain has already begun to erode

The audit that catches it

Trace signals, not charts

- Pick the last three out-of-control points
- Demand the dated investigation for each
- No record means the control plan has failed
- This one test finds most sustainment failures

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BLACK BELT ONLY - FINANCE

BLACK BELT ONLY: Benefits are not real

until Finance signs them. Hard, soft and cost-avoidance are booked in three different ways.

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BLACK BELT ONLY - FINANCE

The three benefit categories

HARD benefit

Hits the P&L

- A cost line genuinely falls or revenue genuinely rises
- Scrap, rework, overtime, freight, warranty claims
- Traceable to a general ledger account
- Finance books it into the budget
- Meridian: scrap down SGD 340k per year

SOFT benefit

Capacity or time released

- Hours or machine capacity freed but not yet monetised
- Real only if the freed resource is REDEPLOYED
- Finance will not book it without a redeployment plan
- Claim it separately, never inside the hard number
- Meridian: 2.1 FTE of inspection time released

COST AVOIDANCE

A future cost prevented

- A cost that would have occurred but now will not
- Avoided capital, avoided headcount, avoided penalty
- Never reduces the current budget - nothing to cut
- Requires an agreed counterfactual baseline
- Meridian: SGD 1.2M line avoided by capacity gain

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BLACK BELT ONLY - FINANCE

Why Finance treats them differently

1. **Hard benefits cut a budget**
   A real budget line is reduced and somebody is held to the new number

2. **Soft benefits cut nothing**
   Freed hours stay on the payroll until redeployed or removed

3. **Avoidance never appears**
   There is no line item for a cost that never happened

4. **Double counting is the risk**
   Two projects touching the same scrap line both claim the full saving

5. **Annualisation is disputed**
   Three good months does not entitle you to a twelve-month number

6. **Credibility is the real asset**
   One inflated claim and the programme's numbers are discounted forever

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BLACK BELT ONLY - FINANCE

The benefits validation and sign-off process

1. Agree the baseline and measurement method with Finance BEFORE Improve
2. Measure the post-improvement performance over at least 3 months
3. Apply the agreed standard costs - Finance owns the rates, not you
4. Separate hard, soft and cost-avoidance into three explicit lines
5. Finance controller signs; benefit enters the tracker at the SIGNED figure

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BLACK BELT ONLY - FINANCE

Meridian worked benefit - Penang fill volume project

Hard	**26,000 units x 0.09 mL overfill x SGD 0.14/mL**	SGD 328k per year, scrap line reduced
Soft	**0.8 FTE of rework released**	SGD 46k - claimed ONLY with a redeployment plan
Avoidance	**Recall provision no longer required**	SGD 180k - stated separately, never added in

Report SGD 328k signed hard benefit. Soft and avoidance are shown, labelled, and never merged into the headline.

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CONTROL - CLOSURE

Closure is a transaction

Ownership transfers, sustainment is verified, lessons are harvested - then the project ends.

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The handover sequence

1. Process owner named and formally accepts the control plan
2. Operators trained on the new SOP with a signed training record
3. Charts running in production with a rehearsed reaction plan
4. 30/60/90-day sustainability audits scheduled in the owner's calendar
5. Finance signs the benefit; steering committee formally closes

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CONTROL - CLOSURE

The 30/60/90-day sustainability audit

1

Day 30 - is it running?

Charts plotted, SOP in use, owner engaged. Fix mechanics now, cheaply

2

Day 60 - is it working?

Signals investigated and closed. Reaction plan actually exercised

3

Day 90 - is it holding?

Capability at or above the improved level, benefit rate on plan

4

If it has slipped

Reopen with the sponsor. Do NOT quietly write off the benefit

5

Who does the audit

The Black Belt, not the process owner. Independence is the point

6

The output

A one-page sustainment status per project into the portfolio review

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Control phase - what you must be able to do now

1. **Select and build any chart**
   Continuous or attribute, and justify the choice from first principles

2. **Explain why Shewhart is slow**
   ARL, the lack of memory, and what a 1-sigma shift costs

3. **Design a CUSUM or EWMA**
   Choose k and h, or lambda and L, for a stated shift size

4. **Handle awkward data**
   Autocorrelation, short runs, correlated multivariate characteristics

5. **Govern control plans**
   Audit a portfolio and detect the signals-never-actioned decay mode

6. **Validate the money**
   Split hard, soft and avoidance, and defend attribution to a CFO

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WHAT YOU'LL DO

Hands-On Labs — Control

1

Lab 27 — Statistical Process Control and Chart Selection — In Depth

A correctly selected and constructed control chart for your capstone Y with applied… • Data: lab-27-spc-control-charts.xlsx

2

Lab 28 — CUSUM, EWMA and Short-Run SPC (Black Belt Only)

CUSUM and EWMA charts on your capstone data plus a short-run standardised chart. • Data: lab-28-cusum-ewma-shortrun.xlsx

3

Lab 29 — Multivariate SPC and Control Plan Governance (Black Belt Only)

A Hotelling T-squared chart plus a governed control plan for your capstone. • Data: lab-29-multivariate-spc-control-plan.xlsx

4

Lab 30 — Financial Benefits Validation, Handover and Project Closure

A Finance-validated benefits statement and a signed handover and closure pack. • Data: lab-30-benefit-validation.xlsx

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DMAIC · CONTROL · HANDS-ON

Statistical Process Control and Chart Selection — In Depth

LAB 27

A full re-derivation of SPC — chart anatomy, the selection logic, control limits and the out-of-control rules. You must be able to teach this to Green Belts and audit their chart choices, which is a core Black Belt accountability. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A correctly selected and constructed control chart for your capstone Y with applied run rules.

Tools & techniques: Chart anatomy, control limits, Xbar-R, Xbar-S, I-MR, p, np, c, u charts, Western Electric rules, control vs spec limits

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Statistical Process Control and Chart Selection — In Depth

STEPS 1-6 OF 11

1. Recap SPC purpose in full: distinguish signal from noise over time, so drift is caught before it becomes a defect.
2. Recap chart anatomy: centre line, upper and lower control limits at +/- 3 sigma, and why 3 sigma balances false alarms against missed signals.
3. Establish the critical distinction: CONTROL limits come from the process's own voice (the data); SPECIFICATION limits come from the customer. They are unrelated and must never be drawn on the same axis as if comparable.
4. Learn the chart selection logic in full. Continuous data: subgroup size 1 = I-MR; 2-8 = Xbar-R; 9+ = Xbar-S.
5. Learn the attribute branch: defectives with constant sample size = np; varying = p. Defects with constant opportunity = c; varying = u.
6. Select the correct chart for your capstone Y and write down the justification. Selecting the wrong chart produces wrong limits and false confidence.

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Statistical Process Control and Chart Selection — In Depth

STEPS 7-11 OF 11

7. Collect at least 25 subgroups of baseline data — fewer gives control limits too unstable to be trustworthy.

8. Calculate the control limits using the correct constants (A2, D3, D4 for Xbar-R; equivalent constants for other charts).

9. Plot the chart and apply the Western Electric run rules in full: 1 point beyond 3 sigma; 2 of 3 beyond 2 sigma; 4 of 5 beyond 1 sigma; 8 consecutive on one side; trends; and hugging the centre line.

10. Investigate every signal. If a special cause is found and removed, recalculate the limits excluding that point — but only with a documented assignable cause.

11. Confirm the process is stable before proceeding. An unstable process has no meaningful capability and cannot be controlled by a control plan alone.

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LAB 27 · VERIFY

Statistical Process Control and Chart Selection — In Depth

✅ Check your work

Your chart selection is justified against the selection logic, limits are computed from 25+ subgroups with the correct constants, and all run rules are applied with every signal investigated.

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CUSUM, EWMA and Short-Run SPC (Black Belt Only)

LAB 28

BLACK BELT ONLY. Shewhart charts are poor at detecting small sustained shifts — they may take dozens of subgroups to react to a 1-sigma drift, which is precisely how a process degrades unnoticed after a project closes. CUSUM and EWMA solve this, and short-run methods handle high-mix low-volume processes Shewhart cannot chart at all. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack,

You'll build

CUSUM and EWMA charts on your capstone data plus a short-run standardised chart.

Tools & techniques: ARL, CUSUM, decision interval, V-mask, EWMA, lambda weighting, autocorrelation, short-run standardised charts, DNOM

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CUSUM, EWMA and Short-Run SPC (Black Belt Only)

STEPS 1-5 OF 15

1. Understand Average Run Length (ARL): the expected number of subgroups before a chart signals. For a 1-sigma shift, a Shewhart Xbar chart has an ARL around 44 — far too slow.
2. Understand the structural reason: a Shewhart chart considers only the CURRENT point and has no memory of the recent past.
3. Learn CUSUM: it accumulates deviations from target, so a small persistent bias adds up into a detectable signal rather than being repeatedly dismissed as noise.
4. Set up the CUSUM parameters: the reference value k (typically half the shift you want to detect) and the decision interval h (typically 4 or 5 sigma).
5. Build upper and lower CUSUM statistics for your capstone data and plot them against the decision interval.

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CUSUM, EWMA and Short-Run SPC (Black Belt Only)

STEPS 6-10 OF 15

6. Compare the CUSUM detection point against the Shewhart chart from Lab 27 on the same data — CUSUM typically detects a sustained small shift several subgroups sooner.

7. Learn EWMA: each point is a weighted average of all history, with weights decaying geometrically. Lambda controls the memory.

8. Choose lambda deliberately: small lambda (0.1-0.2) gives long memory and strong small-shift detection; large lambda (0.4+) approaches Shewhart behaviour.

9. Build the EWMA chart with its widening initial control limits, and note that EWMA also handles autocorrelated data far better than Shewhart.

10. Test your capstone data for autocorrelation. Chemical, batch and continuous-flow processes are frequently autocorrelated, which makes Shewhart limits far too narrow and generates constant false alarms.

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CUSUM, EWMA and Short-Run SPC (Black Belt Only)

STEPS 11-15 OF 15

11	Now short-run SPC. Understand the problem: high-mix low-volume processes never accumulate 25 subgroups of any single part number.
12	Learn the deviation-from-nominal (DNOM) approach: plot (measurement - part nominal) so different part numbers share one chart.
13	Learn the standardised approach for parts with differing variances: plot (measurement - nominal) / sigma_part, putting all parts on a common scale.
14	Build a short-run chart for a multi-product step in your capstone, or for Meridian's shared assembly line across pump variants.
15	Decide which chart type belongs in your capstone control plan and justify the choice against shift size, autocorrelation and product mix.

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LAB 28 · VERIFY

CUSUM, EWMA and Short-Run SPC (Black Belt Only)

✅ Check your work

You have built CUSUM and EWMA charts and compared their detection speed against Shewhart on the same data, tested for autocorrelation, and justified your final chart selection for the control plan.

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DMAIC · CONTROL · HANDS-ON

Multivariate SPC and Control Plan Governance (Black Belt Only)

LAB 29

BLACK BELT ONLY. When several correlated characteristics matter, running separate charts inflates false alarms and — more dangerously — misses failures visible only in the correlation structure. Then step up to the portfolio view: the Black Belt audits control plans across projects, not just their own. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on

You'll build

A Hotelling T-squared chart plus a governed control plan for your capstone.

Tools & techniques: Multivariate SPC, Hotelling T-squared, correlation structure, false alarm inflation, control plan, response plan, audit cadence

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Multivariate SPC and Control Plan Governance (Black Belt Only)

STEPS 1-5 OF 13

1. Understand the false alarm problem: 5 separate charts each at alpha = 0.0027 gives a combined false alarm rate over 1.3 percent — five times intended.
2. Understand the more serious failure: two characteristics can each sit within their own limits while their COMBINATION is impossible, and separate charts will never see it.
3. Identify a set of correlated characteristics in your capstone — dimensions on one part, or related cycle times across linked steps.
4. Compute the correlation matrix and confirm the characteristics are genuinely correlated. If they are independent, separate charts are fine.
5. Learn Hotelling's T-squared: it collapses several correlated characteristics into one statistic accounting for the covariance structure.

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Multivariate SPC and Control Plan Governance (Black Belt Only)

STEPS 6-9 OF 13

6 Build the T-squared chart with its upper control limit from the F-distribution, and note there is no lower limit — T-squared is always positive.

7 Understand the interpretation difficulty: a T-squared signal tells you something is wrong but not which variable. Use decomposition or individual charts as a follow-up diagnostic.

8 Now build your capstone control plan in full: for each critical X and the Y — the metric, target, specification, measurement method, sample size, frequency, chart type, owner and reaction plan.

9 Write the reaction plan for each metric as a specific instruction: containment action, escalation path, named owner and time limit. 'Investigate' is not a reaction plan.

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Multivariate SPC and Control Plan Governance (Black Belt Only)

STEPS 10-13 OF 13

10 Add the control plan to the process documentation and update the SOP, training material and visual management boards.

11 Step up to portfolio governance: as Black Belt you audit control plans across multiple projects. Define your audit cadence and criteria.

12 Define what a control plan audit checks: is the chart still being maintained, are signals being actioned, is the owner still in post, and has the process drifted since handover?

13 Identify the most common decay mode — charts maintained but signals never actioned — and write your countermeasure for it.

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LAB 29 · VERIFY

Multivariate SPC and Control Plan Governance (Black Belt Only)

✅ Check your work

Your correlated characteristics are confirmed by a correlation matrix before applying T-squared, every control plan metric has a specific named reaction plan, and you have defined an audit cadence with explicit criteria.

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DMAIC · CONTROL · HANDS-ON

Financial Benefits Validation, Handover and Project Closure

LAB 30

Benefits are not real until Finance signs them. This lab converts your improvement into a validated financial number, transfers ownership to the process owner, and formally closes the project — the step most often skipped, and the reason gains quietly evaporate. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

You'll build

A Finance-validated benefits statement and a signed handover and closure pack.

Tools & techniques: Hard vs soft benefits, cost avoidance, benefit validation, Finance sign-off, handover, sustainability audit, lessons learned, closure

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Financial Benefits Validation, Handover and Project Closure

STEPS 1-5 OF 14

1. Learn the benefit categories precisely. HARD: cost reduction that appears in the P&L. SOFT: capacity or time released without a headcount or cost change. COST AVOIDANCE: costs not incurred that would have been.

2. Understand why the distinction matters: Finance books these differently, and claiming soft benefits as hard is the fastest way to lose credibility with a CFO.

3. Quantify your capstone's benefit against the Lab 3 baseline, using post-improvement data over a defensible period, not a single good week.

4. Verify the improvement is statistically significant, not a favourable run — apply a hypothesis test comparing post-improvement against baseline.

5. Confirm sustained performance with your control chart: the shift must hold for a meaningful period before the benefit can be booked.

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Financial Benefits Validation, Handover and Project Closure

STEPS 6-10 OF 14

6. Separate your benefit into hard, soft and cost-avoidance components and state each explicitly.

7. Take the calculation to Finance and obtain sign-off. Expect challenge on the baseline period, the attribution and whether other changes could explain the result.

8. Prepare your attribution argument: what else changed in the period, and why the improvement is attributable to your intervention rather than to those.

9. Build the handover pack: control plan, updated SOPs, training records, control charts, reaction plans and the named process owner.

10. Conduct the handover meeting with the process owner. They must accept ownership explicitly — an assumed handover is a failed handover.

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Financial Benefits Validation, Handover and Project Closure

STEPS 11-14 OF 14

11 Schedule the sustainability audit at 30, 60 and 90 days post-handover, with a named auditor and defined criteria.

12 Run the lessons learned session: what worked, what did not, what you would do differently, and what transfers to the Green Belts you mentor.

13 Formally close the project with sponsor sign-off, and archive the full DMAIC storyboard.

14 Update the project portfolio from Lab 2 — record the closed project's realised benefit and select the next project.

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LAB 30 · VERIFY

Financial Benefits Validation, Handover and Project Closure

✅ Check your work

Your benefit is separated into hard, soft and cost-avoidance components, validated statistically and signed off by Finance, and the process owner has explicitly accepted handover with a scheduled sustainability audit.

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PHASE RECAP

Recap — Control

- SPC and control charts — in depth — Chart anatomy, control limits, chart selection (Xbar-R, Xbar-S, I-MR, p, np, c, u) and out-of-control rules, in full.
- Capability and control plans — in depth — Cp, Cpk, Pp, Ppk, control limits vs specification limits, control plans, SOPs and response plans re-derived.
- Limits of Shewhart charts — Shewhart charts are slow to detect small sustained shifts — which is exactly what a drifting process does.
- CUSUM charts — Cumulative sum charts accumulate small deviations, detecting a sustained 1-sigma shift far faster.
- EWMA charts — Exponentially weighted moving average — tunable memory, strong for small shifts and autocorrelated data.
- Short-run SPC — High-mix low-volume processes never gather 25 subgroups per part. Standardise and chart across parts.

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CAP

DMAIC · CAPSTONE

Capstone Project & Steering Committee Presentation

A3 storyboard · Consolidating your DMAIC artifacts · Presenting to sponsors · Defending the analysis · Programme leadership

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CAPSTONE · 10% OF THE COURSE

Key Concepts — Capstone

1. **The A3 storyboard**
   One page tells the whole project: background, current state, analysis, countermeasures, results, next steps.

2. **Consolidating your artifacts**
   Every lab from Day 1 produced one piece of your capstone — now assemble them into one narrative.

3. **Telling the data story**
   Lead with the business impact, support it with the statistics — never the other way round.

4. **Presenting to a steering committee**
   Sponsors want the decision, the risk and the money in the first two minutes.

5. **Defending the analysis**
   Expect challenge on sample size, assumptions and confounding — prepare the evidence for each.

6. **Handover and sustainability**
   Show who owns the process now, how it is monitored, and what triggers escalation.

7. **Lessons learned**
   Harvest what transfers to the next project and to the Green Belts you will mentor.

8. **Your Black Belt journey**
   Certification, project logging, mentoring obligations and continuing practice.

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CAPSTONE - DAY 5

Two things happen on Day 5

You sit the WSQ assessment, AND you present the capstone you have been building since Lab 1.

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How this course is assessed

THE WSQ ASSESSMENT

Required for certification

- Written Assessment (WA) - Short-Answer Questions (SAQ)
- Case Study (CS) - an applied DMAIC case study
- Both are OPEN BOOK - slides, Learner Guide, approved materials
- WA covers the knowledge (K1, K2); CS covers the abilities (A1-A5)
- 75 percent attendance is required to be eligible

THE CAPSTONE

Additional course deliverable

- The CAPSTONE PROJECT REPORT - your consolidated DMAIC storyboard
- The STEERING COMMITTEE PRESENTATION - 15 minutes plus defence
- Built from the artifacts you produced in every lab since Lab 1
- On TOP of the WSQ papers, not a replacement for them

WHY BOTH

Certification and capability

- The papers evidence the competency standard for WSQ
- The capstone evidences that you can actually run a project
- No employer asks for your exam score alone
- Defending an analysis under challenge is the real skill

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CAPSTONE - YOUR ARTIFACTS

You have been building this since Day 1

Every lab in this course advanced YOUR project by exactly one artifact.

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CAPSTONE - YOUR ARTIFACTS

DEFINE - the artifacts you already hold

1. **Project charter**
   Problem, goal, scope, business case, sponsor, team - re-baselined at each tollgate

2. **Y = f(X) statement**
   Your business Y named, with the candidate Xs you set out to test

3. **Baseline sigma level**
   DPMO and sigma computed from real data, before any change

4. **VOC and Kano analysis**
   Customer needs captured and classified basic, performance, delighter

5. **CTQ flowdown**
   Business Y cascaded to sub-Ys and then to controllable process Xs

6. **SIPOC and stakeholder plan**
   Process boundaries agreed, and a move planned per stakeholder

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CAPSTONE - YOUR ARTIFACTS

MEASURE - the artifacts you already hold

1. **Value stream map**
   Current state with cycle time, wait time and the process efficiency ratio

2. **Gage R&R study**
   Continuous MSA with percent study variation and the accept/reject decision

3. **Attribute MSA and Kappa**
   Inspector agreement quantified beyond chance, judged against 0.75

4. **Baseline capability**
   Cp, Cpk, Pp and Ppk with the within-versus-total variation gap explained

5. **Data collection plan**
   Operational definitions, sampling scheme and the sample size calculation

6. **The measurement verdict**
   Written evidence that your data was trustworthy BEFORE you analysed it

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ANALYZE - the artifacts you already hold

1. **Pareto analysis**
   The vital few categories carrying most of the defect or cost

2. **Fishbone and 5 Whys**
   Structured cause hypotheses by 5M+E, driven down to root

3. **Hypothesis tests**
   Named test, assumptions checked, p-value, and the business conclusion

4. **Multiple regression model**
   Coefficients, adjusted R-squared, VIF and residual diagnostics

5. **ANOVA results**
   Group means compared without inflating alpha, interactions examined

6. **Multi-vari study**
   Positional, cyclical and temporal variation separated before spending money

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CAPSTONE - YOUR ARTIFACTS

IMPROVE and CONTROL - the artifacts you already hold

1

DOE design and results

Factors, levels, design matrix, resolution, main effects and interactions

2

RSM optimum

Optimal settings for the vital few Xs, with the confirmation run

3

FMEA before and after

RPN scored, mitigated, and re-scored to prove the risk actually fell

4

Pilot results

The improvement confirmed at small scale before the full roll-out

5

SPC and CUSUM/EWMA

The chart now running in production, with its design justified

6

Control plan and benefit

Owner, reaction plan, audit schedule, and the Finance-validated number

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Assembling the artifact set into a narrative

1. Lay out all 24 artifacts in DMAIC order on one table
2. Mark each one: main deck, appendix, or working file only
3. Check every claim in the main deck has an artifact behind it
4. Find the gaps - a claim with no artifact is a claim you cannot defend
5. Build the A3 from the main-deck set only

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CAPSTONE - A3

The A3 storyboard

One page. If it does not fit on one page, you have not finished thinking.

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The seven A3 boxes

1

1. Background

Why this problem, why now, and which business objective it serves

2

2. Current state

The baseline with data - process map, capability, cost of the problem

3

3. Goal

The target condition, measurable, with a date. Specific enough to fail

4

4. Root cause analysis

The causal chain, and the statistical evidence that established it

5

5. Countermeasures

What you changed, why that change, and how it was validated

6

6. Results

Before versus after, with the Finance-signed benefit figure

7

7. Follow-up

Owner, control plan, audit dates, and the replication targets

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The 60-second visual test

The test

Hand it to a stranger

- Give the A3 to someone outside the project
- Give them 60 seconds, then take it back
- Ask: what was the problem, and what fixed it?
- If they cannot say, the A3 has failed

Why it fails

The usual causes

- Too much text and not enough charts
- Charts with no annotation of the point they make
- No visible before-and-after comparison
- The benefit figure buried in a paragraph

The fix

Make it visual

- One clear chart per box, annotated with its conclusion
- Before and after side by side, same scale
- The money in large type where the eye lands
- Words only where a chart genuinely cannot carry it

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CAPSTONE - A3

Meridian A3 - the seven boxes filled in

1

Background

Penang line 2 overfill costing SGD 41k/month; OTIF at risk from rework

2

Current state

Fill mean 10.09 mL against 10.00 target; Cpk 0.81; 8.4 percent rework

3

Goal

Fill mean within 0.02 mL of target and Cpk above 1.33 by end of Q3

4

Root cause

DOE: pump reseal torque x dwell time interaction drives the mean shift

5

Countermeasures

Torque spec tightened, dwell time fixed at RSM optimum, poka-yoke fitted

6

Results

Mean 10.005 mL, Cpk 1.41, rework 1.2 percent, SGD 328k signed hard benefit

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CAPSTONE - PRESENTING

The steering committee is not your team

They have twenty minutes, three other projects, and one question: what do you need from us?

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Three audiences in one room

The SPONSOR

Owns the money and the mandate

- Wants: the benefit, the risk, and the ask
- Fears: an overclaim they will have to defend upward
- Give them: the signed number and the decision required

The PROCESS OWNER

Lives with it after you leave

- Wants: to know exactly what changes for their team
- Fears: inheriting something unworkable or unresourced
- Give them: the control plan, the training, the support

FINANCE

Guards the credibility of the number

- Wants: baseline, method, attribution, and the split
- Fears: soft benefits smuggled into the headline
- Give them: hard, soft and avoidance, clearly separated

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CAPSTONE - PRESENTING

The two-minute rule

Decision, benefit and ask in the first two minutes. Everything after that is support.

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CAPSTONE - PRESENTING

Two openings for the same project

The wrong opening

Chronological

- Good morning, I will start with some background
- Slide 4: our SIPOC diagram
- Slide 9: the Gage R&R results
- Slide 22: and so the benefit is...
- By slide 9 half the room is on email

The right opening

Decision first

- I need approval to roll this to Singapore and Suzhou
- SGD 328k signed hard benefit, verified over 3 months
- The ask is 2 engineer-months and a 4-week line window
- One risk: Suzhou uses a different pump. Mitigation follows
- The evidence is next. Full analysis in the appendix

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The 8 to 10 slide main deck

1. **1. The decision and the ask**
   One slide. The whole point of the meeting
2. **2. Problem and business impact**
   What it costs, tied to a business objective
3. **3. Baseline with data**
   Capability or defect rate, before any change
4. **4. The root cause established**
   One chart that proves the causal X
5. **5. What we changed**
   The countermeasures, in plain language
6. **6. Results and the benefit**
   Before versus after, plus the signed number
7. **7. How it is sustained**
   Owner, control chart, reaction plan, audit dates
8. **8. Risks and next steps**
   Named risks, mitigations, and the replication plan

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Projector legibility - the rules that get broken

1. **Minimum 18 point**
   Anything smaller is unreadable from the back of a boardroom

2. **Never paste a spreadsheet**
   Screenshot of a 12-column table equals zero communicated

3. **Label the axes in words**
   Fill volume (mL), not Var1. The room does not know your variables

4. **Annotate the conclusion**
   Put the takeaway ON the chart, not only in your narration

5. **Test on the actual projector**
   Pale greys and thin lines vanish on a washed-out screen

6. **One idea per slide**
   If it needs two headlines, it is two slides

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CAPSTONE - PRESENTING

Stating statistics in business language

What you would write

Correct but unusable

- p = 0.003, so we reject the null hypothesis
- Adjusted R-squared = 0.78, VIF all below 3
- Cpk improved from 0.81 to 1.41
- The DOE showed a significant AB interaction

What you should say

Same evidence, business language

- The difference is real - under 1 chance in 300 it is luck
- These two factors explain about 78 percent of the variation
- Defect risk fell from roughly 1 in 100 to under 1 in 10,000
- Torque only matters at the longer dwell time - they interact

The rule

Numbers in the appendix

- Say the business meaning out loud
- Have the statistic on the slide in small type
- Have the full output in the appendix, indexed
- Never make the room do the translation

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CAPSTONE - APPENDIX

The appendix is where you win the challenge

The main deck makes the case. The appendix proves it - on demand, in seconds.

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What belongs in the appendix

1. **Full statistical output**
   Regression tables, ANOVA tables, test statistics, confidence intervals

2. **The complete MSA study**
   Gage R&R and Kappa results with the accept/reject decision

3. **The full DOE**
   Design matrix, alias structure, all effects, the confirmation run

4. **Assumption checks**
   Normality, equal variance, independence, residual plots

5. **The benefit calculation**
   Baseline, rates, the Finance method, and the sign-off

6. **Alternatives rejected**
   The countermeasures considered and why they were not chosen

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CAPSTONE - DEFENCE

Expect to be challenged

Answer with evidence, never with assertion. Assertion is what a weak project sounds like.

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CAPSTONE - DEFENCE

Challenge 1 - sample adequacy and power

The challenge

Was 30 really enough?

- How did you choose that sample size?
- Could you have missed a real effect?
- Is that sample representative of all shifts?

The weak answer

Assertion

- It is a standard sample size
- We collected what we could get
- The p-value was significant, so it was enough

The strong answer

Evidence

- Sized for a 0.05 mL effect at 80 percent power, alpha 0.05 - n = 28, we took 32
- Stratified across all three shifts and both operators
- Appendix A4 shows the power calculation

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Challenge 2 - measurement system validity

The challenge

How do you know the data is real?

- Was the gauge capable of seeing this difference?
- Do your inspectors even agree with each other?
- Could the improvement be a measurement artifact?

The weak answer

Assertion

- The gauge is calibrated
- We have always measured it this way
- Calibration is not the same as capability

The strong answer

Evidence

- Gage R&R at 11.4 percent study variation - acceptable, appendix A2
- Attribute Kappa 0.83 across four inspectors, above the 0.75 threshold
- MSA was run and passed BEFORE any analysis data was collected

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Challenge 4 - confounding and attribution

The challenge

How do you know it was YOUR change?

- Volume changed in the same period
- A new supplier was qualified at the same time
- Could this just be seasonality?

The weak answer

Assertion

- The metric improved after we implemented
- Nothing else changed as far as we know
- This is correlation offered as cause

The strong answer

Evidence

- The DOE established the causal mechanism directly, not just an association
- Singapore line 1 held unimproved as a control - it did not move
- Volume normalised in the before-after test; effect persists

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Challenge 5 - DOE resolution and aliasing

The challenge

What did your fraction give away?

- You ran 16 runs for 7 factors - what is confounded?
- Could that main effect actually be an interaction?
- Did you confirm the optimum?

The weak answer

Assertion

- The software chose the design
- We only care about main effects anyway
- This is exactly the answer that loses the room

The strong answer

Evidence

- Resolution IV: main effects clear of 2-factor interactions, but 2FIs alias in pairs
- AB is aliased with CE - we ran 4 fold-over runs to separate them, appendix A7
- Confirmation run at the optimum matched prediction within 0.01 mL

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Challenge 6 - sustainability and benefit type

The challenge

Will it hold, and is the money real?

- What stops this drifting back in six months?
- How much of that benefit is soft?
- Has Finance actually signed it?

The weak answer

Assertion

- We have a control plan
- The team is committed to sustaining it
- Roughly SGD 500k in total benefit

The strong answer

Evidence

- EWMA chart live in production, named owner, reaction plan rehearsed
- 30/60/90 audits already in the owner's calendar with dates
- SGD 328k HARD, signed by the controller. Soft and avoidance shown separately

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Acknowledging a genuine weakness

Why defensiveness fails

It costs you the room

- Every project has a weakness. They will find it
- Denying it makes them doubt everything else
- It turns a technical point into a credibility problem

The strong pattern

Name, bound, mitigate

- NAME it before they do
- BOUND it - say how much it could change the conclusion
- MITIGATE it - say what you did or will do about it

Worked example

Meridian, honestly stated

- Weakness: only 3 months of post-data, so annualisation is an estimate
- Bound: even at 70 percent hold rate the benefit exceeds SGD 230k
- Mitigation: benefit re-verified at the 6-month audit before full annualisation

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CAPSTONE - LESSONS

Two kinds of lesson - keep them separate

PROJECT-SPECIFIC

Useful to this process only

- The reseal torque spec was wrong in the drawing
- Line 2 operators were trained on the old dwell time
- Value: fix it, log it, replicate to sister lines
- It does not generalise beyond this equipment

TRANSFERABLE

Useful to every future project

- We skipped MSA and lost three weeks to bad data
- The sponsor changed at month 2 and scope drifted immediately
- Screening DOE first would have saved 40 runs
- Value: this changes how you and your Green Belts work

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The coaching question set

1

What does your DATA show?

Not what you believe - what the data shows. Redirects opinion to evidence

2

What have you RULED OUT?

Forces them to show the alternatives they tested and eliminated

3

What ASSUMPTION is that resting on?

Surfaces the normality, independence or causality they assumed

4

How would you know if you were WRONG?

The single most valuable question. Makes the claim falsifiable

5

What would the sponsor ask?

Trains them to anticipate challenge before the tollgate

6

Ask, do not tell

The answer you hand over is forgotten. The one they find, they keep

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Where Green Belts most need intervention

1

Scope creep

The project quietly grows until it cannot finish. The commonest failure by far

2

Wrong test selection

A t-test on non-normal data, or repeated t-tests instead of ANOVA

3

Correlation read as causation

A scatter plot presented as proof of cause. Intervene immediately

4

Skipping MSA

Analysing data from an unvalidated gauge. Everything after it is worthless

5

Solution jumping

The countermeasure was chosen on day 2, and Analyze is being back-filled

6

No baseline

No before number, so no improvement can ever be demonstrated

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When to intervene, and when to let them learn

INTERVENE now

The error is unrecoverable

- MSA skipped - all downstream data is worthless
- Scope has grown past what can be finished
- A causal claim about to be made to the sponsor
- Anything that wastes weeks or damages credibility

LET THEM LEARN

The error is cheap and instructive

- A fishbone that turns out too broad
- A first-draft charter that the sponsor will refine
- A hypothesis that the data does not support
- Being wrong here is the lesson, not the failure

The judgement

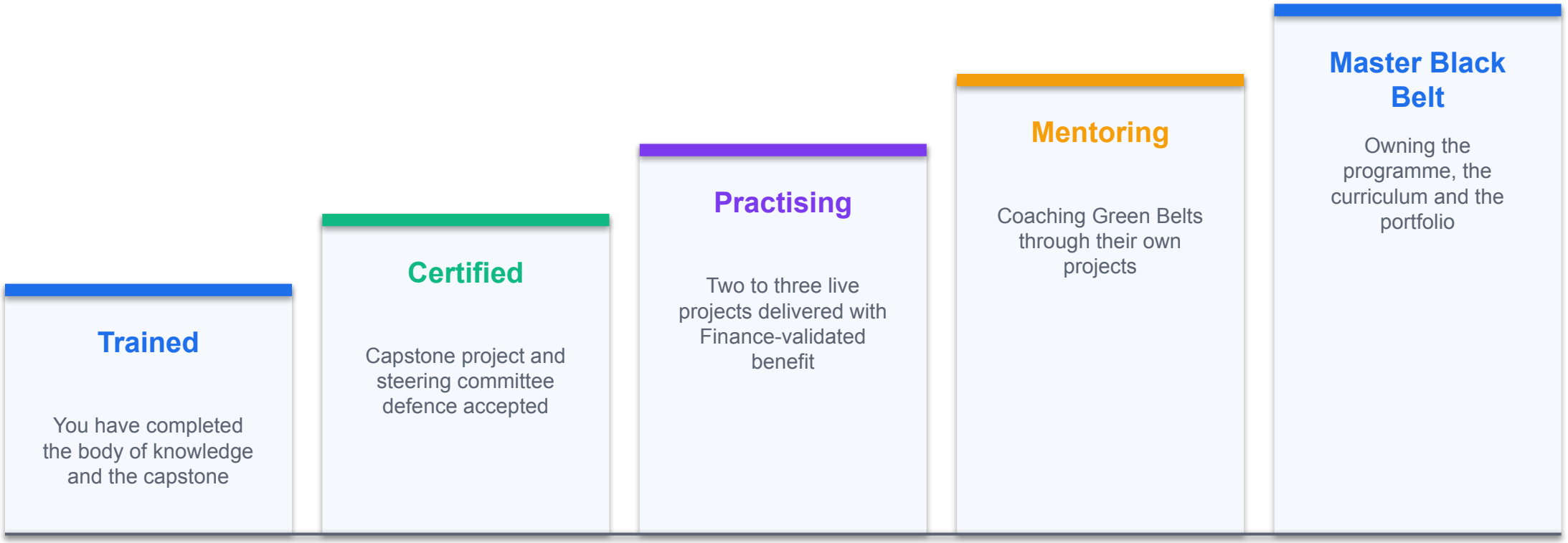
Cost of the mistake

- Expensive or irreversible -> intervene, and explain why
- Cheap and recoverable -> let it run, then debrief
- Always debrief. An unexamined mistake teaches nothing

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CAPSTONE - PATHWAY

Your Black Belt pathway from here



Certification is the start of practice, not the end of learning.

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Your first 90 days back at work

1. Days 1-15: present the capstone to your real sponsor and confirm the mandate
2. Days 16-30: agree the benefit method with Finance BEFORE anything else
3. Days 31-60: run the next tollgate; recruit and scope one Green Belt project
4. Days 61-90: audit the control plan at 30/60/90 and verify the benefit is flowing
5. Ongoing: log every project, mentor by asking, and keep the portfolio honest

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CAPSTONE - DAY 5

Your capstone is your credential

Not a certificate on a wall - a project you can defend in any room, to anyone.

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WHAT YOU'LL DO

Hands-On Labs — Capstone

1

Lab 31 — Consolidating Your Capstone — The A3 Storyboard

A complete A3 storyboard covering your full DMAIC project on a single page. • Data: lab-31-a3-storyboard.xlsx

2

Lab 32 — Building the Steering Committee Presentation

A 15-minute steering committee presentation with an appendix of supporting analysis. • Data: lab-32-steering-committee-deck.xlsx

3

Lab 33 — Capstone Presentation and Defence

A delivered 15-minute presentation with a defended Q&A session. • Data: lab-33-capstone-defence-rubric.xlsx

4

Lab 34 — Lessons Learned, Mentoring Green Belts and Your Black Belt Journey

A lessons-learned record, a Green Belt mentoring plan and a personal development… • Data: lab-34-lessons-learned-mentoring.xlsx

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DMAIC · CAPSTONE · HANDS-ON

Consolidating Your Capstone — The A3 Storyboard

LAB 31

You have been building this project since Lab 1. Now assemble thirty labs' worth of artifacts into ONE page that tells the whole story. The A3 discipline is deliberately constraining — if it does not fit on one page, you do not yet understand which parts matter.

You'll build

A complete A3 storyboard covering your full DMAIC project on a single page.

Tools & techniques: A3 thinking, storyboard structure, artifact selection, visual summary, narrative flow

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Consolidating Your Capstone — The A3 Storyboard

STEPS 1-6 OF 12

1. Retrieve every artifact you have produced since Lab 1 and lay them out in DMAIC order. You should have thirty or more pieces.
2. Learn the A3 structure: Background, Current State, Goal, Root Cause Analysis, Countermeasures, Results, and Follow-up Actions.
3. Write the Background in three sentences: the business context and why this project was selected from the portfolio.
4. Present the Current State using your baseline — sigma level, capability, run chart and the quantified COPQ from Lab 1.
5. State the Goal exactly as chartered, with the SMART target and the deadline.
6. Summarise Root Cause Analysis with your strongest evidence only — the Pareto or stratification, the key hypothesis test result and the regression or ANOVA finding.

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Consolidating Your Capstone — The A3 Storyboard

STEPS 7-12 OF 12

7	Present Countermeasures: the selected solutions, the DOE optimum settings and the FMEA-driven risk mitigations.
8	Present Results: before/after comparison, the statistical significance test, the control chart showing sustained performance, and the Finance-validated benefit.
9	List Follow-up Actions: the control plan owner, audit schedule, open risks and the next project in the portfolio.
10	Now edit ruthlessly to one page. Every element must earn its place — cut anything that does not change the reader's understanding or decision.
11	Apply the visual test: a sponsor should grasp the whole story in 60 seconds from the charts alone, without reading the text.
12	Have a peer read your A3 cold and tell you what the project did. If they get it wrong, the A3 is not yet finished.

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LAB 31 · VERIFY

Consolidating Your Capstone — The A3 Storyboard

✅ Check your work

Your A3 fits on one page, covers all seven sections, carries your real project data throughout, and a peer reading it cold can correctly summarise the project.

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DMAIC · CAPSTONE · HANDS-ON

Building the Steering Committee Presentation

LAB 32

Sponsors are not statisticians. They want the decision, the money and the risk in the first two minutes, with the statistics available underneath as supporting evidence. This lab builds the deck you will deliver and defend in Lab 33.

You'll build

A 15-minute steering committee presentation with an appendix of supporting analysis.

Tools & techniques: Executive communication, data storytelling, chart selection, structuring for decisions, appendix strategy

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Building the Steering Committee Presentation

STEPS 1-6 OF 12

1. Understand your audience: the sponsor wants the business outcome; the process owner wants the operational change; Finance wants the benefit validated.
2. Open with the headline: what the project achieved, what it is worth annually, and what decision you need from the committee.
3. Apply the two-minute rule — if the committee stopped you after two minutes, they should already have everything needed to make a decision.
4. Structure the main body against DMAIC, but at business altitude: the problem, what the data showed, what you changed, what resulted.
5. Select your charts carefully. One before/after comparison, one root cause chart, one control chart showing sustained gain. Resist showing everything you built.
6. Design each chart for a projector at the back of a room: large fonts, direct labels, no unexplained jargon and no default spreadsheet styling.

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Building the Steering Committee Presentation

STEPS 7-12 OF 12

7	State the statistics in business language. 'The difference is statistically significant (p = 0.003)' beats a table of test output.
8	Present the benefit precisely: hard, soft and cost avoidance separated, with the Finance sign-off named explicitly.
9	Present the risks honestly, along with their mitigations. A presentation with no risks reads as naive and invites harder challenge.
10	Close with the specific ask: approve the rollout, release resources, or approve the next project in the portfolio.
11	Build the APPENDIX with your full analysis — regression output, ANOVA tables, DOE design and alias structure, MSA study and control plan. This is where challenges get answered.
12	Rehearse to time. Fifteen minutes means fifteen minutes; overrunning signals you cannot prioritise.

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LAB 32 · VERIFY

Building the Steering Committee Presentation

✅ Check your work

Your presentation delivers the decision, benefit and ask within the first two minutes, charts are projector-legible, statistics are stated in business language, and every likely challenge has an appendix slide behind it.

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DMAIC · CAPSTONE · HANDS-ON

Capstone Presentation and Defence

LAB 33

The capstone assessment. Present your project to a simulated steering committee of peers and the trainer, then defend the analysis under challenge. Defending your methodology under pressure is the skill that separates a Black Belt from a Green Belt.

You'll build

A delivered 15-minute presentation with a defended Q&A session.

Tools & techniques: Presentation delivery, handling challenge, methodological defence, peer assessment, professional judgement

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Capstone Presentation and Defence

STEPS 1-6 OF 12

1. Deliver your 15-minute presentation to the steering committee panel.
2. Hold the two-minute rule under pressure — lead with outcome, benefit and ask.
3. Take questions. Expect challenge on sample adequacy: was your sample size justified, and what power did your tests have?
4. Expect challenge on the measurement system: how do you know the data is trustworthy? Have your Gage R&R or Kappa result ready.
5. Expect challenge on assumptions: did you test normality, equal variance and independence, and what did you do when they failed?
6. Expect challenge on confounding: what else changed during the period, and how do you know your intervention caused the improvement?

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Capstone Presentation and Defence

STEPS 7-12 OF 12

7	Expect challenge on DOE: what was your design resolution, what effects were aliased, and did confirmation runs validate the optimum?
8	Expect challenge on sustainability: what stops this from decaying in six months once attention moves elsewhere?
9	Expect challenge on the benefit: is it hard or soft, who in Finance validated it, and over what period?
10	Answer with evidence, not assertion. Go to your appendix. Where you do not know, say so and state how you would find out — this is far stronger than bluffing.
11	Where a challenge reveals a genuine weakness, acknowledge it and state the mitigation. Defensiveness costs more credibility than the weakness itself.
12	Observe the other presentations and record challenges you had not anticipated for your own project.

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LAB 33 · VERIFY

Capstone Presentation and Defence

✅ Check your work

You delivered within time, answered challenges with evidence from your appendix rather than assertion, and acknowledged genuine weaknesses with a stated mitigation.

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DMAIC · CAPSTONE · HANDS-ON

Lessons Learned, Mentoring Green Belts and Your Black Belt Journey

LAB 34

Close the course as you would close a project. Harvest what transfers, then turn to the accountability that most defines the role: mentoring Green Belts. A Black Belt is measured not only by their own projects but by the belts they develop.

You'll build

A lessons-learned record, a Green Belt mentoring plan and a personal development roadmap.

Tools & techniques: Lessons learned, knowledge transfer, mentoring model, coaching questions, certification pathway, continuing practice

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Lessons Learned, Mentoring Green Belts and Your Black Belt Journey

STEPS 1-6 OF 12

1. Run your own lessons learned: what worked in your capstone, what did not, and what you would do differently next time.
2. Separate lessons that are project-specific from those that transfer to any project. Only the transferable ones belong in the organisational knowledge base.
3. Record the two analytical techniques from this course you found most difficult, and plan how you will build fluency in them.
4. Now the mentoring role. Understand the core principle: a Black Belt coaches Green Belts through their analysis rather than performing it for them.
5. Learn the coaching question set: what does your data show, what have you ruled out, what assumption is that resting on, and how would you know if you were wrong?
6. Identify where Green Belts most commonly need intervention — scope creep, wrong test selection, confusing correlation with causation, and skipping MSA entirely.

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Lessons Learned, Mentoring Green Belts and Your Black Belt Journey

STEPS 7-12 OF 12

7	Plan your tollgate review approach: what evidence you require at each gate before allowing a Green Belt project to proceed.
8	Decide when to intervene directly versus letting a Green Belt learn from a recoverable mistake. Both have costs.
9	Draft a mentoring plan for one Green Belt in your organisation: their project, their development gaps, your review cadence.
10	Map your certification pathway: project logging requirements, the certification body's criteria and any remaining evidence you need.
11	Identify your next portfolio project from the Lab 2 map, and set a start date.
12	Write your 90-day plan: complete or hand over the capstone, begin mentoring, and select the next project.

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LAB 34 · VERIFY

Lessons Learned, Mentoring Green Belts and Your Black Belt Journey

✅ Check your work

Your lessons are separated into project-specific and transferable, you have a named Green Belt mentoring plan with a review cadence, and your 90-day plan has dated commitments.

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PHASE RECAP

Recap — Capstone

- The A3 storyboard — One page tells the whole project: background, current state, analysis, countermeasures, results, next steps.
- Consolidating your artifacts — Every lab from Day 1 produced one piece of your capstone — now assemble them into one narrative.
- Telling the data story — Lead with the business impact, support it with the statistics — never the other way round.
- Presenting to a steering committee — Sponsors want the decision, the risk and the money in the first two minutes.
- Defending the analysis — Expect challenge on sample size, assumptions and confounding — prepare the evidence for each.
- Handover and sustainability — Show who owns the process now, how it is monitored, and what triggers escalation.

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WRAP-UP

Course Summary & Next Steps

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YOUR CAPSTONE PACKAGE

What You Achieved — The Full DMAIC Journey



Define

- Chartered your capstone
- Built the CTQ flowdown
- Mapped stakeholders
- Planned the change

Measure

- Proved the gage (R&R)
- Ran attribute MSA / Kappa
- Handled non-normal data
- Baselined capability

Analyze

- Fitted multiple regression
- Ran ANOVA & interactions
- Decomposed multi-vari
- Proved root cause

Improve

- Designed full factorials
- Screened with fractionals
- Optimised with RSM
- De-risked with FMEA

Control

- Built CUSUM & EWMA
- Applied T-squared
- Validated the benefit
- Handed over

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WHAT YOU BUILT

Your Integrated Capstone Package

1. **Portfolio & charter**
   Weighted selection matrix, business case and governed charter.
2. **Customer requirements**
   VOC log, Kano classification and CTQ flowdown to controllable Xs.
3. **Process baseline**
   SIPOC, swimlane map, VSM, takt time and quantified waste.
4. **Valid measurement**
   Gage R&R, attribute Kappa study and a nested gauge design.
5. **Performance baseline**
   RTY, DPMO, sigma level, Cpk and Ppk with normality handled.
6. **Proven root causes**
   Regression model, ANOVA, multi-vari and chi-square evidence.
7. **Optimised improvements**
   DOE design, alias structure, RSM optimum and FMEA re-scoring.
8. **Sustained control**
   CUSUM/EWMA charts, control plan and Finance-validated benefit.

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BEFORE YOUR PRESENTATION

Capstone Readiness Checklist

1. Can you defend your project selection against the portfolio criteria?
2. Can you prove your measurement system is valid for your data type?
3. Can you state the power of your tests and justify your sample size?
4. Can you interpret your regression coefficients in business language?
5. Can you explain your DOE resolution and what effects were aliased?
6. Can you show sustained performance on a control chart after handover?

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BEFORE YOUR PRESENTATION

Capstone Readiness Checklist (2)

1. Can you separate hard, soft and cost-avoidance benefits?
2. Can you name who in Finance validated your benefit, and over what period?
3. Can you answer a confounding challenge with evidence rather than assertion?
4. Can you state what stops this improvement decaying in six months?
5. Can you coach a Green Belt through the analysis you just performed?

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NEXT STEPS

Continuing Your Black Belt Journey

1

Complete your capstone

Take the project you started here through to real closure.

2

Mentor Green Belts

Coach their analysis rather than performing it for them.

3

Master Black Belt

The next step — leads programmes and develops Black Belts.

4

Log your projects

Certification requires documented, benefit-validated projects.

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ASSESSMENT

Final Assessment

Sit the Written Assessment (SAQ, 1 hour) then the Case Study (90 minutes) — both open book. Your capstone project and presentation are delivered in addition.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ, 1 hour), then the Case Study (90 minutes)
4. Submit your capstone report and present it to the steering committee
5. Sign the Assessment Summary Record

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Digital Attendance (Assessment)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR code with your phone camera
3. Key in your NRIC/FIN and submit
4. Attendance must be recorded before you begin the papers

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YOUR FEEDBACK

TRAQOM Survey

1. Open the TRAQOM survey link on the LMS
2. Key in the last four characters of your NRIC/FIN
3. Key in the six-digit course run ID
4. Complete and submit — your feedback shapes this course

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AFTER THE COURSE

Certificate & Support

- Two e-certificates are awarded on demonstrating competency and achieving at least 75% attendance.
- A SkillsFuture Statement of Attainment (SOA) is issued for the WSQ assessment.
- Email: enquiry@tertiaryinfotech.com
- Tel / WhatsApp: +65 6100 0613

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END OF COURSE

Thank You!

Now go and finish the project you started here, then mentor the next Green Belt through theirs — that is what a Black Belt is for.

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